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THE COLLECTED

MATHEMATICAL PAPERS

OF

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ADVERTISEMENT.

THE present volume contains 64 papers numbered 159 to 222 originally published in the years 1857 to 1862: the chronological order is slightly departed from for the sake of including a series of papers in the Memoirs and the Monthly Notices of the Royal Astronomical Society; there are thus some earlier papers which have been left over for the next volume.

As papers are referred to by their Numbers only it will be convenient to give in each volume a Table such as the following:

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159.

ON SOME INTEGRAL TRANSFORMATIONS.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 4–6.]

SUPPOSE that x, a, b, c and x', a', b', c' have the same anharmonic ratios, or what is the same thing, let these quantities satisfy the equation

$$\begin{vmatrix} 1, & 1, & 1, & 1 \\ x, & a, & b, & c \\ x', & a', & b', & c' \\ xx', & aa', & bb', & cc' \end{vmatrix} = 0;$$

this equation may be represented under a variety of different forms, which are obtained without difficulty; thus, if for shortness

$$K = a(b' - c')(x' - a') + b(c' - a')(x' - b') + c(a' - b')(x' - c'),$$

then

$$\begin{aligned} Kx &= -\{bc(b-c)(x'-a') + ca(c-a)(x'-b') + ab(a-b)(x'-c')\}, \\ K(x-a) &= (c-a)(a-b)(b'-c')(x'-a'), \\ K(x-b) &= (a-b)(b-c)(c'-a')(x'-b'), \\ K(x-c) &= (b-c)(c-a)(a'-b')(x'-c'). \end{aligned}$$

Consider x, x' as variables; then

$$K^2\, dx = (b-c)(c-a)(a-b)(b'-c')(c'-a')(a'-b')\, dx';$$

let d, d' be any corresponding values of x, x'; then

$$\begin{vmatrix} 1, & 1, & 1, & 1 \\ a, & b, & c, & d \\ a', & b', & c', & d' \\ aa', & bb', & cc', & dd' \end{vmatrix} = 0$$

and we have

$$K(x-d) = D(x'-d');$$

where

$$D = (c'-a')(a'-b')(b'-c')\,\Lambda,$$

and

$$\Lambda = \frac{(a-d)(b-c)}{(a'-d')(b'-c')} = \frac{(b-d)(c-a)}{(b'-d')(c'-a')} = \frac{(c-d)(a-b)}{(c'-d')(a'-b')}.$$

Suppose $\alpha+\beta+\gamma+\delta=-2$; then

$$(x-a)^\alpha(x-b)^\beta(x-c)^\gamma(x-d)^\delta\,dx = J\,(x'-a')^\alpha(x'-b')^\beta(x'-c')^\gamma(x'-d')^\delta\,dx',$$

where

$$J = (b-c)^{\beta+\gamma+1}(c-a)^{\gamma+\alpha+1}(a-b)^{\alpha+\beta+1}(b'-c')^{\beta+\gamma+1}(c'-a')^{\gamma+\alpha+1}(a'-b')^{\alpha+\beta+1}\,D^\delta.$$

We may in particular take for a', b', c', d' the systems b, a, d, c; c, d, a, b and d, c, b, a respectively; this gives, writing successively y, z, w instead of x,

$$\begin{aligned}(x-a)^\alpha(x-b)^\beta(x-c)^\gamma(x-d)^\delta\,dx &= M\,(y-a)^\beta(y-b)^\alpha(y-c)^\delta(y-d)^\gamma\,dy\\ &= N\,(z-a)^\gamma(z-b)^\delta(z-c)^\alpha(z-d)^\beta\,dz\\ &= P\,(w-a)^\delta(w-b)^\gamma(w-c)^\beta(w-d)^\alpha\,dw,\end{aligned}$$

where

$$\begin{aligned}M &= (a-b)^{\alpha+\beta+1}(c-d)^{\gamma+\delta+1},\\ N &= (-)^{\alpha+1}(a-c)^{\alpha+\gamma+1}(b-d)^{\beta+\delta+1},\\ P &= (a-b)^{\alpha+\beta+1}(a-c)^{\alpha+\gamma+1}(b-d)^{\beta+\delta+1}(c-d)^{\gamma+\delta+1};\end{aligned}$$

the relations between the variables x, y, z, w being

$$\begin{aligned}x &= \frac{(c+d)\,ab-(a+b)\,cd-(ab-cd)\,y}{ab-cd-(a+b-c-d)\,y}\\ &= \frac{(b+d)\,ac-(a+c)\,bd-(ac-bd)\,z}{ac-bd-(a+c-b-d)\,z}\\ &= \frac{(a+d)\,bc-(b+c)\,ad-(ad-bc)\,w}{ad-bc-(a+d-b-c)\,w}.\end{aligned}$$

These are, in fact, the formulæ in my note, "On an Integral Transformation," *Camb. and Dubl. Math. Jour.* t. III. (1848), p. 286 [62], which was suggested to me by Gudermann's transformation for elliptic functions, (*Crelle*, t. XXIII. (1846), p. 330).

Suppose now that the values of a', b', c', d' are 0, 1, ∞, ζ, we have in this case

$$x = \frac{a\,(b-c)+c\,(a-b)\,y}{(b-c)+(a-b)\,y},$$

and representing the denominator by K, then

$$\begin{aligned} K(x-a) &= -(a-b)(a-c)\,y, \\ K(x-b) &= (a-b)(b-c)(1-y), \\ K(x-c) &= (a-c)(b-c), \\ K(x-d) &= (a-b)(c-d)\,(y-\zeta), \end{aligned}$$

where

$$\zeta = \frac{(a-d)(b-c)}{(a-b)(c-d)},$$

and we have

$$K^2\,dx = (a-b)(a-c)(b-c)\,dy,$$

whence

$$\begin{aligned} &(x-a)^\alpha(x-b)^\beta(x-c)^\gamma(x-d)^\delta\,dx \\ &\qquad = -(-)^\alpha(a-b)^{\alpha+\beta+\delta+1}(a-c)^{\alpha+\gamma+1}(b-c)^{\beta+\gamma+1}(c-d)^\delta\,y^\alpha(1-y)^\beta(y-\zeta)^\delta\,dy. \end{aligned}$$

It is easy, by means of this equation, to generalise a remarkable formula given by M. Serret in his memoir, "Sur la Représentation géométrique des Fonctions elliptiques et ultra-elliptiques," *Liouville*, t. XI. and XII. [1846 and 1847], and *Recueil des Savans étrangers*, t. XI. [1851][1]. In fact, suppose that the indices α, β, γ, δ are integers, and that two of these indices, e.g. γ, δ, are negative, the remaining two indices being positive, then writing $-\gamma$, $-\delta$ instead of γ, δ the integral

$$\int \frac{(x-a)^\alpha(x-b)^\beta\,dx}{(x-c)^\gamma(x-d)^\delta},$$

where $\gamma+\delta = \alpha+\beta+2$, depends on the integral

$$\int \frac{y^\alpha(1-y)^\beta\,dy}{(y-\zeta)^\delta}.$$

Suppose that the fraction under the integral sign is resolved into simple fractions, each of these fractions will be integrable algebraically, except the fraction having for its denominator the simple power $y-\zeta$, the integral of which is a logarithm. The coefficient of this fraction is at once found by writing in the numerator $\zeta+(y-\zeta)$ for y; and expanding in ascending powers of $y-\zeta$ and equating this coefficient to zero, we have

$$\left(\frac{d}{d\zeta}\right)^{\delta-1}\zeta^\alpha(1-\zeta)^\beta = 0,$$

which [observing that $(\gamma-1)+(\delta-1) = \alpha+\beta$] is easily seen to be equivalent to

$$\left(\frac{d}{d\zeta}\right)^{\gamma-1}\zeta^\beta(1-\zeta)^\alpha = 0.$$

1–2

[1] M. Serret has reproduced the theorem in his very interesting and instructive treatise, "Cours d'Algèbre supérieure," deuxième édition, Paris, 1854, [quatrième édition, Paris, 1877].

Hence if the function $\zeta = \dfrac{(a-b)(b-c)}{(a-b)(c-d)}$ satisfy this condition, the indefinite integral

$$\int \frac{(x-a)^{\alpha}(x-b)^{\beta}\, dx}{(x-c)^{\gamma}(x-d)^{\delta}},$$

where $\gamma + \delta = \alpha + \beta + 2$, will be expressible as a rational algebraical fraction.

It may be noticed, that in the general case, observing that $x = a$, $x = b$ give $y = 0$, $y = 1$, the integral

$$\int_a^b (x-a)^{\alpha}(x-b)^{\beta}(x-c)^{\gamma}(x-d)^{\delta}\, dx$$

depends on

$$\int_0^1 y^{\alpha}(1-y)^{\beta}(\zeta - y)^{\delta}\, dy,$$

or, putting $\zeta = \dfrac{1}{u}$, upon

$$\int_0^1 y^{\alpha}(1-y)^{\beta}(1-uy)^{\delta}\, dy,$$

which is expressible by means of a hypergeometric series having u for its argument or fourth element.

2, Stone Buildings, Lincoln's Inn, Feb. 1854.

160.

On a Theorem Relating to Reciprocal Triangles.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 7–10.]

The following theorem is, I assume, known; but the analytical demonstration of it depends upon a formula in determinants which is not without interest. The theorem referred to may be thus stated:

"A triangle and its reciprocal are in perspective," where by the reciprocal of a triangle is meant the triangle the sides of which are the polars of the angles of the first-mentioned triangle with respect to a conic; and triangles are in perspective when the three lines forming the corresponding angles meet in a point, or what is the same thing, when the three points of intersection of the corresponding sides lie in a line.

Let the equation of the conic be

$$x^2 + y^2 + z^2 = 0,$$

and take (α, β, γ), $(\alpha', \beta', \gamma')$, $(\alpha'', \beta'', \gamma'')$ for the coordinates of the angles of the triangle, then if K be the determinant, and (A, B, C) (A', B', C') (A'', B'', C'') the inverse system, i.e. if

$$\begin{aligned} KA &= (\beta'\gamma'' - \beta''\gamma'), & KB &= \gamma'\alpha'' - \gamma''\alpha', & KC &= \alpha'\beta'' - \alpha''\beta', \\ KA' &= (\beta''\gamma - \beta\gamma''), & KB' &= \gamma''\alpha - \gamma\alpha'', & KC' &= \alpha''\beta - \alpha\beta'', \\ KA'' &= (\beta\gamma' - \beta'\gamma), & KB'' &= \gamma\alpha' - \gamma'\alpha, & KC'' &= \alpha\beta' - \alpha'\beta, \end{aligned}$$

equations which may be represented in the notation of matrices by the single equation

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix}^{-1} = \begin{vmatrix} A, & A', & A'' \\ B, & B', & B'' \\ C, & C', & C'' \end{vmatrix},$$

then the equations of the sides of the triangle are

$$\begin{aligned} A\,x + B\,y + C\,z &= 0, \\ A'\,x + B'\,y + C'\,z &= 0, \\ A''\,x + B''\,y + C''\,z &= 0, \end{aligned}$$

and the coordinates of the angles of the reciprocal triangle may be taken to be (A, B, C) (A', B', C') (A'', B'', C''); the equations of the lines joining the corresponding angles of the two triangles are therefore

$$\begin{aligned} (B\gamma - C\beta)\,x + (C\alpha - A\gamma)\,y + (A\beta - B\alpha)\,z &= 0, \\ (B'\gamma' - C'\beta')\,x + (C'\alpha' - A'\gamma')\,y + (A'\beta' - B'\alpha')\,z &= 0, \\ (B''\gamma'' - C''\beta'')\,x + (C''\alpha'' - A''\gamma'')\,y + (A''\beta'' - B''\alpha'')\,z &= 0; \end{aligned}$$

the condition that these lines may meet in a point is therefore

$$\begin{vmatrix} B\gamma - C\beta, & C\alpha - A\gamma, & A\beta - B\alpha \\ B'\gamma' - C'\beta', & C'\alpha' - A'\gamma', & A'\beta' - B'\alpha' \\ B''\gamma'' - C''\beta'', & C''\alpha'' - A''\gamma'', & A''\beta'' - B''\alpha'' \end{vmatrix} = 0,$$

an equation which is satisfied identically when A, B, C; A', B', C'; A'', B'', C'' are replaced by their values. To prove this I transform the different quantities which enter into the determinant as follows: putting

$$\begin{aligned} F &= \alpha'\alpha'' + \beta'\beta'' + \gamma'\gamma'', \\ G &= \alpha''\alpha + \beta''\beta + \gamma''\gamma, \\ H &= \alpha\alpha' + \beta\beta' + \gamma\gamma'; \end{aligned}$$

we have

$$\begin{aligned} K(B\gamma - C\beta) &= \gamma\,(\gamma''\alpha' - \gamma'\alpha'') - \beta\,(\alpha'\beta'' - \alpha''\beta') \\ &= \alpha''(\beta\beta' + \gamma\gamma') - \alpha'(\beta\beta'' + \gamma\gamma'') \\ &= \alpha''(\alpha\alpha' + \beta\beta' + \gamma\gamma') - \alpha'(\alpha\alpha'' + \beta\beta'' + \gamma\gamma'') \\ &= \alpha''H - \alpha'G, \\ &\text{\&c.;} \end{aligned}$$

and the equation becomes

$$\begin{vmatrix} \alpha''H - \alpha'G, & \beta''H - \beta'G, & \gamma''H - \gamma'G \\ \alpha F - \alpha''H, & \beta F - \beta''H, & \gamma F - \gamma''H \\ \alpha'G - \alpha F, & \beta'G - \beta F, & \gamma'G - \gamma F \end{vmatrix} = 0.$$

Now the minor $(\beta F - \beta''H)(\gamma'G - \gamma F) - (\gamma F - \gamma''H)(\beta'G - \beta F)$ is equal to

$$GH(\beta'\gamma'' - \beta''\gamma') + HF(\beta''\gamma - \beta\gamma'') + FG(\beta\gamma' - \beta'\gamma),$$

i.e. to

$$K\,(GHA + HFA' + FGA'');$$

and expressing the other minors in a similar form, the equation to be proved is

$$\begin{aligned} &(GHA + HFA' + FGA'')(B\gamma - C\beta) \\ +&(GHB + HFB' + FGB'')(C\alpha - A\gamma) = 0, \\ +&(GHC + HFC' + FGC'')(A\beta - B\alpha) \end{aligned}$$

i.e.

$$HF \begin{vmatrix} A', & B', & C' \\ A, & B, & C \\ \alpha, & \beta, & \gamma \end{vmatrix} + FG \begin{vmatrix} A'', & B'', & C'' \\ A, & B, & C \\ \alpha, & \beta, & \gamma \end{vmatrix} = 0.$$

The first determinant is

$$-\{\alpha(B'C - BC') + \beta(C'A - CA') + \gamma(A'B - AB')\} = -\frac{1}{K}(\alpha\alpha'' + \beta\beta'' + \gamma\gamma'') = -\frac{1}{K}G,$$

and the second determinant is

$$\{\alpha(B''C - BC'') + \beta(C''A - CA'') + \gamma(A''B - AB'')\} = \frac{1}{K}(\alpha\alpha' + \beta\beta' + \gamma\gamma') = \frac{1}{K}H,$$

and we have therefore identically

$$HF\,(-G) + FG\,(H) = 0.$$

The corresponding theorem in geometry of three dimensions is that a tetrahedron and its reciprocal have to each other a certain relation, viz. the four lines joining the corresponding angles are generating lines of a hyperboloid, or, what is the same thing, the four lines of intersection of corresponding faces are generating lines of a hyperboloid. The demonstration would show how the theorem in determinants is to be generalised.

2, Stone Buildings, Lincoln's Inn, February, 1855.

161.

A Problem in Permutations.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), p. 79.]

The game called Mousetrap gives rise to a singular problem in permutations. A set of cards, ace, two, three, &c., say up to thirteen, are arranged in a circle with their faces upwards—you begin at any card, and count one, two, three, &c., and if upon counting suppose the number five, you arrive at the card five, that card is thrown out; and beginning again with the next card, you count one, two, three, &c., throwing out if the case happen a new card as before, and so on until you have counted up to thirteen, without coming to a card which ought to be thrown out. It is easy to see that, whatever the number of the cards is, they may be so arranged as to be all thrown out in the order of their numbers; but that it is not possible in general to arrange the cards so that all the cards, or any specified cards, may be thrown out in a given order. Thus, if all the cards are to be thrown out in the order of their numbers, the arrangements in the case of a single card, two, three &c. cards, are

$$\begin{array}{llllllll}
1 \\
1 & 2 \\
1 & 3 & 2 \\
1 & 4 & 2 & 3 \\
1 & 3 & 2 & 5 & 4 \\
1 & 4 & 2 & 5 & 6 & 3 \\
1 & 5 & 2 & 7 & 4 & 3 & 6 \\
1 & 6 & 2 & 4 & 5 & 3 & 7 & 8 \\
 & & & \&c.
\end{array}$$

It is required to investigate the general theory.

162.

Two Letters on Cubic Forms.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 85–87 and 90–91.]

Cher Mons. Hermite,

Il y a longtemps que j'ai voulu vous écrire, mais je n'ai été empêché je ne sais comment; j'ai assez à vous dire par rapport aux covariants, mais à présent je vais vous parler des formes cubiques à deux indéterminées. Il me semble que l'on peut simplifier la théorie de Eisenstein, et l'étendre au cas d'un déterminant négatif quelconque, de la manière que voici.

Soit $(a, b, c, d \between x, y)^3$ une forme cubique, je représente par Hess. $(a, b, c, d \between x, y)^2$ la forme quadratique dérivée $(ac - b^2,\ \frac{1}{2}(ad - bc),\ bd - c^2 \between x, y)^2$. Cela étant, soit (A, B, C) une forme représentative (réduite et proprement primitive) au déterminant $-D$; à moins que $(A, B, C)^p = (A, -B, C)$, c'est-à-dire, à moins qu'on n'ait, ou que la nouvelle forme laquelle par sa triplication produit la forme principale, il n'existe pas de forme cubique (a, b, c, d) telle que $b^2 - ac = A$, $bc - ad = 2B$, $c^2 - bd = C$; mais en supposant que l'on ait $(A, B, C)^p = (A, -B, C)$ on peut trouver une seule forme cubique qui satisfait à l'équation dont il s'agit. J'écarte, cela va sans dire, l'une ou l'autre des deux formes (a, b, c, d) et $(-a, -b, -c, -d)$.

En effet on a identiquement

$$\begin{aligned}&(b^2 - ac,\ \tfrac{1}{2}(bc - ad),\ c^2 - bd \between bx + cy,\ cx + dy)^2 \\ &\qquad = (b^2 - ac,\ \tfrac{1}{2}(bc - ad),\ c^2 - bd \between y, (b^2 - ac,\ \tfrac{1}{2}(bc - ad),\ c^2 - bd \between x',\ y')^2,\end{aligned}$$

donc, en supposant que $b^2 - ac = A$, $bc - ad = 2B$, $c^2 - bd = C$, il s'ensuit que $(A, B, C)^p = (A, -B, C)$.

Je suppose donc $(A, B, C)^p = (A, -B, C)$, et je dis qu'il ne peut pas y avoir deux formes cubiques, (a, b, c, d) et (a_1, b_1, c_1, d_1), qui aient la propriété dont il s'agit; car, en écrivant

$$\left.\begin{aligned} \xi &= bxx' + cxy' + cx'y + dyy', \\ \eta &= axx' + bxy' + bx'y + cyy', \end{aligned}\right\} \qquad \left.\begin{aligned} \xi_1 &= b_1xx' + c_1xy' + c_1x'y + d_1yy', \\ \eta_1 &= a_1xx' + b_1xy' + b_1x'y + c_1yy', \end{aligned}\right\}$$

on trouverait

$$(A, B, C \between \xi, \eta)^2 = (A, -B, C \between \xi_1, \eta_1)^2,$$

ce qui implique d'abord que ξ_1, η_1 soient des fonctions linéaires de ξ, η. Mais $(A, -B, C)$ étant une forme réduite et proprement primitive au déterminant $-D$, il n'existe pas de transformation de la forme quadratique en elle-même, hormis $\xi = \xi$, $\eta = \eta$. Le cas $D = 1$ doit se traiter à part; dans ce cas particulier il n'y a que la forme cubique $(0, 1, 0, 1)$. Donc &c. Enfin, si $(A, B, C)^p = (A, -B, C)$, il existe une forme cubique (a, b, c, d) telle que $b^2 - ac = A$, &c.; car en cherchant par la méthode de Gauss les valeurs des coefficients p, p', p'', p''' et q, q', q'', q''' qui donnent cette transformation, on obtient l'identité $p^2 = aq$, $pp' = bq$, $p'p'' = cq$, $p''^2 = dq$. On peut donc représenter ces coefficients par b_1, c_1, c_1, d_1; a_1, b_1, b_1, c_1; savoir, en peut trouver a, b, c, d, de manière que

$$\begin{aligned}(A, -B, C \between bxx' + cxy' + cx'y + dyy',\ axx' + bxy' + bx'y + cyy')^2 \\ = (A, B, C \between x, y) \cdot (A, B, C \between x', y').\end{aligned}$$

Cela étant, les équations de Gauss donnent

$$\begin{aligned} A &= bb_1 - ac_1, & A &= b^2 - ac, \\ 2B &= cc_1 - ad_1, & 2B &= cb_1 + ad_1 - 2bc_1, \\ C &= cc_1 - bd_1, & C &= c_1^2 - b_1d_1, \end{aligned}$$

et de là on obtient

$$\begin{aligned} b\,(b - b_1) - a\,(c - c_1) &= 0, \\ c\,(b - b_1) - b\,(c - c_1) &= 0, \\ c_1(b - b_1) - b_1(c - c_1) &= 0, \\ d_1(b - b_1) - c_1(c - c_1) &= 0, \end{aligned}$$

c'est-à-dire, ou $\dfrac{a}{b} = \dfrac{b}{c} = \dfrac{b_1}{c_1} = \dfrac{c_1}{d_1}$, ce qui n'est pas vrai (car cela donnerait $A = 0$, $B = 0$, $C = 0$), ou $b - b_1 = 0$, $c - c_1 = 0$; donc $b_1 = b$, $c_1 = c$; et en écrivant d au lieu de d_1, on voit que l'équation de transformation devient

$$(A, -B, C \between bxx' + cxy' + cx'y + dyy',\ axx' + bxy' + bx'y + cyy')^2 = (A, B, C \between x, y) \cdot (A, B, C \between x', y'),$$

où

$$A = b^2 - ac, \quad 2B = bc - ad, \quad C = c^2 - bd. \qquad \text{C. } \eta. \text{ f. à d.}$$

Je ne sais pas si je vous ai mentionné que j'ai calculé les formes quadratiques pour les treize numéros -307, &c. aux déterminants irréguliers. Pour $D=-307$ ces formes sont:

Ordre P. P., 1 genre, 9 classes, c'est-à-dire,

$$\left.\begin{matrix} 1, & \delta, & \delta^2 \\ \delta, & \delta\delta, & \delta\delta_1 \\ \delta_1^2, & \delta\delta_1^2, & \delta\delta_1^4 \end{matrix}\right| \cdot \frac{m}{307} \left| \begin{matrix} (1,\ 0,\ 307),\ (7,\ 1,\ 44),\ (7,\ -1,\ 44), \\ (4,\ 1,\ 77),\ (11,\ -1,\ 28),\ (17,\ 4,\ 19), \\ (4,\ -1,\ 77),\ (17,\ -4,\ 19),\ (11,\ 1,\ 28), \end{matrix}\right.$$

où $\delta^3=1$, $\delta_1^3=1$.

Ordre I. P., 1 genre, 3 classes, c'est-à-dire,

$$\alpha,\quad \alpha\delta,\quad \alpha\delta^2 \;\Big|\cdot\Big|(2,1,154),\ (14,1,22),\ (14,-1,22).$$

A chaque forme de l'ordre P. P. correspond donc une forme et une seule forme cubique au déterminant -1228. Ces formes sont

$$\begin{matrix} (0,1,\ 0,\ -307), & (1,\ -6,\ 8), & (1,\ -1,\ -6,\ -8), \\ (0,2,\ 1,\ -38), & (1,\ -3,\ -2,\ 8), & (4,\ 1,\ -4,\ -3), \\ (0,2,\ -1,\ -38), & (4,\ -1,\ -4,\ 3), & (1,\ 3,\ -2,\ 8). \end{matrix}$$

Je serai bien aise d'avoir de vos nouvelles, je vous prie de me croire votre très-dévoué

A. Cayley.

CHER MONS. HERMITE,

On démontre sans peine la proposition avancée dans ma dernière lettre, savoir qu'en supposant

$$\begin{aligned} \xi &= bxx' + cxy' + cx'y + dyy', \\ \eta &= axx' + bxy' + bx'y + cyy', \\ \xi_1 &= b_1xx' + c_1xy' + c_1x'y + d_1yy', \\ \eta_1 &= a_1xx' + b_1xy' + b_1x'y + c_1yy', \\ (A,-B,C \between \xi,\eta)^2 &= (A,B,C \between \xi_1,\eta_1)^2, \end{aligned}$$

on doit avoir

$$\begin{aligned} \xi_1 &= \alpha\xi + \beta\eta, \\ \eta_1 &= \gamma\xi + \delta\eta. \end{aligned}$$

2–2

où α, β, γ, δ sont des entiers. En effet on trouve d'abord en éliminant par ex. xy' et $x'y$,

$$\begin{aligned} \xi_1 &= \alpha\xi + \beta\eta + \lambda xx' + \mu yy', \\ \eta_1 &= \gamma\xi + \delta\eta + \nu xx' + \rho yy', \end{aligned}$$

où α, β, &c., λ, μ, &c., sont des quantités rationnelles. En substituant ces valeurs de ξ_1, η_1, on aura $(A, -B, C \between \lambda, \nu)^2 = (A, B, C \between \mu, \rho)^2 = 0$. Donc le déterminant n'étant pas un carré, $\lambda = 0$, $\nu = 0$, $\mu = 0$, $\rho = 0$, et

$$\begin{aligned} \xi_1 &= \alpha\xi + \beta\eta, \\ \eta_1 &= \gamma\xi + \delta\eta, \end{aligned}$$

où α, β, γ, δ sont des quantités rationnelles. Donc

$$\begin{aligned} b_1 &= \alpha b + \beta a, & a_1 &= \gamma b + \delta a, \\ c_1 &= \alpha c + \beta b, & b_1 &= \gamma c + \delta b, \\ d_1 &= \alpha d + \beta c, & c_1 &= \gamma d + \delta c, \end{aligned}$$

donc α, β seront des quantités rationnelles ayant pour dénominateur l'une quelconque à volonté des trois quantités $b^2 - ac$, $c^2 - bd$, $ad - bc$, c'est-à-dire, des quantités A, B, C; et, puisque (A, B, C) est une forme P. P., cela ne peut arriver à moins que α, β ne soient des entiers; de même, γ, δ seront des entiers. Je remarque de plus les équations

$$\begin{aligned} \alpha\beta + b\,(\alpha - \delta) - c\gamma &= 0, \\ b\delta + c\,(\alpha - \delta) - d\gamma &= 0, \end{aligned}$$

lesquelles donnent

$$\beta : \alpha - \delta : -\gamma = bd - c^2 : bc - ad : ac - b^2 = C : -2B : A.$$

cela fait voir que la transformation en elle-même de la forme $(A, -B, C \between \xi, \eta)^2$ à moyen de $\xi_1 = \alpha\xi + \beta\eta$, $\eta_1 = \gamma\xi + \delta\eta$ est une transformation propre. J'aime cependant mieux le raisonnement dont vous vous êtes servi pour déduire d'une forme cubique quadratique. Il toutes les autres formes cubiques qui correspondent à la même forme quadratique. Il serait facile de la même manière, étant donnée une transformation quelconque d'une forme quadratique dans le produit de deux autres formes quadratiques, d'en déduire toutes les autres transformations; car il y a pour la fonction $axx' + \&c.\ldots$ un covariant qui correspond au cubicovariant de la fonction $ax^3 + \&c.\ldots$ Je suis votre très-dévoué

A. Cayley.

2, Stone Buildings, Lincoln's Inn, 6 Mars, 1855.

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On Hansen's Lunar Theory.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 112–125.]

The following paper was written in order to exhibit, in as clear a form as may be, the investigation of the remarkable equations for the motion of the moon established in Hansen's "Fundamenta Nova Investigationis Orbitæ veræ quam Luna perlustrat," &c., Gothae, 1838. I have availed myself for this purpose of the remarks in Jacobi's two letters in answer to a letter of Hansen's, *Crelle*, t. XLII. [1851], p. 12; it may be convenient to remark that the quantity there represented by λ, and which does not occur in Hansen's own investigation, is in this paper represented by Θ.

The position of the moon referred to the earth as centre is determined by

r, the radius vector,
L, the longitude,
Λ, the latitude.

Suppose, moreover, that the attractive force at distance unity, $= \kappa\,(M+E)$, is represented by n^2a^3, then the principal function will be $V = -\dfrac{n^2a^3}{r}$, and the disturbing function R may be represented by $\dfrac{n^2a^3}{r}\Omega$; the expression for the half of the vis viva is

$$T = \tfrac{1}{2}\left\{\left(\frac{dr}{dt}\right)^2 + r^2\cos^2\Lambda\left(\frac{dL}{dt}\right)^2 + r^2\left(\frac{d\Lambda}{dt}\right)^2\right\},$$

and the equations of motion are therefore

$$\begin{aligned}\frac{d\,dr}{dt\,dt} - r\cos^2\Lambda\left(\frac{dL}{dt}\right)^2 - r\left(\frac{d\Lambda}{dt}\right)^2 + \frac{n^2a^3}{r^2} &= n^2a^3\,\frac{d\Omega}{dr}\,,\\ \frac{d}{dt}\left(r^2\cos^2\Lambda\,\frac{dL}{dt}\right) &= n^2a^3\,\frac{d\Omega}{dL}\,,\\ \frac{d}{dt}\left(r^2\frac{d\Lambda}{dt}\right) + r^2\cos\Lambda\sin\Lambda\left(\frac{dL}{dt}\right)^2 &= n^2a^3\,\frac{d\Omega}{d\Lambda}\,,\end{aligned}$$

where Ω is considered as a function of r, L, Λ.

Consider the orbit as an ellipse, then putting

a, the mean distance,
n, the mean motion $= \sqrt{\dfrac{\kappa\,(M+E)}{a^3}}$,
e, the eccentricity,
c, the mean anomaly at epoch,
ω, the distance of perigee from node,
θ, the longitude of node,
i, the inclination,
Φ, the distance from node,
Ψ, the reduced distance from node, $= L - \theta$,
U, the eccentric anomaly,
f, the true anomaly, $= \Phi - \omega$,

the elements of the orbit are a, e, c, ω, θ, i, and we have from the theory of elliptic motion,

$$nt + c = U - e \sin U,$$

$$f = \tan^{-1} \frac{\sqrt{(1-e^2)} \sin U}{\cos U - e},$$

$$r = a\,(1 - e \cos U) = \frac{a\,(1-e^2)}{1 + e \cos f}.$$

Moreover i is the angle at the base of a right-angled spherical triangle, the base, perpendicular, and hypothenuse of which are Ψ, Λ, Φ, hence

$$\begin{aligned} \tan \Psi &= \cos i \, \tan \Phi, \\ \sin \Lambda &= \sin i \, \sin \Phi, \\ \sin \Psi &= \cot i \, \tan \Lambda, \\ \cos \Phi &= \cos \Lambda \, \cos \Psi. \end{aligned}$$

Considering the elements as constant, we have

$$\begin{aligned} \frac{dr}{dt} &= \frac{nae \sin f}{\sqrt{(1-e^2)}}, \\ \frac{df}{dt} &= \frac{na^2 \sqrt{(1-e^2)}}{r^2}, \\ \frac{d\Phi}{dt} &= \frac{na^2 \sqrt{(1-e^2)}}{r^2}, \\ \frac{d\Psi}{dt} &= \frac{\cos i}{\cos^2 \Lambda} \, \frac{na^2 \sqrt{(1-e^2)}}{r^2}, \\ \frac{dL}{dt} &= \frac{\cos i}{\cos^2 \Lambda} \, \frac{na^2 \sqrt{(1-e^2)}}{r^2}, \\ \frac{d\Lambda}{dt} &= \sin i \, \cos \Psi \, \frac{na^2 \sqrt{(1-e^2)}}{r^2}. \end{aligned}$$

Hence also

$$\begin{aligned}
\frac{d}{dt}\left(\frac{dr}{dt}\right) &= \frac{n^2a^4e\cos f}{r^2}\,,\\
\frac{d}{dt}\left(r^2\cos^2\Lambda\,\frac{dL}{dt}\right) &= 0,\\
\frac{d}{dt}\left(r^2\frac{d\Lambda}{dt}\right) &= -\frac{\cos^2 i\,\sin\Lambda}{\cos^2\Lambda}\,\frac{n^2a^4(1-e^2)}{r^2}\,,\\
\frac{d}{dt}\left(\frac{df}{dt}\right) &= -\frac{2na^2e}{r^3}\,\dot{e}\sin f.
\end{aligned}$$

The foregoing values show that the equations of motion, neglecting the terms which involve the disturbing functions, are satisfied by the elliptic values of r, L, Λ: and in order to satisfy the actual equations of motion, we have only to consider the elements as variable and to write

$$\begin{aligned}
dr &= 0,\\
dL &= 0,\\
d\Lambda &= 0,\\
d\,\frac{dr}{dt} &= n^2a^3\,\frac{d\Omega}{dr}\,dt,\\
d\left(r^2\cos^2\Lambda\,\frac{dL}{dt}\right) &= n^2a^3\,\frac{d\Omega}{dL}\,dt,\\
d\left(r^2\frac{d\Lambda}{dt}\right) &= n^2a^3\,\frac{d\Omega}{d\Lambda}\,dt,
\end{aligned}$$

where the differentiations relate only to the elements, or, what is the same thing, to t in so far only as it enters through the variable elements: the system is at once transformed into

$$\begin{aligned}
dr &= 0,\\
dL &= 0,\\
d\Lambda &= 0,\\
d\,\frac{nae\sin f}{\sqrt{(1-e^2)}} &= n^2a^3\,\frac{d\Omega}{dr}\,dt,\\
d\,na^2\sqrt{(1-e^2)}\,\cos i &= n^2a^3\,\frac{d\Omega}{dL}\,dt,\\
d\,na^2\sqrt{(1-e^2)}\,\sin i\,\cos\Psi &= n^2a^3\,\frac{d\Omega}{d\Lambda}\,dt.
\end{aligned}$$

Now $\Psi = L - \theta$, (or supposing, as before, that the differentiations relate to t, only in so far as it enters through the variable elements) $d\Psi = -d\theta$, and thence $d\theta = \dfrac{\tan\Psi}{\sin i}\,di$; we have also $d\Phi = -\cos i\,d\theta$. The equations containing $\dfrac{d\Omega}{dL}$ and $\dfrac{d\Omega}{d\Lambda}$ give

$$\cos i\,d\,na^2\sqrt{(1-e^2)} - na^2\sqrt{(1-e^2)}\sin i\,di = n^2a^3\,\frac{d\Omega}{dL}\,dt,$$

$$\cos i\,\sin i\,d\,na^2\sqrt{(1-e^2)} + na^2\sqrt{(1-e^2)}(\cos\Psi\,\cos i\,di + \sin i\,\sin\Psi\,d\theta) = n^2a^3\,\frac{d\Omega}{d\Lambda}\,dt;$$

or, expressing $d\theta$ by means of di and reducing, the second of these equations becomes

$$na^2\sqrt{(1-e^2)} = na^2\sqrt{(1-e^2)}\,\frac{\cos i}{\cos^2\Psi}\,di = \frac{1}{\cos\Psi}\,n^2a^3\frac{d\Omega}{d\Lambda}\,dt,$$

and combining this with the first of the two equations, and observing that

$$\frac{\cos^2 i}{\cos^2\Lambda\,\cos^2\Psi} + \sin^2 i = \frac{1}{\cos^2\Psi},$$

we find

$$d\,na^2\sqrt{(1-e^2)} = n^2a^3\left(\frac{\cos i}{\cos^2\Lambda}\,\frac{d\Omega}{dL} + \sin i\,\cos\Psi\,\frac{d\Omega}{d\Lambda}\right)dt,$$

$$di = \frac{na^2}{\sqrt{(1-e^2)}}\left(-\sin i\,\cos\Psi\,\frac{d\Omega}{dL} + \frac{\cos i}{\cos\Lambda\,\cos^2\Psi}\,\frac{d\Omega}{d\Lambda}\right)dt.$$

Now, considering Ω as a function of r, θ, i, Φ, then Λ, L are given as functions of θ, i, Φ by the equations $\sin\Lambda = \sin i\,\sin\Phi$, $\tan\Psi = \cos i\,\tan\Phi$, $\Psi = L - \theta$, and after some simple reductions,

$$\begin{aligned}
\frac{d\Omega}{dr} &= \frac{d\Omega}{dr}\,,\\
\frac{d\Omega}{d\theta} &= \frac{d\Omega}{dL}\,,\\
\frac{d\Omega}{di} &= \tan\Phi\left(\frac{\cos i\,\cos\Psi}{\cos\Lambda}\,\frac{d\Omega}{dL} + \cos i\,\cos\Psi\,\frac{d\Omega}{d\Lambda}\right),\\
\frac{d\Omega}{d\Phi} &= \left(\frac{\cos i}{\cos\Lambda}\,\frac{d\Omega}{dL} + \cos i\,\cos\Psi\,\frac{d\Omega}{d\Lambda}\right);
\end{aligned}$$

whence also

$$\frac{d\Omega}{d\theta} = \cos i\,\frac{d\Omega}{d\Phi} - \sin i\,\cot\Phi\,\frac{d\Omega}{di}\,.$$

We have therefore

$$\begin{aligned}
d\, na^2\sqrt{(1-e^2)} &= n^2a^3\,\frac{d\Omega}{d\Phi}\,dt,\\
di &= \frac{na\cot\Phi}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{di}\,dt,\\
d\theta &= \frac{na}{\sqrt{(1-e^2)}\,\sin i}\,\frac{d\Omega}{di}\,dt,\\
d\phi &= \frac{-na\cot i}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{di}\,dt,\\
dr &= 0,\\
d\,\frac{nae\sin f}{\sqrt{(1-e^2)}} &= n^2a^3\,\frac{d\Omega}{dr}\,dt.
\end{aligned}$$

Suppose now that we have

ρ, a radius vector, τ for t,
ϕ, a true anomaly,
u, an eccentric anomaly, &c.,

i.e. let ρ, ϕ, u, be what the radius vector, the true anomaly and the eccentric anomaly become when the time t, in so far as it enters directly, and not through the variable elements, is replaced by a new variable τ. We have

$$n\tau + c = u - e\sin u,$$

$$\phi = \tan^{-1}\frac{\sqrt{(1-e^2)}\sin u}{\cos u - e},$$

$$\rho = a\,(1-e\cos u) = \frac{a\,(1-e^2)}{1+e\cos\phi};$$

and of course the differential coefficients of ρ, ϕ with respect to τ may be at once deduced from the corresponding expressions for the differential coefficients of r, f with respect to t, the elements being considered as constant. Now, using l to denote a logarithm, and supposing that the differentiations affect only the elements, we have

$$dl\rho = \frac{da}{a} - \frac{2ede}{1-e^2} - \frac{\rho\cos\phi\, de}{a\,(1-e^2)} + \frac{\rho e\sin\phi\, d\phi}{a\,(1-e^2)},$$

$$dlr = \frac{da}{a} - \frac{2ede}{1-e^2} - \frac{r\cos f\, de}{a\,(1-e^2)} + \frac{re\sin f\, df}{a\,(1-e^2)};$$

and putting for shortness

$$X_r = dlr - \frac{\rho e\sin\phi\, d\phi}{a\,(1-e^2)},$$

$$X = dlr - \frac{re\sin f\, df}{a\,(1-e^2)},$$

we find

$$X_r = \frac{\rho\sin\phi}{r\sin f}\,X = \left(1 - \frac{\rho\sin\phi}{r\sin f}\right)\left(\frac{da}{a} - \frac{2ede}{1-e^2}\right) - \frac{\rho}{a\,(1-e^2)}\left(\cos\phi - \frac{\sin\phi\cos r}{\sin f}\right)de,$$

or reducing

$$X_r - \frac{\rho\sin\phi}{r\sin f}\,X = \frac{1}{\sin f\,(1+e\cos\phi)}\left\{\big(\sin f - \sin\phi + e\sin(f-\phi)\big)\left(\frac{da}{a} - \frac{2ede}{1-e^2}\right) - \sin(f-\phi)\,de\right\}.$$

Write for a moment

$$P = na^2\sqrt{(1-e^2)},\quad Q = \frac{nae\sin f}{\sqrt{(1-e^2)}},\quad R = \frac{a\,(1-e^2)}{1+e\cos f},$$

so that

$$a\,(1-e^2) = \frac{P^2}{n^2a^3},$$

$$e^2 = \frac{1}{n^4a^6}\,P^2Q^2 + \frac{1}{n^2a^2}\,\frac{P^2}{R^2} - \frac{2}{n^2a^4}\,\frac{P^2}{R} + 1.$$

We have therefore

$$\frac{da}{a} = \frac{2ede}{1-e^2} = \frac{2dP}{P} = \frac{2}{na^2\sqrt{(1-e^2)}}\,dP,$$

$$ede = \left(\frac{1}{n^4a^6}\,P^2Q\right)dQ + \left(\frac{P^2}{n^4a^6\,R} - \frac{2}{n^2a^4}\,\frac{P^2}{R}\right)dP + \frac{P}{n^4a^6}\,\bar{Q}\,dQ + \left(-\frac{1}{n^2a^2}\,\frac{P^2}{R^2} + \frac{2}{n^2a^4}\,\frac{P^2}{R}\right)dR,$$

which, after reduction, becomes

$$de = \frac{1}{na^2\,(1-e^2)}\big(e\sin f + 2\,(\cos f + e\cos^2 f)\big)\,dP + \frac{na}{e}\,\sqrt{(1-e^2)}\,\sin f\,dQ - \frac{(1+e\cos f)^2\cos f}{a\,(1-e^2)}\,dR,$$

and substituting these values,

$$\begin{aligned} X_r - \frac{\rho\sin\phi}{r\sin f}\,X = \frac{1}{1+e\cos\phi}\Bigg\{&\big(2 - 2\cos(f-\phi) + e\sin f\sin(f-\phi)\big)\frac{1}{na^2\sqrt{(1-e^2)}}\,dP \\ &-a\,(1-e^2)\sin(f-\phi)\,\frac{1}{na^2\sqrt{(1-e^2)}}\,dQ \\ &+\frac{(1+e\cos f)^2\cos f\sin(f-\phi)}{\sqrt{(1-e^2)}}\,\frac{1}{na^2\sqrt{(1-e^2)}}\,dR\Bigg\}, \end{aligned}$$

or substituting for X, X_r, P, Q, R, their values

$$dl\rho - \frac{\rho\sin\phi}{r\sin f}\,dlr + \frac{\rho e\sin\phi}{a\,(1-e^2)}\,(df - d\phi)$$

$$\begin{aligned} = \frac{\rho}{a\,(1-e^2)}&\big(2 - 2\cos(f-\phi) + e\sin f\sin(f-\phi)\big)\,\frac{1}{na^2\sqrt{(1-e^2)}}\,d\,na^2\sqrt{(1-e^2)} \\ &-\rho\sin(f-\phi)\,\frac{1}{na^2\sqrt{(1-e^2)}}\,d\,\frac{nae\sin f}{\sqrt{(1-e^2)}} \\ &+\frac{\rho\,(1-e^2)}{r^2}\,\cot f\sin(f-\phi)\,\frac{1}{na^2\sqrt{(1-e^2)}}\,dr. \end{aligned}$$

Now

$$-\left\{\frac{\rho}{r}\cos(f-\phi)-1+\frac{\rho}{a\,(1-e^2)}(\cos(f-\phi)-1)\right\}$$
$$=\frac{\rho}{a\,(1-e^2)}(2-2\cos(f-\phi)+e\sin f\sin(f-\phi))\,;$$

therefore

$$\begin{aligned}
&dl\rho-\frac{\rho\sin\phi}{r\sin f}\,dlr+\frac{\rho e\sin\phi}{a\,(1-e^2)}\,(df-d\phi)\\
&\quad=-\left\{\frac{\rho}{r}\cos(f-\phi)-1+\frac{\rho}{a\,(1-e^2)}(\cos(f-\phi)-1)\right\}\frac{1}{na^2\sqrt{(1-e^2)}}\,d\,na^2\sqrt{(1-e^2)}\\
&\quad\quad-\rho\sin(f-\phi)\,\frac{1}{na^2\sqrt{(1-e^2)}}\,d\,\frac{nae\sin f}{\sqrt{(1-e^2)}}\\
&\quad\quad+\frac{\rho a\,(1-e^2)}{r^2}\,\cot f\sin(f-\phi)\,\frac{1}{na^2\sqrt{(1-e^2)}}\,dr.
\end{aligned}$$

So far the variations of the elements have, in fact, been treated as independent; but if we substitute for $d\,na^2\sqrt{(1-e^2)}$, $d\,\dfrac{nae\sin f}{\sqrt{(1-e^2)}}$, dr, their values in the disturbed motion, the equation becomes

$$\begin{aligned}
dl\rho+\frac{\rho e\sin\phi}{a\,(1-e^2)}\,(df-d\phi)&=-\left\{\frac{\rho}{r}\cos(f-\phi)-1+\frac{\rho}{a\,(1-e^2)}(\cos(f-\phi)-1)\right\}\frac{na}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{d\Phi}\,dt\\
&\quad-\rho\sin(f-\phi)\,\frac{na}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dr}\,dt.
\end{aligned}$$

Consider now the point in which the orbit is intersected by any orthogonal trajectory to the successive positions of the orbit, or to fix the ideas, the orthogonal trajectory passing through ♈, the point in question may, for want of a recognised name, be called the "departure point;" and the angular distances in the orbit measured from this point may be termed "departures;" the expression, "the departure," is to be understood as meaning the departure of the moon. Write now

χ, the departure of the perigee,
v, the departure, $=f+\chi$,
σ, the departure of the node, $=\chi-\omega$,
Θ, the longitude in orbit of departure point, $=\theta-\sigma$.

It should be remarked that χ is not properly an element, i.e. it is not a function of a, e, c, ω, θ, i without t, and in like manner σ and Θ (which also depend upon χ) are not elements.

We have from the geometrical definition

$$d\chi=d\omega+\cos i\,d\theta;$$

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and therefore

$$d\sigma = \cos i \, d\theta,$$
$$d\Theta = (1 - \cos i) \, d\theta.$$

Moreover $v_1 = \Phi + \sigma$, which gives (assuming that the differentiations are performed with respect to t, only in so far as it enters through the variable elements) $dv_1 = d\Phi + d\sigma = d\Phi + \cos i \, d\theta$, i.e. $dv_1 = 0$, an equation which might have been assumed for the purpose of defining the departure point; the equation, in fact, expresses that the departure v_1 is measured from a point not actually fixed, but such that the increment of v_1 in the interval of time dt is the angular distance between two consecutive positions of the moon.

We have, as above noticed, $d\sigma = \cos i \, d\theta$, and thence and from what has preceded

$$di = \frac{na \cot \Phi}{\sqrt{(1-e^2)}} \frac{d\Omega}{di} \, dt,$$
$$d\sigma = \frac{na \cot i}{\sqrt{(1-e^2)}} \frac{d\Omega}{di} \, dt.$$

Now the position of the moon can be determined by means of the quantities r, v_1, Θ, σ, i; hence Ω (which has been considered as a function of r, Φ, θ, i) may, if we please, be considered as a function of r, v_1, Θ, σ, i and from the differential relations

$$dr = dr,$$
$$dv_1 = d\Phi + \cos i \, d\theta,$$
$$d\Theta = (1 - \cos i) \, d\theta,$$
$$d\sigma = d\omega + \cos i \, d\theta,$$
$$di = di,$$

we find

$$\frac{d\Omega}{dr} = \frac{d\Omega}{dr},$$
$$\frac{d\Omega}{d\Phi} = \frac{d\Omega}{dv_1},$$
$$\frac{d\Omega}{d\theta} = \cos i \left(\frac{d\Omega}{dv_1} + \frac{d\Omega}{d\sigma} \right) + (1 - \cos i) \frac{d\Omega}{d\Theta},$$
$$\frac{d\Omega}{di} = \frac{d\Omega}{di},$$

we have therefore

$$\frac{d\Omega}{d\theta} = \frac{1 - \cos i}{\cos i} \frac{d\Omega}{d\Theta} = \frac{d\Omega}{d\Phi} + \frac{1}{\cos i} \frac{d\Omega}{d\sigma},$$

or in virtue of a preceding equation

$$\frac{d\Omega}{d\sigma} + \frac{1 - \cos i}{\cos i} \frac{d\Omega}{d\Theta} = -\tan i \, \cot \Phi \, \frac{d\Omega}{di};$$

and effecting the substitutions, and collecting the results,

$$\begin{aligned}
d\,na^2\sqrt{(1-e^2)} &= n^2a^3\,\frac{d\Omega}{dv_1}\,dt,\\
dr &= 0,\\
d\,\frac{nae\sin f}{\sqrt{(1-e^2)}} &= n^2a^3\,\frac{d\Omega}{dr}\,dt,\\
di &= \frac{na\cot i}{\sqrt{(1-e^2)}}\left(\frac{d\Omega}{d\sigma}+\frac{1-\cos i}{\cos i}\,\frac{d\Omega}{d\Theta}\right)dt,\\
d\sigma &= \frac{na\cot i}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{di}\,dt,
\end{aligned}$$

where Ω is considered as a function of r, v_1, Θ, σ, i.

Instead of σ, i we may introduce the new quantities p, q defined by the equations

$$p=\sin i\,\sin\sigma,\quad q=\sin i\,\cos\sigma;$$

this gives $\sin^2 i=p^2+q^2$, $\sigma=\tan^{-1}\dfrac{p}{q}$ and retaining in the formulæ the sine and cosine of i, to avoid the introduction of irrational functions of p^2+q^2, we have

$$d\Theta=(1-\cos i)\,d\theta=\frac{1-\cos i}{\cos i}\,d\sigma=\frac{1-\cos i}{\sin i\,\sin^2 i}\,(qdp-pdq),$$

i.e.

$$d\Theta=\frac{qdp-pdq}{\cos i\,(1+\cos i)}\,,$$

which determines Θ by means of p and q. We have moreover

$$\begin{aligned}
dp &= \sin i\,\cos\sigma\,d\sigma+\cos i\,\sin\sigma\,di,\\
dq &= -\sin i\,\sin\sigma\,d\sigma+\cos i\,\cos\sigma\,di,\\
\frac{d\Omega}{dp} &= \frac{\sin\sigma}{\cos i}\,\frac{d\Omega}{di}+\frac{\cos\sigma}{\sin i}\,\frac{d\Omega}{d\sigma}\,,\\
\frac{d\Omega}{dq} &= \frac{\cos\sigma}{\cos i}\,\frac{d\Omega}{di}-\frac{\sin\sigma}{\sin i}\,\frac{d\Omega}{d\sigma}\,,
\end{aligned}$$

from which equations and the foregoing values of di and $d\sigma$ we find the values of dp and dq; the other equations of the system remain unaltered, and we have therefore

$$\begin{aligned}
d\,na^2\sqrt{(1-e^2)} &= n^2a^3\,\frac{d\Omega}{dv_1}\,dt,\\
dr &= 0,\\
d\,\frac{nae\sin f}{\sqrt{(1-e^2)}} &= n^2a^3\,\frac{d\Omega}{dr}\,dt,
\end{aligned}$$

$$dp = \frac{na\cos^2 i}{\sqrt{(1-e^2)}}\left(\frac{d\Omega}{dq} - \frac{p}{\cos i\,(1+\cos i)}\,\frac{d\Omega}{d\Theta}\right)dt,$$
$$dq = -\frac{na\cos^2 i}{\sqrt{(1-e^2)}}\left(\frac{d\Omega}{dp} + \frac{q}{\cos i\,(1+\cos i)}\,\frac{d\Omega}{d\Theta}\right)dt,$$

where Ω is considered as a function of r, v_1, Θ, p, q. The symbols $\frac{d\Omega}{dp}$, $\frac{d\Omega}{dq}$, as employed by Hansen, mean that the differentiations are to be performed as if Θ was a function of p, q, such that

$$d\Theta = \frac{d\Theta}{dp}\,dp + \frac{d\Theta}{dq}\,dq = \frac{qdp - pdq}{\cos i\,(1+\cos i)}\,;$$

the last two equations being therefore nothing else than what Hansen represents by

$$dp = \frac{na\cos^2 i}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dq}\,dt,$$
$$dq = -\frac{na\cos^2 i}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dp}\,dt.$$

Write now

$$\lambda,\ \text{the departure},\ \tau \text{ for } t,$$

i.e. λ is what v_1 becomes when t, in so far as it enters explicitly, and not through the variable elements, is replaced by the new variable τ; so that, in fact $\lambda = \phi + \chi$, The values of r, v_1 could be at once found from those of ρ, λ by changing τ into t; and to determine the values of ρ, λ, Hansen proceeds as follows: writing

$$\lambda = \Pi\,(\zeta, t),$$
$$l\rho = \Gamma\,(\zeta, t) + \beta,$$

where ζ and β are new variables functions of τ and t, Π, Γ are arbitrary functional symbols; so that if z, w are what ζ, β become when τ is changed into t, we should have

$$v_1 = \Pi\,(z, t),$$
$$lr = \Gamma\,(z, t) + w;$$

then the foregoing equations give

$$\frac{d\lambda}{d\tau} = \Pi'\,(\zeta, t)\,\frac{d\zeta}{d\tau}\,,$$
$$\frac{d\lambda}{dt} = \Pi'\,(\zeta, t)\,\frac{d\zeta}{dt} + \Pi_{,}\,(\zeta, t)\,,$$
$$\frac{dl\rho}{d\tau} = \Gamma'\,(\zeta, t)\,\frac{dl\rho}{d\tau} + \frac{d\beta}{d\tau}\,,$$
$$\frac{dl\rho}{dt} = \Gamma''\,(\zeta, t)\,\frac{d\zeta}{dt} + \Gamma_{,}\,(\zeta, t) + \frac{d\beta}{dt}\,;$$

[where the accents and strokes denote differentiation in regard to ζ, t respectively].

Hence eliminating $\Pi'(\zeta, t)$ and $\Gamma'(\zeta, t)$ we have

$$\frac{d\lambda}{d\tau}\frac{d\zeta}{dt} - \frac{d\lambda}{dt}\frac{d\zeta}{d\tau} = -\Pi_,(\zeta, t)\frac{d\zeta}{d\tau},$$
$$\frac{dl\rho}{d\tau}\frac{d\beta}{dt} - \frac{dl\rho}{dt}\frac{d\beta}{d\tau} = \frac{d\beta}{d\zeta}\frac{d\zeta}{dt}\frac{d\beta}{d\tau} + \frac{d\beta}{d\zeta}\frac{d\zeta}{d\tau} - \Gamma_,(\zeta, t)\frac{d\zeta}{d\tau},$$

or, what is the same thing,

$$\frac{\dfrac{d\zeta}{dt}}{\dfrac{d\zeta}{d\tau}} - \frac{d\lambda}{dt} = -\Pi_,(\zeta, t)\frac{1}{\dfrac{d\zeta}{d\tau}},$$

$$\frac{d\beta}{dt} - \frac{\dfrac{d\zeta}{dt}}{\dfrac{d\zeta}{d\tau}}\frac{dl\rho}{d\tau} = \frac{dl\rho}{dt} - \frac{dl\rho}{d\tau}\frac{\dfrac{d\zeta}{dt}}{\dfrac{d\zeta}{d\tau}} + \Pi_,(\zeta, t)\frac{dl\rho}{d\zeta}\frac{d\zeta}{d\tau} - \Gamma_,(\zeta, t),$$

or writing

$$T = \frac{d\lambda}{dt} - \frac{\dfrac{d\zeta}{dt}}{\dfrac{d\zeta}{d\tau}}, \quad R = \frac{dl\rho}{dt} - \frac{d\beta}{dt}\frac{\dfrac{d\zeta}{dt}}{\dfrac{d\zeta}{d\tau}},$$

we have

$$T = \frac{1}{\dfrac{d\zeta}{d\tau}}\left\{\frac{d\lambda}{d\tau}\left(\frac{d\zeta}{dt}\right)\frac{d\lambda}{dt} + \Pi_,(\zeta, t)\left(\frac{d\zeta}{d\tau}\right)\right\} - \Pi_,(\zeta, t)\frac{\dfrac{d\zeta}{dt}}{\dfrac{d\zeta}{d\tau}},$$

$$R = \frac{dl\rho}{dt} - \frac{\dfrac{d\zeta}{dt}}{\dfrac{d\zeta}{d\tau}}\frac{d\lambda}{d\tau} + \Pi_,(\zeta, t)\frac{dl\rho}{d\zeta}\frac{d\zeta}{d\tau} - \Gamma_,(\zeta, t).$$

Now from the equation $\lambda = \phi + \chi$ where χ is independent of τ, the differential coefficients of λ and ρ with respect to τ are at once deduced from those of f, r with respect to t, and we have,

$$\frac{d^2\lambda}{d\tau^2} = -\frac{2n^2a^4}{\rho^3}e\sin\phi,$$
$$\frac{d\lambda}{d\tau} = \frac{na^2\sqrt{(1-e^2)}}{\rho^2},$$
$$\frac{dl\rho}{d\tau} = \frac{nae\sin\phi}{\rho\sqrt{(1-e^2)}}.$$

Consequently

$$\begin{aligned}
T = {} & \frac{\rho^2}{na^2\sqrt{(1-e^2)}}\left\{\frac{d}{dt}\,\frac{na^2\sqrt{(1-e^2)}}{\rho^2} + \frac{2\rho e\sin\phi}{a\,(1-e^2)}\,\frac{d\lambda}{dt}\right.\\
& \left. -\frac{2\rho e\sin\phi}{a\,(1-e^2)}\,\Pi_,(\zeta,t) - \frac{\rho^2}{na^2\sqrt{(1-e^2)}}\,\Pi'_,(\zeta,t)\,\frac{d\zeta}{dt}\right\}\\
= {} & -2\,\frac{dl\rho}{dt} + \frac{2\rho e\sin\phi}{a\,(1-e^2)}\,\frac{d\lambda}{dt} + \frac{1}{na^2\sqrt{(1-e^2)}}\,\frac{d}{dt}\,na^2\sqrt{(1-e^2)}\\
& -\frac{2\rho e\sin\phi}{a\,(1-e^2)}\,\Pi_,(\zeta,t) - \frac{\rho^2}{na^2\sqrt{(1-e^2)}}\,\Pi'_,(\zeta,t)\,\frac{d\zeta}{dt}\,,
\end{aligned}$$

$$R = \frac{dl\rho}{dt} - \frac{\rho e\sin\phi}{a\,(1-e^2)}\,\frac{d\lambda}{dt} + \frac{\rho e\sin\phi}{a\,(1-e^2)}\,\Pi_,(\zeta,t) - \Gamma_,(\zeta,t),$$

and substituting in these equations the values of

$$\frac{dl\rho}{dt} = \frac{\rho e\sin\phi}{a\,(1-e^2)}\,\frac{d\lambda}{dt} \quad\text{and}\quad d\,na^2\sqrt{(1-e^2)}$$

we find

$$\begin{aligned}
T = {} & \left\{2\,\frac{\rho}{r}\cos(v_1-\lambda) - 1 + \frac{2\rho}{a\,(1-e^2)}(\cos(v_1-\lambda)-1)\right\}\frac{na}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dv_1} + 2\,\frac{\rho}{r}\sin(v_1-\lambda)\,\frac{na}{\sqrt{(1-e^2)}}\,r\,\frac{d\Omega}{dr}\\
& - 2\Pi_,(\zeta,t) + \frac{\rho e\sin\phi}{a\,(1-e^2)}\,\Pi_,(\zeta,t)\,\frac{na}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dv_1} - 2\Pi_,(\zeta,t) - \frac{\rho^2}{na^2\sqrt{(1-e^2)}}\,\Pi'_,(\zeta,t)\,\frac{d\zeta}{dt}\,,\\
R = {} & -\left\{\frac{\rho}{r}\cos(v_1-\lambda) - 1 + \frac{\rho}{a\,(1-e^2)}(\cos(v_1-\lambda)-1)\right\}\frac{na}{\sqrt{(1-e^2)}}\,\frac{d\Omega}{dv_1} - \frac{\rho}{r}\sin(v_1-\lambda)\,\frac{na}{\sqrt{(1-e^2)}}\,r\,\frac{d\Omega}{dr}\\
& + \Pi_,(\zeta,t)\,\frac{\rho e\sin\phi}{a\,(1-e^2)}\,\frac{d\lambda}{dt} - \Gamma_,(\zeta,t),
\end{aligned}$$

which are Hansen's values, except that $\dfrac{d\lambda}{d\tau}$ in the coefficient of $\Pi_,(\zeta,t)$ has been replaced by its value.

2, Stone Buildings, 31st March, 1855.

164.

On Gauss' Method for the Attraction of Ellipsoids.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 162–166.]

The following is the method employed in Gauss' Memoir "Theoria Attractionis Corporum Sphæroidicorum ellipticorum homogæneorum methodo novo tractata," 1813. *Gauss. Gott. recent.*, t. II. [and *Werke* t. V. pp. 1–22]. I have somewhat developed the geometrical considerations upon which the method depends.

The attraction of the ellipsoid is found by means of the following theorems, which apply generally to the case of a homogeneous solid bounded by a closed surface:— M denotes the attracted point, P a point of the surface, PQ is the normal (lying outside the surface) at the point P, dS is the element of the surface at this point, MQ, QX, and MX denote angles at the point P, viz. MQ the $\angle MPQ$, and QX and MX the inclinations of QP and MP respectively to a line PX drawn in a direction assumed as that of the axis of X, MP denotes the distance between the points M and P. And X is the attraction in the direction opposite to that of the axis of x; the integrations extend over the entire surface.

Theorem. The integral

$$\iint \frac{dS \cos MQ}{MP^n}$$

has for its value

$$0, \quad -2\pi, \quad \text{or} \quad -4\pi,$$

according as M is exterior to, upon, or interior to, the surface.

This is obviously a purely geometrical theorem.

Theorem. The attraction is given by the formula

$$X = \iint \frac{dS \cos QX}{MP}.$$

THEOREM. The attraction is also given by the formula

$$X = -\iint \frac{dS \cos MQ \cos MX}{MP^2}.$$

Consider now an ellipsoid, the semi-axes of which are A, B, C, and putting a, b, c for the coordinates of the attracted point M, and x, y, z for the coordinates of P, assume

$$\begin{aligned} x &= A\xi, \\ y &= B\eta, \\ z &= C\zeta, \end{aligned}$$

so that ξ, η, ζ are the coordinates of a point P' on a sphere, radius unity, corresponding in a definite manner to the point P on the ellipsoid; and let $d\sigma$ be the corresponding element of the spherical surface, we have

$$dS = \frac{ABC}{p}\, d\sigma,$$

where p denotes the perpendicular let fall from the centre of the ellipsoid upon the tangent plane at P.

Moreover,

$$\cos QX = \frac{p\xi}{A}, \quad \cos MX = \frac{a-x}{MP} = \frac{a-A\xi}{MP};$$

and therefore

$$dS \cos QX = \frac{BC\xi\, d\sigma}{MP}.$$

The second theorem gives therefore

$$A\,\frac{X}{ABC} = \iint \frac{\xi\, d\sigma}{MP},$$

where the integration is extended over the surface of the sphere, and the third theorem gives

$$\frac{X}{ABC} = -\iint \frac{(a-A\xi)\cos MQ\, dS}{MP^3},$$

where the integration is extended over the surface of the ellipsoid.

Suppose now a confocal ellipsoid, the semi-axes of which are $A+\delta A$, $B+\delta B$, $C+\delta C$, and let P' be the point on this new ellipsoid which *corresponds* to the point P on the original ellipsoid, i.e. let P' be the point whose coordinates are $(A+\delta A)\xi$, $(B+\delta B)\eta$, $(C+\delta C)\zeta$; the decrement of MP will be equal to the normal distance δN between the two ellipsoids at the point P, multiplied into the cosine of the angle MQ, and we have, by a property of confocal ellipsoids, $A\delta A = B\delta B = C\delta C = p\delta N$; we have therefore

$$\delta\,\overline{MP} = -\frac{A\delta A \cos MQ}{p}.$$

which gives

$$d\sigma \,.\, \delta MP = \frac{A\delta A}{ABC}\, dS \cos MQ.$$

Now from the equation

$$A\,\frac{X}{ABC} = \iint \frac{\xi\, d\sigma}{MP},$$

we find

$$A\delta\,\frac{X}{ABC} + \frac{X}{ABC}\,\delta A = \iint \frac{\xi\, d\sigma\, \delta\overline{MP}}{MP^2} = -\frac{\delta A}{ABC} \iint \frac{A\xi \cos MQ\, dS}{MP^2}.$$

But

$$-\frac{X}{ABC}\,\delta A = \frac{\delta A}{ABC} \iint \frac{(a - A\xi)\cos MQ\, dS}{MP^2},$$

and consequently

$$A\delta\,\frac{X}{ABC} = \frac{a\delta A}{ABC} \iint \frac{\cos MQ\, dS}{MP^2}.$$

Hence, by the first theorem,

In the case of an exterior point, we have

$$\delta\,\frac{X}{ABC} = 0,$$

i.e. the attractions, in the directions of the axes, of confocal ellipsoids vary as the masses; which is Maclaurin's theorem for the attractions of ellipsoids upon an exterior point.

In the case of an interior point, we have

$$\delta\,\frac{X}{ABC} = -4\pi a\,\frac{a\delta A}{A^2BC},$$

or, taking α, β, γ as the semi-axes of an ellipsoid confocal with the ellipsoid (A, B, C), but exterior to it, and supposing that (X) refers to the ellipsoid (α, β, γ), we have

$$\delta\left(\frac{(X)}{\alpha\beta\gamma}\right) = -4\pi a\,\frac{\delta\alpha}{\alpha^2\beta\gamma}.$$

Now introducing instead of α the new variable θ, such that $\alpha^2 = A^2 + \theta$, we have

$\dfrac{\delta\alpha}{\alpha^2} = \dfrac{\delta\theta}{(A^2+\theta)^{\frac{3}{2}}}$, $\beta = (B+\theta)^{\frac{1}{2}}$, $\gamma = (C^2+\theta)^{\frac{1}{2}}$, and consequently writing d for δ,

$$d\left(\frac{(X)}{\alpha\beta\gamma}\right) = -4\pi a\,\frac{d\theta}{(A^2+\theta)^{\frac{3}{2}}(B^2+\theta)^{\frac{1}{2}}(C^2+\theta)^{\frac{1}{2}}}.$$

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and thence, effecting the integration,

$$\frac{X_{\prime}}{(A^2+\theta_{\prime})^{\frac{1}{2}}(B+\theta_{\prime})^{\frac{1}{2}}(C^2+\theta_{\prime})^{\frac{1}{2}}} - \frac{X}{ABC} = -4\pi a \int_{\theta}^{\theta_{\prime}} \frac{d\theta}{(A^2+\theta)^{\frac{3}{2}}(B^2+\theta)^{\frac{1}{2}}(C^2+\theta)^{\frac{1}{2}}},$$

where $X_{\prime}$ refers to the ellipsoid whose semi-axes are $(A^2+\theta_{\prime})^{\frac{1}{2}}$, $(B^2+\theta_{\prime})^{\frac{1}{2}}$, $(C^2+\theta_{\prime})^{\frac{1}{2}}$. In the case where $\theta_{\prime} = \infty$, we have

$$\frac{X_{\prime}}{(A^2+\theta_{\prime})^{\frac{1}{2}}(B+\theta_{\prime})^{\frac{1}{2}}(C^2+\theta_{\prime})^{\frac{1}{2}}} = 0,$$

and consequently

$$X = 4\pi a\, ABC \int_0^{\infty} \frac{d\theta}{(A^2+\theta)^{\frac{3}{2}}(B^2+\theta)^{\frac{1}{2}}(C^2+\theta)^{\frac{1}{2}}},$$

which is the expression for the attraction, in the direction opposite to that of the axis of x, of the ellipsoid (A, B, C) upon an interior point, the coordinates of which are (a, b, c).

In the case of an exterior point, let $A_{\prime}$, $B_{\prime}$, $C_{\prime}$ be the semi-axes of the confocal ellipsoid passing through the attracted point; so that putting $A_{\prime} = \sqrt{(A^2+\eta)}$, $B_{\prime} = \sqrt{(B^2+\eta)}$, $C_{\prime} = \sqrt{(C^2+\eta)}$, we have

$$\frac{a^2}{A^2+\eta} + \frac{b^2}{B^2+\eta} + \frac{c^2}{C^2+\eta} = 1,$$

the attraction is equal to $\dfrac{ABC}{A_{\prime}B_{\prime}C_{\prime}} \times$ attraction of the ellipsoid which passes through the point, i.e.

$$X = 4\pi a\, ABC \int_{\eta}^{\infty} \frac{d\theta}{(A_{\prime}^2+\theta)^{\frac{3}{2}}(B_{\prime}^2+\theta)^{\frac{1}{2}}(C_{\prime}^2+\theta)^{\frac{1}{2}}},$$

or, putting $\theta - \eta$ instead of θ,

$$X = 4\pi a\, ABC \int_{\eta}^{\infty} \frac{d\theta}{(A^2+\theta)^{\frac{3}{2}}(B^2+\theta)^{\frac{1}{2}}(C^2+\theta)^{\frac{1}{2}}},$$

which is the expression for the attraction upon an exterior point. The formulæ coincide, as they ought to do, in the case of a point upon the surface.

2, Stone Buildings, 9th April, 1855.

165.

On some Geometrical Theorems Relating to a Triangle Circumscribed about a Conic.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 169–175.]

The following investigations were suggested to me by Sir F. Pollock's interesting paper "On a Geometrical Theorem relating to an Equilateral Triangle circumscribed about a Circle," [*Quart. Math. Jour.* t. I. (1857), pp. 167–169].

If on the sides of a triangle ABC, there be taken points α, β, γ, such that $A\alpha$, $B\beta$, $C\gamma$ meet in a point O; and if on each side of the triangle there be taken two points forming with the two angles on the same side an involution having the first-mentioned point on the same side for a double point; then if three of the six points lie in a line, the two lines are said to be harmonically related with respect to the triangle ABC and point O. Call these the lines (r), (s).

The triangle ABC and point O give rise to a determinate conic; viz. the conic touching the sides at the points α, β, γ.

The harmonic of the point O with respect to the triangle is a line (o), which is also the polar of O with respect to the conic. The conic meets this line in two points (as the figure is drawn, imaginary ones), I, J.

Suppose now that the harmonic lines (r), (s) are harmonically related to the points I, J, (i.e. let the lines (r), (s) and the lines through their point of intersection and I, J, be a harmonic pencil, (or, what is the same thing, let I, J, and the points in which the line of junction meets the lines (r), (s) be a harmonic range), then,

Theorem, the point of intersection of the lines (r), (s) lies on the conic.

In order to prove this, take

$$x = 0, \quad y = 0, \quad z = 0, \quad \text{for the equations of } BC,\ CA,\ AB,$$

$$x = y = z, \quad \text{for the point } O,$$

the equations of the harmonic lines will be

$$ax + by + cz = 0,$$

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 0,$$

the equation of the conic is

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0;$$

the equation of the line (o) is

$$x + y + z = 0.$$

The coordinates of the points I, J, are

$$x : y : z = 1 : \omega : \omega^2,$$

and

$$x : y : z = 1 : \omega^2 : \omega,$$

where ω is an imaginary cube root of unity.

The equations of the lines joining the point of intersection of (r), (s) with the points I, J, are

$$\frac{ax + by + cz}{a + b\omega + c\omega^2} = \frac{bcx + cay + abz}{bc + ca\omega + ab\omega^2}$$

and

$$\frac{ax + by + cz}{a + b\omega^2 + c\omega} = \frac{bcx + cay + abz}{bc + ca\omega^2 + ab\omega},$$

and these will form with the lines

$$ax + by + cz = 0,$$

$$bcx + cay + abz = 0,$$

a harmonic pencil, if

$$(a+b\omega+c\omega^2)(bc+ca\omega^2+ab\omega)+(a+b\omega^2+c\omega)(bc+ca\omega+ab\omega^2)=0,$$

i.e. if

$$6abc-(bc^2+b^2c+ca^2+c^2a+ab^2+a^2b)=0;$$

which is, therefore, the condition in order that (r), (s), I, J may be harmonically related.

The coordinates of the point of intersection of (r), (s) are

$$x:y:z=a(b^2-c^2):b(c^2-a^2):c(a^2-b^2);$$

and substituting these values in

$$x^2+y^2+z^2-2yz-2zx-2xy,$$

we see, *à priori*, that the result contains the factor

$$2abc+bc^2+b^2c+ca^2+c^2a+ab^2+a^2b=(b+c)(c+a)(a+b);$$

in fact $b+c=0$ gives $x:y:z=0:b(b^2-a^2):b(a^2-b^2)$, i.e. $x=0$, $y-z=0$, which makes $x^2+y^2+z^2-2yz-2zx-2xy$ vanish; and so for the other factors.

Effecting the substitution, we find

$$x^2+y^2+z^2-2yz-2zx-2xy=(2abc+bc^2+b^2c+ca^2+c^2a+ab^2+a^2b)\times(-6abc+bc^2+b^2c+ca^2+c^2a+ab^2+a^2b),$$

which equals 0, on account of the second factor.

Hence the point of intersection of the lines (r), (s) lies upon the conic. Q.E.D.

Consider, next, a side of the triangle: then, *Theorem*, the following four points on a side of the triangle, viz. the point of contact, the point of intersection with the line (o), and the points of intersection with the harmonic lines (r), (s), form a harmonic range.

In fact, for the side $x=0$ the points in question are given by means of the equations

$$y-z=0,\quad y+z=0,\quad by+cz=0,\quad \frac{y}{b}+\frac{z}{c}=0,$$

and forming the equations

$$y^2-z^2=0,$$
$$y^2+\left(\frac{b}{c}+\frac{c}{b}\right)yz+z^2=0,$$

the theorem is at once seen to be true. Q.E.D.

It is interesting to give the analytical expressions for some other of the points and lines of the figure. The harmonic lines (r), (s), meet in a point of the conic given by the equations

$$\xi : \eta : \zeta = a(b^2 - c^2) : b(c^2 - a^2) : c(a^2 - b^2).$$

This point may be spoken of as the common point of intersection of the harmonic lines with the conic. The foregoing values give

$$\begin{aligned} \xi'' &= \eta - \zeta'' = a\xi\left(\frac{b}{c} + \frac{c}{b}\right), \\ \eta'' - \zeta'' - \xi'' &= b\eta\left(\frac{c}{a} + \frac{a}{c}\right), \\ \zeta'' - \xi'' - \eta'' &= c\zeta\left(\frac{a}{b} + \frac{b}{a}\right), \end{aligned}$$

which may also be written

$$\begin{aligned} \xi'' = (\eta - \zeta)^2 &= a\xi\,\tfrac{1}{bc}(b + c)^2 = QR\xi, \\ \eta'' = (\zeta - \xi)^2 &= b\eta\,\tfrac{1}{ca}(c + a)^2 = RP\eta, \\ \zeta'' = (\xi - \eta)^2 &= c\zeta\,\tfrac{1}{ab}(a + b)^2 = PQ\zeta, \end{aligned}$$

if for shortness

$$\begin{aligned} -\xi + \eta + \zeta &= P, \\ \xi - \eta + \zeta &= Q, \\ \xi + \eta - \zeta &= R, \end{aligned}$$

formulæ which will be presently useful.

Each harmonic line meets the conic in another point, which may be termed the simple point of intersection. For the line $ax + by + cz = 0$, the coordinates of the simple point of intersection are given by $x : y : z = \frac{1}{a}\xi(b + c) : \frac{1}{b}\eta(c + a) : \frac{1}{c}\zeta(a + b)$; or, what is the same thing, by

$$x : y : z = \frac{1}{Pa} : \frac{1}{Qb} : \frac{1}{Rc},$$

and, in like manner, the coordinates of the simple point of intersection of the line

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 0$$

are given by

$$x : y : z = \frac{a}{P} : \frac{b}{Q} : \frac{c}{R}.$$

Hence the equation of the line joining the two simple points of intersection is

$$\begin{vmatrix} x, & y, & z \\ \dfrac{1}{Pa}, & \dfrac{1}{Qb}, & \dfrac{1}{Rc} \\ \dfrac{a}{P}, & \dfrac{b}{Q}, & \dfrac{c}{R} \end{vmatrix} = 0,$$

or expanding and reducing

$$Px + Qy + Rz = 0.$$

The equation of the tangent to the conic at the common point of intersection is evidently

$$Px + Qy + Rz = 0.$$

The last-mentioned lines, together with the harmonic lines (r), (s), viz. the lines

$$ax + by + cz = 0,$$

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 0,$$

may be considered as the sides of an inscribed quadrilateral; the equation of the conic must therefore be expressible in a form in which this is put in evidence: to do this, I first form the equation

$$(ax + by + cz)\Big(\frac{x}{a} + \frac{y}{b} + \frac{z}{c}\Big) = x^2 + y^2 + z^2 + 2\Big(1 - \frac{QR}{bc}\Big)yz + 2\Big(1 - \frac{RP}{ca}\Big)zx + 2\Big(1 - \frac{PQ}{ab}\Big)xy,$$

which may also be written

$$(ax + by + cz)\Big(\frac{x}{a} + \frac{y}{b} + \frac{z}{c}\Big) = x^2 + y^2 + z^2 - 2yz - 2zx - 2xy,$$

where

$$\lambda = \frac{a^2 + b^2 - c^2}{2ab}, \qquad \mu = \frac{b^2 + c^2 - a^2}{2bc}, \qquad \nu = \frac{c^2 + a^2 - b^2}{2ca},$$

and then putting

$$\Delta = P^2 + Q^2 + R^2 - 2QR - 2RP - 2PQ,$$

an equation which gives

$$\begin{aligned} \Delta &= P^2 - 4QR\xi, \\ &= Q^2 - 4RP\eta, \\ &= R^2 - 4PQ\zeta, \end{aligned}$$

and

$$-PQR - 8\xi\eta\zeta = (\xi + \eta + \zeta)\,\Delta,$$

it is easy by means of these relations to verify the identical equation

$$M(ax+by+cz)\Big(\frac{x}{a}+\frac{y}{b}+\frac{z}{c}\Big)+(P\xi x+Q\eta y+R\zeta z)\,(Px+Qy+Rz) = (P^2+Q^2+R^2+2QR+2RP+2PQ)(x^2+y^2+z^2-$$

or, writing for Δ its value, $\Delta = 0$, the equation of the conic takes the form

$$M(ax + by + cz)\Big(\frac{x}{a} + \frac{y}{b} + \frac{z}{c}\Big) - (P\xi x + Q\eta y + R\zeta z)(Px + Qy + Rz) = 0,$$

which is as it should be.

It may be added, that the common point of intersection and the points in which the harmonic lines (r), (s) meet a side of the triangle lie in a conic passing through the points I, J, such that with respect to this conic the point of contact is the pole of the line (o). Thus for the side $x = 0$ the equation of the conic in question is

$$\theta x(y + z) + (1 - \tfrac{1}{2}\theta)\, x^2 + (y + z)\Big(\frac{y}{b} + \frac{z}{c}\Big) = 0,$$

where

$$\theta = \frac{a(b-c)^2(b+c) + 2bc}{bc(a-b)(a-c)} \cdot \frac{(b+c)^2}{2bc}.$$

and similarly for the other two conics.

In the case where the triangle is an equilateral triangle and the line (o) is the line at ∞, the conic becomes a circle, the points I, J, are the circular points at infinity, and lines harmonic in respect to the points I, J, become lines at right angles to each other: the foregoing results agree, therefore, with Sir F. Pollock's Theorem.

166.

NOTE ON THE HOMOLOGY OF SETS.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), p. 178.]

LET L denote a set of any four elements a, b, c, d, and in like manner Λ, L_1, &c. sets of the four elements α, β, γ, δ; a_1, b_1, c_1, d_1, &c.; then we may establish a relation of homology between four sets L, L_1, L_2, L_3, and four other sets Λ, Λ_1, Λ_2, Λ_3; viz. considering the corresponding anharmonic ratios of the different sets, we may suppose a relation of homology between these ratios. Thus considering the set L, to write

$$\begin{aligned} x &= (a-b)(c-d), \\ y &= (a-c)(d-b), \\ z &= (a-d)(b-c), \end{aligned}$$

then $x+y+z=0$ and the anharmonic ratios of the set are $x:y:z$; we may, if we please, take $x:y$ as the anharmonic ratio of the set. And in like manner taking $\xi:\eta$ as the anharmonic ratio of the set α, β, γ, δ, &c., the assumed relation between the sets L, L_1, L_2, L_3 and the sets Λ, Λ_1, Λ_2, Λ_3 will be

$$\begin{vmatrix} x\xi, & x\eta, & y\xi, & y\eta \\ x_1\xi_1, & x_1\eta_1, & y_1\xi_1, & y_1\eta_1 \\ x_2\xi_2, & x_2\eta_2, & y_2\xi_2, & y_2\eta_2 \\ x_3\xi_3, & x_3\eta_3, & y_3\xi_3, & y_3\eta_3 \end{vmatrix} = 0;$$

and it is to be observed, that this relation is independent of the particular ratio $x:y$ which has been chosen as the anharmonic ratio of the set; in fact, if we write $x=-y-z$, $x_1=-y_1-z_1$, &c., then reducing the result by means of an elementary property of determinants, the equation will preserve its original form, but will contain the ratios $y:z$; $\eta:\zeta$, &c., instead of the ratios $x:y$; $\xi:\eta$, &c.

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167.

Apropos of Partitions.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 183–184.]

Let $\Pi(1-x^a)(1-x^b)\ldots(1-x^k)$ (κ factors) and assume that $\left[\dfrac{1}{\Pi(1-x^a)}\right]$ is the part of $\dfrac{1}{\Pi(1-x^a)}$ which involves negative powers of $1-x$, then

$$\text{Coefficient } x^n \text{ in } \left[\frac{1}{\Pi(1-x^a)}\right] = \text{coefficient } x^{n+1} \text{ in } \left(\frac{1}{(1-x)^{n+1}}\Pi\frac{1-x}{1-x^a}\right),$$

which suggests the question of the expansion in powers of z of the function

$$\frac{1}{(1-z)^a}\,\Pi\,\frac{1-z}{1-(1-z)^a}.$$

Now by the definition of Bernoulli's numbers

$$\frac{1}{e^t-1} = \frac{1}{t} - \frac{1}{2} + B_1\,\frac{t}{1.2.3} - B_2\,\frac{t^3}{1.2.3.4.5.6} + B_3\,\frac{t^5}{\ldots} - \&\text{c.},$$

from which it is easy to deduce

$$\frac{t}{1-e^{-t}} = 1 + \tfrac{1}{2}t + \tfrac{1}{12}t^2 - \tfrac{1}{720}t^4 + \&\text{c.},$$

and, writing in this formula $t = a\log(1-z)$, we have

$$\frac{-a\log(1-z)}{1-(1-z)^a} = e^{\frac{1}{2}a\log(1-z)+\frac{1}{12}a^2\log^2(1-z)-\frac{1}{720}a^4\log^4(1-z)+\&\text{c.}},$$

i.e.

$$\frac{z}{1-(1-z)^a} = \frac{1}{a}\,\frac{-z}{\log(1-z)} \cdot e^{-\frac{1}{2}a\log(1-z)+\frac{1}{12}a^2\log^2(1-z)-\frac{1}{720}a^4\log^4(1-z)+\&c.},$$

and putting p_κ for $abc\ldots$ and S_1, $S_2\ldots$ for the sums of the powers, we have, taking the product

$$\Pi\,\frac{z}{1-(1-z)^a} = \frac{1}{p_\kappa}\,e^{\kappa\log\frac{-z}{\log(1-z)}-\frac{1}{2}S_1\log(1-z)+\frac{1}{12}S_2\log^2(1-z)-\frac{1}{720}S_4\log^4(1-z)+\&c.};$$

whence also

$$\frac{1}{(1-z)^q}\,\Pi\,\frac{z}{1-(1-z)^a} = \frac{1}{p_\kappa}\,e^{\kappa\log\frac{-z}{\log(1-z)}-(q+\frac{1}{2}S_1)\log(1-z)+\frac{1}{12}S_2\log^2(1-z)-\frac{1}{720}S_4\log^4(1-z)+\&c.},$$

from which the development may be found.

The index of e is

$$\begin{aligned}
&\left(q+\tfrac{1}{2}-\tfrac{1}{2}S_1\right)z\\
&+\left(\tfrac{1}{2}q+\tfrac{1}{4}-\tfrac{1}{4}S_1+\tfrac{1}{6}S_2-\tfrac{1}{12}\kappa\right)z^2\\
&+\left(\tfrac{1}{3}q+\tfrac{1}{6}-\tfrac{1}{6}S_1+\tfrac{1}{6}S_2-\tfrac{1}{12}\kappa\right)z^3\\
&+\&c.
\end{aligned}$$

and developing the exponential,

1°. The coefficient of z is

$$q+\tfrac{1}{2}S_1-\tfrac{1}{2}(\kappa-2).$$

2°. The coefficient of z^2 is

$$\tfrac{1}{2}\left[q^2+q\{\tfrac{1}{2}S_1-\tfrac{1}{2}(\kappa-3)\}+\tfrac{1}{4}S_1^2-\tfrac{1}{6}S_2-\tfrac{1}{2}(\kappa-3)S_1+\tfrac{1}{12}(\kappa-3)(\kappa-4)\right]$$

and so on.

The peculiarity is the appearance of the factors $\kappa-2$, $\kappa-3$, &c. If we neglect these terms, and consider as well q as a, b, $c\ldots$ to be each of them of the dimension unity, the coefficients will be homogeneous.

168.

A Demonstration of the Fundamental Property of Geodesic Lines.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 185–6.]

This is the translation of a paragraph in M. Bertrand's Memoir "Démonstration géométrique de quelques théorèmes relatifs à la théorie des Surfaces," Liouv. t. XIII. (1848), p. 73, in which he shows that the theorem that the geodesic lines on a surface have at each point their osculating plane normal to the surface is an immediate consequence of Meunier's theorem.

169.

EISENSTEIN'S GEOMETRICAL PROOF OF THE FUNDAMENTAL THEOREM FOR QUADRATIC RESIDUES. (TRANSLATED FROM THE ORIGINAL MEMOIR, CRELLE, T. XXVIII. (1844), WITH AN ADDITION, BY A. CAYLEY.)

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 186–191.]

LET p be a positive odd prime, a the aggregate of all the even numbers $< p$ and > 0, viz. $a = 2, 4, 6 \ldots p-1$; and let q be any integer not divisible by the modulus p; then if r denote the general term of the residues of the multiples qa in respect to the modulus p, it is clear that the numbers of the series the general term of which is $(-)^r r$ will coincide to multiples of p près with the numbers of the series $(-)^a a$; so that we shall have the two congruences

$$q^{\frac{1}{2}(p-1)}\,\Pi a \equiv \Pi r \pmod{p}, \quad \text{and} \quad \Pi a \equiv (-)^{\Sigma r}\Pi r \pmod{p},$$

from which it follows that

$$q^{\frac{1}{2}(p-1)} \equiv (-)^{\Sigma r} \pmod{p}, \quad \text{and therefore} \quad \left(\frac{q}{p}\right) = (-)^{\Sigma r}.$$

Let $E\left\{\dfrac{x}{y}\right\}$ denote the greatest whole number contained in the fraction $\dfrac{x}{y}$, then it is clear that $\Sigma qa = p\Sigma E\left(\dfrac{qa}{p}\right) + \Sigma r$; and since all the a's are even, and $p \equiv 1 \pmod{2}$ it follows that $\Sigma r \equiv \Sigma E\left(\dfrac{qa}{p}\right) \pmod{2}$; and we have, therefore,

$$\left(\frac{q}{p}\right) = (-)^{\Sigma E(qa/p)}.$$

When $q = 2$, the formula gives at once the value of $\left(\dfrac{2}{p}\right)$; when, on the other hand, q is odd, and therefore $q-1$ even, we find, by a simple transformation,

$$\Sigma E\left(\frac{qa}{p}\right) = \Sigma E\left(\frac{a}{p}\right)+\Sigma E\left(\frac{2a}{p}\right)+\Sigma E\left(\frac{(q-1)a}{p}\right)-\tfrac{1}{2}\,E\left(\frac{\frac{1}{2}(q-1)a}{p}\right) \equiv \Sigma E\left(\frac{\frac{1}{2}(q-1)a}{p}\right)\ (\bmod.\ 2);$$

and if the last sum be represented by μ, then also

$$\left(\frac{q}{p}\right) = (-)^{\mu}.$$

Imagine now in a plane, a rectangular system of coordinates (x, y) and the whole plane divided by lines parallel to the axes at distances $= 1$ from each other into squares of the dimension $= 1$. And let the angles which do not lie on the axes of coordinates be called "lattice points."

Take, now, upon any vertical parallel a point corresponding to the ordinate y; then $E(y)$ will denote the number of lattice points which lie between this point and the horizontal axis; and, in like manner, taking upon any horizontal parallel a point corresponding to the abscissa x, then $E(x)$ will denote the number of lattice points which lie between this point and the vertical axis. If, therefore, we draw in the plane a curve the equation of which is $y = \varphi x$, then the sum

$$E\varphi 1 + E\varphi 2 + E\varphi 3 + \&\text{c.}$$

will denote the number of lattice points which lie between the curve and the axis of x, including any below points which lie upon the curve.

Suppose now, to return to the subject, that AB represents the straight line which has for its equation $y = \dfrac{q}{p}x$, where p and q are now assumed to be both of them positive odd primes; and let $AD = FB = \frac{1}{2}p$, $AF = DB = \frac{1}{2}q$, $AC = EG = \frac{1}{2}(p-1)$, $AE = CG = \frac{1}{2}(q-1)$.

Then if μ denote the number of the lattice points between AB and AD as far as the ordinate CG inclusively (these lattice points are distinguished in the figure by the mark [$*$]), by what precedes $\left(\dfrac{q}{p}\right) = (-)^{\mu}$. But since the equation of the straight line AB may also be written $x = \dfrac{p}{q}y$, we have in the same way, if ν denote the number of the lattice points which lie between AB and AF as far as the abscissa EG inclusively (these are distinguished in the figure by the mark [$\circ$]), $\left(\dfrac{p}{q}\right) = (-)^{\nu}$. But the lattice points marked with [$*$] and [$\circ$] taken together, i.e. all the lattice points to the right, and all the lattice points to the left of AB, make up the entire system of lattice points of the rectangle $AEGC$, the number of which is $\frac{1}{2}(p-1)\cdot\frac{1}{2}(q-1)$; consequently, $\mu + \nu = \frac{1}{2}(p-1)\cdot\frac{1}{2}(q-1)$, and therefore

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-)^{\mu+\nu} = (-)^{\frac{p-1}{2}\cdot\frac{q-1}{2}},$$

which is the theorem in question.

It maybe noticed that the foregoing transformation $\Sigma E\left(\dfrac{qa}{p}\right) \equiv \mu$ (mod. 2) may itself be proved by the simple geometrical consideration that $\Sigma E\left(\dfrac{qa}{p}\right)$ is nothing else than the number of lattice points which lie on the even ordinates (those corresponding to $x = 2,\ 4,\ 6 \ldots p-1$) between AB and AD as far as BD, and that each ordinate, between the axes AD and BF exclusively, contains $(q-1)$, i.e. an even number of lattice points; and besides, that the two triangles BAD and ABF are congruent, and that the latter of them stands in the same relation to BF and BD as the former does to AD and AF; the completion of which is left to the reader.

Remark. There are figures for which simple formulæ may be obtained for the number of the interior lattice points. Imagine, for example, a circle, the centre of which lies in the axis of coordinates, and the radius of which is $\sqrt{m}$, then the number S of the lattice points which lie within the circle, including those which lie on the axes, is given by the following formula,

$$S = 1 + 4\{E(m) - E(\tfrac{1}{3}m) + E(\tfrac{1}{5}m) - \&c.\}$$

continued until the series stops of itself. It is easy to see that the equation expresses a relation between the number of lattice points of the circle and the number of lattice points of a segment included between two rectangular hyperbolas. Writing in the formula

$$\frac{S}{m} = \frac{1}{m} + 4\left\{\frac{Em}{m} - \frac{E\frac{1}{3}m}{m} + \frac{E\frac{1}{5}m}{m} - \&c.\right\}$$

$m = \infty$, the left-hand side becomes equal to π, and the right-hand side becomes equal to $4(1 - \frac{1}{3} + \frac{1}{5} - \&c.)$, which gives the formula of Leibnitz. There are similar formulæ for the number of lattice points of a system of sectors of ellipses or hyperbolas; and similar relations exist also in space, and in cases of more than three dimensions. We shall return to this important subject, which has the closest connection with the properties of the higher forms, upon another occasion.

(*Addition by the Translator.*) Eisenstein is now, alas! dead; too soon for the complete development of his various and profound researches in elliptic functions and the theory of numbers; and the promise at the conclusion of the foregoing memoir has not, I believe, been fulfilled. The formula in the Remark must, I think, have been established by geometrical considerations, and would have served to give the number of decompositions of a number into the sum of two squares; but, as I do not perceive how this is to be done, I shall follow a reverse course, and establish the theorem from considerations founded on the theory of numbers. I remark, first, that the number of lattice points in a quadrant of the circle, inclusive of those on the vertical axis, but exclusive of those on the horizontal axis and of the centre, is equal to $E\sqrt{m} + E\sqrt{(m-1)} + E\sqrt{(m-4)} + E\sqrt{(m-9)} + \&c.$; and that this sum, multiplied by four and increased by unity, gives precisely the number of lattice points of the circle, including those on the horizontal and vertical axes. The formula to be established is therefore

$$E\sqrt{m}+E\sqrt{(m-1)}+E\sqrt{(m-4)}+E\sqrt{(m-9)}+\&c. = E(m)-E(\tfrac{1}{3}m)+E(\tfrac{1}{5}m)-E(\tfrac{1}{7}m)+\&c.$$

where, as already noticed, the left-hand side denotes the number of lattice points of a quadrant of the circle, inclusive of those on the vertical axis, but exclusive of those on the horizontal axis and of the centre.

Let X_n be the number of ways in which the integer number n can be expressed as the sum of two squares. {If $n = \alpha^2 + \beta^2$, then if α and β are unequal, and neither of them is zero, this counts as two decompositions, viz. $x = \alpha$, $y = \beta$ or $x = \beta$, $y = \alpha$; but if $\alpha = \beta$, this counts only as a single decomposition; or if either of the numbers α, β, e.g. α, is zero, then, since $y = -\beta$ is excluded, this counts as a single decomposition, $x = 0$, $y = \beta$.} X_n will denote the number of lattice points on the quadrant of the circle radius $\sqrt{n}$. Suppose also that $E\left\{\dfrac{1}{k}\,n\right\}$ stands for unity or zero, according as $\dfrac{1}{k}\,n$ is or is not an integer. Then as m passes through the integer number n, i.e. from a value between $n-1$ and n to a value between n and $n+1$, the left-hand side of the equation is increased by X_n, and the right-hand side of the equation is increased by $E(n) - E(\tfrac{1}{3}n) + E(\tfrac{1}{5}n) - \&c.$ We ought therefore to have

$$X_n = E(n) - E(\tfrac{1}{3}n) + E(\tfrac{1}{5}n) - E(\tfrac{1}{7}n) + \&c.,$$

and conversely from this equation, the original equation will at once follow. The right-hand side denotes, it should be observed, the number of factors of n of the form $\equiv 1$ (mod. 4), less the number of factors of the form $\equiv 3$ (mod. 4). Let $n = 2^\lambda f^\alpha f'^{\alpha'} \ldots g^\beta g'^{\beta'} \ldots$, where f, $f' \ldots$ are odd primes $\equiv 1$ (mod. 4) and g, $g' \ldots$ are odd primes $\equiv 3$ (mod. 4). Consider first the factors g, $g' \ldots$, and forming the product

$$(1 + g + g^2 + \ldots + g^\beta)(1 + g' + g'^2 + \ldots + g'^{\beta'}) \ldots$$

it is easy to see that if all or any one or more of the indices β, $\beta' \ldots$ are odd, then the number of terms of the product which are $\equiv 1$ (mod. 4) is equal to the number of terms of the product which are $\equiv 3$ (mod. 4); but if all the indices β, $\beta' \ldots$ are even, then the number of terms of the first form is greater by unity than the number of terms of the second form. Now the terms of the product $(1 + f + \ldots + f^\alpha)(1 + f' + \ldots + f'^{\alpha'}) \ldots$ are all $\equiv 1$ (mod. 4), and the number of terms is $(1 + \alpha)(1 + \alpha') \ldots$. Hence if all or any one or more of the indices β, $\beta' \ldots$ are odd, then the number of factors of n of the first form less the number of factors of the second form is zero; but if the indices β, $\beta' \ldots$ are all even, the number of factors of the first form less the number of factors of the second form is $(1 + \alpha)(1 + \alpha') \ldots$, i.e. we have

$$E(n) - E(\tfrac{1}{3}n) + E(\tfrac{1}{5}n) - \&\text{c.} = 0, \quad \text{or} \quad = (1 + \alpha)(1 + \alpha') \ldots,$$

according as β, $\beta' \ldots$ are all or any one or more of them odd, or according as they are all of them even. Now it is well known that the number $n = 2^\lambda f^\alpha f'^{\alpha'} \ldots g^\beta g'^{\beta'} \ldots$ does not, in the case of all or any one or more of the indices β, $\beta' \ldots$ being odd, admit of decomposition into two squares, i.e. in this case $X_n = 0$; but if the indices β, $\beta' \ldots$ are all even, then the number n will admit of precisely as many decompositions into two squares as the number $n' = 2^\lambda f^\alpha f'^{\alpha'} \ldots$ (in fact, the only decompositions of n are those obtained from the decompositions of n' by multiplying the roots into the common factor $g^{\beta/2} g'^{\beta'/2} \ldots$), and the number of decompositions of n' is moreover equal to the number of decompositions of $n'' = f^\alpha f'^{\alpha'} \ldots$, which last number is in fact the product of the numbers of decompositions of f^α, $f'^{\alpha'} \ldots$; the number of decompositions of n into two squares (estimated according to the foregoing convention) is thus shown to be, in the case of β, $\beta' \ldots$, all of them even, equal to $(1 + \alpha)(1 + \alpha') \ldots$. And we have therefore in every case

$$X_n = E(n) - E(\tfrac{1}{3}n) + E(\tfrac{1}{5}n) - \&\text{c.},$$

and the principal theorem is thus shown to be true.

6–2

170.

On Schellbach's Solution of Malfatti's Problem.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 222–226.]

The following elegant solution of Malfatti's Problem as applied to spherical triangles is given by Dr Schellbach (*Crelle*, t. XLV. (1853), p. 186); for the reason which will be mentioned I have made a change of notation.

In a spherical triangle ABC to describe three small circles, each of them touching the other two, and also two sides of the triangle.

Let the sides of the triangle be a, b, c, and let x, y, z, be the distances of the points of contact from the adjacent angles of the triangle. Then writing

$$a+b+c=2s,$$

$$a-x=\xi,\quad b-y=\eta,\quad c-z=\zeta,$$

whence

$$a+b+c=2s,$$

and putting also $a=\frac{1}{2}s-l$, $b=\frac{1}{2}s-m$, $c=\frac{1}{2}s-n$,

$$l+m+n=\tfrac{1}{2}s,$$

it is easy to obtain

$$\frac{\cos l\cos\xi\cos\eta}{\cos\frac{1}{2}s}=\frac{\sin l\sin\xi\sin\eta}{\sin\frac{1}{2}s}=1,$$

$$\frac{\cos m\cos\eta\cos\zeta}{\cos\frac{1}{2}s}=\frac{\sin m\sin\eta\sin\zeta}{\sin\frac{1}{2}s}=1,$$

$$\frac{\cos n\cos\zeta\cos\xi}{\cos\frac{1}{2}s}=\frac{\sin n\sin\zeta\sin\xi}{\sin\frac{1}{2}s}=1,$$

from which equations the unknown quantities ξ, η, ζ are to be determined. And the equations may be solved without assuming the existence of the relation $l+m+n=\frac{1}{2}s$.

To solve the equations, let the subsidiary angles λ, μ, ν, be determined by the conditions

$$\frac{\cos\lambda\cos m\cos n}{\cos\frac{1}{2}s} = \frac{\sin\lambda\sin m\sin n}{\sin\frac{1}{2}s} = 1,$$

$$\frac{\cos\mu\cos n\cos l}{\cos\frac{1}{2}s} = \frac{\sin\mu\sin n\sin l}{\sin\frac{1}{2}s} = 1,$$

$$\frac{\cos\nu\cos l\cos m}{\cos\frac{1}{2}s} = \frac{\sin\nu\sin l\sin m}{\sin\frac{1}{2}s} = 1,$$

then it may be shown that

$$\cos(\eta+\zeta) = \frac{\cos\frac{1}{2}(\lambda+\eta-\zeta)}{\cos\frac{1}{2}(\lambda+l)}, \quad \cos(\zeta-\eta) = \frac{\cos\frac{1}{2}(\lambda-\eta+\zeta)}{\cos\frac{1}{2}(\lambda+l)},$$

$$\cos(\zeta+\xi) = \frac{\cos\frac{1}{2}(\mu+\zeta-\xi)}{\cos\frac{1}{2}(\mu+m)}, \quad \cos(\xi-\zeta) = \frac{\cos\frac{1}{2}(\mu-\zeta+\xi)}{\cos\frac{1}{2}(\mu+m)},$$

$$\cos(\xi+\eta) = \frac{\cos\frac{1}{2}(\nu+\xi-\eta)}{\cos\frac{1}{2}(\nu+n)}, \quad \cos(\xi-\eta) = \frac{\cos\frac{1}{2}(\nu-\xi+\eta)}{\cos\frac{1}{2}(\nu+n)}.$$

If we write

$$\tan\tfrac{1}{2}\phi = \tan u\tan v\cot l,$$

$$\tan\tfrac{1}{2}\chi = \tan u\tan v\cot m,$$

$$\tan\tfrac{1}{2}\psi = \tan u\tan v\cot n,$$

then

$$\cos(\lambda-\phi) = \frac{\cos\frac{1}{2}\phi}{\cos u\cos v}\,\lambda,$$

$$\cos(\mu-\chi) = \frac{\cos\frac{1}{2}\chi}{\cos u\cos v}\,\chi,$$

$$\cos(\nu-\psi) = \frac{\cos\frac{1}{2}\psi}{\cos u\cos v}\,\psi,$$

equations which give the values of λ, μ, ν, from which ξ, η, ζ are determined as above.

If we suppose that the sides become indefinitely small, we have the case of a plane triangle, and the equations then are

$$\eta^2+\zeta^2+\frac{2l}{s}\,\eta\zeta = l^2 - l\xi^2,$$

$$\zeta^2+\xi^2+\frac{2m}{s}\,\zeta\xi = m^2 - m\eta^2,$$

$$\xi^2+\eta^2+\frac{2n}{s}\,\xi\eta = n^2 - n\zeta^2.$$

We have here

$$(\eta+\zeta)^2 = \tfrac{1}{s}(s-a)\,yz = (\tfrac{1}{2}s-l)^2 - l\big\{(\tfrac{1}{2}s-y)+(\tfrac{1}{2}s-z)\big\}^2 - \tfrac{1}{s}\big\{(\tfrac{1}{2}s-y)-(\tfrac{1}{2}s-z)\big\}^2$$

$$(\eta-\zeta)^2 = (\tfrac{1}{2}s-y-\tfrac{1}{2}s+z)^2 = (y-z)^2,$$

and consequently

$$\eta^2+\zeta^2=\tfrac{1}{2}\,l^2-2lxl,\quad \eta\zeta=(l-l)(l-l),$$

where if

$$\phi=\frac{2mn}{s},\quad (\lambda-\phi)l=\tfrac{1}{2}l+\tfrac{1}{2}\sqrt{s^2-4m^2}\,\sqrt{s^2-4n^2}-4mn,$$

i.e.

$$\begin{aligned}\lambda&=\frac{2mn}{s}+\frac{1}{s}\sqrt{l^2+\frac{4m^2n^2}{l^2}-m^2-n^2}\\&=\frac{2mn}{s}+\frac{1}{2sl}\sqrt{(s^2-4m^2)(s^2-4n^2)}.\end{aligned}$$

Hence

$$\begin{aligned}\eta^2+\zeta^2&=2nl-\frac{4lmn}{s}-sl+\tfrac{1}{2}\sqrt{s^2-4m^2}\,\sqrt{s^2-4n^2},\\ \eta\zeta&=2nl-sn+\tfrac{1}{2}\sqrt{s^2-4l^2}\,\sqrt{s^2-4n^2},\\ \zeta^2+\xi^2&=2ml-\frac{4lmn}{s}-sm+\tfrac{1}{2}\sqrt{s^2-4n^2}\,\sqrt{s^2-4l^2},\\ \zeta\xi&=2ml-sl+\tfrac{1}{2}\sqrt{s^2-4m^2}\,\sqrt{s^2-4n^2},\\ \xi^2+\eta^2&=2ml-\frac{4lmn}{s}-sn+\tfrac{1}{2}\sqrt{s^2-4l^2}\,\sqrt{s^2-4m^2},\\ \xi\eta&=2ml-sm+\tfrac{1}{2}\sqrt{s^2-4l^2}\,\sqrt{s^2-4m^2},\end{aligned}$$

which is in fact at once deducible from the formulæ in my paper "On a System of Equations connected with Malfatti's Problem and on another Algebraical System," (*Camb. and Dubl. Math. Journ.* t. IV. (1849), p. 270 [79]).

Write now for l, m, n, ξ, η, ζ, their values in terms of a, b, c, x, y, z. We have

$$(\tfrac{1}{2}s-y)^2+(\tfrac{1}{2}s-z)^2+\tfrac{2}{s}(\tfrac{1}{2}s-a)(\tfrac{1}{2}s-y)(\tfrac{1}{2}s-z)=(\tfrac{1}{2}s-a)^2-(\tfrac{1}{2}s-a)x^2,$$

i.e.

$$y^2+z^2-\frac{2}{s}(s-a)yz-2a(y+z)+a^2=0,$$

or reducing

$$\big(y+z-a\big)^2=4\Big(1-\frac{a}{s}\Big)yz,$$

and we have thus three equations:

$$y+z+2\sqrt{1-\frac{a}{s}}\,\sqrt{yz}=a,$$

$$z+x+2\sqrt{1-\frac{b}{s}}\,\sqrt{zx}=b,$$

$$x+y+2\sqrt{1-\frac{c}{s}}\,\sqrt{xy}=c,$$

which are given by Schellbach in the same volume, p. 29; and it was for the sake of facilitating the comparison that the notation has been altered: in the case of the spherical triangle. To solve the system, Schellbach writes

$$a = s\sin^2\phi, \quad b = s\sin^2\chi, \quad c = s\sin^2\psi,$$

reducing the equations to

$$\begin{aligned} y + z + 2\sqrt{yz}\,\cos\phi &= s\sin^2\phi, \\ z + x + 2\sqrt{zx}\,\cos\chi &= s\sin^2\chi, \\ x + y + 2\sqrt{xy}\,\cos\psi &= s\sin^2\psi, \end{aligned}$$

whence, putting

$$\phi + \chi + \psi = 2\sigma,$$

the equations are satisfied by

$$x = s\sin^2(\sigma - \phi), \quad y = s\sin^2(\sigma - \chi), \quad z = s\sin^2(\sigma - \psi),$$

which leads to a simple geometrical construction. And if we substitute for ϕ, χ, ψ, σ, their values, it is easy to obtain

$$x = \tfrac{1}{2}\Big[1 - \sqrt{\Big(1 - \frac{a}{s}\Big)\Big(1 - \frac{b}{s}\Big)\Big(1 - \frac{c}{s}\Big)}\,\Big]$$

$$+ \sqrt{1 - \frac{b}{s}}\,\sqrt{\tfrac{a}{s}}\,\sqrt{\tfrac{b}{s}}\,\sqrt{1 - \tfrac{c}{s}} + \sqrt{1 - \tfrac{c}{s}}\,\sqrt{\tfrac{a}{s}}\,\sqrt{\tfrac{c}{s}}\,\sqrt{1 - \tfrac{b}{s}},$$

$$y = \tfrac{1}{2}\Big[1 - \sqrt{\Big(1 - \frac{a}{s}\Big)\Big(1 - \frac{b}{s}\Big)\Big(1 - \frac{c}{s}\Big)}\,\Big] - \sqrt{1 - \frac{a}{s}}\,\sqrt{1 - \tfrac{b}{s}}\,\sqrt{\tfrac{a}{s}}\,\sqrt{\tfrac{c}{s}}\,\sqrt{1 - \tfrac{c}{s}}\ \ldots,$$

$$z = \tfrac{1}{2}\Big[1 - \sqrt{\Big(1 - \frac{a}{s}\Big)\Big(1 - \frac{b}{s}\Big)\Big(1 - \frac{c}{s}\Big)}\,\Big] - \sqrt{1 - \frac{a}{s}}\,\sqrt{1 - \tfrac{b}{s}}\,\sqrt{1 - \tfrac{c}{s}}\ \ldots,$$

values which are also at once obtained from the formulæ in my paper above referred to. It may be remarked that the above equations for the determination of x, y, z (the distances of the points of contact from the adjacent angles of the triangle) are very similar in form to those given in the same paper for the determination of X, Y, Z, the radii of the inscribed circles.

171.

NOTE ON MR SALMON'S EQUATION OF THE ORTHOTOMIC CIRCLE.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 242–244.]

LET $U_1 = 0$, $U_2 = 0$, $U_3 = 0$ be the equations of three circles, and let V be the functional determinant of U_1, U_2, U_3, the functions being in the first instance made homogeneous by the introduction of a variable z, which is ultimately replaced by unity; then the equation of the circle cutting at right angles the three given circles, or, as it may be called, the orthotomic circle, is $V = 0$. This elegant theorem of Mr Salmon's is connected with the theory developed by Hesse in the memoir, "Über die Wendepuncte der Curven dritter Ordnung," *Crelle*, t. XXVIII. (1844), p. 97).

In fact, let $U_1 = 0$, $U_2 = 0$, $U_3 = 0$ be the equations of three conics, the locus of a point such that its polars with respect to each of these conics, or indeed with respect to any conic having for its equation $\lambda U_1 + \mu U_2 + \nu U_3$ (where λ, μ, ν are arbitrary), pass through the same point, is a curve of the third order $V = 0$, where V is the functional determinant of U_1, U_2, U_3.

Conversely, if the curve of the third order $V = 0$ be given, and U be a function of the third order, such that the functional determinant of $\frac{dU}{dx}$, $\frac{dU}{dy}$, $\frac{dU}{dz}$, or what is the same thing, the "Hessian" of the function U is equal to V, a condition which may be written $V = H(U)$, then we may take for the conics any three conics the equations of which are of the form $\lambda\frac{dU}{dx} + \mu\frac{dU}{dy} + \nu\frac{dU}{dz} = 0$. The equation $V = H(U)$ affords the means of determining U; in fact, we shall have $U = aV + bH(V)$, where a and b are constants to be determined. This gives $H(U) = H(aV + bH(V)) = AV + BH(V)$, where A and B are given functions of a, b (a practical method of determining these functions was first given in Aronhold's memoir, "Zur Theorie der homogenen Functionen dritten Grades von zwei Variabeln," *Crelle*, t. XXXIX. (1850), pp. 140–159); and we have therefore

$V = AV + BH(V)$, i.e. $A = 1$, $B = 0$: the latter equation determines, what is alone important, the ratio $b : a$; the equation is of the third order, so that there are in general three distinct solutions $U = aV + BH(V) = 0$.

In the particular case in which the curves of the third order $V = 0$ is made up of a line $P = 0$ and a conic $W = 0$, i.e. where $V = PW = 0$, the curve $H(V) = 0$ is made up of the same line $P = 0$ and of a conic having double contact with the conic $W = 0$ at the point of intersection with the line $P = 0$, i.e. $H(PW) = P(lW + mP^2)$. And $U = aPW + bH(PW)$ is consequently a function of the same form, i.e. the cubic $U = 0$ is made up of the line $P = 0$ and of a conic having double contact with the conic $W = 0$ at its points of intersection with the line $P = 0$. We may therefore write $U = P(fW + gP^2)$, and forming with this value the equation $V = H(U)$ it may be noticed that, owing to the occurrence of a special factor which may be rejected, the resulting equation $G = 0$ gives only a single value for the ratio $f : g$. Forming from the value $U = P(fW + gP^2)$, the equation $\lambda\frac{dU}{dx} + \mu\frac{dU}{dy} + \nu\frac{dU}{dz} = 0$, the equation thus obtained will be of the form $W + 2PQ = 0$, which is the equation of a conic passing through the points of intersection of the line and conic $P = 0$, $W = 0$, and besides intersecting the conic $W = 0$ in two other points. And it may also be shown that the four points of intersection, (i.e. the points given by the equations $W = 0$, $W + 2PQ = 0$), the pole of the line $P = 0$ with respect to the conic $W = 0$, and the pole of this same line with respect to the conic $W + 2PQ = 0$, lie all six in the same conic. We see, therefore, that, given a curve of the third order, the aggregate of a line $P = 0$ and a conic $W = 0$, as the locus of the point such that its polars, with respect to three several conics (or a system depending on three conics), meet in a point, each conic of the system is a conic passing through the points of intersection of the line $P = 0$ and conic $W = 0$, and, moreover, such that the four points of intersection with the conic $W = 0$, and the poles of the line $P = 0$ with respect to the conic of the system, and with respect to the conic $W = 0$, lie all six in the same conic. In the particular case in which the line and conic $P = 0$, $W = 0$ are the line at ∞ and a circle, each conic of the system is a circle such that its points of intersection with the circle $W = 0$ and the centres of the two circles lie in a circle, i.e. the conics are circles cutting at right angles the circle $W = 0$, which agrees with Mr Salmon's theorem.

To verify the assumed theorems in the case of the curve of the third order $V = PW = 0$, we may take

$$P = \alpha x + \beta y + \gamma z = 0$$

for the equation of the line, and

$$W = x^2 + y^2 + z^2 = 0$$

for the equation of the conic. I write, for greater convenience, $U = P(\frac{1}{2}fW + \frac{1}{2}gP^2)$; the Hessian of this is

$$\begin{vmatrix} f(P+2\alpha x) + g\alpha^2 P, & f(\beta x + \alpha y) + g\alpha\beta P, & f(\alpha z + \gamma x) + g\gamma\alpha P \\ f(\beta x + \alpha y) + g\alpha\beta P, & f(P + 2\beta y) + g\beta^2 P, & f(\gamma y + \beta z) + g\beta\gamma P \\ f(\alpha z + \gamma x) + g\gamma\alpha P, & f(\gamma y + \beta z) + g\beta\gamma P, & f(P + 2\gamma z) + g\gamma^2 P \end{vmatrix},$$

which is equal to

$$f^3P\big(4P^2-(\alpha^2+\beta^2+\gamma^2)\,W\big)+f^2gP^3(\alpha^2+\beta^2+\gamma^2)P^2,$$

i.e. we must have $4f+g(\alpha^2+\beta^2+\gamma^2)=0$, or putting $g=-24$ and $\therefore f=6(\alpha^2+\beta^2+\gamma^2)$, we have

$$U=P\big(3(\alpha^2+\beta^2+\gamma^2)\,W-4P^2\big).$$

Forming the function $\lambda\dfrac{dU}{dx}+\mu\dfrac{dU}{dy}+\nu\dfrac{dU}{dz}$, and dividing by the constant factor $3(\alpha^2+\beta^2+\gamma^2)(\lambda\alpha+\mu\beta+\nu\gamma)$, we have for the equation of any one of the conics

$$W+2\Big[\frac{2(\lambda x+\mu y+\nu z)}{\lambda\alpha+\mu\beta+\nu\gamma}-\frac{4P}{\alpha^2+\beta^2+\gamma^2}\Big]\,P=0,$$

which may be written under the form $W+2PQ=0$, where

$$Q=ax+by+cz=\frac{\lambda x+\mu y+\nu z}{\lambda\alpha+\mu\beta+\nu\gamma}-\frac{2(\alpha x+\beta y+\gamma z)}{\alpha^2+\beta^2+\gamma^2}.$$

We have therefore $a\alpha+b\beta+c\gamma=1-2=-1$, i.e. $a\alpha+b\beta+c\gamma+1=0$. And there is no difficulty in showing that, given the two conics

$$x^2+y^2+z^2=0,$$

$$x^2+y^2+z^2+2(\alpha x+\beta y+\gamma z)(ax+by+cz)=0,$$

the condition in order that the four points of intersection and the poles, with respect to each conic, of the line $\alpha x+\beta y+\gamma z=0$, may lie in a conic is precisely this equation $a\alpha+b\beta+c\gamma+1=0$.

172.

NOTE ON THE LOGIC OF CHARACTERISTICS.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 257–259.]

THE conditions in order that an equation of the sixth degree

$$(a, b, c, d, e, f, g)\,(x, y)^6 = 0$$

may have five of its roots equal are

$$\begin{aligned}
A &= ae - 4bd + 3c^2 = 0,\\
B &= af - 3be + 2cd = 0,\\
C &= bf - 4ce + 3d^2 = 0,\\
D &= ag - 9ce + 8d^2 = 0,\\
E &= bg - 3cf + 2de = 0,\\
F &= cg - 4df + 3e^2 = 0,
\end{aligned}$$

equivalent of course to four relations between the coefficients: among the connections of these equations are

$$fA - eB - bF + cE = 0,$$

$$(3e^2 - 2df)A - 2deB - ecD + 2cdE + (3c^2 - 2bd)F = 0.$$

The system is one of the tenth order. To verify this, I write first

$$(A, B, C, F) = (A, B, C, F, cE) = (A, B, C, F, c) + (A, B, C, F, E),$$

i.e. the equations $A = 0$, $B = 0$, $C = 0$, $F = 0$ imply (by the first of the connectives) the additional equation $cE = 0$, viz. the system $A = 0$, $B = 0$, $C = 0$, $F = 0$, $cE = 0$, or what is the same thing, one of the systems $A = 0$, $B = 0$, $C = 0$, $F = 0$, $c = 0$, and $A = 0$, $B = 0$, $C = 0$, $F = 0$, $E = 0$.

We have in like manner

$$(A, B, C, F, E) = (A, B, C, F, E, ceD) = (A, B, C, F, E, D)$$

since (A, B, C, F, E, c) and (A, B, C, F, E, e) respectively vanish as being each of them equivalent not to four but to five relations, and therefore as not adding to the order of the system.

Again,

$$\begin{aligned}(A, B, C, c) &= (A, B, C, c, bF)\\ &= (A, B, C, c, b) + (A, B, C, c, F)\\ &= (ae, af, d^2, b, c) + (A, B, C, c, F).\end{aligned}$$

But here

$$\begin{aligned}(ae, af, d^2, b, c) &= (e, af, d^2, b, c) + d^2(a, af, d, b, c)\\ &= d^2(a, af, d, b, c),\end{aligned}$$

(for (e, af, d^2, b, c) vanishes as being equivalent to five relations, and therefore as not adding to the order of the system),

$$= d(a, a, d^2, b, c) + d(a, f, d^2, b, c)$$

$$= d(a, d^2, b, c),$$

since, (a, f, d^2, b, c) vanishes for the above-mentioned reason.

Hence

$$(A, B, C, c) = d(a, d^2, b, c) + (A, B, C, c, F).$$

We have consequently

$$\begin{aligned}(A, B, C, D, E, F) &= (A, B, C, E, F)\\ &= (A, B, C, F) - (A, B, C, F, c)\\ &= (A, B, C, F) - \big\{(A, B, C, c) - (a, b, c, d^2)\big\},\end{aligned}$$

which may be thus interpreted: the system (A, B, C, c) contains the system (a, b, c, d^2), or what is the same thing, contains twice-over the system (a, b, c, d). Discarding this contained system, the remainder of the system (A, B, C, c) is contained in the system (A, B, C, F), and the residue of the last-mentioned system is the system (A, B, C, D, E, F), i.e. the system represented by the equations which express the equality of five roots of the given equation of the sixth degree.

It follows immediately that the order of the system (A, B, C, D, E, F) is $16 - (8 - 2) = 10$, i.e. that the system is, as above stated, one of the tenth order. The preceding is, I think, a good example of the kind of reasoning to be employed in what Mr Sylvester has most happily termed the Logic of Characteristics.

173.

On Laplace's Method for the Attraction of Ellipsoids.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 285–300.]

[The method referred to is that given in Book III. of the *Mécanique Céleste*, Ed. I. 1798.] Let the equation of the surface of the ellipsoid be

$$lx^2 + my^2 + nz^2 - k = 0,$$

and take a, b, c as the coordinates of the attracted point, which is supposed to be exterior to the surface. Imagine an indefinitely thin cone, having the attracted point for vertex, and intersecting the surface of the ellipsoid; let ξ, η, ζ be the direction-cosines of the cone, and dS its spherical angle. Then if for a moment r denote the radius vector corresponding to a point within the ellipsoid, the element of the mass within the cone is $r^2\, dr\, dS$; and we may thence, by an integration with respect to r, find the attractions parallel to the axes (and tending towards the centre) and the potential of the mass within the cone, viz. if r', r'' be the values of r at the surface of the ellipsoid, the attractions are

$$(r'' - r')\,\xi\, dS, \quad (r'' - r')\,\eta\, dS, \quad (r'' - r')\,\zeta\, dS,$$

and the potential is

$$\tfrac{1}{2}\,(r''^2 - r'^2)\, dS.$$

And putting for shortness

$$L = l\xi^2 + m\eta^2 + n\zeta^2,$$

$$I = la\xi + mb\eta + nc\zeta,$$

$$P = la^2 + mb^2 + nc^2 - k,$$

$$R = I^2 - PL,$$

then r', r'' will be the roots of the equation $Lr^2 - 2Ir + P = 0$, and we have consequently $r' = \dfrac{I - \sqrt{R}}{L}$, $r'' = \dfrac{I + \sqrt{R}}{L}$. We have, therefore, for the attractions parallel to the axes (and tending towards the centre) and for the potential of the entire ellipsoid the well-known expressions

$$A = 2\int dS\, \frac{\xi\sqrt{R}}{L},$$

$$B = 2\int dS\, \frac{\eta\sqrt{R}}{L},$$

$$C = 2\int dS\, \frac{\zeta\sqrt{R}}{L},$$

and

$$V = 2\int dS\, \frac{I\sqrt{R}\, dS}{L^2},$$

where the integrations extend over the spherical angle of the circumscribed cone $R = 0$.

We have moreover, by the general theory of attractions,

$$A = -\frac{dV}{da},\quad B = -\frac{dV}{db},\quad C = -\frac{dV}{dc}.$$

Since at the limits of the integration the quantity under the integral sign vanishes, it is easy to see that the first differentials of V, A, B, C with respect to any of the quantities a, b, c, l, m, n, k, may be found by simply differentiating under the integral sign without its being necessary to pay attention to the variation of the limits, so that, for instance, $\dfrac{dV}{da} = 2\int dS\, \dfrac{d}{da}\, \dfrac{I\sqrt{R}}{L}$. It is proper to remark that the expressions thus obtained for the attractions A, B, C, are of a different form from the foregoing expressions for the same quantities. Laplace writes

$$F = aA + bB + cC,$$

and he remarks that it may be shown by differentiation that the quantities B, C, F, V, are connected by an equation which (writing k for k^2, and $\dfrac{m}{l}$, $\dfrac{n}{l}$ for m, n, in the notation of equation of the ellipsoid being with Laplace $x^2 + my^2 + nz^2 = k^2$) becomes, in the notation of the present memoir,

$$\begin{aligned}\{l(a^2+b^2+c^2) - k\}\left(\frac{dV}{dk} - \frac{dF}{dk}\right) + V - F&\\ + \left(1 - \frac{l}{m}\right) b \left(\frac{dF}{db} - \tfrac{1}{2}\,\frac{dV}{db} - B\right)&\\ + \left(1 - \frac{l}{n}\right) c \left(\frac{dF}{dc} - \tfrac{1}{2}\,\frac{dV}{dc} - C\right)&\\ -(m-l)\,\frac{dF}{dm} - (n-l)\,\frac{dF}{dn} = 0.&\end{aligned}$$

I write this equation in the form

$$\begin{aligned}(la^2+mb^2+nc^2-k)\left(\frac{dV}{dk}-\frac{dF}{dk}\right)+V-F\\+(l-l)\left\{-a^2\left(\frac{dV}{dk}-\frac{dF}{dk}\right)+\frac{a}{l}\left(\frac{dF}{da}-\tfrac{1}{2}\frac{dV}{da}-A\right)-\frac{dF}{dl}\right\}\\+(m-l)\left\{-b^2\left(\frac{dV}{dk}-\frac{dF}{dk}\right)+\frac{b}{m}\left(\frac{dF}{db}-\tfrac{1}{2}\frac{dV}{db}-B\right)-\frac{dF}{dm}\right\}\\+(n-l)\left\{-c^2\left(\frac{dV}{dk}-\frac{dF}{dk}\right)+\frac{c}{n}\left(\frac{dF}{dc}-\tfrac{1}{2}\frac{dV}{dc}-C\right)-\frac{dF}{dn}\right\}\\=0;\end{aligned}$$

and I remark that this equation may be broken up into two equations, each of which separately is satisfied; these equations are

$$\begin{aligned}-k\left(\frac{dV}{dk}-\frac{dF}{dk}\right)+(V-F)-l\frac{dF}{dl}-m\frac{dF}{dm}-n\frac{dF}{dn}\\+a\left(\frac{dF}{da}-\tfrac{1}{2}\frac{dV}{da}-A\right)+b\left(\frac{dF}{db}-\tfrac{1}{2}\frac{dV}{db}-B\right)+c\left(\frac{dF}{dc}-\tfrac{1}{2}\frac{dV}{dc}-C\right)=0,\end{aligned}$$

and

$$\begin{aligned}(a^2+b^2+c^2)\left(\frac{dV}{dk}-\frac{dF}{dk}\right)-\frac{dF}{dl}-\frac{dF}{dm}-\frac{dF}{dn}\\+\frac{a}{l}\left(\frac{dF}{da}-\tfrac{1}{2}\frac{dV}{da}-A\right)+\frac{b}{m}\left(\frac{dF}{db}-\tfrac{1}{2}\frac{dV}{db}-B\right)+\frac{c}{n}\left(\frac{dF}{dc}-\tfrac{1}{2}\frac{dV}{dc}-C\right)=0.\end{aligned}$$

It may be added that the functions under the integral signs, and consequently the integrals, are all of them homogeneous of the degree zero in l, m, n, k. The thing to be verified is that the foregoing two equations are satisfied by the functions under the integral signs, independently of the integrations, in fact by the values

$$\left\{\begin{aligned}&A=\frac{\xi\sqrt{R}}{L},\ B=\frac{\eta\sqrt{R}}{L},\ C=\frac{\zeta\sqrt{R}}{L},\\&V=\frac{I\sqrt{R}}{L},\\&F=\frac{(a\xi+b\eta+c\zeta)\sqrt{R}}{L}.\end{aligned}\right.$$

We find by differentiating these values, and after a few obvious reductions,

$$\frac{dV}{da}=l\xi\left(\frac{\sqrt{R}}{L^2}+\frac{I}{L^2\sqrt{R}}\right)+I\sigma\left(\frac{-I}{L\sqrt{R}}\right),$$

$$\frac{dV}{dl}=a\xi\left(\frac{\sqrt{R}}{L^2}+\frac{I}{L^2\sqrt{R}}\right)+a^2\left(\frac{-I}{2L\sqrt{R}}\right)+\mathfrak{F}\left(-\frac{3I\sqrt{R}}{2L^2}-\frac{I^3}{2L^2\sqrt{R}}\right),$$

$$\frac{dV}{dk}=\frac{I}{2L\sqrt{R}},$$

$$\frac{dF}{da}\cdot\frac{\sqrt{R}}{L} = (a\xi+b\eta+c\zeta)\left\{l\xi\left(\frac{I}{L\sqrt{R}}+l\sigma\left(\frac{-1}{\sqrt{R}}\right)\right)\right\},$$

$$\frac{dF}{dl} = (a\xi+b\eta+c\zeta)\left\{a\xi\left(\frac{I}{L\sqrt{R}}+a^2\left(\frac{-1}{2\sqrt{R}}\right)\right)+\mathfrak{F}\left(-\frac{\sqrt{R}}{2L}-\frac{I^2}{2L\sqrt{R}}\right)\right\},$$

$$\frac{dF}{dk} = (a\xi+b\eta+c\zeta)\left\{\frac{1}{2L\sqrt{R}}\right\},$$

values which give, as they should do,

$$l\,\frac{dF}{dl}+m\,\frac{dF}{dm}+n\,\frac{dF}{dn}+k\,\frac{dF}{dk}=0.$$

Hence forming the values of the different parts of the first equation,

$$-k\left(\frac{dV}{dk}-\frac{dF}{dk}\right)+V-F-l\,\frac{dF}{dl}-m\,\frac{dF}{dm}-n\,\frac{dF}{dn}$$

$$=\frac{I\sqrt{R}}{L^2}-\frac{kI}{2L\sqrt{R}}+(a\xi+b\eta+c\zeta)\left\{-\frac{\sqrt{R}}{L}+\frac{k}{L\sqrt{R}}\right\},$$

$$a\left(\frac{dF}{da}-A\right)+b\left(\frac{dF}{db}-B\right)+c\left(\frac{dF}{dc}-C\right)=(a\xi+b\eta+c\zeta)\left\{-\frac{\sqrt{R}}{L}+\frac{k}{L\sqrt{R}}\right\},$$

$$-\tfrac{1}{2}\left(a\,\frac{dV}{da}+b\,\frac{dV}{db}+c\,\frac{dV}{dc}\right)=-\frac{I\sqrt{R}}{L^2}+\frac{kI}{2L\sqrt{R}},$$

which satisfy the first equation.

And in like manner for the second equation,

$$-(a^2+b^2+c^2)\left(\frac{dV}{dk}-\frac{dF}{dk}\right)=-(a^2+b^2+c^2)\,\frac{I}{2L\sqrt{R}}+(a^2+b^2+c^2)(a\xi+b\eta+c\zeta)\,\frac{1}{2L\sqrt{R}},$$

$$\frac{a}{l}\left(\frac{dF}{da}-A\right)+\frac{b}{m}\left(\frac{dF}{db}-B\right)+\frac{c}{n}\left(\frac{dF}{dc}-C\right)$$

$$=(a\xi+b\eta+c\zeta)\,\frac{I}{L\sqrt{R}}-(a^2+b^2+c^2)(a\xi+b\eta+c\zeta)\,\frac{1}{2\sqrt{R}},$$

$$-\tfrac{1}{2}\left(\frac{a}{l}\,\frac{dV}{da}+\frac{b}{m}\,\frac{dV}{db}+\frac{c}{n}\,\frac{dV}{dc}\right)=-(a^2+b^2+c^2)(a\xi+b\eta+c\zeta)\,\frac{I}{L\sqrt{R}}+(a\xi+b\eta+c\zeta)\left(\frac{\sqrt{R}}{2L^2}+\frac{I^2}{2L^2\sqrt{R}}\right),$$

$$-\frac{dF}{dl}-\frac{dF}{dm}-\frac{dF}{dn}=-(a\xi+b\eta+c\zeta)\,\frac{I}{2L\sqrt{R}}+(a^2+b^2+c^2)(a\xi+b\eta+c\zeta)\,\frac{1}{2\sqrt{R}}$$

$$+(a\xi+b\eta+c\zeta)\left(\frac{\sqrt{R}}{2L^2}+\frac{I^2}{2L^2\sqrt{R}}\right),$$

which satisfy the second equation.

Now considering V, F, A, B, C as standing for the definite integrals, we may replace A, B, C by the differential coefficients of V, and retaining F for shortness to stand for

$$F = -a\,\frac{dV}{da} - b\,\frac{dV}{db} - c\,\frac{dV}{dc}\ ;$$

the first equation becomes

$$-k\left(\frac{dV}{dk} - \frac{dF}{dk}\right) + V - F - l\,\frac{dF}{dl} - m\,\frac{dF}{dm} - n\,\frac{dF}{dn}$$

$$+a\left(\frac{dF}{da} + \tfrac{1}{2}\,\frac{dV}{da}\right) + b\left(\frac{dF}{db} + \tfrac{1}{2}\,\frac{dV}{db}\right) + c\left(\frac{dF}{dc} + \tfrac{1}{2}\,\frac{dV}{dc}\right) = 0;$$

and the second equation becomes

$$-(a^2+b^2+c^2)\left(\frac{dV}{dk} - \frac{dF}{dk}\right) - \frac{dF}{dl} - \frac{dF}{dm} - \frac{dF}{dn}$$

$$+\frac{a}{l}\left(\frac{dF}{da} + \tfrac{1}{2}\,\frac{dV}{da}\right) + \frac{b}{m}\left(\frac{dF}{db} + \tfrac{1}{2}\,\frac{dV}{db}\right) + \frac{c}{n}\left(\frac{dF}{dc} + \tfrac{1}{2}\,\frac{dV}{dc}\right) = 0.$$

If we put as before $V = \dfrac{I\sqrt{R}}{L}$, the preceding values of the differential coefficients of V give $F = \dfrac{2I\sqrt{R}}{L^2} + \dfrac{kI}{L\sqrt{R}}$, or as we may write it $F = -2V + W$, where $W = \dfrac{kI}{L\sqrt{R}}$.

I put then for the moment

$$\begin{cases} V = \dfrac{I\sqrt{R}}{L}, \quad W = \dfrac{kI}{L\sqrt{R}}, \\ F = -2V + W. \end{cases}$$

It should be remarked that there is nothing in what has preceded which tends to show that these values must satisfy the differential equations. The definite integrals must, of course, satisfy as before the equations, but it does not follow that the equations are satisfied by the elements separately. And in fact only the first equation is so satisfied; the second equation is not satisfied. To verify this I form the differential coefficients

$$\frac{dW}{da} = l\xi\left(\frac{k}{L\sqrt{R}} - \frac{kl}{LR\sqrt{R}}\right) + l\sigma\left(\frac{kl}{R\sqrt{R}}\right),$$

$$\frac{dW}{dk} = \frac{I}{L\sqrt{R}} - \frac{kl}{2LR\sqrt{R}},$$

$$\frac{dW}{dl} = a\xi\left(\frac{k}{L\sqrt{R}} - \frac{kl}{LR\sqrt{R}}\right) + \mathfrak{F}\left(\frac{-3kl}{2L^2\sqrt{R}} + \frac{kI^2}{2L^2R\sqrt{R}}\right) + a^2\,\frac{kl}{2R\sqrt{R}}.$$

We have as before

$$l\,\frac{dF}{dl} + m\,\frac{dF}{dm} + n\,\frac{dF}{dn} + k\,\frac{dF}{dk} = 0,$$

and forming the expressions for the different parts of the first equation,

$$-k\left(\frac{dV}{dk} - \frac{dF}{dk}\right) + V - F - l\,\frac{dF}{dl} - m\,\frac{dF}{dm} - n\,\frac{dF}{dn}$$

$$= -5k\,\frac{dV}{dk} + 5k\,\frac{dW}{dk} + 3V - W = -\frac{3I\sqrt{R}}{L^2} - \frac{3kl}{2L\sqrt{R}} - \frac{kI}{L\sqrt{R}},$$

$$a\left(\frac{dF}{da} + \tfrac{1}{2}\,\frac{dV}{da}\right) + \&c. =$$

$$-\tfrac{3}{2}\left(a\,\frac{dV}{da} + b\,\frac{dV}{db} + c\,\frac{dV}{dc}\right) + \left(a\,\frac{dW}{da} + b\,\frac{dW}{db} + c\,\frac{dW}{dc}\right) = -\frac{3I\sqrt{R}}{L^2} + \frac{3kl}{2L\sqrt{R}},$$

and

$$-\tfrac{1}{2}\left(a\,\frac{dV}{da} + b\,\frac{dV}{db} + c\,\frac{dV}{dc}\right) = -\frac{kl}{L\sqrt{R}},$$

which show that the first equation is satisfied.

Proceeding in the same manner with the second equation,

$$-(a^2+b^2+c^2)\left(\frac{dV}{dk} - \frac{dF}{dk}\right) =$$

$$-(a^2+b^2+c^2)\left(3\,\frac{dV}{dk} - \frac{dW}{dk}\right) = (a^2+b^2+c^2)\left(-\frac{I}{L\sqrt{R}} - \frac{kl}{2RL\sqrt{R}}\right),$$

$$\frac{a}{l}\left(\frac{dF}{da} - A\right) + \frac{b}{m}\left(\frac{dF}{db} - B\right) + \frac{c}{n}\left(\frac{dF}{dc} - C\right) =$$

$$-\tfrac{1}{2}\left(\frac{a}{l}\,\frac{dV}{da} + \frac{b}{m}\,\frac{dV}{db} + \frac{c}{n}\,\frac{dV}{dc}\right) + \left(\frac{a}{l}\,\frac{dW}{da} + \frac{b}{m}\,\frac{dW}{db} + \frac{c}{n}\,\frac{dW}{dc}\right) = (a^2+b^2+c^2)\left(-\frac{3\sqrt{R}}{2L^2} - \frac{3I^2}{2L^2\sqrt{R}}\right);$$

$$-\frac{dF}{dl} - \frac{dF}{dm} - \frac{dF}{dn} = 2\left(\frac{dV}{dl} + \frac{dV}{dm} + \frac{dV}{dn}\right) - \left(\frac{dW}{dl} + \frac{dW}{dm} + \frac{dW}{dn}\right),$$

and

$$2\left(\frac{dV}{dl}+\frac{dV}{dm}+\frac{dV}{dn}\right)=(a^2+b^2+c^2)\left(\frac{-1}{L\sqrt{R}}\right)+(a\xi+b\eta+c\zeta)\left(\frac{2\sqrt{R}}{L^2}+\frac{2I^2}{L^2\sqrt{R}}\right)-\frac{3I\sqrt{R}}{L^2}-\frac{I^2}{L^2\sqrt{R}},$$

$$-\left(\frac{dW}{dl}+\frac{dW}{dm}+\frac{dW}{dn}\right)=(a^2+b^2+c^2)\left(\frac{-kI}{2RL\sqrt{R}}\right)+(a\xi+b\eta+c\zeta)\left(\frac{-k}{L\sqrt{R}}+\frac{kI^2}{LR\sqrt{R}}\right)$$

$$+\frac{3kl}{2L\sqrt{R}}+\frac{kI^2}{2L\sqrt{R}\,R};$$

and the value of the left-hand side of the second equation is therefore,

$$(a\xi+b\eta+c\zeta)\left(\frac{\sqrt{R}}{2L^2}+\frac{3I^2}{2L^2\sqrt{R}}\right)-\frac{3I\sqrt{R}}{L^2}-\frac{I^2}{2L^2\sqrt{R}}+\frac{3kl}{2L\sqrt{R}}-\frac{kl^2}{2L\sqrt{R}\,R},$$

which is not equal to zero, or the second equation is not satisfied.

I consider again V, F as denoting the definite integrals, and I eliminate $\frac{dV}{dl}$, $\frac{dF}{dl}$ by means of the equations

$$l\frac{dV}{dl}+m\frac{dV}{dm}+n\frac{dV}{dn}+k\frac{dV}{dk}=0,$$

$$l\frac{dF}{dl}+m\frac{dF}{dm}+n\frac{dF}{dn}+k\frac{dF}{dk}=0.$$

The first equation thus becomes

$$l\frac{dV}{dl}=\frac{dV}{dm}+n\frac{dV}{dn}-2\left(l\frac{dF}{dl}+m\frac{dF}{dm}+n\frac{dF}{dn}\right)+V-F$$

$$+\tfrac{1}{2}\left(a\frac{dV}{da}+b\frac{dV}{db}+c\frac{dV}{dc}\right)+\left(a\frac{dF}{da}+b\frac{dF}{db}+c\frac{dF}{dc}\right)=0,$$

and the second equation becomes

$$(a^2+b^2+c^2)\frac{1}{l}\left(l\frac{dV}{dl}+m\frac{dV}{dm}+n\frac{dV}{dn}\right)-(a^2+b^2+c^2)\frac{1}{l}\left(l\frac{dF}{dl}+m\frac{dF}{dm}+n\frac{dF}{dn}\right)$$

$$+\tfrac{1}{2}\left(\frac{a}{l}\frac{dV}{da}+\frac{b}{m}\frac{dV}{db}+\frac{c}{n}\frac{dV}{dc}\right)+\left(\frac{a}{l}\frac{dF}{da}+\frac{b}{m}\frac{dF}{db}+\frac{c}{n}\frac{dF}{dc}\right)-\left(\frac{dF}{dl}+\frac{dF}{dm}+\frac{dF}{dn}\right)=0\,;$$

and it will be remembered that in these equations

$$F=-a\frac{dV}{da}-b\frac{dV}{db}-c\frac{dV}{dc}.$$

The first equation (it is easy to perceive) shows merely that V is made up of terms separately homogeneous in a, b, c, and in l, m, n, and such that the degrees in the two sets respectively being κ, λ, then $\lambda=\frac{1}{2}(\kappa-2)$. In fact V being a function of the form in question, if we attend only to the term the degrees of which are κ, $\frac{1}{2}\lambda$,

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then, by the properties of homogeneous functions, $F = -V$, and the first equation is satisfied if only

$$\lambda + 2\lambda\kappa + 1 + \kappa + \tfrac{1}{2}\kappa - \kappa^2 = 0,$$

i.e. if $\lambda(2\kappa+1) = \kappa^2 - \frac{3}{2}\kappa - 1 = \frac{1}{2}(\kappa-2)(2\kappa+1)$ or $\lambda = \frac{1}{2}(\kappa-2)$. Or, what is the same thing, we may say that the first equation shows that V is made up of terms the degrees of which in a, b, c and in l, m, n, are κ, $-2i-1$, and i, $-i-\frac{1}{2}$ respectively. Attending henceforth only to the second equation, I write $\dfrac{l}{k} = \dfrac{1}{\alpha+\theta}$, $\dfrac{m}{k} = \dfrac{1}{\beta+\theta}$, $\dfrac{n}{k} = \dfrac{1}{\gamma+\theta}$; so that $\alpha+\theta$, $\beta+\theta$, $\gamma+\theta$ denote the squared semiaxes of the ellipsoid. We have

$$\frac{d}{dl} = -\frac{(\alpha+\theta)^2}{k}\,\frac{d}{d\alpha}, \quad \&c.,$$

and the equation becomes

$$\begin{aligned}
&-(a^2+b^2+c^2)\left((\alpha+\theta)\frac{dV}{d\alpha} + (\beta+\theta)\frac{dV}{d\beta} + (\gamma+\theta)\frac{dV}{d\gamma}\right)\\
&+(a^2+b^2+c^2)\left((\alpha+\theta)\frac{dF}{d\alpha} + (\beta+\theta)\frac{dF}{d\beta} + (\gamma+\theta)\frac{dF}{d\gamma}\right)\\
&+\tfrac{1}{2}\left(a(\alpha+\theta)\frac{dV}{da} + b(\beta+\theta)\frac{dV}{db} + c(\gamma+\theta)\frac{dV}{dc}\right)\\
&+\left(a(\alpha+\theta)\frac{dF}{da} + b(\beta+\theta)\frac{dF}{db} + c(\gamma+\theta)\frac{dF}{dc}\right)\\
&+\left((\alpha+\theta)^2\frac{dF}{d\alpha} + (\beta+\theta)^2\frac{dF}{d\beta} + (\gamma+\theta)^2\frac{dF}{d\gamma}\right) = 0.
\end{aligned}$$

Put for shortness $\Theta = (\alpha+\theta)(\beta+\theta)(\gamma+\theta)$, and write

$$V = v\sqrt{\Theta}, \quad F = f\sqrt{\Theta},$$

($\sqrt{\Theta}$ is to a constant factor *près* the mass of the ellipsoid) then v, f are connected by the equation

$$f = -a\frac{dv}{da} - b\frac{dv}{db} - c\frac{dv}{dc};$$

and observing that

$$(\alpha+\theta)\frac{d\sqrt{\Theta}}{d\alpha} + (\beta+\theta)\frac{d\sqrt{\Theta}}{d\beta} + (\gamma+\theta)\frac{d\sqrt{\Theta}}{d\gamma} = 3\Theta\,\frac{1}{2\sqrt{\Theta}} = \tfrac{3}{2}\sqrt{\Theta},$$

$$\begin{aligned}
&(\alpha+\theta)^2\frac{d\sqrt{\Theta}}{d\alpha} + (\beta+\theta)^2\frac{d\sqrt{\Theta}}{d\beta} + (\gamma+\theta)^2\frac{d\sqrt{\Theta}}{d\gamma}\\
&= (\alpha+\beta+\gamma+3\theta)\,\Theta\,\frac{1}{2\sqrt{\Theta}} = \tfrac{1}{2}(\alpha+\beta+\gamma+3\theta)\sqrt{\Theta},
\end{aligned}$$

the equation becomes

$$\begin{gathered}
-(a^2+b^2+c^2)\left((\alpha+\theta)\frac{dv}{d\alpha}+(\beta+\theta)\frac{dv}{d\beta}+(\gamma+\theta)\frac{dv}{d\gamma}\right)\\
+(a^2+b^2+c^2)\left((\alpha+\theta)\frac{df}{d\alpha}+(\beta+\theta)\frac{df}{d\beta}+(\gamma+\theta)\frac{df}{d\gamma}\right)\\
-\tfrac{1}{2}(a^2+b^2+c^2)\,v+\tfrac{1}{2}(a^2+b^2+c^2)\,f\\
+\tfrac{1}{2}\left(a(\alpha+\theta)\frac{dv}{da}+b(\beta+\theta)\frac{dv}{db}+c(\gamma+\theta)\frac{dv}{dc}\right)\\
+\left(a(\alpha+\theta)\frac{df}{da}+b(\beta+\theta)\frac{df}{db}+c(\gamma+\theta)\frac{df}{dc}\right)\\
+\left((\alpha+\theta)^2\frac{df}{d\alpha}+(\beta+\theta)^2\frac{df}{d\beta}+(\gamma+\theta)^2\frac{df}{d\gamma}\right)\\
+\tfrac{1}{2}(\alpha+\beta+\gamma+3\theta)\,f=0.
\end{gathered}$$

Now $v\left(=\dfrac{1}{\sqrt{\Theta}}V\right)$ may be expanded in the form

$$v=U_0+U_1+U_2+\cdots,$$

where U_i is of the degree $2i$ in the semiaxes $\sqrt{(\alpha+\theta)}$, $\sqrt{(\beta+\theta)}$, $\sqrt{(\gamma+\theta)}$, and of the degree $-2i-1$ in the coordinates a, b, c. And it is easy to see that the first term of the expansion is

$$U_0=\frac{4\pi}{3}\cdot\frac{1}{\sqrt{(a^2+b^2+c^2)}}.$$

The preceding value of v gives

$$f=U_0+3U_1+5U_2+\cdots+(2i+1)\,U_i+\cdots,$$

and substituting the values of v, f in the differential equation, and attending only to the terms which are of the same dimensions as $(a^2+b^2+c^2)\,U_{i+1}$, we have

$$\begin{gathered}
\left\{-(i+1)+(i+1)(2i+3)-\tfrac{1}{2}+\tfrac{1}{2}(2i+3)\right\}(a^2+b^2+c^2)\,U_{i+1}\\
+\left\{\alpha x\frac{dU_i}{d\alpha}+b\beta\frac{dU_i}{d\beta}+c\gamma\frac{dU_i}{d\gamma}-(2i+1)\,\theta U_i\right\}\left\{\tfrac{1}{2}+2i+1\right\}\\
+\left\{a^2\frac{dU_i}{d\alpha}+\beta^2\frac{dU_i}{d\beta}+\gamma^2\frac{dU_i}{d\gamma}+2i\theta U_i-\theta^2\left(\frac{dU_i}{d\alpha}+\frac{dU_i}{d\beta}+\frac{dU_i}{d\gamma}\right)\right\}(2i+1)\\
+\tfrac{1}{2}(\alpha+\beta+\gamma+3\theta)(2i+1)\,U_i=0;
\end{gathered}$$

or, reducing,

$$\begin{aligned}(i+1)(2i+5)(a^2+b^2+c^2)\,U_{i+1}&\\ +(2i+\tfrac{1}{2})\left(\alpha a\,\frac{dU_i}{da}+b\beta\,\frac{dU_i}{db}+c\gamma\,\frac{dU_i}{dc}\right)&\\ +(2i+1)\left(\alpha^2\,\frac{dU_i}{d\alpha}+\beta^2\,\frac{dU_i}{d\beta}+\gamma^2\,\frac{dU_i}{d\gamma}+\tfrac{1}{2}(\alpha+\beta+\gamma)\,U_i\right)&\\ -(2i+1)\left(\frac{dU_i}{d\alpha}+\frac{dU_i}{d\beta}+\frac{dU_i}{d\gamma}\right)\theta^2=0.&\end{aligned}$$

Now U_i is a function of $\alpha+\theta$, $\beta+\theta$, $\gamma+\theta$; if, therefore, for any particular value of i, U_i is independent of θ, it is clear that U_i must be a function of the differences of these quantities, and consequently we shall have $\frac{dU_i}{d\alpha}+\frac{dU_i}{d\beta}+\frac{dU_i}{d\gamma}=0$; and this being so, the equation of differences, and consequently U_{i+1}, will be independent of θ. But U_0 is independent of θ, hence U_1, U_2, &c. are all of them independent of θ, or v is independent of θ; i.e. for ellipsoids having the same foci for their principal sections, and acting on the same external point, the potentials, and therefore the attractions, are proportional to the masses, which is Maclaurin's theorem for the attractions of ellipsoids upon an external point.

The foregoing equation, omitting the term which vanishes, gives

$$U_{i+1}=-\frac{(2i+1)\left\{\alpha a\,\frac{dU_i}{da}+b\beta\,\frac{dU_i}{db}+c\gamma\,\frac{dU_i}{dc}\right\}+(2i+1)\left(\alpha^2\,\frac{dU_i}{d\alpha}+\beta^2\,\frac{dU_i}{d\beta}+\gamma^2\,\frac{dU_i}{d\gamma}+\tfrac{1}{2}(\alpha+\beta+\gamma)\,U_i\right)}{(i+1)(2i+5)(a^2+b^2+c^2)}.$$

It may be remarked that this equation, with the assistance of the equation

$$\frac{dU_i}{d\alpha}+\frac{dU_i}{d\beta}+\frac{dU_i}{d\gamma}=0,$$

gives

$$(i+1)(2i+5)\left(\frac{dU_{i+1}}{d\alpha}+\frac{dU_{i+1}}{d\beta}+\frac{dU_{i+1}}{d\gamma}\right)=-(2i+\tfrac{1}{2})\left(a\,\frac{dU_i}{da}+b\,\frac{dU_i}{db}+c\,\frac{dU_i}{dc}\right)$$

$$=(2i+1)\left(2\left(\alpha\,\frac{dU_i}{d\alpha}+\beta\,\frac{dU_i}{d\beta}+\gamma\,\frac{dU_i}{d\gamma}\right)+\tfrac{3}{2}U_i\right)=\left((2i+\tfrac{1}{2})(2i+1)-(2i+1)(2i+\tfrac{1}{2})\right)U_i=0,$$

which is as it should be.

Write

$$U_i=\frac{Q_i}{(a^2+b^2+c^2)^{i+\frac{1}{2}}}.$$

where U_i is of the degree i in α, β, γ, and of the degree $2i$ in a, b, c. We have

$$a\alpha\,\frac{dU_i}{d\alpha}+b\beta\,\frac{dU_i}{d\beta}+c\gamma\,\frac{dU_i}{dc}=(a^2+b^2+c^2)^{-i-\frac{1}{2}}\left(\alpha a\,\frac{dQ_i}{d\alpha}+b\beta\,\frac{dQ_i}{d\beta}+c\gamma\,\frac{dQ_i}{dc}\right)$$

$$-(4i+1)(a^2+b^2+c^2)^{-i-\frac{3}{2}}(a^2\alpha+b^2\beta+c^2\gamma)\,Q_i,$$

$$\alpha^2\,\frac{dU_i}{d\alpha}+\beta^2\,\frac{dU_i}{d\beta}+\gamma^2\,\frac{dU_i}{d\gamma}+\tfrac{1}{2}(\alpha+\beta+\gamma)\,U_i$$

$$=(a^2+b^2+c^2)^{-i-\frac{1}{2}}\left(\alpha^2\,\frac{dQ_i}{d\alpha}+\beta^2\,\frac{dQ_i}{d\beta}+\gamma^2\,\frac{dQ_i}{d\gamma}\right)+\tfrac{1}{2}(\alpha+\beta+\gamma)\,Q_i.$$

Hence, putting in like manner

$$(i+1)(2i+5)\,Q_{i+1}=-(2i+\tfrac{1}{2})\left\{(a^2+b^2+c^2)\left(\alpha a\,\frac{dQ_i}{d\alpha}+b\beta\,\frac{dQ_i}{d\beta}+c\gamma\,\frac{dQ_i}{dc}\right)\right.$$

$$\left.-(4i+1)(a^2\alpha+b^2\beta+c^2\gamma)\,Q_i\right\}$$

$$-(2i+1)(a^2+b^2+c^2)\left(\alpha^2\,\frac{dQ_i}{d\alpha}+\beta^2\,\frac{dQ_i}{d\beta}+\gamma^2\,\frac{dQ_i}{d\gamma}\right)+\tfrac{1}{2}(\alpha+\beta+\gamma)\,Q_i,$$

i.e.

$$(i+1)(2i+5)\,Q_{i+1}=-(a^2+b^2+c^2)\left\{(2i+\tfrac{1}{2})\left(\alpha a\,\frac{dQ_i}{d\alpha}+b\beta\,\frac{dQ_i}{d\beta}+c\gamma\,\frac{dQ_i}{dc}\right)\right.$$

$$\left.+(2i+1)\left(\alpha^2\,\frac{dQ_i}{d\alpha}+\beta^2\,\frac{dQ_i}{d\beta}+\gamma^2\,\frac{dQ_i}{d\gamma}+\tfrac{1}{2}(\alpha+\beta+\gamma)\,Q_i\right)\right\}+(2i+1)(4i+1)(a^2\alpha+b^2\beta+c^2\gamma)\,Q_i,$$

from which the functions Q_i may be calculated successively.

We may, it is clear, write

$$U_i=\frac{4\pi}{3}\cdot\frac{1}{2^i\cdot 1.2.3\cdots i\cdot 5.7\cdots 2i+3}\,K_i,$$

and we shall then have

$$(a^2+b^2+c^2)\,K_{i+1}+(4i+3)\left(\alpha a\,\frac{dK_i}{d\alpha}+b\beta\,\frac{dK_i}{db}+c\gamma\,\frac{dK_i}{dc}\right)$$

$$+(4i+2)\left(\alpha^2\,\frac{dK_i}{d\alpha}+\beta^2\,\frac{dK_i}{d\beta}+\gamma^2\,\frac{dK_i}{d\gamma}\right)+(2i+1)(\alpha+\beta+\gamma)\,K_i=0,$$

where

$$K_0=\frac{1}{\sqrt{(a^2+b^2+c^2)}}.$$

I proceed to show that

$$K_i=\left(\alpha\,\frac{d^2}{da^2}+\beta\,\frac{d^2}{db^2}+\gamma\,\frac{d^2}{dc^2}\right)^i\frac{1}{\sqrt{(a^2+b^2+c^2)}}.$$

In fact, assuming this equation for any particular value of i, we find first

$$\left(\alpha^2\frac{d}{d\alpha}+\beta^2\frac{d}{d\beta}+\gamma^2\frac{d}{d\gamma}\right)K_i = i\left(\alpha\frac{d^2}{da^2}+\beta\frac{d^2}{db^2}+\gamma\frac{d^2}{dc^2}\right)^{i-1}\left(\alpha^2\frac{d^2}{da^2}+\beta^2\frac{d^2}{db^2}+\gamma^2\frac{d^2}{dc^2}\right)\frac{1}{\sqrt{(a^2+b^2+c^2)}}$$

$$= i\left(\alpha\frac{d^2}{da^2}+\beta\frac{d^2}{db^2}+\gamma\frac{d^2}{dc^2}\right)^{i-1}\left(\alpha\frac{d^2}{da^2}+\beta\frac{d^2}{db^2}+\gamma\frac{d^2}{dc^2}\right)\frac{1}{\sqrt{(a^2+b^2+c^2)}}$$

$$= i\left(\alpha\frac{d^2}{da^2}+\beta\frac{d^2}{db^2}+\gamma\frac{d^2}{dc^2}\right)^{i}K_{i-1}.$$

Now putting for shortness

$$\Delta = \alpha\frac{d^2}{da^2}+\beta\frac{d^2}{db^2}+\gamma\frac{d^2}{dc^2},$$

we find, replacing ΔK_{i+1} and ΔK_i by their values K_{i+2} and K_{i+1},

$$\Delta\left((a^2+b^2+c^2)K_{i+1}\right) = (a^2+b^2+c^2)K_{i+2}$$

$$+4\left(\alpha a\frac{d}{da}+b\beta\frac{d}{db}+c\gamma\frac{d}{dc}\right)K_{i+1}$$

$$+2(\alpha+\beta+\gamma)K_{i+1},$$

$$\Delta\left(\alpha a\frac{d}{da}+b\beta\frac{d}{db}+c\gamma\frac{d}{dc}\right)K_i = \left(\alpha a\frac{d}{da}+b\beta\frac{d}{db}+c\gamma\frac{d}{dc}\right)K_{i+1}$$

$$+2\left(\alpha^2\frac{d^2}{da^2}+\beta^2\frac{d^2}{db^2}+\gamma^2\frac{d^2}{dc^2}\right)K_i,$$

$$\Delta\left(\alpha^2\frac{d}{d\alpha}+\beta^2\frac{d}{d\beta}+\gamma^2\frac{d}{d\gamma}\right)K_i = \left(\alpha^2\frac{d}{d\alpha}+\beta^2\frac{d}{d\beta}+\gamma^2\frac{d}{d\gamma}\right)K_{i+1}$$

$$-\left(\alpha^2\frac{d^2}{da^2}+\beta^2\frac{d^2}{db^2}+\gamma^2\frac{d^2}{dc^2}\right)K_i,$$

$$\Delta(\alpha+\beta+\gamma)K_i = (\alpha+\beta+\gamma)K_{i+1};$$

hence operating on the equation of differences with the symbol Δ, we obtain

$$(a^2+b^2+c^2)K_{i+2}+(4i+7)\left(\alpha a\frac{d}{da}+b\beta\frac{d}{db}+c\gamma\frac{d}{dc}\right)K_{i+1}$$

$$+(4i+2)\left(\alpha^2\frac{d}{d\alpha}+\beta^2\frac{d}{d\beta}+\gamma^2\frac{d}{d\gamma}\right)K_{i+1}$$

$$+\{2(4i+3)-(4i+2)\}\left(\alpha^2\frac{d^2}{da^2}+\beta^2\frac{d^2}{db^2}+\gamma^2\frac{d^2}{dc^2}\right)K_i$$

$$+(2i+3)(\alpha+\beta+\gamma)K_{i+1} = 0,$$

the third line of which is

$$4(i+1)\left(\alpha^2\frac{d^2}{da^2}+\beta^2\frac{d^2}{db^2}+\gamma^2\frac{d^2}{dc^2}\right)K_i=4i\left(\alpha^2\frac{d}{d\alpha}+\beta^2\frac{d}{d\beta}+\gamma^2\frac{d}{d\gamma}\right)K_{i+1},$$

by a foregoing equation, and the assumed equation of difference thus leads to

$$\begin{aligned}(a^2+b^2+c^2)\,K_{i+2}+(4i+7)\left(\alpha a\frac{d}{da}+b\beta\frac{d}{db}+c\gamma\frac{d}{dc}\right)K_{i+1}\\ +(4i+6)\left(\alpha^2\frac{d}{d\alpha}+\beta^2\frac{d}{d\beta}+\gamma^2\frac{d}{d\gamma}\right)K_{i+1}\\ +(2i+3)\,(\alpha+\beta+\gamma)\,K_{i+1}=0,\end{aligned}$$

which is the assumed equation, writing $i+1$ instead of i. The equation, if true for i, is therefore true for $i+1$, and it is easily seen to be true for $i=0$; hence it is true generally, or the value

$$K_i=\left(\alpha\frac{d^2}{da^2}+\beta\frac{d^2}{db^2}+\gamma\frac{d^2}{dc^2}\right)^i\frac{1}{\sqrt{(a^2+b^2+c^2)}}$$

satisfies the equation obtained by Laplace's method, and gives, moreover, the proper value for K_0. We have thus the value of K_i; and remembering that

$$V=\sqrt{(\alpha+\theta)(\beta+\theta)(\gamma+\theta)}\,v,$$

and observing that the symbol Δ may be replaced by

$$\Delta=(\alpha+\theta)\frac{d^2}{da^2}+(\beta+\theta)\frac{d^2}{db^2}+(\gamma+\theta)\frac{d^2}{dc^2},$$

the value of V is

$$V=\frac{4\pi}{3}\sqrt{(\alpha+\theta)(\beta+\theta)(\gamma+\theta)}\sum_0^\infty\left(\frac{1}{2^i\cdot1\cdot2\cdots i\cdot5\cdot7\cdots2i+3}\,\Delta^i\,\frac{1}{\sqrt{(a^2+b^2+c^2)}}\right),$$

which is in fact the value which I have found by a much more simple method in the *Cambridge Mathematical Journal*, t. III. p. 69 [2].

174.

ON THE OVAL OF DESCARTES.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 320–328.]

THIS is an extract of the passage in the Geometry (*La Géométrie de M. Descartes*, 12^{mo} Paris, 1705, pp. 79–92) in which he applies his method of coordinates to this now well-known curve: the marginal reference is "*Explication de quatre nouveaux genres d'Ovales qui servent à l'Optique*." The paper was intended to be introductory to a discussion in some detail of the history and properties of the Curve, but no continuation was written.

175.

On the Porism of the In-and-Circumscribed Triangle.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 344–354.]

The porism of the in-and-circumscribed triangle in its most general form relates to a triangle the angles of which lie in fixed curves, and the sides of which touch fixed curves; but at present I consider only the case in which the angles lie in one and the same fixed curve, which for greater simplicity I assume to be a conic. We have therefore a triangle ABC, the angles of which lie in a fixed conic $\mathfrak{S}$, and the sides of which touch the fixed curves $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$; the points of contact may be represented by α, β, γ. And if we consider the conic $\mathfrak{S}$ and the curves $\mathfrak{A}$, $\mathfrak{B}$ as given, the curve $\mathfrak{C}$ will be the envelope of the side AB; to construct this side we have only to take at pleasure a point C on the curve $\mathfrak{S}$ and to draw through this point tangents to the curves $\mathfrak{B}$, $\mathfrak{A}$ respectively meeting the conic $\mathfrak{S}$ in the points A and B; the line joining these points is the required side AB. I may notice that in the case supposed of the curve $\mathfrak{S}$ being a conic, the lines $A\alpha$, $B\beta$, $C\gamma$ meet in a point— which gives at once a construction for γ, the point of contact of AB with the curve $\mathfrak{C}$. For the sake however of exhibiting the reasoning in a form which may be modified so as to be applicable to a curve $\mathfrak{S}$ of any order, instead of the conic $\mathfrak{S}$, I shall dispense with the employment of the property just mentioned, which is peculiar to the case of the conic.

Suppose for a moment (figs. 1 and 1 *bis*) that the curves $\mathfrak{A}$, $\mathfrak{B}$ are points, and let the line through M, meet $\mathfrak{A}$, $\mathfrak{B}$ the conic $\mathfrak{S}$ in the points M, N. If we take the point N for the angle C of the triangle, the points A, B will each of them coincide with M, and the side AB will be the tangent at M to the conic $\mathfrak{S}$: call this tangent T. Consider next a point C in the neighbourhood of N, we shall have two points A, B in the neighbourhood of M, and the point in which T intersects AB will be the point of contact γ of T with the curve $\mathfrak{C}$. To find this point, suppose that $M\mathfrak{A} = a$, $N\mathfrak{A} = a'$, $M\mathfrak{B} = b$, $N\mathfrak{B} = b'$ and let the distance of C from N be ds; the distances of A, B from the line MN parallel to T will be proportional to $\dfrac{b}{b'}\,ds$, $\dfrac{a}{a'}\,ds$, and

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the perpendicular distances of these points from T are consequently proportional to $\frac{b^2}{b'^2}\,ds^2$, $\frac{a^2}{a'^2}\,ds^2$. The inclination of AB to T is therefore proportional to

$$\left(\frac{b^2}{b'^2}-\frac{a^2}{a'^2}\right)ds^2=\left(\frac{b}{b'}-\frac{a}{a'}\right)ds,\ \text{i.e. to}\ \left(\frac{b}{b'}+\frac{a}{a'}\right)ds,$$

which is of the same order as ds, and it is at once seen that the point γ will coincide with M, i.e. that the curve $\mathfrak{C}$ touches the conic $\mathfrak{S}$ at the point M. If, however, $\frac{a}{a'}+\frac{b}{b'}=0$, i.e. if the points M, N are harmonically related to $\mathfrak{A}$, $\mathfrak{B}$, then the inclination of AB to T is in general of the order ds^2, and the point γ will be at a finite distance from M; moreover T is in this case a stationary tangent (i.e. a tangent at a point of inflexion) of the curve $\mathfrak{C}$. Now reverting to the general case of $\mathfrak{A}$, $\mathfrak{B}$ being any two curves, then if there be a common tangent touching these curves in the points α, β, and meeting the curve $\mathfrak{S}$ in the points M, N, the like reasonings apply to this case. Hence

FIRST LEMMA. If a common tangent to the curves $\mathfrak{A}$, $\mathfrak{B}$ touch these curves in α, β and meet the conic $\mathfrak{S}$ in the points M, N; the point N gives rise to a branch of the curve $\mathfrak{C}$ which (except in the case after mentioned) touches the conic $\mathfrak{S}$ at the point M. If however M, N are harmonically related with respect to α, β, then the branch of the curve $\mathfrak{C}$ does not pass through M, but it has for a stationary tangent the tangent M to the conic $\mathfrak{S}$.

Suppose again that the curve $\mathfrak{A}$ (fig. 2) is a point, and let the curve $\mathfrak{B}$ intersect the conic $\mathfrak{S}$ at the point M, and let the tangent to $\mathfrak{B}$ at M meet $\mathfrak{S}$ in a point R, and $R\mathfrak{A}$ meet $\mathfrak{S}$ in a point Q. Then taking the point R for the angle C of the triangle, we shall have A, B coinciding with M, Q respectively, and thence MQ a tangent to the curve $\mathfrak{C}$. To find the point of contact, take C in the neighbourhood of R at a distance from it ds; B will be a point in the neighbourhood of Q and distant from it by an infinitesimal of the order ds, A will be in the neighbourhood of M and distant from it by an infinitesimal of the order ds^2. Hence AB will intersect MQ at the point M, or the curve $\mathfrak{C}$ will pass through M. Reverting to the general case where $\mathfrak{A}$ is a curve, we have only to consider RQ as a tangent to the curve $\mathfrak{A}$ at a point α, and the like reasoning will apply to this case: hence

SECOND LEMMA. If the curve $\mathfrak{B}$ intersect the conic $\mathfrak{S}$ at a point M, and the tangent to $\mathfrak{B}$ at M intersect $\mathfrak{S}$ in R, if moreover a tangent through R to the curve $\mathfrak{A}$ intersect $\mathfrak{S}$ in Q; then to each of the points Q there corresponds a branch through M of the curve $\mathfrak{C}$, viz. a branch touching MQ at the point M.

Suppose as before (fig. 3) that $\mathfrak{A}$ is a point, and let the curve $\mathfrak{B}$ intersect the conic $\mathfrak{S}$ at the point M, and the tangent to $\mathfrak{B}$ at M meet $\mathfrak{S}$ in the point R. And let $M\mathfrak{A}$ meet $\mathfrak{S}$ in the point N. Then taking the point R for the angle C of the triangle we shall have A, B coinciding with R, N respectively, and thence NR a tangent to the curve $\mathfrak{C}$. To find the point of contact, take C in the neighbourhood of R at a distance from it ds, then C will be in the neighbourhood of M and distant from it by an infinitesimal of the order ds^2, and B will be in the neighbourhood of N and distant from it by an infinitesimal of the same order ds^2, and consequently AB will intersect NR at the point N, or the curve $\mathfrak{C}$ passes through N. Reverting to the general case where $\mathfrak{A}$ is a curve, we have only to consider MN as a tangent to the curve $\mathfrak{A}$ at a point α and the like reasoning applies: hence,

THIRD LEMMA. If the curve $\mathfrak{B}$ intersect the conic $\mathfrak{S}$ at a point M, and the tangent to $\mathfrak{B}$ at M intersect $\mathfrak{S}$ in R, if moreover a tangent through M to the curve

$\mathfrak{A}$ meet $\mathfrak{S}$ in N; then to each of the points R there corresponds a branch through N of the curve $\mathfrak{C}$, viz. a branch touching NR at N.

Suppose again (fig. 4) that the curve $\mathfrak{A}$ is a point, and let the curve $\mathfrak{B}$ touch $\mathfrak{S}$ at the point M. And let $M\mathfrak{A}$ meet $\mathfrak{S}$ in the point N. Then taking the point M for the angle C of the triangle, the angle A will coincide with M and the angle B will coincide with N, and consequently we shall have MN a tangent to the curve $\mathfrak{C}$. And MN is in fact a double tangent, for proceeding to find the point of contact, take C in the neighbourhood of M at a distance ds; then from C we may draw to the curve $\mathfrak{B}$ two tangents each of them meeting $\mathfrak{S}$ in a point $\mathfrak{A}$ in the neighbourhood of M and distant from it by an infinitesimal of the order ds; again CA meets $\mathfrak{S}$ in a point B in the neighbourhood of N and distant from it by an infinitesimal of the same order ds; hence AB will meet MN at a point which will be in general at a finite distance from M, or rather (since there are two positions of the point A) the lines AB will meet MN in two points, each of them in general at a finite distance from M. Reverting to the case of $\mathfrak{A}$ being a curve, we must as before consider MN as a tangent to the curve $\mathfrak{A}$ at the point α, and the same reasoning applies: hence

FOURTH LEMMA. If the curve $\mathfrak{B}$ touch the conic $\mathfrak{S}$ at the point M, and if a tangent through M to the curve $\mathfrak{A}$ meet $\mathfrak{S}$ in N, then MN is a double tangent of the curve $\mathfrak{C}$, viz. the line MN has two distinct points of contact with the curve $\mathfrak{C}$.

Suppose (fig. 5) that the curves $\mathfrak{A}$ and $\mathfrak{B}$ meet the conic $\mathfrak{S}$ in one and the same point M, and let the tangent at M to the curve $\mathfrak{A}$ meet $\mathfrak{S}$ in the point P, and the tangent at M to the curve $\mathfrak{B}$ meet $\mathfrak{S}$ in the point N; then if we take the point M for the angle C of the triangle, the angles A and B of the triangle will coincide with N, P respectively, and NP will be a tangent of the curve $\mathfrak{C}$. But NP will be a double tangent, for proceeding to find the point of contact, take C in the neighbourhood of M, and distant from it by an infinitesimal of the second order ds^2; then since from the point C there may be drawn two tangents touching

$\mathfrak{A}$ in the neighbourhood of M, and two tangents touching $\mathfrak{B}$ in the neighbourhood of M, there will be two points A in the neighbourhood of N and distant from it by infinitesimals of the order ds, and in like manner two points B in the neighbourhood of P and distant from it by infinitesimals of the order ds. Call these points A, A'; B, B'; then AB, $A'B'$ will meet NP in one and the same point, and AB', $A'B$ will in like manner meet NP in one and the same point; these two points, which will be in general at finite distances from N, P, will be points of contact of NP with the curve $\mathfrak{C}$: hence,

FIFTH LEMMA. If the curves $\mathfrak{A}$, $\mathfrak{B}$ meet the conic $\mathfrak{S}$ in one and the same point M, and if the tangents at M to the curves $\mathfrak{B}$ and $\mathfrak{A}$ respectively meet $\mathfrak{S}$ in the points N and P, then, joining these points, the line NP will be a double tangent to the curve $\mathfrak{C}$, viz. there will be two distinct points of contact of the line NP with the curve $\mathfrak{C}$.

The double tangents of the fourth and fifth lemmas exist by virtue of the particular relations assumed between the curves $\mathfrak{A}$, $\mathfrak{B}$ and the conic $\mathfrak{S}$, viz. from one or both of the curves $\mathfrak{A}$, $\mathfrak{B}$ touching the conic $\mathfrak{S}$, or from the two curves having a common point of intersection or common points of intersection with $\mathfrak{S}$; there are besides double tangents which exist independently of any such relations, and the theory of which will be presently investigated, but it will be convenient in the first instance to find the class of $\mathfrak{C}$.

The class of $\mathfrak{C}$ is at once determinable from the classes (which I represent by m, n) of the curves $\mathfrak{A}$, $\mathfrak{B}$. In fact, take any point M on the conic $\mathfrak{S}$ for the angle A of the triangle. Through the point M we may draw n tangents to $\mathfrak{B}$, each of which will intersect the conic $\mathfrak{S}$ in a single point, or there will be n positions of the point C, and since from each of these we may draw m tangents to the curve $\mathfrak{A}$, each tangent intersecting $\mathfrak{S}$ in a single point; or we have mn positions of the angle B, i.e. this same number of tangents through M to the curve $\mathfrak{C}$. But if the same point M had been taken as the angle B of the triangle, there would have been in like manner mn positions of the angle A, i.e. this same number of tangents through M to the curve $\mathfrak{C}$. Hence there are in all $2mn$ tangents through M to the curve $\mathfrak{C}$, or the curve $\mathfrak{C}$ is of the class $2mn$.

Considering now the double tangents of the curve $\mathfrak{C}$; imagine a quadrilateral inscribed in the conic $\mathfrak{S}$ of which two opposite sides touch the curve $\mathfrak{A}$, and the other two opposite sides touch the curve $\mathfrak{B}$: suppose $MNPQ$, of which the sides MN and PQ touch $\mathfrak{A}$ and the sides NP and QM touch $\mathfrak{B}$. If we take M for the angle C of the triangle, then the angles A, B will coincide with Q, N, i.e. NQ is a tangent to the curve $\mathfrak{C}$, and the point of contact may be determined as before by considering a point C in the neighbourhood of M; but in like manner if we take P for the angle C of the triangle, then the angles A, B will coincide with N, Q, i.e. QN is again a tangent to the curve $\mathfrak{C}$, and the point of contact may be determined by considering a point C in the neighbourhood of P; QN is therefore a double tangent of the curve $\mathfrak{C}$. But in like manner MP is a double tangent of the curve $\mathfrak{C}$, i.e. the diagonals of the quadrilateral are each of them double tangents of the curve $\mathfrak{C}$, and the number of double tangents is consequently double the number of quadrilaterals.

Imagine a pentilateral inscribed in the conic $\mathfrak{S}$, the first and third sides of which touch the curve $\mathfrak{A}$ and the second and fourth sides of which touch the curve $\mathfrak{B}$, the fifth or closing side will envelope a curve $\mathfrak{C}'$, and in the cases in which the curve $\mathfrak{C}'$ and the conic $\mathfrak{S}$ have a common tangent, the fifth side will vanish, and the pentilateral becomes a quadrilateral of the kind before referred to. Now as $\mathfrak{C}$ was shown to be of the class $2mn$, so it may be shown that $\mathfrak{C}'$ is of the class $4mn - m - n$, hence $\mathfrak{C}'$ and $\mathfrak{S}$ have $4m^2n^2 - 2mn$ common tangents. The quadrilateral may reduce itself to a common tangent of the curves $\mathfrak{A}$ and $\mathfrak{B}$: this gives rise to $2mn$ points of contact of $\mathfrak{C}$ and $\mathfrak{S}$, and the common tangent at a point of contact reckons as two common tangents; the number of the remaining common tangents is therefore $4m^2n^2 - 4mn$. And these are in fact tangents at points of contact of $\mathfrak{C}'$ and $\mathfrak{S}$; i.e. $\mathfrak{C}'$ and $\mathfrak{S}$ touch in $2m^2n^2 - 2mn$ points. And since each angle of the quadrilateral may be taken as the first angle, the number of quadrilaterals is one fourth of this, or $\frac{1}{2}(m^2n^2 - mn)$, and the number of double tangents of the curve $\mathfrak{C}$ from the before- mentioned cause is therefore $m^2n^2 - mn$.

But there is another way in which double tangents arise; we may have a quadrilateral $MNPQ$ inscribed in the conic $\mathfrak{S}$, such that two adjacent sides MN, NP touch the curve $\mathfrak{A}$, and the other two adjacent sides PQ, QM touch the curve $\mathfrak{B}$. In fact in this case one of the diagonals, viz. NQ, is a double tangent of the curve $\mathfrak{C}$; the number of double tangents is therefore equal to the number of quadrilaterals.

Consider a pentilateral inscribed in the conic, and such that the first and second sides touch the curve $\mathfrak{A}$ and the third and fourth sides touch the curve $\mathfrak{B}$, the fifth or closing side will envelope a curve $\mathfrak{C}''$, and in the cases in which the curve $\mathfrak{C}''$ and the conic $\mathfrak{S}$ have a common tangent, the fifth side will vanish, and the pentilateral will become a quadrilateral of the kind last referred to. The curve $\mathfrak{C}''$ is of the class $2mn-(m-1)(n-1)$; hence $\mathfrak{C}''$ and $\mathfrak{S}$ have $4mn-(m-1)(n-1)$ common tangents; but these are tangents at points of contact, or the curves touch in $2mn-(m-1)(n-1)$ points, and the number of quadrilaterals is $mn(m-1)(n-1)$. The number of double tangents of the curve $\mathfrak{C}$ from the cause last referred to is therefore $mn(m-1)(n-1)$, which is equal to $mn(mn-m-n+1)$. The total number of double tangents of the curve $\mathfrak{C}$ is consequently $mn(2mn-m-n)$. And the curve $\mathfrak{C}$ has not in general any stationary tangents or what is the same thing any inflexions. It has been shown that $\mathfrak{C}$ is of the class $2mn$, it is therefore of the order $2mn(2mn-1)-2mn(2mn-m-n)$, which is equal to $2mn(m+n-1)$. Hence

THEOREM. If a triangle ABC be inscribed in a conic $\mathfrak{S}$, and the sides BC, AC touch curves $\mathfrak{A}$, $\mathfrak{B}$ of the classes m, n respectively, the side AB will envelope a curve $\mathfrak{C}$ of the class $2mn$ with in general $mn(2mn-m-n)$ double tangents, but no stationary tangents, and therefore of the order $2mn(m+n-1)$. If the curve $\mathfrak{A}$ touch the conic $\mathfrak{S}$, each point of contact will give rise to n double tangents of the curve $\mathfrak{C}$, and so if the curve $\mathfrak{B}$ touch the conic $\mathfrak{S}$, each point of contact will give rise to m double tangents of the curve $\mathfrak{C}$. And if $\mathfrak{A}$ and $\mathfrak{B}$ intersect on the conic $\mathfrak{S}$, each such intersection will give rise to a double tangent of the curve $\mathfrak{C}$. The curve $\mathfrak{C}$ in general touches the conic $\mathfrak{S}$ in the points in which it is intersected by any common tangent of the curves $\mathfrak{A}$ and $\mathfrak{B}$; but if the points of contact be harmonically situated with respect to the conic $\mathfrak{S}$, then $\mathfrak{C}$ does not pass through the points of intersection, but the tangents to $\mathfrak{S}$ at the points of intersection are stationary tangents of $\mathfrak{C}$. There is of course in the above-mentioned special cases a corresponding reduction in the order of $\mathfrak{C}$.

It sometimes happens that the number of double tangents of $\mathfrak{C}$ becomes infinite, i.e. in fact that $\mathfrak{C}$ is made up of two coincident curves; instances of this will be presently mentioned.

Suppose that the curves $\mathfrak{A}$, $\mathfrak{B}$ are points; $\mathfrak{C}$ is in general a conic having double contact with the conic $\mathfrak{S}$ in the points in which it is intersected by the line joining $\mathfrak{A}$, $\mathfrak{B}$. But if the points $\mathfrak{A}$, $\mathfrak{B}$ are harmonically situated with respect to the conic $\mathfrak{S}$, then $\mathfrak{C}$ does not pass through the points of intersection, but the tangents to $\mathfrak{S}$ at the points of intersection are stationary tangents of $\mathfrak{C}$. This implies that the curve $\mathfrak{C}$ is made up of two coincident points at the point of intersection of the two tangents of $\mathfrak{S}$: call this the point $\mathfrak{C}$, then $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$ are conjugate points of the conic $\mathfrak{S}$, and we have the well-known theorem that in a conic $\mathfrak{S}$ there may be inscribed an infinity of triangles the sides of which pass through three conjugate points of the conic. It should be remarked that for each position of the side AB, there are two positions of the angle C, i.e. each side AB is properly a double tangent of the curve (point) $\mathfrak{C}$.

Let $\mathfrak{A}$ be a point and $\mathfrak{B}$ be a conic, the curve $\mathfrak{C}$ is in general of the class 4, with two double tangents, and therefore of the order 8. It is proper to remark that the double tangents originate in a quadrilateral of the kind first considered, viz. a quadrilateral of which two opposite sides pass through the point $\mathfrak{A}$ and the other two opposite sides touch the conic $\mathfrak{B}$. It is easy in the present case to construct the quadrilateral: consider the point $\mathfrak{A}$ as a pole, and take its polar with respect to the conic $\mathfrak{S}$, take the pole of this polar with respect to the conic $\mathfrak{B}$. Join the two poles, and from the point in which the joining line meets the polar draw tangents to the conic $\mathfrak{B}$, these tangents will meet the conic $\mathfrak{S}$ in four points, lying two and two in lines passing through the point $\mathfrak{A}$: we have thus the quadrilateral, and the diagonals of the quadrilateral (or the other two lines passing through the four points) are the double tangents of the curve $\mathfrak{C}$.

There are two particular cases to be considered. First, where the conic $\mathfrak{B}$ has double contact with the conic $\mathfrak{S}$. Here the lines joining the point $\mathfrak{A}$ with the points of contact are double tangents of the curve $\mathfrak{C}$, which has therefore in all four double tangents, and being a curve of the fourth class it must break up into two curves of the second class, i.e. into two conics. And these are curves having double contact with the conic $\mathfrak{S}$; for the curve $\mathfrak{C}$ touches the conic $\mathfrak{S}$ in the four points in which $\mathfrak{S}$ is intersected by the tangents through $\mathfrak{A}$ to the conic $\mathfrak{B}$. Secondly, where $\mathfrak{A}$ is one of the conjugate points of the conics $\mathfrak{B}$, $\mathfrak{S}$. The general construction for the quadrilateral shows that if from any point of the common polar of $\mathfrak{A}$ with respect to $\mathfrak{B}$ and with respect to $\mathfrak{S}$, we draw tangents to $\mathfrak{B}$, these will meet $\mathfrak{S}$ in four points, lying two and two in lines passing through $\mathfrak{A}$, i.e. that the number of the double tangents of the curve $\mathfrak{C}$ is indefinite; $\mathfrak{C}$ is therefore made up of two coincident curves of the second class, i.e. of two coincident conics. Moreover, $\mathfrak{C}$ passes through the points of intersection of $\mathfrak{B}$, $\mathfrak{S}$; hence, disregarding one of the two coincident conics, we may say that the curve $\mathfrak{C}$ is a conic passing through the points of intersection of the conics $\mathfrak{B}$, $\mathfrak{S}$.

Next, let the curves $\mathfrak{A}$, $\mathfrak{B}$ be each of them a conic. The curve $\mathfrak{C}$ is of the class 8, with in general 16 double tangents, and therefore of the order 24. But there are two particular cases to be considered: first, where the conics $\mathfrak{A}$, $\mathfrak{B}$ have each of them double contact with the conic $\mathfrak{S}$. Here the tangents drawn from the points of contact of either of the conics $\mathfrak{A}$ or $\mathfrak{B}$ with $\mathfrak{S}$ to the other of the conics, $\mathfrak{B}$ or $\mathfrak{A}$, is a double tangent of the curve $\mathfrak{C}$, i.e. there are 8 new double tangents, or in all 24 double tangents of the curve $\mathfrak{C}$, which is therefore of the order 8; and being of the class 8 with 24 double tangents, it must break up into four curves of the second class, i.e. into four conics. And the curve $\mathfrak{C}$ touches the conic $\mathfrak{S}$ in the points in which $\mathfrak{S}$ is intersected by any one of the four common tangents of the conics $\mathfrak{A}$, $\mathfrak{B}$, viz. 8 points in all; hence each of the four conics has double contact with the conic $\mathfrak{S}$. Attending only to one of the four conics of which $\mathfrak{C}$ is made up, we have thus what (in a restricted sense of the expression, the porism of the in-and-circumscribed triangle) I call the porism (homographic) of the in-and- circumscribed triangle, viz.

If a triangle be inscribed in a conic, and two of the sides touch conics having double contact with the circumscribed conic, then will the third side touch a conic having double contact with the circumscribed conic.

Secondly, the conics $\mathfrak{A}$ and $\mathfrak{B}$ may cut the conic $\mathfrak{S}$ in the same four points. Here it may be seen that there are an infinity of inscribed quadrilaterals of the kind first considered, viz. of which two opposite sides touch the conic $\mathfrak{A}$, and the other two opposite sides touch the conic $\mathfrak{B}$. Hence, the curve $\mathfrak{C}$ is made up of two coincident curves of the class 4. But the curve of the class 4 has in fact 4 double tangents, viz. considering each of the points of intersection of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{S}$, and drawing tangents to $\mathfrak{A}$ and $\mathfrak{B}$ meeting $\mathfrak{S}$ in two new points, the line joining these points is a double tangent of the curve in question, which is therefore of the 4th order, and being of the class 4 with 4 double tangents, it must break up into two curves of the second class, i.e. into two conics. Each of these conics passes through the points of intersection of $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{S}$, and touches the four lines last referred to, the conics would of course be completely determined by the condition of passing through the four points and touching one of the four lines. Attending only to one of the two conics, we have thus what I call the porism (allographic) of the in-and-circumscribed triangle, viz.

If a triangle be inscribed in a conic, and two of the sides touch conics meeting the circumscribed conic in the same four points, the remaining side will touch a conic meeting the circumscribed conic in the four points.

The *a posteriori* demonstration of these theorems will form the subject of another paper, [178].

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176.

Note on Jacobi's Canonical Formulæ for Disturbed Motion in an Elliptic Orbit.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 355–356.]

Consider a body (afterwards called the disturbed body) revolving about a central body under the influence of their mutual attraction and of any disturbing forces. Then referring the disturbed body to axes through the central body and parallel to fixed lines in space, write

$x,\ y,\ z,$ the coordinates of the disturbed body,

$r,$ the radius vector, $=\sqrt{(x^2+y^2+z^2)}$,

$M,$ the mass of the disturbed body,

$M',$ the mass of the central body.

Write also

R, the disturbing function, taken negatively, i.e. the sign of R is taken as in the *Mécanique Céleste*.

The equations of motion then are

$$\frac{d^2x}{dt^2}=-\frac{(M+M')\,x}{r^3}-\frac{dR}{dx},$$

$$\frac{d^2y}{dt^2}=-\frac{(M+M')\,y}{r^3}-\frac{dR}{dy},$$

$$\frac{d^2z}{dt^2}=-\frac{(M+M')\,z}{r^3}-\frac{dR}{dz},$$

and the motion may be represented by supposing that the body moves in an ellipse with variable elements, and such that the direction and velocity of motion are always the same as in the actual orbit.

We may, to fix the ideas, take the plane of xy to be the plane of the ecliptic and the axis of x to be the line through the first point of Aries, or origin of longitude.

Jacobi's canonical elements may be taken to be,

First, the constant of vis viva or invariable part of the half-square of the velocity, which is equal to the sum of the masses divided by twice the mean distance and with the sign minus.

Secondly, the constant of areas, which is equal to the square root of the sum of the masses into the half of the latus rectum.

Thirdly, the constant of the reduced area (i.e. of the area described on the plane of the ecliptic), which is equal to the square root of the sum of the masses into the half of the latus rectum into the cosine of the inclination.

Fourthly, the constant attached by addition to the time t, or what is the same thing, the epoch or time of pericentric passage, taken with the sign minus.

Fifthly, the angular distance from node (or argument of latitude) of the pericentre.

Sixthly, the longitude of the node.

(The first and fourth elements are taken by Jacobi with the contrary sign, but this difference is not material.)

Representing the preceding system of canonical elements by $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{F}$, $\mathfrak{G}$, $\mathfrak{H}$, and observing that Jacobi's disturbing function Ω is the same as the disturbing function R of the Mécanique Céleste, except that the sign is reversed (i.e. $\Omega = -R$), the expressions for the variations of the canonical elements are

$$\frac{d\mathfrak{A}}{dt} = -\frac{dR}{d\mathfrak{F}}, \quad \frac{d\mathfrak{B}}{dt} = -\frac{dR}{d\mathfrak{G}}, \quad \frac{d\mathfrak{C}}{dt} = -\frac{dR}{d\mathfrak{H}},$$

$$\frac{d\mathfrak{F}}{dt} = +\frac{dR}{d\mathfrak{A}}, \quad \frac{d\mathfrak{G}}{dt} = +\frac{dR}{d\mathfrak{B}}, \quad \frac{d\mathfrak{H}}{dt} = +\frac{dR}{d\mathfrak{C}}.$$

In the ordinary case in which the disturbing force is the attraction of a third body, write

x', y', z', the coordinates of the disturbing body,

r', the radius vector, $= \sqrt{(x'^2 + y'^2 + z'^2)}$,

M'', the mass of the disturbing body.

Then the expression for the disturbing function is

$$R = M'' \left\{ \frac{xx' + yy' + zz'}{r'^3} - \frac{1}{\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}} \right\}.$$

The preceding formulae form a convenient standard of reference for the various systems of elements which have been made use of by writers upon Physical Astronomy; any such system may be without difficulty derived from the canonical system by expressing the elements adopted in terms of the above-mentioned canonical elements.

177.

SOLUTION OF A MECHANICAL PROBLEM.

[From the *Quarterly Mathematical Journal*, vol. I. (1857), pp. 405–406.]

A HEAVY plane is supported by parallel elastic strings of small extensibility; and the strings are of the same length and extensibility: required the position of equilibrium.

Imagine the plane horizontal, and let n be the number of strings, (a, b), (a', b'), &c. the coordinates of the points of attachment; ξ, η the coordinates of the centre of gravity of the plane; W the weight; let the equation of the horizontal line about which the plane turns be

$$x\cos\alpha + y\sin\alpha - p = 0;$$

and let $\delta\theta$ be the inclination of the plane in its position of equilibrium to the horizontal plane, and $\omega\delta l$ the force generated by an increase δl in the length of one of the strings.

We have for the conditions of equilibrium

$$\Sigma(a\cos\alpha + b\sin\alpha - p)\,\omega\delta\theta - W = 0,$$

$$\Sigma(a\cos\alpha + b\sin\alpha - p)\,a\omega\delta\theta - W\xi = 0,$$

$$\Sigma(a\cos\alpha + b\sin\alpha - p)\,b\omega\delta\theta - W\eta = 0;$$

or putting $\Sigma a = L$, $\Sigma b = M$, $\Sigma a^2 = A$, $\Sigma ab = H$, $\Sigma b^2 = B$, we have

$$L\cos\alpha + M\sin\alpha - np - \frac{W}{\omega\delta\theta} = 0,$$

$$A\cos\alpha + H\sin\alpha - Lp - \frac{W\xi}{\omega\delta\theta} = 0,$$

$$H\cos\alpha + B\sin\alpha - Mp - \frac{W\eta}{\omega\delta\theta} = 0.$$

188.

ON THE SIMULTANEOUS TRANSFORMATION OF TWO HOMOGENEOUS FUNCTIONS OF THE SECOND ORDER.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 192–195.]

In a former paper with this title, *Cambridge and Dublin Math. Journal*, t. IV. [1849], pp. 47–50 [74], I gave (founded on the methods of Jacobi and Prof. Boole) a simple solution of the problem, but the solution may I think be presented in an improved form as follows, where as before I consider for greater convenience the case of three variables only.

Suppose that by the linear transformation[1]

$$(x,\ y,\ z) = \begin{pmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{pmatrix} \begin{pmatrix} x_1, & y_1, & z_1 \end{pmatrix}$$

we have identically

$$(a,\ b,\ c,\ f,\ g,\ h \mid x,\ y,\ z)^2 = (a_1,\ b_1,\ c_1,\ f_1,\ g_1,\ h_1 \mid x_1,\ y_1,\ z_1)^2,$$

$$(A,\ B,\ C,\ F,\ G,\ H \mid x,\ y,\ z)^2 = (A_1,\ B_1,\ C_1,\ F_1,\ G_1,\ H_1 \mid x_1,\ y_1,\ z_1)^2,$$

and write also

$$(\xi,\ \eta,\ \zeta) = \begin{pmatrix} \alpha, & \alpha', & \alpha'' \\ \beta, & \beta', & \beta'' \\ \gamma, & \gamma', & \gamma'' \end{pmatrix} \begin{pmatrix} \xi_1, & \eta_1, & \zeta_1 \end{pmatrix}.$$

[1] I represent in this manner the system of equations

$$x = \alpha x_1 + \beta y_1 + \gamma z_1, \quad \&c.$$

and so in all like cases.

Comparing these with the relations between $(x,\ y,\ z)$ and $(x_1,\ y_1,\ z_1)$, we see that

$$(\xi,\ \eta,\ \zeta\ \mid\ x,\ y,\ z) = (\xi_1,\ \eta_1,\ \zeta_1\ \mid\ x_1,\ y_1,\ z_1),$$

and multiplying the first of the relations between two quadrics by an indeterminate quantity λ, and adding it to the second, we have

$$(\lambda a + A, \ldots \mid x,\ y,\ z)^2 = (\lambda a_1 + A_1, \ldots \mid x_1,\ y_1,\ z_1)^2.$$

We have thus a linear function and a quadric transformed into functions of the same form by means of the linear substitutions, and any invariant of the system will remain unaltered to a factor près, such factor being a power of the determinant of substitution. The invariants are, 1° the discriminant of the quadric; 2° the reciprocant, considered not as a contravariant of the quadric, but as an invariant of the system. And if we write

$$K = \text{Disc.}\,(\lambda a + A, \ldots \mid x,\ y,\ z)^2,$$

$$(\mathfrak{A},\ \mathfrak{B},\ \mathfrak{C},\ \mathfrak{F},\ \mathfrak{G},\ \mathfrak{H}\ \mid\ \xi,\ \eta,\ \zeta)^2 = \text{Recip.}\,(\lambda a + A, \ldots \mid x,\ y,\ z)^2,$$

then K_1, &c. being the analogous expressions for the transformed functions, and the determinant of substitution being represented by Π, we have

$$K_1 = \Pi^2 K,$$

$$(\mathfrak{A}_1, \ldots \mid \xi_1,\ \eta_1,\ \zeta_1)^2 = \Pi^2 (\mathfrak{A}, \ldots \mid \xi,\ \eta,\ \zeta)^2,$$

and substituting for ξ_1, η_1, ζ_1 their values in terms of ξ, η, ζ, the last equation breaks up into six equations, and we have

$$K_1 = \Pi^2 K,$$

$$(\mathfrak{A}_1, \ldots \mid \alpha,\ \alpha',\ \alpha'')^2 = \Pi^2 \mathfrak{A},$$

$$(\mathfrak{A}_1, \ldots \mid \beta,\ \beta',\ \beta'')(\gamma,\ \gamma',\ \gamma'') = \Pi^2 \mathfrak{F},$$

which is the system obtained in a somewhat different manner in my former paper.

Putting $f_1 = g_1 = h_1 = F_1 = G_1 = H_1 = 0$, and writing also (which is no additional loss of generality) $a_1 = b_1 = c_1 = 1$, the formulae become

$$(a,\ b,\ c,\ f,\ g,\ h \mid x,\ y,\ z)^2 = (1,\ 1,\ 1\ \mid\ x_1^2,\ y_1^2,\ z_1^2),$$

$$(A,\ B,\ C,\ F,\ G,\ H \mid x,\ y,\ z)^2 = (A_1,\ B_1,\ C_1 \mid x_1^2,\ y_1^2,\ z_1^2),$$

viz. there are two given quadrics which are to be by the same linear substitution transformed, one of them into the form $x_1^2 + y_1^2 + z_1^2$ and the other into the form $A_1 x_1^2 + B_1 y_1^2 + C_1 z_1^2$, where A_1, B_1, C_1 have to be determined. The solution is contained in the following system of formulae, viz.

$$(A_1 + \lambda)(B_1 + \lambda)(C_1 + \lambda) = \Pi^2 \cdot \text{Disc.}\,(\lambda a + A, \ldots),$$

which gives A_1, B_1, C_1 as the roots of a cubic equation, and gives also

$$1 = \Pi^2 \cdot \text{Disc.}\,(a, \ldots) = \Pi^2\,\kappa,\ \text{ or } \Pi^2 = \frac{1}{\kappa} \text{ suppose,}$$

and we have then, writing for shortness, $(*| X,\ Y,\ Z)$ for

$$((B_1+\lambda)(C_1+\lambda),\ (C_1+\lambda)(A_1+\lambda),\ (A_1+\lambda)(B_1+\lambda) \mid X,\ Y,\ Z),$$

$$(*| \alpha^2,\ \alpha'^2,\ \alpha''^2) = \frac{1}{\kappa}\mathfrak{A},$$

$$(*| \beta^2,\ \beta'^2,\ \beta''^2) = \frac{1}{\kappa}\mathfrak{B},$$

$$(*| \gamma^2,\ \gamma'^2,\ \gamma''^2) = \frac{1}{\kappa}\mathfrak{C},$$

$$(*| \beta\gamma,\ \beta'\gamma',\ \beta''\gamma'') = \frac{1}{\kappa}\mathfrak{F},$$

$$(*| \gamma\alpha,\ \gamma'\alpha',\ \gamma''\alpha'') = \frac{1}{\kappa}\mathfrak{G},$$

$$(*| \alpha\beta,\ \alpha'\beta',\ \alpha''\beta'') = \frac{1}{\kappa}\mathfrak{H},$$

where $(\mathfrak{A},\ \mathfrak{B},\ \mathfrak{C},\ \mathfrak{F},\ \mathfrak{G},\ \mathfrak{H})$ are the coefficients of the reciprocant of $(\lambda a + A, \ldots \mid x,\ y,\ z)^2$. Writing $\lambda = -A_1,\ -B_1$, or $-C_1$, the quadric functions on the left-hand side become mere monomials, and we have the actual values of the squares and products α^2, β^2, &c. of the coefficients of the linear substitutions: thus α^2, β^2, γ^2, $\beta\gamma$, $\gamma\alpha$, $\alpha\beta$ are respectively equal to $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{F}$, $\mathfrak{G}$, $\mathfrak{H}$, each into the common factor

$$\frac{1}{\kappa}\,(B_1 - A_1)(C_1 - A_1),$$

the suffix denoting that we are to write in the expressions for $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{F}$, $\mathfrak{G}$, $\mathfrak{H}$ the value $-A_1$ for λ; and similarly for the sets $(\alpha',\ \beta',\ \gamma')$ and $(\alpha'',\ \beta'',\ \gamma'')$.

2, Stone Buildings, 27th March, 1857.

189.

NOTE ON A FORMULA IN FINITE DIFFERENCES.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 198–201.]

In Jacobi's Memoir "De usu legitimo formulae summatoriae Maclaurinianae," *Crelle*, t. XII. [1834], pp. 263–273 (1834), expressions are given for the sums of the odd powers of the natural numbers 1, 2, $3 \ldots x$ in terms of the quantity

$$u = x(x+1),$$

viz. putting for shortness

$$Sx^r = 1^r + 2^r + \ldots + x^r,$$

the expressions in question are

$$\begin{aligned} Sx^3 &= \tfrac{1}{4}u^2, \\ Sx^5 &= \tfrac{1}{6}u^3\,(u - \tfrac{1}{3}), \\ Sx^7 &= \tfrac{1}{8}u^4\,(u^2 - \tfrac{4}{3}u + \tfrac{2}{3}), \\ Sx^9 &= \tfrac{1}{10}u^2\,(u^4 - 3u^3 + \tfrac{17}{5}u^2 - \tfrac{10}{3}u + \tfrac{4}{3}), \\ Sx^{11} &= \tfrac{1}{12}u^2\,(u^5 - \tfrac{14}{3}u^4 + \tfrac{32}{3}u^3 - \tfrac{72}{5}u^2 + \tfrac{208}{15}u - \tfrac{32}{5}), \\ &\&c., \end{aligned}$$

which, especially as regards the lower powers, are more simple than the ordinary expressions in terms of x.

The expressions are continued by means of a recurring formula, viz. if

$$Sx^{2p-1} = \frac{1}{2p-2}\left[a_0u^{p+1} - a_1u^{p+2} + \ldots + (-)^{p-1}a_{p-1}u^2\right],$$

$$Sx^{2p+1} = \frac{1}{2p}\left[u^p - b_1u^{p-1} + \ldots + (-)^p\, b_p\, u^0\right],$$

then

$$\begin{aligned}
2p(2p-1)\,a_1 &= (2p-2)(2p-3)\,b_1 - p(p-1),\\
2p(2p-1)\,a_2 &= (2p-4)(2p-5)\,b_2 - (p-1)(p-2)\,b_1,\\
2p(2p-1)\,a_3 &= (2p-6)(2p-7)\,b_3 - (p-2)(p-3)\,b_2,\\
&\vdots\\
2p(2p-1)\,a_{p-2} &= 5\cdot 6\,b_{p-2} - 3\cdot 4\,b_{p-3},\\
0 &= 3\cdot 4\,b_{p-1} - 2\cdot 3\,b_{p-2},
\end{aligned}$$

by means of which the coefficients b can be determined when the coefficients a are known.

Jacobi remarks also that the expressions for the sums of the even powers may be obtained from those for the odd powers by means of the formula

$$Sx^{2p} = \frac{1}{2p+2}\,\tfrac{d}{du}\,Sx^{2p+1},$$

which shows that any such sum will be of the form $(2x+1)u$ into a rational and integral function of u: thus in particular

$$Sx^2 = \tfrac{1}{3}(2x+1)\,u.$$

To show *a priori* that Sx^{2p+1} can be expressed as a rational and integral function of u, it may be remarked that $Sx^{2p+1} = \phi_1 x$ where $\phi_1 x$ denotes the summatory integral $\Sigma(x+1)^{2p+1}$, taken so as to vanish for $x=0$: $\phi_1 x$ is a rational and integral function of x of the degree $2p+2$, and which, as is well known, contains x^2 as a factor. Suppose that y is any positive or negative integer less than x, we have

$$\phi_1 x - \phi_1 y = (y+1)^{2p+1} + (y+2)^{2p+1} + \ldots + x^{2p+1},$$

and in particular putting $y = -1-x$,

$$\phi_1 x - \phi_1(-1-x) = (-x)^{2p+1} + (1-x)^{2p+1} + \ldots + x^{2p+1} = 0,$$

since the terms destroy each other in pairs; we have therefore $\phi_1 x = \phi_1(-1-x)$. Now $u = x^2+x$, or writing this equation under the form $x^2 = -x+u$, we see that any rational and integral function of x may be reduced to the form $Px+Q$, where P and Q are rational and integral functions of u. Write therefore $\phi_1 x = Px+Q$: the substitution of $-1-x$ in the place of x leaves u unaltered, and the equation $\phi_1 x = \phi_1(-1-x)$ thus shows that $P=0$; we have therefore $\phi_1 x = Q$, a rational and integral function of u. Moreover $\phi_1 x$ as containing the factor x^2, must clearly contain the factor u^2, and the expressions for Sx^{2p+1} are thus shown to be of the form given by Jacobi.

We may obtain a finite expression for Sx^n in terms of the differences of 0^n as follows: we have

$$Sx^n = 1^n+2^n+\ldots+x^n = \big\{(1+\Delta)+(1+\Delta)^2+\ldots+(1+\Delta)^x\big\}0^n = \frac{1+\Delta}{\Delta}\,\{(1+\Delta)^x-1\}\,0^n,$$

and putting $(1+\Delta)^x = e^{x\log(1+\Delta)}$ and observing that the term independent of x vanishes, and that the terms containing powers higher than x^{n+1} also vanish, we have

$$Sx^n = S_k\left\{\frac{1+\Delta}{\Delta}\log^k(1+\Delta)\right\}0^n\cdot\frac{x^k}{\Pi k},$$

where the summation with respect to k, extends from $k=1$ to $k=n+1$, or what is the same thing (since the term corresponding to $k=1$ in fact vanishes) from $k=2$ to $k=n+1$.

The equation $x^2=-x+u$ gives

$$x^k = P_k x + Q_k,$$

and it is easy to see that writing for shortness

$$M_k = 1+\frac{k-3}{1}u+\frac{k-4\cdot k-5}{1\cdot 2}u^2+\frac{k-5\cdot k-6\cdot k-7}{1\cdot 2\cdot 3}u^3+\ldots,$$

where the series is to be continued to the term $u^{\frac{1}{2}(k-2)}$ or $u^{\frac{1}{2}(k-3)}$ according as k is even or odd, we have

$$P_k = (-)^{k+1}M_{k+1},\quad Q_k = (-)^k u\,M_k,$$

we have consequently

$$Sx^n = xS_k\left\{\frac{1+\Delta}{\Delta}\log^k(1+\Delta)\right\}0^n\cdot\frac{(-)^{k+1}M_{k+1}}{\Pi k}+S_k\left\{\frac{1+\Delta}{\Delta}\log^k(1+\Delta)\right\}0^n\cdot\frac{(-)^k u\,M_k}{\Pi k}.$$

If n is odd, $=2p+1$, then (by what precedes) the first term vanishes, or we have

$$S_k\left\{\frac{1+\Delta}{\Delta}\log^k(1+\Delta)\right\}0^{2p+1}\cdot\frac{(-)^{k+1}M_{k+1}}{\Pi k}=0,\quad (k=1\text{ to }k=2p+2),$$

and the formula becomes

$$Sx^{2p+1} = S_k\left\{\frac{1+\Delta}{\Delta}\log^k(1+\Delta)\right\}0^{2p+1}\cdot\frac{(-)^k u\,M_k}{\Pi k},\quad (k=1\text{ to }k=2p+2),$$

which it may be noticed puts in evidence the factor u but not the factor u^2.

If n is even, $=2p$, then (by what precedes) the coefficient of x is to the constant term in the ratio $2:1$, or we have

$$S_k\left\{\frac{1+\Delta}{\Delta}\log^k(1+\Delta)\right\}0^{2p}\cdot\frac{(-)^{k+1}(M_{k+1}-2uM_k)}{\Pi k}=0,\ (k=1\text{ to }k=2p+1),$$

and the formula becomes

$$Sx^{2p} = (2x+1)\,S_k\left\{\frac{1+\Delta}{\Delta}\log^k(1+\Delta)\right\}0^{2p}\cdot\frac{(-)^k u\,M_k}{\Pi k},\ (k=1\text{ to }k=2p+1).$$

The values of the functions M are as follows:

$$\begin{aligned} M_1 &= 0, \\ M_2 &= 1, \\ M_3 &= 1, \\ M_4 &= 1+u, \\ M_5 &= 1+2u, \\ M_6 &= 1+3u+u^2, \\ M_7 &= 1+4u+3u^2, \\ &\&\text{c.} \end{aligned}$$

As a simple example of the formulae, we have

$$\begin{aligned} Sx^3 = &\left\{\frac{1+\Delta}{\Delta}\log^2(1+\Delta)\right\}0^3 \cdot \tfrac{1}{2}\,u \\ &+ \left\{\frac{1+\Delta}{\Delta}\log^3(1+\Delta)\right\}0^3 \cdot -\tfrac{1}{6}\,u \\ &+ \left\{\frac{1+\Delta}{\Delta}\log^4(1+\Delta)\right\}0^3 \cdot \tfrac{1}{24}\,(u+u^2), \end{aligned}$$

and the coefficients are

$$\begin{aligned} (\Delta - \tfrac{1}{12}\Delta^3)\,0^3 &= 1 - \tfrac{1}{4}\cdot 6 = \tfrac{1}{2}, \\ (\Delta^2 - \tfrac{1}{2}\Delta^3)\,0^3 &= 6 - \tfrac{1}{2}\cdot 6 = 3, \\ \Delta^3\,0^3 &= 6, \end{aligned}$$

and therefore

$$Sx^3 = \tfrac{1}{2}u - \tfrac{1}{2}u + \tfrac{1}{4}\,(u+u^2) = \tfrac{1}{4}u^2,$$

which is right; the example shows however that the calculation for the higher powers would be effected more readily by means of Jacobi's recurring formula.

2, Stone Buildings, 27th Oct., 1857.

190.

ON THE SYSTEM OF CONICS WHICH PASS THROUGH THE SAME FOUR POINTS.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 206–207.]

I CONSIDER the system of conics passing through the same four points; these points may be real or imaginary, but it is assumed that there is a real system of conics, this will in fact be the case if two conics of the system are real. The four points are therefore given as the points of intersection of two real conics, and it will be proper to assume in the first instance that the conics intersect in four separate and distinct points, none of them at infinity. The four points may be all real, or two real and two imaginary, or all imaginary.

First, if the points are all real, we have here two cases, viz. each of the points may lie outside of the triangle formed by the other three, or as this may be expressed, the points may form a convex quadrangle; or else one of the points may be inside the triangle formed by the other three, or as this may be expressed, the points may form a triangle and interior point. In each case the pairs of lines joining the points, two and two together, will be conics (degenerate hyperbolas) forming part of the system of conics. Consider the two cases separately.

[*Fig. A. omitted: four real points forming a convex quadrangle; six lines joined and labelled.*]

Fig. A. Four real points forming a convex quadrangle. The system contains two parabolas, and the pairs of lines and the parabolas divide the plane of the figure into five distinct regions, one of which contains only ellipses, and the other four contain each of them hyperbolas.

Fig. A′. Four real points forming a triangle and interior point. The system does not contain any parabolas, the three pairs of lines divide the plane of the figure into three distinct regions, each of which contains only hyperbolas.

[*Fig. A′. omitted.*]

Next, if the points are two of them real and two of them imaginary. The line joining the two imaginary points will be real and this line may meet the line joining the two real points, in a point outside the two real points, or included between them, i.e. the real centre of the quadrangle may lie outside the real points, or may be included between them; I consider the two cases separately.

Fig. B. Two real and two imaginary points, the real centre of the quadrangle lying outside the real points. The system contains two parabolas, and these with the line joining the two real points and the line joining the two imaginary points divide the plane of the figure into three regions, one of which contains ellipses and the other two contain each of them hyperbolas.

[*Fig. B. omitted.*]

Fig. B′. Two real and two imaginary points, the real centre of the quadrangle lying between the real points. There are no parabolas, and the system contains only hyperbolas.

[*Fig. B′. omitted.*]

Lastly, when the four points are imaginary. We have here only a single case.

Fig. C. Four imaginary points. The points lie on two real lines, there are (besides the point of intersection of these lines) two other real centres of the quadrangle, which lie harmonically with respect to the two lines. The system contains two parabolas and these and the two lines divide the plane of the figure into four regions, two of which contain each of them ellipses, and the other two contain each of them hyperbolas.

[*Fig. C. omitted.*]

191.

NOTE ON THE EXPANSION OF THE TRUE ANOMALY.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 229–232.]

If the true anomaly and the mean anomaly are respectively denoted by u, m, and if e be the eccentricity, then as usual $u - e\sin u = m$; and if we write

$$\lambda = \frac{1-\sqrt{(1-e^2)}}{e},$$

and take c to denote the base of the hyperbolic system of logarithms, we have

$$u = m + 2\Sigma_1^\infty A_r\,\frac{\sin rm}{r},$$

and

$$A_r = \lambda^r\, c^{-\frac{1}{2}r(\lambda-\lambda^{-1})} + \lambda^{-r}\, c^{\frac{1}{2}r(\lambda-\lambda^{-1})},$$

where, after expanding the exponentials, the negative powers of λ are to be rejected and the term independent of λ is to be multiplied by $\frac{1}{2}$ (see *Camb. Math. Journal*, t. I. [1839] p. 228 and t. III. [1843] p. 165, [4]).

It is easily seen that e^r is the lowest power of e which enters into the value of A_r, and the question arises to find the numerical coefficient of the term in question; this is readily obtained from the formula; in fact considering first a term of the form

$$\lambda^{-r}\, c^s\,(\lambda-\lambda^{-1})^s,$$

since λ is itself of the order e, when the negative powers of λ are rejected this is at least of the order e^s and it is consequently to be neglected if $s > r$. But if $s < r$ all the powers of λ are negative and the term is to be rejected. The only case to be considered is therefore that of $s = r$, in which case there is a term containing e^r. We thus obtain from $\lambda^{-r}\, c^{\frac{1}{2}r(\lambda-\lambda^{-1})}$ the term

$$\frac{1}{2^r}\cdot\frac{r^r e^r}{1\cdot 2\cdot 3\ldots r}.$$

In the next place a term of the form $\lambda^r\, c^s\,(\lambda-\lambda^{-1})^s$ is at least of the order e^r if $s > r$, or the terms to be considered are those for which $s = r$ or $r < r$. But in such term the only part of the order e^r is

$$(-)^s\, 2^{-r+s}\, e^r,$$

or, since neglecting higher powers of e we have $\lambda = \frac{1}{2}e$, this is

$$(-)^s\, 2^{-r+s}\, e^r,$$

and the set of terms arising from

$$\lambda^r\, c^{-\frac{1}{2}r(\lambda-\lambda^{-1})}$$

is

$$\frac{e^r}{2^r}\left\{1+\frac{r}{1}+\frac{r^2}{1\cdot 2}+\ldots+\frac{r^{r-1}}{1\cdot 2\ldots(r-1)}+\tfrac{1}{2}\,\frac{r^r}{1\cdot 2\ldots r}\right\},$$

the last term being divided by 2 because arising from a term independent of λ. Hence the first term of A_r is

$$\frac{e^r}{2^r}\left\{1+\frac{r}{1}+\frac{r^2}{1\cdot 2}+\ldots+\frac{r^r}{1\cdot 2\ldots r}\right\},$$

a result which it may be remarked is contained in the general formula given in Hansen's Memoir "Entwickelung des Products u.s.w.," *Leipzig Trans.*, t. II. p. 277 (1853).

The preceding expression is

$$\frac{e^r}{2^r}\, c^r\, \frac{1}{\Gamma(r+1)}\int_r^\infty x^r c^{-x}\, dx,$$

and to find its value when r is large, we have

$$\begin{aligned}\int_r^\infty x^r c^{-x}\, dx &= \int_0^\infty (y+r)^r\, c^{-y-r}\, dy = r^r c^{-r}\int_0^\infty \left(1+\tfrac{y}{r}\right)^r\, c^{-y}\, dy\\ &= r^r c^{-r}\int_0^\infty c^{-y+r\log(1+\frac{y}{r})}\, dy\\ &= r^r c^{-r}\int_0^\infty c^{-\frac{y^2}{2r}+\frac{y^3}{3r^2}-\&\text{c.}}\, dy\\ &= r^r c^{-r}\int_0^\infty \left(1+\frac{y^3}{3r^2}+\ldots\right) c^{-\frac{y^2}{2r}}\, dy\\ &= r^r c^{-r}\sqrt{2r}\int_0^\infty \left(1+\frac{2\sqrt{2}}{3\sqrt{r}}\, z^3+\ldots\right) c^{-z^2}\, dz,\end{aligned}$$

or neglecting all the terms except the first, this is

$$= r^r c^{-r} \sqrt{2r} \int_0^\infty c^{-z^2}\, dz = \sqrt{2\pi r}\, r^r\, c^{-r}.$$

Hence multiplying by $\dfrac{1}{2^r}\, e^r\, c^r\, \dfrac{1}{\Gamma(r+1)}$ and observing that when r is large, we have, by a well-known formula,

$$\Gamma(r+1) = \sqrt{2\pi r}\, r^r\, c^{-r},$$

we obtain finally the result that when r is large the first term of A_r is approximately

$$= \left(\frac{ec}{2}\right)^r.$$

I take the opportunity of mentioning the following somewhat singular theorem, which seems to belong to a more general theory: viz. if $u - e\sin u = m$, then we have

$$\log(1 - e\cos u) = \frac{1}{a}\log(1 - ae\cos\phi),$$

where

$$\phi - \frac{1}{a}\tan\phi = m,$$

provided that the negative powers of a are rejected, and a is then put equal to unity.

To show this, we have by Lagrange's theorem, observing that

$$\frac{d}{dm} F(1 - e\cos m) = e\sin m\, F'(1 - e\cos m),$$

$$\begin{aligned} F(1 - e\cos u) = F(1 - e\cos m) &+ \frac{e^2}{1\cdot 2}\frac{d}{dm}\sin^2 m\, F'(1 - e\cos m) \\ &+ \frac{e^3}{1\cdot 2\cdot 3}\frac{d^2}{dm^2}\sin^3 m\, F''(1 - e\cos m) + \&\text{c.}, \end{aligned}$$

and the coefficient of e^r in $F(1 - e\cos u)$ is

$$\frac{(-)^r}{1\cdot 2\ldots(r-1)}\left\{\tfrac{1}{r}\, F_r \cos^r m + \frac{r-1}{1}\, F_{r-1}\cos^{r-1} m\sin^2 m - \frac{(r-1)(r-2)}{1\cdot 2}\, F_{r-2}\, \tfrac{d}{dm}(\cos^{r-3} m\sin^3 m) + \&\text{c.}\right\},$$

where $F_r = F^{(r)}(1)$.

Hence in particular when $Fx = \log x$, $F_r = (-)^{r-1} \cdot 1 \cdot 2 \ldots (r-1)$ and thence the coefficient of e^r in $\log(1 - e\cos u)$ is

$$-\left\{\tfrac{1}{r}\cos^r m - \frac{1}{1\cdot 2}\cos^{r-2} m\sin^2 m - \frac{1}{1\cdot 2}\frac{d}{dm}(\cos^{r-3} m\sin^3 m) - \&\text{c.}\right\},$$

continued as long as the exponent of $\cos m$ is not negative. Now in the expansion of $\frac{1}{a}\log(1 - ae\cos\phi)$, where $\phi - \frac{1}{a}\tan\phi = m$, the coefficient of e^r is $-\frac{1}{r}a^{r-1}\cos^r\phi$, and this (by Lagrange's theorem) is equal to

$$-\frac{1}{r}a^{r-1}\left\{\cos^r m - \frac{1}{1\cdot 2}r\cos^{r-1} m\sin m\tan m - \frac{1}{1\cdot 2\cdot a}\frac{d}{dm}(r\cos^{r-2} m\sin^2 m\tan^2 m) - \&\text{c.}\right\}$$

$$= -\left\{\tfrac{1}{r}a^{r-1}\cos^r m - \tfrac{1}{1}a^{r-2}\cos^{r-2} m\sin^2 m - \frac{1}{1\cdot 2}a^{r-3}\cos^{r-3} m\sin^2 m - \&\text{c.}\right\},$$

where the series is continued indefinitely; but if we reject the negative powers of a and then put a equal to unity this is precisely equal to the former expression for the coefficient of e^r, and the formula is thus shown to be true.

2, Stone Buildings, W.C., 17th Nov., 1857.

192.

ON THE AREA OF THE CONIC SECTION REPRESENTED BY THE GENERAL TRILINEAR EQUATION OF THE SECOND DEGREE.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 248–253.]

[THE original title was "Direct Investigation of the Question discussed in the Foregoing Paper," viz. a Paper with the present title by N. M. Ferrers (now Dr Ferrers), pp. 247–248. The area S of the conic section represented by the general equation $(A,\ B,\ C,\ A',\ B',\ C' \mid x,\ y,\ z)^2 = 1$, where the coordinates are connected by the equation $x+y+z=1$, was by considerations founded on the form of the function found to be

$$S = \frac{2\pi\,(AA'^2 + BB'^2 + CC'^2 - ABC - 2A'B'C')\,\Delta}{[A'^2 - BC + B'^2 - CA + C'^2 - AB + 2(B'C' - AA') + 2(C'A' - BB') + 2(A'B' - CC')]^{\frac{3}{2}}},$$

where Δ is the area of the fundamental triangle; and it was remarked that a similar method might be applied to determine the area of the conic section when it is defined by the distances of its several tangents from three given points.]

The position of a point P being determined as in the foregoing paper, let α, β, γ denote in like manner the coordinates of a point O, we have

$$\alpha + \beta + \gamma = 1,$$

and consequently if ξ, η, ζ are the relative coordinates $x-\alpha$, $y-\beta$, $z-\gamma$, we have

$$\xi + \eta + \zeta = 0.$$

The expression for the distance of the two points O, P is readily obtained in terms of the relative coordinates, viz. calling this distance r, we have

$$r^2 = L\xi^2 + M\eta^2 + N\zeta^2,$$

where, if l, m, n are the sides of the triangle ABC, we have

$$\begin{aligned} L &= \tfrac{1}{2}(m^2 + n^2 - l^2), \\ M &= \tfrac{1}{2}(n^2 + l^2 - m^2), \\ N &= \tfrac{1}{2}(l^2 + m^2 - n^2); \end{aligned}$$

and it is to be remarked that these values give

$$MN + NL + LM = \tfrac{1}{4}\,(2m^2n^2 + 2n^2l^2 + 2l^2m^2 - l^4 - m^4 - n^4) = 4\Delta^2,$$

if Δ denote the area of the triangle ABC.

Consider now a conic

$$(a,\ b,\ c,\ f,\ g,\ h \mid x,\ y,\ z)^2,$$

and suppose as usual that $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{F}$, $\mathfrak{G}$, $\mathfrak{H}$ are the inverse coefficients and that K is the discriminant, suppose also for shortness

$$P = (\mathfrak{A},\ \mathfrak{B},\ \mathfrak{C},\ \mathfrak{F},\ \mathfrak{G},\ \mathfrak{H} \mid 1,\ 1,\ 1)^2.$$

The coordinates of the centre being α, β, γ, we have

$$\begin{aligned} \alpha &= \frac{1}{P}\,(\mathfrak{A},\ \mathfrak{H},\ \mathfrak{G} \mid 1,\ 1,\ 1), \\ \beta &= \frac{1}{P}\,(\mathfrak{H},\ \mathfrak{B},\ \mathfrak{F} \mid 1,\ 1,\ 1), \\ \gamma &= \frac{1}{P}\,(\mathfrak{G},\ \mathfrak{F},\ \mathfrak{C} \mid 1,\ 1,\ 1), \end{aligned}$$

and writing as before ξ, η, ζ for $x-\alpha$, $y-\beta$, $z-\gamma$, so that ξ, η, ζ are the coordinates of a point P of the conic, in relation to the centre, we have x, y, z respectively equal to $\xi+\alpha$, $\eta+\beta$, $\zeta+\gamma$, and the equation of the conic gives

$$(a, \ldots \mid \xi+\alpha,\ \eta+\beta,\ \zeta+\gamma)^2 = 0,$$

which may be written

$$\begin{aligned} &(a, \ldots \mid \xi,\ \eta,\ \zeta)^2 \\ &\quad + 2\,(a, \ldots \mid \xi,\ \eta,\ \zeta)\,(\xi,\ \eta,\ \zeta \mid \alpha,\ \beta,\ \gamma)_1 \\ &\quad + (a, \ldots \mid \alpha,\ \beta,\ \gamma)^2 = 0. \end{aligned}$$

Now observing the equations

$$(a,\ h,\ g \mid \alpha,\ \beta,\ \gamma) = \tfrac{K}{P},$$
$$(h,\ b,\ f \mid \alpha,\ \beta,\ \gamma) = \tfrac{K}{P},$$
$$(g,\ f,\ c \mid \alpha,\ \beta,\ \gamma) = \tfrac{K}{P},$$

we have

$$(a, \ldots \mid \alpha,\ \beta,\ \gamma)\,(\xi,\ \eta,\ \zeta) = \tfrac{K}{P}\,(\xi+\eta+\zeta) = 0,$$
$$(a, \ldots \mid \alpha,\ \beta,\ \gamma)^2 = \tfrac{K}{P}\,(\alpha+\beta+\gamma) = \tfrac{K}{P}.$$

and the equation of the conic gives therefore

$$(a, \ldots \mid \xi,\ \eta,\ \zeta)^2 + \tfrac{K}{P} = 0,$$

and we have as before

$$\xi+\eta+\zeta = 0.$$

To find the axes we have only to make

$$r^2 = L\xi^2 + M\eta^2 + N\zeta^2,$$

a maximum or minimum, ξ, η, ζ varying subject to the preceding two conditions; this gives

$$(a,\ h,\ g \mid \xi,\ \eta,\ \zeta) + \lambda L\xi + \mu = 0,$$
$$(h,\ b,\ f \mid \xi,\ \eta,\ \zeta) + \lambda M\eta + \mu = 0,$$
$$(g,\ f,\ c \mid \xi,\ \eta,\ \zeta) + \lambda N\zeta + \mu = 0,$$

and multiplying by ξ, η, ζ adding and reducing, we have

$$-\tfrac{K}{P} + \lambda r^2 = 0,$$

which gives

$$\lambda = \tfrac{K}{Pr^2}.$$

Substituting this value, and joining to the resulting three equations the equation

$$\xi+\eta+\zeta = 0.$$

we may eliminate ξ, η, ζ, μ, and the result is

$$\begin{vmatrix} a+\frac{KL}{Pr^2}, & h, & g, & 1 \\ h, & b+\frac{KM}{Pr^2}, & f, & 1 \\ g, & f, & c+\frac{KN}{Pr^2}, & 1 \\ 1, & 1, & 1, & \cdot \end{vmatrix} = 0,$$

which may also be written

$$(\mathfrak{A}',\ \mathfrak{B}',\ \mathfrak{C}',\ \mathfrak{F}',\ \mathfrak{G}',\ \mathfrak{H}' \mid 1,\ 1,\ 1)^2 = 0,$$

where $(\mathfrak{A}', \ldots)$ are what $(\mathfrak{A}, \ldots)$ become when a, b, c are changed into

$$a+\tfrac{KL}{Pr^2}, \quad b+\tfrac{KM}{Pr^2}, \quad c+\tfrac{KN}{Pr^2};$$

we in fact have

$$\mathfrak{A}' = \mathfrak{A} + \tfrac{K}{Pr^2}\,(bN+cM) + \tfrac{K^2}{P^2r^4}\,MN,$$

$$\vdots$$

$$\mathfrak{F}' = \mathfrak{F} - \tfrac{K}{Pr^2}\,Lf,$$

$$\vdots$$

and (observing the value of P) the result consequently is

$$P + \tfrac{K}{Pr^2}\,\{(b+c-2f)L + (c+a-2g)M + (a+b-2h)N\} + \tfrac{K^2}{P^2r^4}\,(MN+NL+LM) = 0,$$

which may also be written

$$Pr^4 + PKr^2\,\{(b+c-2f)L + (c+a-2g)M + (a+b-2h)N\} + 4\Delta^2K^2 = 0.$$

Hence if r_1, r_2 are the two semiaxes, we have

$$r_1^2r_2^2 = \frac{4\Delta^2K^2}{P^2},$$

and the area is πr_1r_2, which is equal to

$$\frac{2\pi K\Delta}{\sqrt{(P^3)}},$$

which agrees with Mr Ferrers' result.

The formula $r^2 = L\xi^2 + M\eta^2 + N\zeta^2$ which is assumed in the preceding investigation may be proved as follows:

Writing a, b, c (instead of l, m, n) for the sides of the fundamental triangle and A, B, C for the angles, the equation in question is

$$r^2 = bc\cos A \cdot \xi^2 + ca\cos B\,\eta^2 + ab\cos C\,\zeta^2.$$

Now writing α, β, γ for the inclinations of the line r to the sides of the triangle, we have

$$A = \beta - \gamma,$$

$$B = \gamma - \alpha,$$

$$C = \pi + \alpha - \beta.$$

Moreover taking for a moment λ, μ, ν to denote the perpendiculars from the angles on the opposite sides, we have

$$\lambda = c\sin B = b\sin C,$$

$$\mu = a\sin C = c\sin A,$$

$$\nu = b\sin A = a\sin B,$$

and

$$\xi = \frac{r\sin\alpha}{\lambda},\quad \eta = \frac{r\sin\beta}{\mu},\quad \zeta = \frac{r\sin\gamma}{\nu};$$

the values of ξ^2, η^2, ζ^2 consequently are

$$\frac{r^2\sin^2\alpha}{bc\sin B\sin C},\quad \frac{r^2\sin^2\beta}{ca\sin C\sin A},\quad \frac{r^2\sin^2\gamma}{ab\sin A\sin B},$$

and the equation to be proved becomes

$$1 = \frac{\cos A\sin^2\alpha}{\sin B\sin C} + \frac{\cos B\sin^2\beta}{\sin C\sin A} + \frac{\cos C\sin^2\gamma}{\sin A\sin B},$$

or, what is the same thing,

$$\sin A\sin B\sin C = \sin A\cos A\sin^2\alpha + \sin B\cos B\sin^2\beta + \sin C\cos C\sin^2\gamma,$$

or again

$$4\sin A\sin B\sin C = \sin 2A\,(1-\cos 2\alpha) + \sin 2B\,(1-\cos 2\beta) + 2C\,(1-\cos 2\gamma),$$

or putting for A, B, C their values in terms of α, β, γ this is

$$\begin{aligned} -4\sin(\beta-\gamma)\sin(\gamma-\alpha)\sin(\alpha-\beta) = {} & \sin(2\beta-2\gamma)\,(1-\cos 2\alpha) \\ & +\sin(2\gamma-2\alpha)\,(1-\cos 2\beta) \\ & +\sin(2\alpha-2\beta)\,(1-\cos 2\gamma), \end{aligned}$$

which is an identical equation; it is most readily proved by writing x, y, z for $\tan\alpha$, $\tan\beta$, $\tan\gamma$; the equation thus becomes

$$\frac{-4}{(1+x^2)(1+y^2)(1+z^2)}\,(y-z)(z-x)(x-y) = \Sigma\,\frac{1}{(1+y^2)(1+z^2)}\,[2y(1-z^2)-2z(1-y^2)]\,\frac{2x^2}{1+x^2},$$

or multiplying out

$$-(y-z)(z-x)(x-y) = \Sigma\,(y-z)(1+yz)\,x^2 = \Sigma x^2(y-z) + xyz\,\Sigma x\,(y-z),$$

that is

$$-(y-z)(z-x)(x-y) = \Sigma x^2(y-z) = x^2(y-z) + y^2(z-x) + z^2(x-y),$$

which is an identity.

[A different investigation of the formula $r^2 = L\xi^2 + M\eta^2 + N\zeta^2$, by Dr Ferrers, was appended to the original Paper.]

193.

ON RODRIGUES' METHOD FOR THE ATTRACTION OF ELLIPSOIDS.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 333–337.]

THE following is in substance the method given in the "Mémoire sur l'attraction des Sphéroïdes," par M. Rodrigues, *Corresp. sur l'École Polyt.*, t. III. pp. 361–385 (1815). It will be seen that the method is very similar to that given two years before by Gauss, see my paper "On Gauss' Method for the Attraction of Ellipsoids," *Journal*, t. I. pp. 162–166 [164]: the solution in fact depends upon the geometrical theorem therein quoted, viz. if M be any point, P a point of a closed surface, PQ the normal (lying outside the surface) at the point P, dS the element of the surface at that point, and if MQ denotes the angle MPQ and $\overline{MP}$ the distance of the points M and P, then, theorem, the integral

$$\iint \frac{dS\,\cos MQ}{\overline{MP}^2}$$

has for its value

$$0, \quad -2\pi \text{ or } -4\pi$$

according as M is exterior to, upon, or interior to the surface.

Suppose that M is the attracted point and taking A, B, C for the semiaxes of the surface of the attracting ellipsoid, or, if we please, for any semiaxes of an arbitrary ellipsoidal surface confocal with the surface of the attracting ellipsoid, let P be a point on the surface of the interior similar ellipsoid whose semiaxes are rA, rB, rC. The coordinates of M are taken to be a, b, c, and those of P are taken to be x, y, z, and the value of the potential is

$$V = \int \frac{dm}{\overline{MP}},$$

where dm is the element of mass.

We may write

$$x = rA\xi, \quad y = rB\eta, \quad z = rC\zeta,$$

and then ξ, η, ζ will be the coordinates of a point P' on the surface of a sphere, radius unity, corresponding in a definite manner to the point P on the surface of the internal similar ellipsoid. And if $d\sigma$ be the element of the spherical surface, then we have

$$dm = ABC\, r^2\, dr\, d\sigma,$$

and therefore

$$\frac{V}{ABC} = \int \frac{r^2\, dr\, d\sigma}{\overline{MP}};$$

where, in order to obtain the value of the potential V for the ellipsoid whose semiaxes are A, B, C, the integrations must be extended over the spherical surface and from $r = 0$ to $r = 1$.

Suppose that dS is the element of the internal similar surface at P, and let p be the perpendicular from the centre upon the tangent plane at P, we have

$$dS = \frac{r^2 ABC}{p}\, d\sigma.$$

Let P_0 be the point on the ellipsoid $(A,\ B,\ rC)$ similarly situated to the point P on the ellipsoid $(rA,\ rB,\ rC)$; the coordinates of P_0 are $A\xi$, $B\eta$, $C\zeta$; and if p_0 be the perpendicular from the centre upon the tangent plane at P_0, then $p = rp_0$, and the preceding equation becomes

$$dS = \frac{r^3 ABC}{p_0}\, d\sigma.$$

Imagine now an ellipsoidal surface confocal with the surface $(A,\ B,\ C)$ and having for its semiaxes

$$A + \delta A, \quad B + \delta B, \quad C + \delta C,$$

and let P_0' be the point on this surface which corresponds with the point P_0 on the surface $(A,\ B,\ C)$; that is, let P_0' be the point whose coordinates are

$$(A + \delta A)\, \xi, \quad (B + \delta B)\, \eta, \quad (C + \delta C)\, \zeta;$$

and let P' be in like manner the point whose coordinates are

$$r\,(A + \delta A)\, \xi, \quad r\,(B + \delta B)\, \eta, \quad r\,(C + \delta C)\, \zeta;$$

the points P, P' will be in like manner *corresponding* points on the surface $(rA,\ rB,\ rC)$ and on the confocal surface $[r\,(A+\delta A),\ r\,(B+\delta B),\ r\,(C+\delta C)]$; and if the normal distance at the point P_0 of the first two surfaces is δN, then the normal distance at the point P of the second two surfaces will be $r\,\delta N$. The decrement of $\overline{MP}$ will be equal to the normal distance $r\,\delta N$ of the two surfaces at the point P multiplied into the cosine of the angle MQ, and we have, by a property of confocal surfaces,

$$A\delta A = B\delta B = C\delta C = p_0\,\delta N = \text{(suppose)}\ \tfrac{1}{2}\delta\theta,$$

so we have therefore

$$d\,\overline{MP} = -\,\frac{\frac{1}{2}r\,\delta\theta}{p_0}\cos MQ.$$

Hence from the equation

$$\frac{V}{ABC} = \int \frac{r^2\,dr\,d\sigma}{\overline{MP}}.$$

We deduce

$$\delta\frac{V}{ABC} = \int r^2\,dr\,d\sigma\,\frac{\frac{1}{2}r\,\delta\theta}{p_0}\,\frac{\cos MQ}{\overline{MP^2}}.$$

But we have

$$\frac{r^2\,d\sigma}{p_0} = \frac{dS}{ABC};$$

and the equation thus becomes

$$\delta\frac{V}{ABC} = \frac{\frac{1}{2}\,\delta\theta}{ABC}\int r\,dr\,\frac{dS\,\cos MQ}{\overline{MP^2}}.$$

It may be proper to remark here by way of recapitulation that the course of the investigation has been as follows: viz. that, with a view to obtaining the potential V of an attracting ellipsoid, we have found the increment of $\dfrac{V}{ABC}$ in passing from the ellipsoidal surface $(A,\ B,\ C)$ to the ellipsoidal surface $(A+\delta A,\ B+\delta B,\ C+\delta C)$, each of them confocal with the surface of the attracting ellipsoid; and that for finding such increment we have had to consider the two surfaces $(rA,\ rB,\ rC)$ and $[r\,(A+\delta A),\ r\,(B+\delta B),\ r\,(C+\delta C)]$ confocal to each other and respectively similar to the first-mentioned two surfaces.

Resuming the formula just obtained, the integral with respect to dS is taken over the entire surface of the internal similar ellipsoid $(rA,\ rB,\ rC)$, and if the attracted point M is external to the ellipsoid $(A,\ B,\ C)$ it will be external to the interior similar ellipsoid $(rA,\ rB,\ rC)$; hence in this case the double integral vanishes for all values of r, or we have

$$\delta\frac{V}{ABC} = 0;$$

that is the function $V \div ABC$, which represents the ratio of the potential to the mass, is not altered in passing from the ellipsoid $(A,\ B,\ C)$ to the confocal ellipsoid

$$(A+\delta A,\ B+\delta B,\ C+\delta C),$$

or, what is the same thing, the potentials (and therefore the attractions) of confocal ellipsoids upon the same external point are proportional to their masses; this is in fact Maclaurin's theorem for the attraction of ellipsoids upon an external point.

But if the attracted point M is interior to the ellipsoid $(A,\ B,\ C)$, then writing

$$\frac{a^2}{A^2}+\frac{b^2}{B^2}+\frac{c^2}{C^2}=r'^2;$$

where r' is less than unity, the double integral is $=0$ from $r=0$ to $r=r'$ and is $=-4\pi$ from $r=r'$ to $r=1$, and we have

$$\begin{aligned}\delta\frac{V}{ABC} &= -\frac{2\pi\,\delta\theta}{ABC}\int_{r'}^{1} r\,dr\\ &= -\frac{\pi\,\delta\theta}{ABC}(1-r'^2)\\ &= -\frac{\pi\,\delta\theta}{ABC}\left(\frac{a^2}{A^2}+\frac{b^2}{B^2}+\frac{c^2}{C^2}-1\right);\end{aligned}$$

that is, the right-hand side of the equation is the increment (or taken with its sign reversed so as to be positive, it is the decrement) of the function $V \div ABC$ in passing from the ellipsoid $(A,\ B,\ C)$ to the confocal ellipsoid

$$(A+\delta A,\ B+\delta B,\ C+\delta C),$$

where

$$\tfrac{1}{2}\delta\theta = A\delta A = B\delta B = C\delta C.$$

The preceding formula gives at once the potential for an interior point; in fact taking α, β, γ for the semiaxes of the ellipsoid and writing

$$A^2=\alpha^2+\theta,\quad B^2=\beta^2+\theta,\quad C^2=\gamma^2+\theta,$$

and using the ordinary symbol d instead of δ, we have

$$\frac{d}{d\theta}\,\frac{V}{\sqrt{(\alpha^2+\theta)(\beta^2+\theta)(\gamma^2+\theta)}}=\frac{\pi}{\sqrt{(\alpha^2+\theta)(\beta^2+\theta)(\gamma^2+\theta)}}\left\{\frac{a^2}{\alpha^2+\theta}+\frac{b^2}{\beta^2+\theta}+\frac{c^2}{\gamma^2+\theta}-1\right\},$$

and integrating from $\theta=0$ to $\theta=\infty$, we have

$$-\frac{V}{\alpha\beta\gamma}=\pi\int_0^\infty\frac{d\theta}{\sqrt{(\alpha^2+\theta)(\beta^2+\theta)(\gamma^2+\theta)}}\left\{\frac{a^2}{\alpha^2+\theta}+\frac{b^2}{\beta^2+\theta}+\frac{c^2}{\gamma^2+\theta}-1\right\},$$

where V is now the potential for the ellipsoid whose semiaxes are α, β, γ; and we have therefore

$$V = -\pi\alpha\beta\gamma \int_0^\infty \frac{d\theta}{\sqrt{(\alpha^2+\theta)(\beta^2+\theta)(\gamma^2+\theta)}} \left\{\frac{a^2}{\alpha^2+\theta} + \frac{b^2}{\beta^2+\theta} + \frac{c^2}{\gamma^2+\theta} - 1\right\}.$$

To find the potential for an external point it is only necessary to remark that by the theorem above demonstrated $V \div \alpha\beta\gamma$ is equal to the corresponding function for the confocal ellipsoid through the attracted point, that is for the ellipsoid whose semiaxes are $\sqrt{(\alpha^2+\theta_1)}$, $\sqrt{(\beta^2+\theta_1)}$, $\sqrt{(\gamma^2+\theta_1)}$, where θ_1 is a positive quantity such that

$$\frac{a^2}{\alpha^2+\theta_1} + \frac{b^2}{\beta^2+\theta_1} + \frac{c^2}{\gamma^2+\theta_1} = 1;$$

hence in the value of $V \div \alpha\beta\gamma$ we have only to write the above values in the place of α, β, γ; and if we then write $\theta - \theta_1$ in the place of θ the limits will be ∞, θ_1, and the expression for the potential is

$$V = -\pi\alpha\beta\gamma \int_{\theta_1}^\infty \frac{d\theta}{\sqrt{(\alpha^2+\theta)(\beta^2+\theta)(\gamma^2+\theta)}} \left\{\frac{a^2}{\alpha^2+\theta} + \frac{b^2}{\beta^2+\theta} + \frac{c^2}{\gamma^2+\theta} - 1\right\};$$

this completes the investigation.

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202. Supplementary Remarks on the Porism of the In-and-Circumscribed Triangle

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 19–30.]

In my former papers (see *Phil. Mag.* August and November, 1853, [115, 116]) I established (as part of a more general one) the following theorem, viz. the condition that there may be inscribed in the conic $U = 0$, an infinity of triangles circumscribed about the conic $V = 0$, is, that if we develope in ascending powers of k the square root of the discriminant of $kU + V$, the coefficient of k^2 in this development must vanish. Thus writing

$$\text{disct.}\,.(kV + U) = (K,\ \Theta,\ \Theta',\ K'\tfrac{1}{2}k,\ 1)^3,$$

the condition in question is found to be

$$3\Theta^2 - 4K\Theta' = 0.$$

The following investigations, although relating only to particular cases of the theorem, are, I think, not without interest.

If the equation of the conic containing the angles is

$$U = 2ayz + 2bzx + 2cxy = 0,$$

and the equation of the conic touched by the sides is

$$V = x^2 + y^2 + z^2 - 2yz - 2zx - 2xy = 0,$$

we have

$$\text{disct.}\,.(k,\ k,\ k,\ a-k,\ b-k,\ c-k \mid x,\ y,\ z)^2 = (K,\ \Theta,\ \Theta',\ K'\tfrac{1}{2}k,\ 1)^3,$$

that is,

$$\begin{aligned} K &= -4, \\ \Theta &= \tfrac{4}{3}(a+b+c), \\ \Theta' &= -\tfrac{4}{3}(a+b+c)^2, \\ K' &= 2abc. \end{aligned}$$

and the equation $3\Theta^2 - 4K\Theta' = 0$ becomes

$$\tfrac{4^2}{3}(a+b+c)^2 - \tfrac{4^2}{3}(a+b+c)^2 = 0,$$

which is satisfied identically. This is as it should be; for it is plain that there exists a triangle, viz. the triangle $(x = 0,\ y = 0,\ z = 0)$, inscribed in the conic $U = 0$, and circumscribed about the conic $V = 0$.

Suppose that the equation of the conic containing the angles is

$$y^2 - 4zx = 0,$$

and the equation of the conic touched by the sides is

$$ax^2 + by^2 + cz^2 = 0,$$

then the tangential equation of the last-mentioned conic is

$$bc\xi^2 + ca\eta^2 + ab\zeta^2 = 0;$$

and if we take for the angles of the triangle $x : y : z = 1 : 2\lambda : \lambda^2$, or $1 : 2\mu : \mu^2$, or $1 : 2\nu : \nu^2$, then the equation of the line joining the angles $(\mu), (\nu)$ is

$$2\mu\nu x - (\mu + \nu)y + z = 0,$$

which will touch the conic $ax^2 + by^2 + cz^2 = 0$ if

$$bc \cdot 4\mu^2\nu^2 + ca(\mu + \nu)^2 + ab \cdot 4 = 0;$$

and it is required to find under what circumstances the equations

$$\begin{aligned} bc \cdot 4\mu^2\nu^2 + ca(\mu + \nu)^2 + ab \cdot 4 = 0, \\ bc \cdot 4\nu^2\lambda^2 + ca(\nu + \lambda)^2 + ab \cdot 4 = 0, \\ bc \cdot 4\lambda^2\mu^2 + ca(\lambda + \mu)^2 + ab \cdot 4 = 0, \end{aligned}$$

become equivalent to two equations only. The condition is of course included in the general formula; and putting

$$\text{disct.}\,.(ka,\ kb + 1,\ kc,\ 0,\ -2,\ 0 \mid x,\ y,\ z)^2 = (K,\ \Theta,\ \Theta',\ K'\tfrac{1}{2}k,\ 1)^3,$$

we must have

$$3\Theta^2 - 4K\Theta' = 0.$$

The discriminant in question is

$$k^3abc + k^2ac - k \cdot 4b - 4 = 0,$$

where $K = \frac{1}{2}ac$, $\Theta = -\frac{4}{3}b$, $K' = -4$; the required condition is therefore $ac + 16b^2 = 0$, or say

$$b = \pm\tfrac{1}{4}i\sqrt{ac}.$$

Substituting this value, the equations become

$$\begin{aligned} c\mu^2\nu^2 + i\sqrt{ca}\,(\mu + \nu)^2 + a = 0, \\ c\nu^2\lambda^2 + i\sqrt{ca}\,(\nu + \lambda)^2 + a = 0, \\ c\lambda^2\mu^2 + i\sqrt{ca}\,(\lambda + \mu)^2 + a = 0; \end{aligned}$$

the first and second of these are

$$A + 2H + B\nu^2 = 0, \qquad A' + 2H'\nu + B'\nu^2 = 0,$$

where

$$\begin{aligned} A = (-i\sqrt{a} + \mu^2\sqrt{c})\,i\sqrt{a}, \quad H = i\sqrt{ca}\,\mu, \quad B = (i\sqrt{a} + \mu^2\sqrt{c})\sqrt{c}, \\ A' = (-i\sqrt{a} + \lambda^2\sqrt{c})\,i\sqrt{a}, \quad H' = i\sqrt{ca}\,\lambda, \quad B' = (i\sqrt{a} + \lambda^2\sqrt{c})\sqrt{c}, \end{aligned}$$

and

$$\begin{aligned} AB' - 2HH' + A'B &= -2i\sqrt{ac}\,(a + c\lambda^2\mu^2) - 2i\sqrt{ac}\,\lambda\mu \cdot i\sqrt{ac}, \\ AB' - HH' &= -i\sqrt{ac}\,(a + i\sqrt{ac}\,\lambda\mu + c\mu^2), \\ H'A' - AH &= -i\sqrt{ac}\,(a - i\sqrt{ac}\,\lambda^2 + c\lambda^4); \end{aligned}$$

and the result of the elimination therefore is

$$(-a + i\sqrt{ac}\,\lambda^2 - c\lambda^4)(-a + i\sqrt{ac}\,\mu^2 - c\mu^4) + (-a + i\sqrt{ac}\,\lambda\mu - c\lambda^2\mu^2)^2 = 0,$$

viz.

$$2\sqrt{ca}\,(\lambda - \mu)^2(c\lambda^2\mu^2 + i\sqrt{ca}\,(\lambda + \mu)^2 + a) = 0;$$

which agrees, as it should do, with the third equation.

To find the condition that it may be possible in the conic

$$x^2 + y^2 + z^2 = 0$$

to inscribe an infinity of triangles, each of them circumscribed about the conic

$$ax^2 + by^2 + cz^2 = 0;$$

let the equations of the sides be

$$\begin{aligned} l\sqrt{a}\,x + m\sqrt{b}\,y + n\sqrt{c}\,z &= 0, \\ l'\sqrt{a}\,x + m'\sqrt{b}\,y + n'\sqrt{c}\,z &= 0, \\ l''\sqrt{a}\,x + m''\sqrt{b}\,y + n''\sqrt{c}\,z &= 0; \end{aligned}$$

then the conditions of circumscription are

$$\begin{aligned} l^2 + m^2 + n^2 &= 0, \\ l'^2 + m'^2 + n'^2 &= 0, \\ l''^2 + m''^2 + n''^2 &= 0; \end{aligned}$$

and the conditions of inscription are

$$\begin{aligned} bc(m'n'' - m''n')^2 + ca(n'l'' - n''l')^2 + ab(l'm'' - l''m')^2 &= 0, \\ bc(m''n - mn'')^2 + ca(n''l - nl'')^2 + ab(l''m - lm'')^2 &= 0, \\ bc(mn' - m'n)^2 + ca(nl' - n'l)^2 + ab(lm' - l'm)^2 &= 0. \end{aligned}$$

Now

$$\begin{aligned} (mn' - m'n)^2 &= (m^2 + n^2)(m'^2 + n'^2) - (mm' + nn')^2 \\ &= -l^2l'^2 - (mm' + nn')^2 \\ &= -mm' - mm' + (ll' + nn')(ll' + nn'); \end{aligned}$$

and making the like change in the analogous expressions, and putting for shortness

$$\begin{aligned} -bc + ca + ab &= \alpha, \\ bc - ca + ab &= \beta, \\ bc + ca - ab &= \gamma, \end{aligned}$$

the conditions in question become

$$\begin{aligned}(l'l''+m'm''+n'n'')(\alpha l'l''+\beta m'm''+\gamma n'n'')&=0,\\(l''l+m''m+n''n)(\alpha l''l+\beta m''m+\gamma n''n)&=0,\\(ll'+mm'+nn')(\alpha ll'+\beta mm'+\gamma nn')&=0.\end{aligned}$$

The proper solution is that given by the system of equations

$$\begin{aligned}l^2+m^2+n^2&=0,\\l'^2+m'^2+n'^2&=0,\\l''^2+m''^2+n''^2&=0,\\\alpha l'l''+\beta m'm''+\gamma n'n''&=0,\\\alpha l''l+\beta m''m+\gamma n''n&=0,\\\alpha ll'+\beta mm'+\gamma nn'&=0;\end{aligned}$$

and by writing $l=\dfrac{f}{\sqrt{\alpha}}$, $l'=\dfrac{f'}{\sqrt{\alpha}}$, $m=\dfrac{g}{\sqrt{\beta}}$, &c., $A=\dfrac{1}{\alpha}$, $B=\dfrac{1}{\beta}$, $C=\dfrac{1}{\gamma}$, these equations become

$$\begin{aligned}Af^2+Bg^2+Ch^2&=0,\\Af'^2+Bg'^2+Ch'^2&=0,\\Af''^2+Bg''^2+Ch''^2&=0,\\f'f''+g'g''+h'h''&=0,\\f''f+g''g+h''h&=0,\\ff'+gg'+hh'&=0.\end{aligned}$$

The first of which systems expresses that the points (f,g,h), (f',g',h'), (f'',g'',h'') are points in the conic

$$Ax^2+By^2+Cz^2=0;$$

and the second condition expresses that each of the points in question is the pole with respect to the conic

$$x^2+y^2+z^2=0$$

of the line joining the other two points, i.e. that the three points are a system of conjugate points with respect to the last-mentioned conic. The problem is thus reduced to the following one:—

To find the condition in order that it may be possible in the conic

$$Ax^2+By^2+Cz^2=0$$

to inscribe an infinity of triangles such that the angles are a system of conjugate points with respect to the conic

$$x^2+y^2+z^2=0.$$

Before going further it is proper to remark that if, instead of assuming

$$\alpha l'l''+\beta m'm''+\gamma n'n''=0,$$

we had assumed

$$l'l''+m'm''+n'n''=0,$$

this, combined with the equations

$$l'^2 + m'^2 + n'^2 = 0, \quad l''^2 + m''^2 + n''^2 = 0,$$

would have given $l' : m' : n' = l'' : m'' : n''$, i.e. two of the angles of the triangle would have been coincident: this obviously does not give rise to any proper solution. Returning now to the system of equations in f, g, h, &c., since the equations give only the ratios $f : g : h$; $f' : g' : h'$; $f'' : g'' : h''$, we may if we please assume

$$f^2 + g^2 + h^2 = 1, \quad f'^2 + g'^2 + h'^2 = 1, \quad f''^2 + g''^2 + h''^2 = 1,$$

which, combined with the second system of equations, gives

$$f + f' + f'' = 1, \quad g + g' + g'' = 1, \quad h + h' + h'' = 1.$$

We have, consequently,

$$\begin{aligned} A + B + C &= A(f^2 + f'^2 + f''^2) + B(g^2 + g'^2 + g''^2) + C(h^2 + h'^2 + h''^2) \\ &= (Af^2 + Bg^2 + Ch^2) + (Af'^2 + Bg'^2 + Ch'^2) + (Af''^2 + Bg''^2 + Ch''^2), \end{aligned}$$

i.e.

$$A + B + C = 0,$$

for the condition that it may be possible in the conic

$$Ax^2 + By^2 + Cz^2 = 0$$

to describe an infinity of triangles the angles of which are conjugate points with respect to the conic $x^2 + y^2 + z^2 = 0$.

The equation of the conic $Ax^2 + By^2 + Cz^2 = 0$ may be written in the form

$$(b^2c^2 - a^2bc + 2a^2bc)\, x^2 + (c^2a^2 - ab^2c + 2ab^2c)\, y^2 + (a^2b^2 - abc^2 + 2abc^2)\, z^2 = 0,$$

which gives the values of A, B, C; or again in the form

$$\begin{gathered} 2(bc + ca + ab)(bcx^2 + cay^2 + abz^2) \\ - (bc + ca + ab)(x^2 + y^2 + z^2) \\ + 4abc(ax^2 + by^2 + cz^2) = 0; \end{gathered}$$

where it should be observed that $bcx^2 + cay^2 + abz^2 = 0$ is the equation of the conic which is the polar of $ax^2 + by^2 + cz^2 = 0$ with respect to $x^2 + y^2 + z^2 = 0$. It is very easy from the last form to deduce the equation of the auxiliary conic, when the conics $ax^2 + by^2 + cz^2 = 0$, $x^2 + y^2 + z^2 = 0$ are replaced by conics represented by perfectly general equations.

The condition $A + B + C = 0$ gives, substituting the values of A, B, C,

$$b^2c^2 + a^2c^2 + a^2b^2 - 2abc(a + b + c) = 0;$$

or in a more convenient form,

$$(bc + ca + ab)^2 - 4abc(a + b + c) = 0,$$

as the condition in order that it may be possible to inscribe in the conic $x^2+y^2+z^2=0$ an infinity of triangles, the sides of which touch the conic $ax^2+by^2+cz^2=0$: this agrees perfectly with the general theorem.

It is convenient to add (as a somewhat more general form of the equation $A+B+C=0$), that the condition in order that it may be possible in the conic $Ax^2+By^2+Cz^2=0$ to inscribe an infinity of triangles the angles of which are conjugate points with respect to the conic $A_1x^2+B_1y^2+C_1z^2=0$, is

$$\frac{A}{A_1}+\frac{B}{B_1}+\frac{C}{C_1}=0.$$

But the problem to find the condition in order that it may be possible in the conic $x^2+y^2+z^2=0$ to inscribe an infinity of triangles the sides of which touch the conic $ax^2+by^2+cz^2=0$, may, by the assistance of the geometrical theorem to be presently mentioned, be at once reduced to the problem:—

To find the condition in order that it may be possible in the conic

$$x^2+y^2+z^2=0$$

to inscribe an infinity of triangles the sides of which are conjugate points with respect to a conic

$$A_1x^2+B_1y^2+C_1z^2=0.$$

The theorem referred to is as follows:—

Theorem. *If the chord PP' of a conic S envelope a conic σ, the points P, P' are harmonics with respect to a conic T which has with S, σ, a system of common conjugate points.*

Take for the equation of S,

$$x^2+y^2+z^2=0,$$

and for the equation of σ,

$$ax^2+by^2+cz^2=0;$$

then if (x_1,y_1,z_1), (x_2,y_2,z_2) are the coordinates of the points P, P' respectively, we have

$$x_1^2+y_1^2+z_1^2=0,\quad x_2^2+y_2^2+z_2^2=0,$$

and the condition in order that the chord may touch the conic σ is

$$bc(y_1z_2-z_1y_2)^2+ca(z_1x_2-x_1z_2)^2+ab(x_1y_2-y_1x_2)^2=0.$$

But we have

$$\begin{aligned}(y_1z_2-y_2z_1)^2 &= (y_1^2+z_1^2)(y_2^2+z_2^2)-(y_1y_2+z_1z_2)^2\\ &= x_1^2x_2^2-(y_1y_2+z_1z_2)^2\\ &= (x_1x_2+y_1y_2+z_1z_2)(x_1x_2-y_1y_2-z_1z_2);\end{aligned}$$

and making the like change in the analogous quantities, and putting for shortness

$$\begin{aligned}\alpha &= -bc+ca+ab,\\ \beta &= bc-ca+ab,\\ \gamma &= bc+ca-ab,\end{aligned}$$

the condition in question becomes

$$(x_1x_2 + y_1y_2 + z_1z_2)(\alpha x_1x_2 + \beta y_1y_2 + \gamma z_1z_2) = 0.$$

But the equation $x_1x_2+y_1y_2+z_1z_2 = 0$ must be rejected, as giving with the equations $x_1^2 + y_1^2 + z_1^2 = 0$, $x_2^2 + y_2^2 + z_2^2 = 0$ the relation $x_1 : y_1 : z_1 = x_2 : y_2 : z_2$; we have therefore

$$\alpha x_1x_2 + \beta y_1y_2 + \gamma z_1z_2 = 0,$$

which implies that the points (x_1, y_1, z_1) and (x_2, y_2, z_2) are harmonics with respect to the conic

$$\alpha x^2 + \beta y^2 + \gamma z^2 = 0,$$

which is a conic having with S, σ, a system of common conjugate points. The equation may also be written

$$(-bc + ca + ab)\, x^2 + (bc - ca + ab)\, y^2 + (bc + ca - ab)\, z^2 = 0;$$

or, as it may also be written,

$$(bc + ca + ab)(x^2 + y^2 + z^2) - 2(bcx^2 + cay^2 + abz^2) = 0;$$

and, as before remarked, $bcx^2 + cay^2 + abz^2 = 0$ is the equation of the conic which is the polar of $ax^2 + by^2 + cz^2 = 0$ with respect to $x^2 + y^2 + z^2 = 0$.

The condition in order that there may be inscribed in the conic $x^2 + y^2 + z^2 = 0$ an infinity of triangles the angles of which are conjugate points with respect to the conic $\alpha x^2 + \beta y^2 + \gamma z^2 = 0$, is

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 0;$$

or writing this equation under the form $\beta\gamma + \gamma\alpha + \alpha\beta = 0$, and substituting for α, β, γ their values, we have the equation already found, as the condition in order that it may be possible in the conic $x^2 + y^2 + z^2 = 0$ to inscribe an infinity of triangles the sides of which touch the conic $ax^2 + by^2 + cz^2 = 0$.

Theorem. *Let*

$$ax^2 + by^2 + cz^2 = 0$$

be the equation of a spherical conic, and let $(\xi : \eta : \zeta)$, *a point on the conic, be the pole of a great circle cutting the conic in two points; the conic intersects upon the great circle an arc given by the equation*

$$\cos\delta = \frac{(a + b + c)\sqrt{\xi^2 + \eta^2 + \zeta^2}}{\sqrt{(a + b + c)^2(\xi^2 + \eta^2 + \zeta^2) - 4(bc\xi^2 + ca\eta^2 + ab\zeta^2)}};$$

hence if $a + b + c = 0$, $\delta = 90°$; *or there may be inscribed in the conic an infinity of triangles having each of their sides equal to* $90°$.

It is worth while, in connexion with the subject, and for the sake of a remark to which they give rise, to reproduce in a short compass some results long ago obtained by Jacobi and Richelot. The following are Jacobi's formulae for the chords of a circle subjected to the condition of touching another circle; viz. if in the figure we put

$$\begin{gathered} CP = R, \\ cp = r, \\ Cc = a, \\ \angle ACP = 2\phi, \\ \angle ACP' = 2\phi', \end{gathered}$$

then it is clear from geometrical considerations that

$$\frac{d\phi}{MA} = \frac{d\phi'}{MA'}.$$

We have

$$\begin{aligned} \overline{MA}^2 = \overline{cA}^2 - \overline{cM}^2 &= a^2 + R^2 + 2aR\cos 2\phi - r^2 \\ &= (a+R)^2 - r^2 - 4aR\sin^2\phi \\ &= [(a+R)^2 - r^2](1 - k^2\sin^2 2\phi), \end{aligned}$$

or

$$\frac{d\phi}{\sqrt{1-k^2\sin^2\phi}} = \frac{d\phi'}{\sqrt{1-k^2\sin^2\phi'}},$$

where

$$k^2 = \frac{4aR}{(a+R)^2 - r^2},$$

and therefore also

$$k'^2 = \frac{(R-a)^2 - r^2}{(R+a)^2 - r^2}.$$

It will be convenient for comparison with the formulae of Richelot to write $\angle ACQ = 2\psi$; this gives $2\psi = \pi - 2\phi$, and the differential equation thus becomes

$$\frac{d\psi}{\sqrt{a^2+R^2-r^2-2aR\cos 2\psi}} = \frac{d\psi'}{\sqrt{a^2+R^2-r^2-2aR\cos 2\psi'}},$$

i.e.

$$\frac{d\psi}{\sqrt{m-n\cos 2\psi}} = \frac{d\psi'}{\sqrt{m-n\cos 2\psi'}},$$

or if

$$\tan\psi = \tfrac{m-n}{m+n}\tan\theta,$$

and therefore

$$\cos 2\psi = \frac{1 - \frac{(m-n)^2}{(m+n)^2}\tan^2\theta}{1 + \frac{(m-n)^2}{(m+n)^2}\tan^2\theta},$$

$$m - n\cos 2\psi = \frac{(m-n)\cos^2\theta + (m+n)^{-1}(m-n)^2\sin^2\theta}{\cos^2\theta + \frac{(m-n)^2}{(m+n)^2}\sin^2\theta},$$

if $g^2 = \frac{m-n}{m+n}$, then this is

$$= \frac{m-n}{\cos^2\theta + \frac{m-n}{m+n}\sin^2\theta},$$

and we have also

$$d\psi = \frac{\sqrt{\frac{m-n}{m+n}}\,d\theta}{1 - \frac{2n}{m+n}\sin^2\theta},$$

and thence

$$\frac{d\psi}{\sqrt{m - n\cos 2\psi}} = \frac{d\theta}{\sqrt{m+n}\sqrt{1 - \frac{2n}{m+n}\sin^2\theta}},$$

that is,

$$\frac{d\psi}{\sqrt{m - n\cos 2\psi}} = \frac{d\theta}{\sqrt{m+n}\sqrt{1 - k^2\sin^2\theta}},$$

where $k^2 = \dfrac{2n}{m+n}$ has the same value as before. Hence the relation between θ, θ' is

$$\frac{d\theta}{\sqrt{1 - k^2\sin^2\theta}} = \frac{d\theta'}{\sqrt{1 - k^2\sin^2\theta'}},$$

which is identical with that between ϕ and ϕ'; and in fact the equation between θ, ϕ is

$$\tan\theta\tan\phi = g',$$

which, if $\phi = \operatorname{am} u$, gives $\theta = \operatorname{am}(K' - u)$.

The differential equation contains only a single arbitrary parameter; hence the same differential equation might have been obtained from different values of a, R, r, the parameters which determine the circle enveloped by the moveable chord. The condition for this of course is $\dfrac{m'}{n'} = \dfrac{m}{n}$, that is

$$\frac{a^2 + R^2 - r^2}{2aR} = \frac{a'^2 + R'^2 - r'^2}{2a'R'},$$

or as the equation may be written

$$(aa' - RR')(a - a') - R(a(r'^2 - ar'^2)) = 0;$$

this implies that the enveloped circles intersect the other circle in the same two points, or that all the circles have a common chord.

Suppose for $\psi = 0$, we have $\psi' = \beta$, then it is easy to see geometrically that

$$\tan^2\beta = \frac{(R-a)^2 - r^2}{r^2};$$

let the corresponding value of θ' be $\theta' = \alpha$, i.e. suppose that for $\theta = 0$, we have $\theta' = \alpha$, then

$$\tan^2\alpha = \frac{(R+a)^2 - r^2}{(R-a)^2 - r^2} \cdot \frac{(R-a)^2 - r^2}{r^2},$$

i.e.

$$\tan^2\alpha = \frac{(R+a)^2 - r^2}{r^2};$$

or, what is the same thing,

$$\sec\alpha = \frac{R}{r} + \frac{a}{r};$$

and α having this value, we have for the finite relation between θ, θ',

$$F\theta' = F\theta + F\alpha.$$

Richelot has shown, by precisely similar reasoning, that for circles of the sphere we have

$$\frac{d\psi}{\sqrt{\cos^2 r - (\cos R\cos a + \sin R\sin a\cos 2\psi)^2}} = \frac{d\psi'}{\sqrt{\cos^2 r - (\cos R\cos a + \sin R\sin a\cos 2\psi')^2}},$$

which is of the form

$$\frac{d\psi}{\sqrt{1 - (\lambda + \mu\cos 2\psi)^2}} = \frac{d\psi'}{\sqrt{1 - (\lambda + \mu\cos 2\psi')^2}},$$

where

$$\lambda = \frac{\cos R\cos a}{\cos r},$$
$$\mu = \frac{\sin R\sin a}{\cos r}.$$

It is very important to remark, that this equation contains the two parameters λ, μ, so that the same equation cannot be obtained with any new values of the parameters a, r; or the formulae in *plano* for three or more circles do not apply to circles of the sphere: the geometrical reason for this is as follows, viz. in the plane a circle is a conic passing through two fixed points (the circular points at ∞), and consequently any number of circles having a common chord are in fact to be considered as conics each of which passes through the same four points. But circles of the sphere are not spherical conics passing through two fixed points, but are merely spherical conics having a double contact with an imaginary spherical conic (viz. the curve of intersection of the sphere with a sphere radius zero); hence circles of the sphere having a common spherical chord are not spherical conics passing through the same four points. I am not sure whether this remark as to the ground of the distinction between the theory of circles in *plano* and that of circles on the sphere has been explicitly made in any of the treatises on spherical geometry.

To reduce the equation, write

$$\tan\psi = \sqrt{\frac{1-(\lambda+\mu)}{1-(\lambda-\mu)}}\,\tan\theta,$$

then, after a simple reduction,

$$\frac{d\psi}{\sqrt{1-(\lambda+\mu\cos 2\psi)^2}} = \frac{d\theta}{\sqrt{(1+\mu)^2-\lambda^2}\sqrt{1-\frac{4\mu}{(1+\mu)^2-\lambda^2}\sin^2\theta}},$$

or the relation between the two values of θ is

$$\frac{d\theta}{\sqrt{1-k^2\sin^2\theta}} = \frac{d\theta'}{\sqrt{1-k^2\sin^2\theta'}},$$

where

$$k^2 = \frac{4\mu}{(1+\mu)^2-\lambda^2},$$

that is

$$k^2 = \frac{4\,\dfrac{\tan R}{\tan r}\cdot\dfrac{\sin a}{\cos R\sin r}}{\left(\dfrac{\tan R}{\tan r}+\dfrac{\sin a}{\cos R\sin r}\right)^2-1}.$$

Suppose that for $\psi=0$, we have $\psi'=\beta$, it is easy to see that

$$\tan^2\beta = \frac{\sin^2(R-a)-\sin^2 r}{\cos^2 r};$$

let the corresponding value of θ' be $\theta'=\alpha$, i.e. suppose that for $\theta=0$, we have $\theta'=\alpha$, then

$$\begin{aligned}
\tan^2\alpha &= \frac{1-\dfrac{\cos(R+a)}{\cos r}}{1-\dfrac{\cos(R-a)}{\cos r}}\cdot\frac{\sin^2(R-a)-\sin^2 r}{\cos^2 r}\\
&= \frac{\cos r-\cos(R+a)}{\cos r-\cos(R-a)}\cdot\frac{\cos^2 r-\cos^2(R-a)}{\cos^2 r}\\
&= \frac{(\cos r-\cos(R+a))(\cos r+\cos(R-a))}{\cos^2 r}\\
&= \frac{\sin R\sin a+\cos r\cos a-\cos^2 R\cos^2 a}{\cos^2 R\sin^2 r}\\
&= \frac{(\cos r+\sin R\sin a)^2-\cos^2 R\cos^2 a}{\cos^2 R\sin^2 r},
\end{aligned}$$

i.e.

$$\tan^2\alpha = \left(\frac{\tan R}{\tan r}+\frac{\sin a}{\cos R\sin r}\right)^2-1,$$

whence

$$\sec\alpha = \left(\frac{\tan R}{\tan r}+\frac{\sin a}{\cos R\sin r}\right);$$

and α having this value, the finite relation between θ,θ' is

$$F\theta' = F\theta + F\alpha.$$

By comparing with the corresponding formula in *plano*, we arrive at Richelot's conclusion, that the formulae for the sphere may be deduced from those in *plano* by writing in the place of $\dfrac{R}{r}, \dfrac{a}{r}$, the functions $\dfrac{\tan R}{\tan r}, \dfrac{\sin a}{\cos R\sin r}$, respectively.

2, Stone Buildings, October 1, 1856.

203. On the Theory of the Analytical Forms called Trees

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 172–176.]

A symbol such as $A\,dx + B\,dy + \cdots$, where $A, B,$ &c. contain the variables $x, y,$ &c. in respect to which the differentiations are to be performed, partakes of the natures of an operand and operator, and may be therefore called an Operandator. Let $P, Q, R, \ldots$ be any operandators, and let U be a symbol of the same kind, or to fix the ideas, a mere operand; PU denotes the result of the operation P performed on U, and QPU denotes the result of the operation Q performed on PU; and generally in such combinations of symbols, each operation is considered as affecting the operand denoted by means of all the symbols on the right of the operation in question. Now considering the expression QPU, it is easy to see that we may write

$$QPU = (Q \times P)U + (QP)U,$$

where on the right-hand side $(Q \times P)$ and (QP) signify as follows: viz. $Q \times P$ denotes the mere algebraical product of Q and P, while QP (consistently with the general notation as before explained) denotes the result of the operation Q performed upon P as operand; and the two parts $(Q \times P)U$ and $(QP)U$ denote respectively the results of the operations $(Q \times P)$ and (QP) performed each of them upon U as operand. It is proper to remark that $(Q \times P)$ and $(P \times Q)$ have precisely the same meaning, and the symbol may be written in either form indifferently. But without a more convenient notation, it would be difficult to find the corresponding expressions for $RQPU$, &c. This, however, can be at once effected by means of the analytical forms called trees (see figs. 1, 2, 3), which contain all the trees which can be formed with one branch, two branches, and three branches respectively.

The inspection of these figures will at once show what is meant by the term in question, and by the terms *root*, *branches* (which may be either main branches, intermediate branches, or free branches), and *knots* (which may be either the root itself, or proper knots, or the extremities of the free branches). To apply this to the question in hand, PU consists of a single term represented by fig. 1 (*bis*); QPU consists, as above, of two terms represented by the two parts of fig. 2 (*bis*), viz. the first part represents the term $(Q \times P)U$, and the second part represents the term $(QP)U$. And it is obvious that fig. 2 (*bis*) is at once formed from the figure 1 (*bis*) by adding on a branch terminated by Q at each of the knots of the single part of fig. 1 (*bis*). In like manner $RQPU$ consists of six terms represented by the six parts of fig. 3 (*bis*), and this figure is at once formed from fig. 2 (*bis*) by adding on a branch terminated by R at each knot of each part of fig. 2 (*bis*). It is hardly necessary to remark that the first part of fig. 3 (*bis*) denotes what, in the notation first explained, would be denoted by $(R \times Q \times P)U$, the second term what would in like manner be denoted by $(RQ \times P)U$, and so on; the last part being the term which would be denoted by $((RQ)P)U$, viz. R operates upon Q, giving the operandator RQ, which operates upon P, giving the operandator $(RQ)P$, which finally operates upon U.

The figures 1 (*bis*), 2 (*bis*), &c. contain the same trees as are contained in the corresponding figures 1, 2, &c.; only, on account of the different modes of filling up, trees are considered as so many distinct trees in a figure of the second set which are considered as one and the same tree in the corresponding figure of the first set. A difference in the number of trees first occurs in the figures 3 and 3 (*bis*), the first of which contains only four, while the latter contains six trees, viz. the first tree, the second, third and fourth trees, the fifth tree and the sixth tree of fig. 3 (*bis*) correspond respectively to the first tree, the second tree, the third tree, and the fourth tree of fig. 3. To derive fig. 3 (*bis*) from fig. 3, we must fill up the trees of fig. 3 with U at the root and R, Q, P at the other knots in every possible manner, subject only to the restriction, that, reckoning up from the extremity of a free branch to the root, there must not be any transposition in the order of the symbols RQP, and taking care to admit only distinct trees. Thus the first tree of fig. 3 might be filled up in six ways; but the trees so obtained are considered as one and the same tree, and we have only the first tree of fig. 3 (*bis*). Again, on account of the restriction, the fourth tree of fig. 3 can be filled up in one way only, and we have thus the sixth tree of fig. 3 (*bis*). And thus, in general, each figure of the second set can be formed at once from the corresponding figure of the first set; or when the first set of figures is given, the expression for $YX \dots QPU$ can be formed directly without the assistance of the expression for the preceding symbol $X \dots QPU$; the number of terms for the nth figure of the second set is obviously $1.2.3 \dots n$, and consequently it is only necessary to count the terms in order to ascertain that no admissible mode of filling up has been omitted.

The number of parts in any one of the figures of the first set is much smaller than the number of parts in the corresponding figure of the second set; and the law for the number of parts, i.e. for the number n of the trees with n branches, is a very singular one. To obtain this law, we must consider how the trees with n branches can be formed by means of those of a smaller number of branches. A tree with n branches has either a single main branch, or else two main branches, three main branches, &c.,... to n main branches. If the tree has one main branch, it can only be formed by adding on to this main branch a tree with $(n-1)$ branches, i.e. A_n contains a part A_{n-1}. If the tree has two main branches, then $p+q$ being a partition of $n-2$, the tree can be formed by adding on to one main branch a tree of p branches, and to the other main branch a tree of q branches; the number of trees so obtained is $A_p\,A_q$: this, however, assumes that the parts p and q are unequal; if they are equal, it is easy to see that the number of trees is only $\frac{1}{2}A_p(A_p+1)$. Hence $p+q$ being any partition of $n-2$, A_n contains the part $A_p\,A_q$ if p and q are unequal, and the part $\frac{1}{2}A_p(A_p+1)$ if p and q are equal. In like manner, considering the trees with three main branches, then if $p+q+r$ is any partition of $n-3$, A_n contains the part $A_p\,A_q\,A_r$ if p, q, r are unequal; but if two of these numbers, e.g. p and q, are equal, then the part $\frac{1}{2}A_p(A_p+1)\,A_r$; and if p, q, r are all equal, then the part $\frac{1}{6}A_p(A_p+1)(A_p+2)$; and so on, until lastly we have a single tree with n main branches, or A_n contains the part unity. A little consideration will show that the preceding rule for the formation of the number A_n is completely expressed by the equation

$$(1-x)^{-A_1}(1-x^2)^{-A_2}(1-x^3)^{-A_3}\cdots = 1 + A_1x + A_2x^2 + A_3x^3 + \text{ \&c.},$$

and consequently that we may, by means of this equation, calculate successively for the different values of n the number A_n of the trees with n branches. The calculation may be effected very easily as follows: [the table as originally printed contained at the end of it some errors of calculation which were corrected, B.A. Report for 1875, p. 258].

$A_1 =$	1	(1)	1	1	1	1	1	1	1	1	1	1	1	1	
			1	1	1	1	1	1	1	1	1	1	1	1	
					1	1	1	1	1	1	1	1	1	1	
							1	1	1	1	1	1	1	1	
									1	1	1	1	1	1	
$A_2 =$	1	1	(2)	2	3	3	4	4	5	5	6	6	7	7	
				2	2	4	4	6	6	8	8	10	10	12	1
						3	3	6	6	9	9	12	12	15	1
								4	4	8	8	12	12	16	1
$A_3 =$	1	1	2	(4)	3	7	11	13	17	23	27	38	47	61	7
				1	4	8	16	20	28	38	50	68	88	116	14
						4	8	16	20	28	40	50	68	88	11
								10	10	20	30	40	60	80	11
$A_4 =$	1	1	2	4	(9)	11	19	29	47	66	91	109	167	231	29
					9	9	18	27	45	54	81	99	162	207	29
$A_5 =$	1	1	2	4	9	(20)	26	47	83	142	235	357	561	873	132
						20	20	40	60	100	140	200	280	400	56
$A_6 =$	1	1	2	4	9	20	(48)	67	123	222	415	752	1284	2228	374
							48	48	96	144	240	336	528	720	105
$A_7 =$	1	1	2	4	9	20	48	(115)	171	318	607	1144	2104	3838	692
								115	115	230					
$A_8 =$	1	1	2	4	9	20	48	115	(286)	433	837	1626	3138	5910	1099
									286	286					
$A_9 =$	1	1	2	4	9	20	48	115	286	(719)	1123	2230	4424	8716	1696
										719					
$A_n =$	1	2	4	9	20	48	115	286	719	1842					
for $r =$	1	2	3	4	5	6	7	8	9	10					

I have had occasion, for another purpose, to consider the question of finding the number of trees with a given number of free branches, bifurcations at least. Thus, when the number of free branches is three, the trees of the form in question are those in the annexed figure, and the number is therefore two. It is not difficult to see that we have in this case (B_r being the number of such trees with r free branches),

$$(1-x)^{-B_1}(1-x^2)^{-B_2}(1-x^3)^{-B_3}\cdots = 1 + x + 2B_1x^3 + 2B_2x^4 + 2B_3x^5 + \&\text{c.};$$

and a like process of development gives:

$B_r =$	1	2	5	12	33	90
for $r =$	2	3	4	5	6	7

I may mention, in conclusion, that I was led to the consideration of the foregoing theory of trees by Professor Sylvester's researches on the change of the independent variables in the differential calculus.

2, Stone Buildings, January 2, 1856.

204. On a Problem in the Partition of Numbers

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 245–248.]

It is required to find the number of partitions into a given number of parts, such that the first part is unity, and that no part is greater than twice the preceding part.

Commencing to form the partitions in question, these are

$$\begin{array}{c|c|c|cccccccc}
1 & 1 & 1\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1\ \text{\&c.;} \\
  & 2 & 2 & 1 & 1 & 2 & 2 & 2 & 2 & & \\
  &   &   & 1 & 2 & 1 & 2 & 3 & 4 & &
\end{array}$$

and if we were to proceed to the 4-partitions, each 3-partition ending in 1 would give rise to two such partitions; each 3-partition ending in 2 to four such partitions; each 3-partition ending in 3 to six partitions; and each 3-partition ending in 4 to eight such partitions. We form in this manner the Table:

	ending in																
Number of	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Totals
1-partitions	1																1
2-partitions	1	1															2
3-partitions	2	2	1	1													6
4-partitions	6	6	4	4	2	2	1	1									26
5-partitions	26	26	20	20	14	14	10	10	6	6	4	4	2	2	1	1	166
&c.																	

and we are thus led to the series

$$\begin{array}{l}
1, \\
1, 2, \\
1, 2, 4, 6, \\
1, 2, 4, 6, 10, 14, 20, 26, \\
\text{\&c.;}
\end{array}$$

where, considering 0 as the first term of each series, the first differences of any series are the terms twice repeated of the next preceding series: thus the differences of the fourth series are $1, 1, 2, 2, 4, 4, 6, 6$. It is moreover clear that the first half of each series is precisely the series which immediately precedes it; we need, in fact, only consider a single infinite series, $1, 2, 4, 6$, &c. It is to be remarked, moreover, that in the column of totals, the total of any line is precisely the first number in the next succeeding line.

Consider in general a series A, B, C, D, E, &c., and a series A', B', C', D', E', &c. derived from it as follows:

$$\begin{aligned}
A' &= 1A, \\
B' &= 2A, \\
C' &= 2A + B, \\
D' &= 2A + 2B, \\
E' &= 2A + 2B + C, \\
F' &= 2A + 2B + 2C, \\
&\&c.;
\end{aligned}$$

viz. the first differences of the series $0, A', B', C', D', E'$, &c. are A, A, B, B, C, C, &c. Then multiplying by $1, x, x^2$, &c. and adding, we have

$$\begin{aligned}
A' + B'x + C'x^2 + \&c. &= (1 + 2x + 2x^2 + \cdots)(A + Bx^2 + Cx^4 + \&c.) \\
&= \frac{1+x}{1-x}(A + Bx^2 + Cx^4 + \&c.);
\end{aligned}$$

and if we form in a similar manner from A', B', C', D', &c. a series A'', B'', C'', D'', &c. and so on, we have

$$\begin{aligned}
A'' + B''x + C''x^2 + \&c. &= \frac{1+x}{1-x}(A' + B'x^2 + C'x^4 + \&c.) \\
&= \frac{1+x}{1-x} \cdot \frac{1+x^2}{1-x^2}(A + Bx^4 + Cx^8 + \&c.),
\end{aligned}$$

and so on. Write $A = 1$, and suppose that the process is repeated an indefinite number of times, we have

$$1 + \mathfrak{B}x + \mathfrak{C}x^2 + \mathfrak{D}x^3 + \&c. = \frac{1+x}{1-x} \cdot \frac{1+x^2}{1-x^2} \cdot \frac{1+x^4}{1-x^4} \cdot \&c.,$$

and the coefficients $1, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$, &c. are precisely those of the infinite series $1, 2, 4, 6$, &c. We have more simply

$$1 + \mathfrak{B}x + \mathfrak{C}x^2 + \mathfrak{D}x^3 + \&c. = \frac{1}{(1-x)^2(1-x^2)(1-x^4)(1-x^8)\,\&c.},$$

which gives rise to the following very simple algorithm for the calculation of the coefficients:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	0	1	2	4	6	9	12	16	20	25	30	36	42	49	56
1	2	4	6	9	12	16	20	25	30	36	42	49	56	64	72
0	0	0	0	1	2	4	6	10	14	20	26	35	44	56	68
1	2	4	6	10	14	20	26	35	44	56	68	84	100	120	140
0	0	0	0	0	0	0	0	1	2	4	6	10	14	20	26
1	2	4	6	10	14	20	26	36	46	60	74	94	114	140	166
&c.															

The last line is marked off into periods of (reckoning from the beginning) $1, 2, 4, 8$, &c.; and by what has preceded, the series which gives the number of 1-partitions, 2-partitions, 3-partitions, &c. is found by summing to the end of each period and doubling the results; we thus, in fact, obtain $(1), 2, 6, 26, 166, 1626$, &c.: and the same series is also given by means of the last terms of the several periods.

The preceding expression for $1 + \mathfrak{B}x + \mathfrak{C}x^2 +$ &c. shows that $\mathfrak{B}, \mathfrak{C}$, &c. are the number of partitions of $1, 2, 3, 4, 5, 6$, &c. respectively into the parts $1, 1', 2, 4, 8$, &c.; and we are thus led to

Theorem. *The number of λ-partitions (first part unity, no part greater than twice the preceding one) is equal to the number of partitions of $2^{\lambda-1}-1$ into the parts $1, 1', 2, 4, \ldots 2^{\lambda-2}$. Or, again, it is equal to twice the sum of the number of partitions of $0, 1, 2, \ldots 2^{\lambda-1} - 1$ respectively into the parts $1, 1', 2, 4, \ldots 2^{\lambda-3}$ (where the number of partitions of 0 counts for 1).*

For example, the partitions of $0, 1, 2, 3$, &c. with the parts $1, 1', 2, \ldots$ are

$$\begin{array}{l}
(\,)\\
1,\ 1'\\
1+1,\ 1+1',\ 1'+1',\ 2\\
1+1+1,\ 1+1+1',\ 1+1'+1',\ 1'+1'+1',\ 2+1,\ 2+1',
\end{array}$$

the numbers of which are $1, 2, 4, 6$. Hence, by the first part of the theorem, the number of 3-partitions is 6, and by the second part of the theorem, the number of 4-partitions is

$$2(1 + 2 + 4 + 6) = 26.$$

2, Stone Buildings, March 17, 1857.

205. Note on the Summation of a Certain Factorial Expression

[From the *Philosophical Magazine*, vol. XIII. (1857), pp. 419–423.]

Mr Kirkman some months ago communicated to me a formula for the double summation of a factorial expression, to which formula he had been led by his researches on the partition of polygons. The formula in a slightly altered form is as follows: viz.

$$\Sigma_x\Sigma_y\,\frac{[x+y+2]^y}{[y+1]^{y+1}}\cdot\frac{[x]^x}{[y]^y}\cdot\frac{[r+k-x-y]^{r-1-x-y}}{[k-y]^{k-y}}\cdot\frac{[r-1-x]^{x-1-x}}{[k-1-y]^{k-1-y}}=\frac{2k}{r+3}\cdot\frac{[r+k+2]^k\,[r]^r}{[k+1]^{k+1}\,[k]^k},$$

the summation extending from $x=0$ to $x=r-1$, and $y=0$ to $y=k-1$. In the particular case when $k=r$, then all the terms of the series except those in which $y=x$ vanish; and putting therefore $k=r$ and $y=x$, and making a slight change in the form of the right-hand side, the formula becomes

$$\Sigma\,\frac{[2x+2]^x}{[x+1]^{x+1}}\cdot\frac{(2r-2x)^{r-1-x}}{[r-x]^{r-x}}=4\,\frac{[2r+1]^{r-1}}{[r-1]^{r-1}},$$

the summation extending from $x=0$ to $x=r-1$.

We have, in the notation of Gauss, $[m]^n=m(m-1)\cdots2\cdot1=\Pi m$, and a factorial $[m]^n$ is expressed in terms of the function Π by the formula $[m]^n=\Pi m\div\Pi(m-n)$. Write also

$$\Pi_1(m-\tfrac{1}{2})=(m-\tfrac{1}{2})(m-\tfrac{3}{2})\cdots\tfrac{3}{2}\cdot\tfrac{1}{2},$$

we have

$$\begin{gathered}\Pi(2m)=2^{2m}\,\Pi m\,\Pi_1(m-\tfrac{1}{2}),\\ \Pi(2m+1)=2^{2m+1}\,\Pi m\,\Pi_1(m+\tfrac{1}{2});\end{gathered}$$

and transforming the factorials by these formulae, the series becomes

$$\Sigma\,\frac{\Pi_1(x+\frac{1}{2})}{\Pi_1(\frac{1}{2})}\cdot\frac{\Pi_1(r-x-\frac{1}{2})}{\Pi_1(r-\frac{1}{2})}\cdot\frac{\Pi2}{\Pi(x+2)\,\Pi(r-x+1)}=\frac{8r(r+\frac{1}{2})}{(r+2)(r+3)}\cdot\frac{\Pi(r+1)}{\Pi(r+3)},$$

the summation, as before, from $x=0$ to $x=r-1$. This may be written

$$\Sigma\,\frac{4(r+\frac{1}{2})}{r+2}\cdot\frac{-8(r+\frac{1}{2})(r+\frac{1}{2})}{(r+2)(r+3)}+\frac{\Pi2}{(x-2)}\cdot\frac{\Pi(r+1)}{(r-2)(r-3)}=\frac{8r(r+\frac{1}{2})}{(r+2)(r+3)},$$

the summation from $x=0$ to $x=r-1$. The general term does not vanish for $x=r$ or $x=r+1$, but it vanishes for all greater values of x; hence if we add to the right-hand side the two terms corresponding to $x=r$ and $x=r+1$, the summation may be extended from $x=0$ to $x=r+1$, or what is the same thing, from $x=0$ indefinitely. The two terms in question are

$$\frac{r+2}{r+2}+\frac{(r+2)(r+3)}{(r+2)(r+3)}-\frac{4r(r+\frac{1}{2})}{(r+2)(r+3)},$$

and the resulting equation is

$$\Sigma\,\frac{\Pi_1(x+\frac{1}{2})\,\Pi_1(r-x-\frac{1}{2})\,\Pi2}{\Pi_1(\frac{1}{2})\,\Pi_1(r-\frac{1}{2})\,\Pi(x+2)\,\Pi(r-x+1)}=\frac{4r(r+\frac{1}{2})}{\Pi(r+1)},$$

the summation from $x=0$ indefinitely; or substituting for the functions Π and Π_1 their values, the formula is

$$\frac{1}{3\cdot(r+\frac{3}{2})}-\frac{3\cdot(r-\frac{1}{2})}{3\cdot4\cdot(r+\frac{3}{2})(r+\frac{5}{2})}-\frac{3\cdot4\cdot5\cdot(r-\frac{1}{2})(r-\frac{3}{2})}{3\cdot4\cdot5\cdot(r+\frac{3}{2})(r+\frac{5}{2})(r+\frac{7}{2})}+\cdots=\frac{4r(r+\frac{1}{2})}{(r+2)(r+3)},$$

which is a formula obviously belonging to the theory of the hypergeometric series

$$F(\alpha,\beta,\gamma,x)=1+\frac{\alpha\cdot\beta}{1\cdot\gamma}x+\frac{\alpha\cdot\alpha+1\cdot\beta\cdot\beta+1}{1\cdot2\cdot\gamma\cdot\gamma+1}x^2+\&\text{c.};$$

but the formula applicable to the case in hand has probably not been given. It may be proved as follows, premising that I disregard all difficulties arising from infinite values of the functions in the definite integrals, convergency, &c. We have

$$\int_0^1\theta^{\alpha-1}(1-\theta)^{-\gamma-\alpha+\beta-1}(1-x\theta)^{-\beta}\,d\theta=\frac{\Pi(\alpha-1)\,\Pi(-\gamma-\alpha+\beta-1)}{\Pi(-\gamma-1)}\left(1+\frac{\alpha\cdot\beta}{1\cdot\gamma}x+\frac{\alpha\cdot\alpha+1\cdot\beta\cdot\beta+1}{1\cdot2\cdot\gamma\cdot\gamma+1}\right.$$

Now we have

$$\int_0^1dx\int_0^1dx\,(1-x\theta)^{-\beta}=\frac{1}{(\beta+1)(\beta+2)\theta^2}\,(1-x\theta)^{-\beta+2}-\frac{1}{(\beta+1)(\beta+2)}\cdot\frac{1}{\theta^2}+\frac{x}{\theta(\beta+1)},$$

and hence multiplying by $d\alpha$ and integrating from $x=0$, and again multiplying by $d\alpha$ and integrating from $x=0$ to $x=1$, we find

$$\int_0^1d\theta\,\theta^{\alpha-3}(1-\theta)^{-\gamma-\alpha+\beta-1}=\int_0^1d\theta\,\theta^{\alpha-1}(1-\theta)^{-\gamma-\alpha+\beta-1}+(\beta+2)\int_0^1d\theta\,\theta^{\alpha-1}(1-\theta)^{-\gamma-\alpha+\beta-1}\cdot\tfrac{1}{2}S,$$

if for shortness

$$S=1+\frac{\alpha\cdot\beta}{3\cdot\gamma}+\frac{\alpha\cdot\alpha+1\cdot\beta\cdot\beta+1}{3\cdot4\cdot\gamma\cdot\gamma+1}+\&\text{c.}$$

Substituting for the definite integrals their values,

$$\frac{\Pi(\alpha-3)\,\Pi(-\gamma+\beta-1)}{\Pi(-\gamma+\beta-3)}=\frac{\Pi(\alpha-1)\,\Pi(-\gamma-\alpha-2)}{\Pi(-\gamma-1)}+(\beta+2)\frac{\Pi(\alpha-1)\,\Pi(-\gamma-\alpha-2)}{\Pi(-\gamma-1)}\cdot\tfrac{1}{2}S,$$

whence

$$\tfrac{1}{2}(\beta+1)(\beta+2)S=\frac{\Pi(\alpha-3)\,\Pi(-\alpha-\gamma+\beta+1)}{\Pi(\alpha-1)\,\Pi(-\alpha-\gamma-1)}\cdot\frac{\Pi(-\gamma-1)}{\Pi(-\gamma+\beta-1)}-\frac{\Pi(\alpha-3)\,\Pi(-\gamma-1)}{\Pi(\alpha-1)\,\Pi(-\gamma-2)}+(\beta+2)\,\frac{\Pi(\alpha-3}{\Pi(\alpha-1}$$

The second and third terms are

$$-\frac{1}{(\alpha-1)(\alpha-2)}(\gamma+1)(\gamma+2)-(\beta+2)\frac{1}{\alpha-1}(\gamma+1),$$

which are

$$=-\frac{\gamma+1}{(\alpha-1)(\alpha-2)}(\alpha\beta+2\alpha+2\beta+\gamma-2).$$

For the reduction of the first term we have

$$\Pi(-\alpha-\gamma+\beta-1)=[\beta-\gamma-1]^\beta\,\Pi(-\alpha-\gamma-1);$$

$$\Pi(\beta-\gamma-1) = [\beta-\gamma-1]^\beta\,\Pi(-\gamma-1);$$

and we thus find

$$\tfrac{1}{2}(\beta+1)(\beta+2)(\alpha-1)(\alpha-2)S = \frac{[\beta-\gamma-1]^\beta\,\Pi(-\alpha-\gamma+\beta-1)\,\Pi(-\gamma-1)}{[\beta-\gamma-1]^\beta\,\Pi(-\alpha-\gamma-1)} - (\gamma+1)(\alpha\beta+2\alpha+2\beta+\gamma-2),$$

where, as before,

$$S = 1 + \frac{\alpha\cdot\beta}{3\cdot\gamma} + \frac{\alpha\cdot\alpha+1\cdot\beta\cdot\beta+1}{3\cdot4\cdot\gamma\cdot\gamma+1} + \&\text{c.};$$

this is the formula in hypergeometric series required for the present purpose, and it is certainly true when the series is finite.

Write now

$$\alpha = \tfrac{1}{2}, \quad \beta = r+1, \quad \gamma = r-\tfrac{1}{2};$$

then the first term is $[1]^{r+2} \div [\frac{1}{2}]^{r+1}$, which vanishes on account of the numerator, and the second term is $-\frac{1}{2}r(r+\frac{1}{2})$, and we have consequently

$$3\Pi_1 = -\tfrac{1}{2}r(r+2)(r+\tfrac{3}{2})\cdot\tfrac{1}{2}S = -\tfrac{1}{2}r(r+\tfrac{1}{2}),$$

which gives

$$S = \frac{4r(r+\frac{1}{2})}{(r+2)(r+3)},$$

S being here the series in r, the sum of which was required, and the particular case of Mr Kirkman's formula is thus verified. It is probable that the general case might be treated in an analogous manner by first grouping together the terms which correspond to a given difference $x-y$, and ultimately summing the sums of these partial series; but I have not examined this question.

2, Stone Buildings, W.C., April 18, 1857.

The Collected Mathematical Papers of Arthur Cayley, Vol. III

PDF Pages 326–350 (printed pp. 304–328)

Transcribed from the Cambridge University Press edition

Papers 213 (conclusion) and 214 (opening)

Contents

213. On the Development of the Disturbing Function in the Lunar Theory *(continued)*

Tables for the lunar part of the disturbing function (continued)

[PDF p. 326; printed p. 304.] We have

$$\Omega_2 = m' \frac{a^4}{a'^5}\left(\tfrac{3}{8} - \tfrac{33}{8}\eta^2 + \tfrac{75}{8}\eta^4\right) \text{ multiplied into}$$

Lunar part *infra.*		
$\frac{73}{6}e''^6$	cos	$-4g'\}$
$\frac{55}{8}e''^5$		$-3g'$
$3e''^4 + \frac{11}{4}e''^6$		$-2g'$
$1 + 2e''^2$		$-g'$
$*\, e'' + \frac{8}{3}e''^3$		$0g' \quad -\mathfrak{C}'$
$1 + \frac{5}{2}e''^2$		$+g'$
$\frac{11}{8}e''^2$		$+2g'$
$\frac{55}{12}e''^3$		$+3g'\}$
* Exact value is $e'(1-e'^2)^{-\frac{3}{2}}$.		
$\frac{7}{128}e^8$	Solar part *suprà.*	$-3g$
$\frac{1}{6}e^6$		$-2g$
$\frac{11}{8}e^5 + \frac{7}{48}e^7$		$-g$
$-\frac{5}{2}e - \frac{15}{8}e^3$ (exact)		$0g$
$1 + 2e^2 + \frac{61}{64}e^4$		$+g \quad +\mathfrak{C}$
$-\frac{1}{2}e + e^3$		$+2g$
$-\frac{3}{8}e^2 + \frac{11}{16}e^4$		$+3g$
$-\frac{7}{24}e^3$		$+4g$
$-\frac{95}{384}e^4$		$+5g\}$

$g - g' + \mathfrak{C} - \mathfrak{C}'$ parallactic equation.

[PDF p. 327; printed p. 305.] We have

$$\Omega_3 = m' \frac{a^4}{a'^5} \left(\tfrac{5}{8} - \tfrac{15}{8}\eta^2 + \tfrac{15}{8}\eta^4 \right) \text{ multiplied into}$$

Lunar part *infra.*		
$\frac{165}{4}e''^6$	cos	$-6g'\}$
$\frac{177}{8}e''^5$		$-5g'$
$5e''^4 + \frac{22}{1}e''^6$		$-4g'$
$1 + 6e''^2$		$-3g' \quad -3\mathfrak{C}'$
$-e'' + \frac{8}{3}e''^3$		$-2g'$
$\frac{1}{8}e''^2$		$-g'$
0 (exact)		$0g'\}$
$\frac{75}{128}e^8$	Solar part *suprà.*	$-g$
$-\frac{35}{8}e^7$ (exact)		$0g$
$\frac{57}{8}e^6 - \frac{65}{16}e^8$		$+g$
$-\frac{9}{2}e^5 + \frac{35}{4}e^7$		$+2g$
$1 - 6e^2 + \frac{593}{64}e^4$		$+3g \quad +3\mathfrak{C}$
$\frac{3}{2}e^3 + \frac{57}{8}e^5$		$+4g$
$\frac{15}{8}e^4 - \frac{135}{16}e^6$		$+5g$
$\frac{9}{4}e^5$		$+6g$
$\frac{343}{128}e^6$		$+7g\}$

[PDF p. 328; printed p. 306.] We have

$$\Omega_4 = m'\frac{a^4}{a'^5}\left(\tfrac{9}{4}\eta^2 - \tfrac{15}{2}\eta^4\right) \text{ multiplied into}$$

Lunar part *infra.*		
$\frac{23}{12}e''^2$	cos	$-2g'\}$
$\frac{11}{8}e''^3$		$-g'$
$*\,e'' + \frac{8}{3}e''^3$		$0g' \quad -\mathfrak{C}'$
$1 + 2e''^2$		$+g'$
$3e''^2 + \frac{11}{4}e''^4$		$+2g'$
$\frac{55}{8}e''^3$		$+3g'$
$\frac{77}{6}e''^4$		$+4g'\}$
* Exact value is $e'(1-e'^2)^{-\frac{3}{2}}$.		
$\frac{7}{128}e^8$	Solar part *suprà.*	$-3g$
$\frac{5}{6}e^6$		$-2g$
$\frac{11}{8}e^5 + \frac{7}{48}e^7$		$-g$
$-\frac{5}{2}e - \frac{15}{8}e^3$ (exact)		$0g$
$1 + 2e^2 + \frac{61}{64}e^4$		$+g \quad +\mathfrak{C}$
$-\frac{1}{2}e + e^3$		$+2g$
$-\frac{3}{8}e^2 + \frac{11}{16}e^4$		$+3g$
$-\frac{7}{24}e^3$		$+4g$
$-\frac{95}{384}e^4$		$+5g\}$

[PDF p. 329; printed p. 307.] We have

$$\Omega_5 = m'\frac{a^4}{a'^5}\left(\tfrac{15}{8}\eta^2 - \tfrac{15}{4}\eta^4\right) \text{ multiplied into}$$

Lunar part *infra.*		
$\frac{77}{6}e''^4$	cos	$-4g'\}$
$\frac{55}{8}e''^3$		$-3g'$
$3e''^2 + \frac{11}{4}e''^4$		$-2g'$
$1 + 2e''^2$		$-g'$
$* \, e'' + \frac{8}{3}e''^3$		$0g' \quad -\mathfrak{C}'$
$\frac{11}{8}e''^3$		$+g'$
$\frac{23}{12}e''^2$		$+2g'\}$
* Exact value is $e'(1-e'^2)^{-\frac{3}{2}}$.		
$\frac{75}{128}e^8$	Solar part *suprà.*	$-g$
$-\frac{35}{8}e^7$ (exact)		$0g$
$\frac{57}{8}e^6 - \frac{65}{16}e^8$		$+g$
$-\frac{9}{2}e^5 + \frac{35}{4}e^7$		$+2g$
$1 - 6e^2 + \frac{593}{64}e^4$		$+3g \quad +3\mathfrak{C}$
$\frac{3}{2}e^3 + \frac{57}{8}e^5$		$+4g$
$\frac{15}{8}e^4 - \frac{135}{16}e^6$		$+5g$
$\frac{9}{4}e^5$		$+6g$
$\frac{343}{128}e^6$		$+7g\}$

[PDF p. 330; printed p. 308.] We have

$$\Omega_6 = m'\frac{a^4}{a'^5}\left(\tfrac{15}{8}\eta^2 - \tfrac{15}{4}\eta^4\right) \text{ multiplied into}$$

Lunar part *infra.*		
$\frac{165}{4}e''^6$	cos	$-6g'\}$
$\frac{177}{8}e''^5$		$-5g'$
$5e''^4 + 22e''^6$		$-4g'$
$1 + 6e''^2$		$-3g' \quad -3\mathfrak{C}'$
$-e'' + \frac{8}{3}e''^3$		$-2g'$
$\frac{1}{8}e''^2$		$-g'$
0 (exact)		$0g'\}$
$\frac{7}{128}e^8$	Solar part *suprà.*	$-3g$
$\frac{1}{6}e^6$		$-2g$
$\frac{11}{8}e^5 + \frac{7}{48}e^7$		$-g$
$-\frac{5}{2}e - \frac{15}{8}e^3$ (exact)		$0g$
$1 + 2e^2 + \frac{61}{64}e^4$		$+g \quad +\mathfrak{C}$
$-\frac{1}{2}e + e^3$		$+2g$
$-\frac{3}{8}e^2 + \frac{11}{16}e^4$		$+3g$
$-\frac{7}{24}e^3$		$+4g$
$-\frac{95}{384}e^4$		$+5g\}$

[PDF p. 331; printed p. 309.] The arrangement of the tables hardly requires any explanation; each line of the upper half of a table is to be read in combination with each line of the lower half of the same table. Thus a term of Ω_1 is

$$m'\frac{a^4}{a'^5}(\tfrac{3}{4}\eta^2+\tfrac{9}{4}\eta^4)(-\tfrac{47}{4}e''^4)(1-\tfrac{3}{2}e''^2+\tfrac{1}{16}e''^4)\cos(-g-2g'+2\mathfrak{C}-2\mathfrak{C}').$$

It is to be observed that in the table for Ω_1 (where the arguments depend only on the mean anomalies), the several terms (other than the constant term) occur in pairs of equal terms, having respectively the arguments $ig+i'g'$ and $-ig-i'g'$; such equal terms are to be united together, or, what is the same thing, the coefficients must be multiplied by 2. Thus we have the two terms

$$m'\frac{a^4}{a'^5}(\tfrac{3}{4}\eta^2+\tfrac{9}{4}\eta^4)(-\tfrac{47}{4}e''^4)(1-\tfrac{3}{2}e''^2+\tfrac{1}{16}e''^4)2\cos(-6g+5g'),$$

$$\text{Ditto}\qquad\qquad \cos(\;\;6g+5g'),$$

or, what is the same thing, the term is

$$m'\frac{a^4}{a'^5}(\tfrac{3}{4}\eta^2+\tfrac{9}{4}\eta^4)(-\tfrac{47}{4}e''^4)(1-\tfrac{3}{2}e''^2+\tfrac{1}{16}e''^4)2\cos(\;\;6g+5g').$$

This is the case even when either i or i' vanishes; but, as already noticed, it is not the case for the constant term where i and i' both vanish. There are not any such equal pairs in the tables for Ω_2, &c., where all the arguments contain a part independent of the mean anomalies. The quantity $\mathfrak{C}$ occurs in the upper or solar half of the table; but it is to be recollected that $\mathfrak{C}\,(=\varpi-\theta)$ involves θ, which is an element of the moon's orbit.

The peculiar form in which the coefficients are exhibited, viz. as the product of a term depending on the inclination, a term depending on the eccentricity of the moon's orbit, and a term depending on the eccentricity of the sun's orbit, is not to be considered as a want of completion of the development; it is, on the contrary, an important advantage.

[The Errata in the Tables, noticed *Mem. R. Ast. Soc.* vol. XXVIII. p. 216, have been corrected. The values of P'_1, Q'_1, Q'_2, &c. used in the construction of the Tables *ante* p. 298 may also be obtained from the Tables of the Developments of Functions in *The Theory of Elliptic Motion*, p. 216.]

Addition.

I deduce from the preceding an expression for the disturbing function in a form similar to and easily comparable with the forms given by Sir J. W. Lubbock in his work *On the Theory of the Moon* &c. (London, 1834), pp. 30–35, and by Pontécoulant in the *Théorie Analytique du Système du Monde*, t. IV. (Paris, 1846), pp. 58–61. The several notations are as follows, viz.:

[PDF p. 332; printed p. 310.]

Lubbock.	Pontécoulant.	Supra.
ξ,	ϕ,	g
ξ',	ϕ',	g'
τ,	ξ,	$g - g' + \mathfrak{C} - \mathfrak{C}'$
η,	η,	$g + \mathfrak{C}$
γ,	γ,	$2\eta\sqrt{1-\eta^2} \div (1 - 2\eta^2)$

(γ, the tangent of the inclination, is employed by the above-named two authors instead of η, the sine of the semi-inclination, the relation between these two quantities gives to γ^2 or η^4

$$\gamma^2 = 4\eta^2 + 12\eta^4,$$

and conversely,

$$\eta^2 = \tfrac{1}{4}\gamma^2 - \tfrac{1}{16}\gamma^4,$$
$$\eta^4 = \tfrac{1}{16}\gamma^4,$$

which are useful for replacing one of these quantities by the other). The disturbing function is taken by Lubbock with the contrary sign, and he includes in the before-mentioned omitted term depending only on the sun's radius vector, that is, R (Lubbock) $= -m'\dfrac{1}{r'} - \Omega$; he uses also, in reference to the sun, subscript strokes instead of accents (so that, in referring to his notation, $\xi_{|}$ should properly be here used in the place of ξ', but this difference is obviously immaterial). Pontécoulant's disturbing function is taken with the same sign as in the present memoir, or we have R (Pontécoulant) $= \Omega$. Lubbock and Pontécoulant give in the principal terms the part involving $m'\dfrac{a^4}{a'^5}$, which depends on the square of the parallax, and which was disregarded in the preceding development. I have in the sequel inserted these terms from Lubbock's expression. We have, in fact:

[PDF p. 333; printed p. 311.]

$$\Omega = m'\frac{a^2}{a'^3} \text{ multiplied into}$$

	Lubbock.		Pontécoulant.		Arguments ut *suprà.*
	Arguments.	Nos.	Arguments.	Nos.	
$\frac{1}{4}+\frac{3}{4}e^2+\frac{9}{64}e^4+\frac{9}{2}e'^2+\frac{15}{32}e'^4+\cdots$		0		0	
$-\frac{3}{2}e-\frac{27}{16}e^3-\frac{15}{64}e^5+\cdots$	τ	1	2ξ	30	$2g-2g'+2\mathfrak{C}-2\mathfrak{C}'$
$-\frac{1}{2}e+\frac{1}{16}e^3-\frac{5}{4}ee'^2+\cdots$	ξ	2	ϕ	1	g
$\frac{15}{8}e^2-\frac{45}{32}e^4-\frac{15}{4}ee'+\cdots$	$\tau-\xi$	3	$2\xi-\phi$	31	$g-2g'$
$-\frac{3}{4}ee'-\frac{45}{16}e^3e'+\cdots$	$\tau+\xi$	4	$2\xi+\phi$	32	$3g-2g'$
$\frac{3}{4}e^2-\frac{15}{32}e^4\cdots$	ξ'	5	ϕ'	6	
$\frac{3}{8}e^2-\frac{105}{16}e^4+\cdots$	$\tau-\xi'$	7	$2\xi-\phi'$	33	$2g-3g'$
$\frac{15}{8}ee'+\cdots$	$\tau+\xi'$	8	$2\xi+\phi'$	34	$2g-g'$
$-\frac{3}{4}ee'\cdots$	$2\xi'$	9	$2\phi'$	2	$2g'$
$\frac{15}{16}e^2-\frac{15}{4}ee'^2\cdots$	$\tau-2\xi'$	9	$2\xi-2\phi'$	35	$-2g'$
$\frac{1}{2}ee'\cdots$	$\tau+2\xi'$	10	$2\xi+2\phi'$	36	$4g-2g'$
$\frac{3}{16}e^3+\frac{9}{4}ee'^2\cdots$	$\xi-\xi'$	11	$\phi+\phi'$	9	$g-g'$

(Table continues in original; arguments to No. 41 with full coefficients.)

* Variation. † Evection.

[PDF p. 334; printed p. 312.]

$$\Omega\,(\text{continued}) = m'\frac{a^2}{a'^3}\text{ multiplied into}$$

	Lubbock.		Pontécoulant.		Arguments
	Arguments.	Nos.	Arguments.	Nos.	ut *suprà.*
$-\frac{1}{16}e^3$	3ξ	20	3ϕ	3	$3g$
$-\frac{3}{32}e^2$	$\tau-3\xi$	21	$2\xi-3\phi$	43	$-g-2g'$
$\frac{15}{32}e^2$	$\tau+3\xi$	22	$2\xi+3\phi$	44	$5g-2g'$
$\frac{3}{4}ee'$	$2\xi+\xi'$	23	$2\phi+\phi'$	11	$2g+g'$
$\frac{105}{16}e^2e'$	$\tau-2\xi+\xi'$	24	$2\xi-2\phi+\phi'$	45	g'
$-\frac{3}{8}ee'$	$2\xi-\xi'$	26	$2\phi-\phi'$	12	$2g-g'$
$\frac{3}{4}e^2e'$	$\tau-2\xi-\xi'$	28	$2\xi-2\phi-\phi'$	46	$-g'$
$-\frac{15}{16}ee'^2$	$\tau+2\xi-\xi'$	28	$2\xi+2\phi-\phi'$	47	$4g-g'$
$\frac{9}{16}e^3$	$\xi+2\xi'$	29	$\phi+2\phi'$	13	$g+2g'$
$\frac{45}{32}e^2e'$	$\tau-\xi-2\xi'$	30	$2\xi-\phi-2\phi'$	49	$g-4g'$
$-\frac{9}{16}ee'^2$	$\xi-2\xi'$	32	$\phi-2\phi'$	14	$g-2g'$
$\frac{3}{32}e^3$	$\tau-\xi+2\xi'$	34	$2\xi-\phi+2\phi'$	51	$3g-4g'$

(Remainder of table continues numbering through No. 53.)

[PDF p. 335; printed p. 313.]

$$\Omega\,(\text{continued}) = m'\frac{a^2}{a'^3}\ \text{multiplied into}$$

	Lubbock.		Pontécoulant.		Arguments
	Arguments.	Nos.	Arguments.	Nos.	ut *suprà.*
$-\frac{15}{16}e^2e'$	$\tau+3\xi-\xi'$	43			$5g-3g'+2\mathfrak{C}-2\mathfrak{C}'$
$\frac{3}{32}e^3$	$3\xi-\xi'$	44	$3\phi-\phi'$	16	$3g-g'$
$\frac{3}{32}e^3$	$\tau-3\xi-\xi'$	45	$2\xi-3\phi+\phi'$	54	$-g-3g'+2\mathfrak{C}-2\mathfrak{C}'$
$\frac{175}{64}e^2$	$\tau+3\xi-\xi'$	46			$5g-3g'$
$\frac{15}{16}e^4$	$2\xi+2\xi'$	47	$2\phi+2\phi'$	15	$2g+2g'$
$-\frac{255}{16}e^4e'$	$\tau-2\xi+2\xi'$	48	$2\xi-2\phi+2\phi'$	55	$4g'$
$-\frac{3}{8}e^4e'$	$2\xi-2\xi'$	49	$2\phi-2\phi'$	14	$2g-2g'$
$\frac{45}{8}e^4e'$	$\tau+2\xi-2\xi'$	52	$2\xi+2\phi-2\phi'$	56	$4g-2g'$
$\frac{535}{64}e^4e'$	$\tau-2\xi-2\xi'$	53			$-3g'+2\mathfrak{C}-2\mathfrak{C}'$
$\frac{8}{64}ee'^3$	$\xi'-3\xi'$	54			$-5g'$
$\frac{45}{8}e^4e'$	$\xi+3\xi'$	56			$g-5g'$
$\frac{77}{64}e^4$		59			$g-3g'$
$-\frac{545}{64}e^4e'$	$\tau-4\xi'$	61			$2g-6g'$
$-\frac{1}{32}ee'$	$\tau+4\xi'$	62			$2g+2g'$

(Numbering continues to the last argument of the Lunar half.)

[PDF p. 336; printed p. 314.]

$$\Omega\,(\text{continued}) = m'\frac{a^2}{a'^3}\ \text{multiplied into}$$

	Lubbock.		Pontécoulant.		Arguments
	Arguments.	Nos.	Arguments.	Nos.	ut *suprà.*
$-\frac{3}{4}\eta^2 ee'$	$\tau - \xi - 2\eta$ rev.	67	$2\xi - \phi - 2\eta$ rev.	59	$g + 2g'$
$-\frac{3}{4}\eta^2 ee'$	$\tau + \xi - 2\eta$ rev.	69	$2\xi + \phi - 2\eta$ rev.	61	$-g + 2g'$
$\frac{9}{4}\eta^2 e'$	$\xi' - 2\eta$ rev.	71	$\phi' - 2\eta$ rev.	21	$2g - g'$
$\frac{9}{4}\eta^2 e'$	$\xi' + 2\eta$ rev.	72	$\phi' + 2\eta$ rev.	22	$2g + g'$
$-\frac{3}{4}\eta^2 e^2 e'$	$\tau - \xi' - 2\eta$ rev.	73	$2\xi - \phi' - 2\eta$ rev.	63	$3g'$
$\frac{3}{4}\eta^2 e^2 e'$	$\tau + \xi' - 2\eta$ rev.	75	$2\xi + \phi' - 2\eta$ rev.	65	g'
$\frac{15}{4}\eta^2$	$2\xi - 2\eta$	77	$2\phi - 2\eta$	24	$0g$
$\frac{15}{4}\eta^2$	$2\xi + 2\eta$	78	$2\phi + 2\eta$	24	$4g$
$\frac{3}{4}\eta^2 ee'$	$\tau - \xi - 2\eta$ rev.	79	$2\xi - 2\phi - 2\eta$ rev.	67	$2g + 2g'$
$-\frac{3}{4}\eta^2 ee'$	$\tau + \xi - 2\eta$ rev.	81	$2\xi + 2\phi - 2\eta$ rev.	69	$2g - 2g'$
$\frac{27}{4}\eta^2 ee'$	$\xi + \xi' - 2\eta$	83	$\phi - \phi' - 2\eta$ rev.	77	$g - g'$
$\frac{9}{4}\eta^2 ee'$	$\xi + \xi' + 2\eta$	84	$\phi + \phi' + 2\eta$	28	$3g + g'$

(Numbering reaches 97 at end of this page.)

[PDF p. 337; printed p. 315.]

$$\Omega\,(\text{continued}) = m'\frac{a^2}{a'^3}\ \text{multiplied into}$$

*(Lunar half of Ω ends with No. 100, table preceded by * Parallactic equation.)*					
	Lubbock.		Pontécoulant.		Arguments
	Arguments.	Nos.	Arguments.	Nos.	ut *suprà*.
$*\frac{3}{4}+\frac{3}{4}e^2+\frac{3}{4}e'^2-\frac{33}{8}\eta^2$	τ	101	ξ	70	$g-g'$
$-\frac{15}{16}e$	$\tau-\xi$	102	$\xi-\phi$	71	$-g'$
$-\frac{3}{16}e$	$\tau+\xi$	103	$\xi+\phi$	72	$g-g'$
$\frac{6}{8}e'$	$\tau-\xi'$	104	$\xi-\phi'$	73	$g-2g'$
$\frac{6}{8}e'$	$\tau+\xi'$	105	$\xi+\phi'$	74	g
$-\frac{135}{64}ee'$	$\tau-2\xi'$	106	$\xi-2\phi$	75	$-g-g'$
$\frac{9}{64}ee'$	$\tau+2\xi'$	108	$\xi-\phi-\phi'$	76	$3g-g' \quad \mathfrak{C}-\mathfrak{C}'$
$-\frac{45}{16}ee'^2$	$\tau-\xi'+\xi'$	109	$\xi+\phi-\phi'$	77	$2g$
$-\frac{3}{16}ee'^2$	$\tau+\xi'-\xi'$	110	$\xi+\phi+\phi'$	79	$g+g'$
$\frac{9}{16}ee'^2$	$\tau-\xi'-\xi'$	111	$\xi-\phi+\phi'$	78	$g+0g$
$\frac{135}{64}ee'^2$	$\tau+\xi'+\xi'$	112	$\xi-2\phi+\phi'$		$g+g'$
$\frac{33}{64}\eta^2$	$\tau-2\eta$	113			g
$\frac{9}{4}\eta^2$	$\tau+2\eta$	114			g

(Solar half continues to No. 130 in the original.)

[PDF p. 338; printed p. 316.]

$$\Omega\,(\text{continued}) = m'\frac{a^2}{a'^3} \text{ multiplied into}$$

	Lubbock.		Pontécoulant.		Arguments
	Arguments.	Nos.	Arguments.	Nos.	ut *suprà.*
$\frac{75}{64}e^2$	$3\tau+3\xi$	122			$5g-3g'$
$-\frac{225}{16}e^2$	$3\tau-\xi-\xi'$	123			$2g-4g'$
$-\frac{15}{16}ee'^2$	$3\tau+\xi+\xi'$	124			$4g-2g'$
$\frac{45}{16}e^2e'$	$3\tau-\xi+\xi'$	125			$2g-2g' \quad +2\mathfrak{C}-2\mathfrak{C}'$
$\frac{75}{16}e^2e'$	$3\tau-\xi-\xi'$	126			$4g-4g'$
$\frac{635}{64}\eta^2$	$3\tau-2\xi$	127			$3g-5g'$
$\frac{3}{64}\eta^2$	$3\tau+2\xi$	128			$3g-g'$
$\frac{15}{8}\eta^2$	$3\tau-2\eta$	129			$g-5g' \quad +\mathfrak{C}-3\mathfrak{C}'$

where the abbreviation *rev.* denotes that the argument to which it is attached has its sign reversed in the third column of arguments: thus Arg. 63 (Lubbock), it is $-(2\tau-2\eta)$ which is equal to $2\eta+2\mathfrak{C}$. As the formula contains only cosines, there is, of course, no change in the sign of the coefficient.

On comparing with Lubbock's value, I find some differences, which are as follows:–It will be recollected that R (Lubbock) $= -m'\frac{1}{r'} - \Omega$, so that the signs of the coefficients of Ω are to be reversed in order to deduce the corresponding coefficients of R. The Nos. refer to Lubbock's arguments, and the exterior factor $m'\frac{a^2}{a'^3}$ or $m'\frac{a^4}{a'^5}$ is disregarded [1].

Arg. 1. Lubbock's coefficient (viz. the coefficient in R) is

$$-\tfrac{3}{2}\Big(1-\tfrac{5}{8}e^2-\tfrac{5}{8}e''^4+\tfrac{15}{4}e^2+\tfrac{3}{4}e'^2+\tfrac{1}{2}\tfrac{e^4}{e'^4}+\tfrac{1}{12}\tfrac{a^2}{a'^2}\Big)\cos^4\tfrac{1}{2}i.$$

[1] I have been favoured with a note from Sir J. W. Lubbock, confirming my values of the coefficients for the arguments 8, 18, 58, 101 and 123.—Added 12th Feb. 1859, A. C.

[PDF p. 339; printed p. 317.] which, substituting for $\cos^4 \frac{1}{2}i$ its value $1 - \frac{1}{2}\gamma^2 + \frac{7}{16}\gamma^4 -$ &c., and developing to the fourth order, gives

$$-\tfrac{3}{2}\Big(1 - \tfrac{5}{8}e^2 - \tfrac{5}{8}e''^4 + \tfrac{15}{4}e^2e''^2 + \tfrac{15}{16}e^4e''^4 - \tfrac{1}{2}\gamma^2 + \tfrac{1}{2}\gamma^2e^2 + \tfrac{1}{2}\gamma^2e''^2 + \tfrac{7}{16}\gamma^4 + \tfrac{1}{12}\tfrac{a^2}{a'^2}\Big)e^2,$$

which, in fact, agrees with the value given *suprà*. I have only referred to this term in order to make the reduction.

Arg. 8, Lubbock's coefficient is

$$-\tfrac{1}{8}\big(1 - \tfrac{5}{8}e^2 + \tfrac{3}{4}e''^2 - \tfrac{3}{4}\gamma^2\big)e^2;$$

the exterior sign should be + instead of −. The term is given with the correct sign, Pontécoulant, Arg. 2.

Arg. 18, Lubbock's coefficient is

$$-\tfrac{15}{4}\big(1 - \tfrac{3}{2}e^2 - \tfrac{105}{4}e^4e''^2 - \tfrac{1}{2}\gamma^2\big)\cos^4 \tfrac{1}{2}i,$$

which, developed to γ^4, would be

$$-\tfrac{15}{4}\big(1 - \tfrac{3}{2}e^2 - \tfrac{105}{4}e^4e''^2 - \gamma^2\big).$$

I make it

$$-\tfrac{15}{4}\big(1 - \tfrac{3}{2}e^2 - \tfrac{105}{4}e^4e''^2 - \tfrac{1}{2}\gamma^2\big).$$

The remaining differences are

No. of Arg. Lubbock.	Lubbock's coefficient.	Coefficient from Development *suprà*.
58	$+\frac{45}{64}e^4e'^2$	$-\frac{845}{64}e^4e'^2$
59	$+\frac{591}{64}e^4$	$-\frac{77}{32}e^4$
60	$-\frac{7453}{128}e^4$	$-\frac{27}{32}e^4e'^2$
61	$+\frac{741}{128}e^4$	$-\frac{1}{32}e^4e'^2$
*62	$-\frac{3}{8}\big(1 - \frac{5}{2}e^2 + \frac{3}{2}e^4 - \gamma^2\big)\gamma^2$	$-\frac{3}{8}\big(1 - \frac{5}{2}e^2 + \frac{3}{2}e^4 - \gamma^2\big)\gamma^2$

[PDF p. 340; printed p. 318.]

No. of Arg. Lubbock.	Lubbock's coefficient.	Coefficient from Development *suprà.*
*63	$-\frac{3}{8}\left(1+\frac{3}{2}e^2-\frac{5}{2}e''^2+\frac{5}{2}\gamma^2\right)\gamma^2$	$-\frac{3}{8}\left(1+\frac{3}{2}e^2-\frac{5}{2}e''^2-\gamma^2\right)\gamma^2$
95	$-\frac{27}{16}\gamma^2 e''^3$	$-\frac{27}{32}\gamma^2 e''^3$
96	$-\frac{27}{16}\gamma^2 e''^3$	$-\frac{27}{32}\gamma^2 e''^3$
*101	$+\frac{3}{8}\left(1+3e''^2+3e^2-\frac{11}{4}\gamma^2\right)\gamma^2$	$-\frac{3}{8}\left(1+2e''^2+2e^2-\frac{11}{4}\gamma^2\right)\gamma^2$
115	$-\frac{15}{128}\gamma^2$	$-\frac{15}{32}\gamma^2$
123	$-\frac{225}{16}ee'$	$+\frac{225}{16}ee'$

The greater part of the discordant terms do not occur in Pontécoulant's development, which is not carried so far, and the only differences which I find in the coefficients of Pontécoulant $R\,(=\Omega)$ are as regards the arguments 18, 57, 70, corresponding respectively to Lubbock's arguments 62, 63, 101, included in the preceding table, and for which Pontécoulant's coefficients, correcting for the change of sign, correspond with those given by Lubbock. But I see no room for a mistake in the preceding investigation as regards the coefficients of these three terms; the names in η of the coefficients of 62 and 63 are simply the quantity $(\frac{1}{2}\eta^2-\frac{1}{8}\eta^4)$, which forms the exterior factor of Ω_3 and Ω_6 respectively, and which, putting for η^2 and η^4 their values, is equal to $\frac{1}{8}(1-\gamma^2)\gamma^2$, and as regards the coefficient of 101, the question being to obtain by the mere multiplication of the factors $1+2e^2+2e''^2$ &c., and $1+2e''^2$ set opposite to $g+\mathfrak{C}$ and $0g'-\mathfrak{C}'$ respectively in the table for Ω_2.

2, *Stone Buildings, W.C., 7th July*, 1858.

[PDF p. 341; printed p. 319.]

214. The First Part of a Memoir on the Development of the Disturbing Function in the Lunar and Planetary Theories

[From the *Memoirs of the Royal Astronomical Society*, vol. XXVIII. (1860), pp. 187–215. Read November 10, 1858.]

THE development, as is well known, depends upon that of the reciprocal of the distance of the two planets; and Hansen's Memoir "Entwickelung der negativen und ungeraden Potenzen der Quadratwurzel der Function $r^2 + r'^2 - 2rr'(\cos U \cos U' + \sin U \sin U' \cos J)$," *Abh. der K. Sächs. Ges. zu Leipzig*, t. II., pp. 286–376 (1854), contains a formula which is truly fundamental, viz. the expression of the coefficient of the general term

$$\frac{r^n}{r'^{n+1}} \cos(jU + j'U')$$

of the development of the reciprocal of the distance as expressed in the above-mentioned form, where r, r' are the radius vectors of the inferior and superior planets respectively, and U, U' are the angular distances from the mutual node. In the lunar theory, where the higher powers of $\frac{r}{r'}$ are neglected, we have in this manner a small number of terms each of which is to be separately developed in multiple cosines of the mean anomalies. This can be effected as in the *Fundamenta Nova*, and my "Memoir on the Development of the Disturbing Function in the Lunar Theory," *R. Ast. Soc. Mem.*, t. XXVII., 1859, [213], a mere completion of which is Hansen's process [1]. In fact if f, f' are the true anomalies, and $\mathfrak{C}$, $\mathfrak{C}'$ the distances

[1] I take the opportunity of mentioning the memoir of Hansen's which immediately precedes that above referred to, viz. "Entwickelung des Products einer Potenz des Radius Vectors mit dem Sinus oder Cosinus eines Vielfachen der wahren Anomalie in Reihen die nach den Sinussen oder Cosinussen der Vielfachen der wahren der excentrischen oder mittleren Anomalie fortschreiten," t. II., pp. 183–281 (1853).

[PDF p. 342; printed p. 320.] of the pericentres from the mutual node, then we have $U = f + \mathfrak{C}$, $U' = f' + \mathfrak{C}'$, and the general term is

$$\frac{r^n}{r'^{n+1}} \cos(jf + j'f' + j\mathfrak{C} + j'\mathfrak{C}')$$

where r, f are given functions of the mean anomaly g, and r', f' are the like functions of the mean anomaly g'. And the development depends upon those of

$$r^n {}^{\cos}_{\sin} jf, \qquad r'^{-n-1} {}^{\cos}_{\sin} jf',$$

which (if we consider as well negative as positive values of the index n) are each of the form

$$r^n {}^{\cos}_{\sin} jf,$$

and when the developments of these expressions are known, we obtain at once by the mere addition and subtraction of the coefficients of the cosines and sines of the different multiples of g and g', the development of

$$\frac{r^n}{r'^{n+1}} \cos(jf + j'f' + j\mathfrak{C} + j'\mathfrak{C}')$$

in the tabular form employed in my memoir just referred to. In the planetary theory, we must unite together the terms containing the different powers of $\frac{r}{r'}$ so as to form the entire coefficient $D(j, j')$ of $\cos(jU + j'U')$; if then we write

$$r = a(1 + x), \qquad r' = a'(1 + x'),$$

and develope the coefficient in powers of x, x', we have the general term

$$x^\alpha x'^{\alpha'} \cos(jU + j'U')$$

which admits of development in multiple cosines of the mean anomalies, in the same manner precisely as the before-mentioned general term

$$\frac{r^n}{r'^{n+1}} \cos(jU + j'U')$$

in the lunar theory. It is proper to remark that this method is really identical with that commonly made use of in the planetary theory: the only difference is, that by Hansen's fundamental formula, we have the complete expression of the coefficient $D(j, j')$ developed in powers of sin or of tan $\frac{1}{2}\phi$, instead of (as in the ordinary methods) the first two or three terms of this development.

[PDF p. 343; printed p. 321.] The required development of

$$x^{\alpha}x'^{\alpha'}\cos(jU+j'U')$$

or, what is the same thing,

$$x^{\alpha}x'^{\alpha'}\cos(jf+j'f'+j\mathfrak{C}+j'\mathfrak{C}')$$

depends on the developments of

$$x^{\alpha}{}^{\cos}_{\sin}jf, \qquad x'^{\alpha'}{}^{\cos}_{\sin}jf'.$$

These are functions of the same form, and we may consider only

$$x^{\alpha}{}^{\cos}_{\sin}jf.$$

The value of x is $\left(\frac{r}{a}-1\right)$ and we could of course calculate

$$\left(\frac{r}{a}-1\right)^{\alpha}{}^{\cos}_{\sin}jf$$

by the methods of the *Fundamenta Nova* or the memoir of Hansen's referred to in the foot-note. But if we write $f=g+y$ (y is the equation of the centre), then the required expressions depend on

$$x^{\alpha}{}^{\cos}_{\sin}jy$$

which are actually calculated as far as e^7 for $\alpha=0,1,2,\dots,7$, and j an undetermined symbol, by Leverrier in the *Annales de l'Observatoire de Paris*, t. I. pp. 346–348 (1855). Hence, by the mere substitution, in Leverrier's formula, of the numerical values of j and j', and by the addition and subtraction of the coefficients of the cosines or sines of the different multiples of g and g', we may obtain the development of

$$x^{\alpha}x'^{\alpha'}\cos(jf+j'f'+j\mathfrak{C}+j'\mathfrak{C}')$$

in a tabular form similar to that employed in my memoir already referred to.

I have thought it desirable to put together the various results above referred to, and to investigate by a different process the expression for Hansen's coefficient, which in his memoir is obtained by means of a long series of transformations which it is not very easy to follow, and is not exhibited in quite the most simple form. And this is what I have done in the present first part of a memoir on the development of the disturbing function. My object has been to exhibit, in as complete a form as possible, the preliminary development in multiple cosines of the true anomalies; and to indicate the process of the ulterior development in multiple cosines of the mean anomalies.

[PDF p. 344; printed p. 322.]

I.

Let $\mathfrak{M}$ be the inferior, $\mathfrak{M}'$ the superior of the two planets (in the lunar theory $\mathfrak{M}$ is the moon, $\mathfrak{M}'$ the sun) and let the quantities relating to the two bodies be for $\mathfrak{M}$,

r, the radius-vector,
U, the distance from node,
$\mathfrak{C}$, the distance of pericentre from node,
f, the true anomaly,
g, the mean anomaly,
a, the mean distance,
e, the eccentricity,

and for $\mathfrak{M}'$ the accented letters, r', &c., in the like significations.

The node referred to is the ascending node of the orbit of $\mathfrak{M}$ upon that of $\mathfrak{M}'$, and I write also

Φ, the inclination of the orbit of $\mathfrak{M}$ to that of $\mathfrak{M}'$,
η, $=\sin\frac{1}{2}\Phi$.

The disturbing function is in the first instance given as a function of r, r', H, where

$$\cos H = \cos U \cos U' + \sin U \sin U' \cos\Phi,$$

that is, as a function of r, r', U, U', Φ, or, what is the same thing, of r, r', U, U', η. And the preliminary development is a development in multiple cosines of U, U'. We have then

$$U = f + \mathfrak{C},$$
$$U' = f' + \mathfrak{C}',$$

and finally

$$r = a\ \mathrm{elqr}(e, g),$$
$$f = \mathrm{elta}(e, g),$$
$$r' = a'\ \mathrm{elqr}(e', g'),$$
$$f' = \mathrm{elta}(e', g'),$$

and the ulterior development is a development in multiple cosines of g, g', $\mathfrak{C}$, $\mathfrak{C}'$, the coefficients involving, as before, η, and also a, e, a', e'. But as usual it is not attempted to carry the development further, by introducing in the coefficients in place of the relative quantities $\mathfrak{C}$, $\mathfrak{C}'$, η, the remaining elements of the two orbits, which, if it were necessary to use them, would be

for $\mathfrak{M}$,

θ, the longitude of node,
σ, the departure of node,
ϕ, the inclination,
ϖ, the departure of pericentre,

and for $\mathfrak{M}'$ the like accented quantities, the orbit of reference being any fixed or moveable orbit whatever.

[PDF p. 345; printed p. 323.]

II.

The expression for the disturbing function on $\mathfrak{M}$ (that is, when the superior planet disturbs the inferior) is

$$\mathfrak{M}'\left\{-\frac{r\cos H}{r'^2}+\frac{1}{\sqrt{r^2+r'^2-2rr'\cos H}}\right\}$$

and that for the disturbing function on $\mathfrak{M}'$ (that is, when the inferior planet disturbs the superior) is

$$\mathfrak{M}\left\{-\frac{r'\cos H}{r^2}+\frac{1}{\sqrt{r^2+r'^2-2rr'\cos H}}\right\}$$

where the disturbing function is taken with Lagrange's sign ($=-R$, if R be the disturbing function of the *Mécanique Céleste*).

But we may in the first instance consider the development of the reciprocal of the distance of the two planets

$$=\frac{1}{\sqrt{r^2+r'^2-2rr'\cos H}}.$$

The preceding expression for $\cos H$ may be written

$$\cos H=\cos U\cos U'+\sin U\sin U'\,(1-2\eta^2),$$

or in either of the two forms

$$\cos H=\cos(U-U')-2\eta^2\sin U\sin U',$$
$$\cos H=(1-\eta^2)\cos(U-U')+\eta^2\cos(U+U').$$

Now imagine the function developed in ascending powers of $\frac{r^n}{r'^{n+1}}$, the coefficient of $\frac{r^n}{r'^{n+1}}$ will contain $\cos^n H$, $\cos^{n-2} H$, … to $\cos H$ or 1 according as n is even or odd; and if we then substitute for $\cos H$ the given last expression, and express the different powers of $\cos H$ in multiple cosines of $U-U'$ and $U+U'$, and make the final expression contain the cosines of opposite arguments each with the same coefficient, it is easy to see that the form of the general term is

$$\frac{r^n}{r'^{n+1}}\,C_n(j,j')\cos(jU+j'U'),$$

where j, j' each of them extend through the values n, $n-2$, $\ldots-n$, and where

$$C_n(-j,-j')=C_n(j,j').$$

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[PDF p. 346; printed p. 324.] Thus in particular for $n = 0, 1, 2, 3$, the combinations (j, j') belonging to the several arguments are,

$0,0$	$1,1$ $1,-1$	$-1,1$ $-1,-1$	$2,2$ $2,0$ $2,-2$	$0,2$ $0,0$ $0,-2$	$3,3$ $3,1$ $3,-1$ $3,-3$	$1,3$ $1,1$ $1,-1$ $1,-3$

But as the coefficients C satisfy the condition $C(-j, -j') = C(j, j')$, the two terms $C(j, j')\cos(jU+j'U')$ and $C(-j, -j')\cos(-jU - j'U')$ are equal to each other, and they may be combined together into a single term. The general term may consequently be written

$$\frac{r^n}{r'^{n+1}}\, 2C_n(j, j')\cos(jU + j'U'),$$

where j has only the values $n, n-2, \ldots 1$ or 0, and j' has as before the values $n, n-2, \ldots, -n$; except (which occurs only when n is even) for $j = 0$, when j' has only the values $n, n-2, \ldots 0$: and in the particular case $j = j' = 0$, the last-mentioned expression for the general term must be multiplied by $\frac{1}{2}$, or, what is the same thing, the factor 2 must be omitted. In particular for $n = 0, 1, 2, 3, 4$, the combinations (j, j') belonging to the several arguments are

$0,0$	$1,1$ $1,-1$	$2,2$ $2,0$ $2,-2$	$0,2$ $0,0$	$3,3$ $3,1$ $3,-1$ $3,-3$	$1,3$ $1,1$ $1,-1$ $1,-3$	$4,4$ $4,2$ $4,0$ $4,-2$ $4,-4$	$2,4$ $2,2$ $2,0$ $2,-2$ $2,-4$

I remark that in a series $K(j, j')\genfrac{}{}{0pt}{}{\cos}{\sin}(jU + j'U')$, where each argument occurs positively and negatively, and $K(-j, -j') = \pm K(j, j')$ according as the series is one of cosines

[PDF p. 347; printed p. 325.] or of sines, we may say that the *discrete* general term is $K(j,j')^{\cos}_{\sin}(jU+j'U')$, or that $K(j,j')$ is the *discrete* coefficient of the cosine or sine, but if we unite together the terms with opposite arguments so as to form the general term $2K(j,j')^{\cos}_{\sin}(jU+j'U')$, then this may be called the *concrete* general term, and $2K(j,j')$ may be said to be the concrete coefficient of the cosine or sine. In a sine series the term corresponding to the argument zero vanishes; in a cosine series this is not in general the case, and the concrete term corresponding to the argument zero must be multiplied by $\frac{1}{2}$.

Returning to the question in hand, from the symmetry of the expression to be developed, we have

$$C_n(j,j') = C_n(j',j),$$

and it follows that the only coefficients which need be calculated are those for which j is not negative, and not less in absolute magnitude than j'; the remaining coefficients are respectively equal to coefficients which satisfy these conditions. Thus for $n = 0,1,2,3,4$, the combinations (j,j') corresponding to the coefficients in question are,

$0,0$	$1,1$ $1,-1$	$2,2 \quad 0,0$ $2,0$ $2,-2$	$3,3 \quad 1,1$ $3,1 \quad 1,-1$ $3,-1$ $3,-3$	$4,4 \quad 2,2$ $4,2 \quad 2,0$ $4,0 \quad 2,-2$ $4,-2 \quad 0,0$ $4,-4 \quad 2,-4$

and we have, for instance, $C(2,4) = C(4,2)$, $C(-2,-4) = C(2,4) = C(4,2)$, &c. Under the preceding restriction, viz. j not negative, and not less in absolute magnitude than j', the expression for $C_n(j,j')$ (deduced from the formulæ of Hansen's Memoir) is as follows; viz. putting as usual $\Pi\alpha = 1.2.3\ldots\alpha$, and also $\Pi_1(x-\frac{1}{2}) = \frac{1}{2}\cdot\frac{3}{2}\cdot\frac{5}{2}\ldots(x-\frac{1}{2})$, and representing the hypergeometric series

$$1 + \frac{\alpha\cdot\beta}{1\cdot\gamma}x + \frac{\alpha\cdot\alpha+1\cdot\beta\cdot\beta+1}{1\cdot2\cdot\gamma\cdot\gamma+1}x^2 + \&c.$$

by $F(\alpha,\beta,\gamma,x)$, we have

$$C(j,j') = \frac{2^{j+j'}\,\Pi_1(\frac{1}{2}(n+j)-\frac{1}{2})\,\Pi_1(\frac{1}{2}(n+j')-\frac{1}{2})}{\Pi\frac{1}{2}(n-j)\,\Pi\frac{1}{2}(n-j')\,\Pi(j+j')}$$

$$\times\eta^{j+j'}(1-\eta^2)^{\frac{1}{2}(j-j')}F(-n+j+1,\, n+j+1,\, j+j'+1,\, \eta^2)$$

[PDF p. 348; printed p. 326.] which, it is to be noticed, is a rational and integral function of η, the highest power being η^n, and the lowest power or order of the coefficient being $\eta^{j+j'}$. I reserve the demonstration of this formula for a separate paragraph.

Let the discrete general term involving $\cos(jU + j'U')$ be represented by

$$D(j, j') \cos(jU + j'U');$$

we have, in like manner as for the coefficients C, $D(-j, -j') = D(j, j')$, $D(j', j) = D(j, j')$, and consequently $D(j, j')$ will be known, if we know its value when the before-mentioned conditions are satisfied; viz. if j be not negative and not less in absolute magnitude than j'; and collecting together the different terms which involve the cosine in question, we find at once, the conditions being satisfied,

$$D(j, j') = \frac{r^j}{r'^{j+1}} C_j(j, j') + \frac{r^{j+2}}{r'^{j+3}} C_{j+2}(j, j') + \&\text{c.},$$

so that the value of $D(j, j')$ is known. A transformation of this expression will be given in the sequel.

III.

The concrete general term is

$$2D(j, j') \cos(jU + j'U');$$

in particular the concrete terms involving the arguments $U - U'$, $U + U'$, are

$$2D(1, -1) \cos(U - U'), \qquad 2D(1, 1) \cos(U + U'),$$

or, substituting for the D coefficients their values, the terms are

$$2\left\{\frac{r}{r'^2} C_1(1, -1) + \frac{r^3}{r'^4} C_3(1, -1) + \cdots\right\} \cos(U - U'),$$

$$2\left\{\frac{r}{r'^2} C_1(1, 1) + \frac{r^3}{r'^4} C_3(1, 1) + \cdots\right\} \cos(U + U').$$

So far I have considered the reciprocal of the radius vector; but if we consider, instead, the disturbing functions, the only difference will be that we must add the term

$$-\frac{r}{r'^2} \cos H,$$

or

$$-\frac{r'}{r^2} \cos H,$$

[PDF p. 349; printed p. 327.] according as the superior planet disturbs the inferior one, or the inferior planet the superior one. The value of $\cos H$ is

$$(1-\eta^2)\cos(U-U') + \eta^2\cos(U+U'),$$

$$= 2C_1(1,-1)\cos(U-U') + 2C_1(1,1)\cos(U+U'),$$

observing that

$$C_1(1,-1) = \tfrac{1}{2}(1-\eta^2), \qquad C_1(1,1) = \tfrac{1}{2}\eta^2;$$

and the terms to be added are consequently, when the superior planet disturbs the inferior,

$$-2\,\frac{r}{r'^2}\,C_1(1,-1)\cos(U-U'),$$

$$-2\,\frac{r}{r'^2}\,C_1(1,1)\cos(U+U'),$$

the effect of which is simply to destroy the same terms contained with the opposite sign in the reciprocal of the distance;

and when the inferior planet disturbs the superior one, the terms to be added are

$$-2\,\frac{r'}{r^2}\,C_1(1,-1)\cos(U-U'),$$

$$-2\,\frac{r'}{r^2}\,C_1(1,1)\cos(U+U'),$$

which are not equal to any terms in the reciprocal of the distance. We may write as follows:

[PDF p. 350; printed p. 328.]

Reciprocal of Distance is,

Disturbing function + Mass of Disturbing Planet is,

	U	$+U'$
$D(0,0)$	cos	0 0
$+2\,D(1,-1)$		1 $-$ 1
(1) $\left[-2\,\frac{r}{r'^2}\,C_1(1,-1)\right]$		1 $-$ 1
(2) $\left[-2\,\frac{r'}{r^2}\,C_1(1,-1)\right]$		1 $-$ 1
$+2\,D(1,1)$		1 1
(1) $\left[-2\,\frac{r}{r'^2}\,C_1(1,1)\right]$		1 1
(2) $\left[-2\,\frac{r'}{r^2}\,C_1(1,1)\right]$		1 1
$+2\,D(2,-2)$		2 $-$ 2
$+2\,D(2,0)$		2 0
$+2\,D(0,2)$		0 2
$+2\,D(2,2)$		2 2
$+2\,D(3,-3)$		3 $-$ 3
$+2\,D(3,-1)$		3 $-$ 1
$+2\,D(1,-3)$		1 $-$ 3
$+2\,D(3,1)$		3 1
$+2\,D(1,3)$		1 3
$+2\,D(3,3)$		3 3
$+2\,D(4,-4)$		4 $-$ 4
$+2\,D(4,-2)$		4 $-$ 2
$+2\,D(2,-4)$		2 $-$ 4
$+2\,D(4,0)$		4 0
$+2\,D(0,4)$		0 4
$+2\,D(4,2)$		4 2
$+2\,D(2,4)$		2 4
$+2\,D(4,4)$		4 4
+&c., &c.		

[1] Only to be inserted for the disturbing function when the superior planet disturbs the inferior, and having the effect of destroying portions of the coefficients of the next preceding terms.
[2] Only to be inserted for the disturbing function when the inferior planet disturbs the superior one.

[End of pp. 326–350 chunk; paper 214 continues in next chunk.]

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Papers 214 (conclusion), 215, and 216 (opening)

Contents

214. On the Development of the Disturbing Function in the Lunar and Planetary Theories *(continued)*

[PDF p. 351; printed p. 329.] For the lunar theory, to the extent to which it is necessary to carry the development, and to which it is carried in Hansen's *Fundamenta Nova*, and my memoir before referred to, we might simply write,

Disturbing function ÷ Sun's Mass is

		U	$+U'$
	$\frac{a^2}{a'^3}\,C_2(0,0)$	cos	$0\ \ 0$
(1)	$+\frac{a^3}{a'^4}\,C_3(0,0)$		$0\ \ 0$
(2)	$+2\,\frac{a^3}{a'^4}\,C_3(1,-1)$		$1\ -1$
(3)	$+2\,\frac{a^3}{a'^4}\,C_3(1,1)$		$1\ \ 1$
(4)	$+2\,\frac{a^3}{a'^4}\,C_3(2,-2)$		$2\ -2$
(5)	$+2\,\frac{a^3}{a'^4}\,C_3(2,0)$		$2\ \ 0$
(6)	$+2\,\frac{a^3}{a'^4}\,C_3(0,2)$		$0\ \ 2$
(7)	$+2\,\frac{a^3}{a'^4}\,C_3(2,2)$		$2\ \ 2$
(8)	$+2\,\frac{a^3}{a'^4}\,C_3(3,-3)$		$3\ -3$
(9)	$+2\,\frac{a^3}{a'^4}\,C_3(3,-1)$		$3\ -1$
(10)	$+2\,\frac{a^3}{a'^4}\,C_3(1,-3)$		$1\ -3$
	$+2\,\frac{a^3}{a'^4}\,C_3(3,1)$		$3\ \ 1$
	$+2\,\frac{a^3}{a'^4}\,C_3(1,3)$		$1\ \ 3$

where the prefixed numbers are those of Hansen's two parts Ω_1, Ω_2,... Ω_6; or omitting the first term which depends only on the moon's radius vector, and the last two terms which we ultimately neglected, and substituting for the coefficients C their values we here, [PDF p. 352; printed p. 330.] Disturbing function ÷ Sun's Mass is

		U	$+U'$
(1)	$\dfrac{a}{a'} - \frac{3}{4}\eta^2\dfrac{a^3}{a'^4} + \frac{3}{8}\eta^4\dfrac{a^3}{a'^4}$	cos	0 0
(2)	$+\dfrac{a^3}{a'^4}\,\frac{1}{4}\eta^2$		1 −1
(3)	$+\dfrac{a^3}{a'^4}\,\frac{1}{4}\eta^2$		1 1
(4)	$+\dfrac{a^3}{a'^4}\,\frac{15}{8}\eta^4$		2 −2
(5)	$-\frac{3}{4}\eta^2\dfrac{a^3}{a'^4} + \frac{15}{8}\eta^4\dfrac{a^3}{a'^4}$		2 0
(6)	$-\frac{3}{4}\eta^2\dfrac{a^3}{a'^4} + \frac{15}{8}\eta^4\dfrac{a^3}{a'^4}$		0 2
(7)	$+\frac{15}{8}\eta^4\dfrac{a^3}{a'^4}$		2 2
(8)	$+\frac{15}{8}\eta^4\dfrac{a^3}{a'^4}$		3 −3
(9)	$+\frac{15}{8}\eta^4\dfrac{a^3}{a'^4}$		3 −1
(10)	$+\frac{15}{8}\eta^4\dfrac{a^3}{a'^4}$		1 −3

If the same form be adopted for the planetary theory, the expressions for the leading coefficients will be,

$$
\begin{array}{lll}
C_2(0,0), \quad D(0,0) = \frac{1}{2}, & & \\
\qquad\qquad 2 & + \dfrac{a^2}{a'^2}\left(1 - 6\,\eta^2 + 6\,\eta^4\right), & \\
\qquad\qquad 4 & + \dfrac{a^4}{a'^4}\,\frac{9}{32}\left(1 - 20\,\eta^2 + 90\,\eta^4 - 140\,\eta^6 + 70\,\eta^8\right), & \\
\qquad\qquad 6 & + \dfrac{a^6}{a'^6}\,\frac{25}{12\cdot 32}\left(1 - 42\,\eta^2 + 420\,\eta^4 - 1680\,\eta^6 + 3150\,\eta^8 \ldots\right), & \\
\qquad\qquad 8 & + \dfrac{a^8}{a'^8}\,\frac{1225}{2\cdot 16384}\left(1 - 72\,\eta^2 + 1260\,\eta^4 - 9240\,\eta^6 + 34650\,\eta^8 \ldots\right), & \\
\qquad\qquad 2\rho & + \dfrac{a^{2\rho}}{a'^{2\rho}}\,\dfrac{\Pi_1(\rho-1)\,\Pi_1(\rho-\frac{1}{2})}{\Pi_1\rho\,\Pi\rho}\,F\left(-2\rho,\, 2\rho+1,\, 1,\, \eta^2\right); &
\end{array}
$$

[PDF p. 353; printed p. 331.]

$$\begin{array}{lll}
C_1(1,-1), \quad D(1,-1) = & \dfrac{a}{a'}\,\tfrac{1}{2}(1-\eta^2) \\
3 & +\dfrac{a^3}{a'^3}\,\tfrac{3}{16}(1-10\,\eta^2+\ldots) \\
5 & +\dfrac{a^5}{a'^5}\,\tfrac{15}{128}(1-28\,\eta^2+168\,\eta^4-336\,\eta^6+210\,\eta^8\ldots) \\
7 & +\dfrac{a^7}{a'^7}\,\tfrac{105}{1024}(1-54\,\eta^2+675\,\eta^4-3100\,\eta^6+7425\,\eta^8\ldots) \\
2\rho+1 & +\dfrac{a^{2\rho+1}}{a'^{2\rho+1}}\,\dfrac{\Pi_{\frac{1}{2}}(2\rho+\frac{1}{2})\,\Pi_{\frac{1}{2}}(\rho-\frac{1}{2})}{\Pi_{\frac{1}{2}}\rho\,\Pi_{\frac{1}{2}}(\rho+1)}\,(1-\eta^2)\,F(-2\rho,\,2\rho+3,\,3,\,\eta^2);
\end{array}$$

$$\begin{array}{lll}
C_1(1,1), \quad D(1,1) = & \dfrac{a}{a'}\,\tfrac{1}{2}\eta^2 \\
3 & +\dfrac{a^3}{a'^3}\,\tfrac{1}{8}\eta^2(1-7\,\eta^2+\tfrac{15}{2}\eta^4) \\
5 & +\dfrac{a^5}{a'^5}\,\tfrac{15}{128}\eta^2(1-14\,\eta^2+\tfrac{63}{2}\eta^4-\ldots) \\
7 & +\dfrac{a^7}{a'^7}\,\tfrac{105}{1024}\eta^2(1-18\,\eta^2+\ldots) \\
2\rho+1 & +\dfrac{a^{2\rho+1}}{a'^{2\rho+1}}\,\dfrac{\Pi_{\frac{1}{2}}(\rho+\frac{1}{2})\,\Pi_{\frac{1}{2}}(\rho-\frac{1}{2})}{\Pi_{\frac{1}{2}}\rho\,\Pi_{\frac{1}{2}}(\rho-1)}\,\eta^2\,F(-2\rho,\,2\rho+3,\,3,\,\eta^2);
\end{array}$$

$$\begin{array}{lll}
C_2(2,-2), \quad D(2,-2) = & \dfrac{a^2}{a'^2}\,\tfrac{3}{8}(1-\eta^2)^2 \\
4 & +\dfrac{a^4}{a'^4}\,\tfrac{5}{16}(1-\eta^2)^2(1-14\,\eta^2+24\,\eta^4) \\
6 & +\dfrac{a^6}{a'^6}\,\tfrac{105}{512}(1-\eta^2)^2(1-36\,\eta^2+360\,\eta^4-660\,\eta^6+495\,\eta^8) \\
\vdots & \\
2\rho & +\dfrac{a^{2\rho}}{a'^{2\rho}}\,\dfrac{\Pi_1(\rho+\frac{1}{2})\,\Pi_1(\rho-\frac{1}{2})}{\Pi_1(\rho+1)\,\Pi_1(\rho-1)}\,(1-\eta^2)^2\,F(-2\rho+2,\,2\rho+2,\,5,\,\eta^2);
\end{array}$$

$$\begin{array}{lll}
C_2(2,0), \quad D(2,0) = & \dfrac{a^2}{a'^2}\,\tfrac{1}{2}\eta^2(1-\eta^2) \\
4 & +\dfrac{a^4}{a'^4}\,\tfrac{5}{8}\eta^2(1-\eta^2)(1-6\,\eta^2+\ldots) \\
6 & +\dfrac{a^6}{a'^6}\,\tfrac{105}{32}\eta^2(1-\eta^2)(1-12\,\eta^2+45\,\eta^4-\ldots) \\
\vdots & \\
2\rho & +\dfrac{a^{2\rho}}{a'^{2\rho}}\,\dfrac{\Pi_1(\rho+\frac{1}{2})\,\Pi_1(\rho-\frac{1}{2})}{\Pi_1\rho\,\Pi_1(\rho-1)}\,\eta^2(1-\eta^2)\,F(-2\rho+2,\,2\rho+2,\,3,\,\eta^2).
\end{array}$$

[PDF p. 354; printed p. 332.]

$$C_2(0,2) = C_2(2,0), \qquad D(0,2) = D(2,0);$$

$$\begin{array}{rl}
C_2(2,2), \quad D(2,2) = & \dfrac{a^2}{a'^2}\,\tfrac{3}{8}\eta^4 \\
4 \qquad & + \dfrac{a^4}{a'^4}\,\tfrac{5}{16}\eta^4(1 - 7\,\eta^2 + \tfrac{63}{8}\eta^4) \\
6 \qquad & + \dfrac{a^6}{a'^6}\,\tfrac{105}{512}\eta^4(1 - 17\,\eta^2 + 18\,\eta^4) \\
\vdots \qquad & \\
2\rho \qquad & + \dfrac{a^{2\rho}}{a'^{2\rho}}\,\dfrac{\Pi_1(\rho+\frac{1}{2})\,\Pi_1(\rho-\frac{1}{2})}{\Pi_1(\rho+1)\,\Pi_1(\rho-1)}\,\eta^4\,F(-2\rho+2,\,2\rho+2,\,5,\,\eta^2);
\end{array}$$

$$\begin{array}{rl}
C_3(3,-3), \quad D(3,-3) = & \dfrac{a^3}{a'^3}\,\tfrac{5}{16}(1-\eta^2)^3 \\
5 \qquad & + \dfrac{a^5}{a'^5}\,\tfrac{35}{128}(1-\eta^2)^3(1 - 18\,\eta^2 + \ldots) \\
\vdots \qquad & \\
2\rho+1 \qquad & + \dfrac{a^{2\rho+1}}{a'^{2\rho+1}}\,\dfrac{\Pi_{\frac{1}{2}}(\rho+\frac{3}{2})\,\Pi_{\frac{1}{2}}(\rho-\frac{1}{2})}{\Pi_{\frac{1}{2}}(\rho+2)\,\Pi_{\frac{1}{2}}(\rho-1)}\,(1-\eta^2)^3\,F(-2\rho+2,\,2\rho+3,\,7,\,\eta^2);
\end{array}$$

$$\begin{array}{rl}
C_3(3,-1), \quad D(3,-1) = & \dfrac{a^3}{a'^3}\,\tfrac{15}{8}\eta^2(1-\eta^2)^2 \\
5 \qquad & + \dfrac{a^5}{a'^5}\,\tfrac{105}{32}\eta^2(1-\eta^2)^2(1 - 6\,\eta^2 + \ldots) \\
\vdots \qquad & \\
2\rho+1 \qquad & + \dfrac{a^{2\rho+1}}{a'^{2\rho+1}}\,\dfrac{\Pi_{\frac{1}{2}}(\rho+\frac{3}{2})\,\Pi_{\frac{1}{2}}(\rho-\frac{1}{2})}{\Pi_{\frac{1}{2}}(\rho+1)\,\Pi_{\frac{1}{2}}(\rho-1)}\,\eta^2(1-\eta^2)^2\,F(-2\rho+2,\,2\rho+3,\,5,\,\eta^2);
\end{array}$$

$$C_3(1,-3) = C_3(3,-1), \qquad D(1,-3) = D(3,-1);$$

$$\begin{array}{rl}
C_3(3,1), \quad D(3,1) = & \dfrac{a^3}{a'^3}\,\tfrac{15}{8}\eta^4(1-\eta^2) \\
5 \qquad & + \dfrac{a^5}{a'^5}\,\tfrac{105}{32}\eta^4(1-\eta^2)(1 - \ldots) \\
\vdots \qquad & \\
2\rho+1 \qquad & + \dfrac{a^{2\rho+1}}{a'^{2\rho+1}}\,\dfrac{\Pi_{\frac{1}{2}}(\rho+\frac{3}{2})\,\Pi_{\frac{1}{2}}(\rho-\frac{1}{2})}{\Pi_{\frac{1}{2}}\rho\,\Pi_{\frac{1}{2}}(\rho-1)}\,\eta^4(1-\eta^2)\,F(-2\rho+2,\,2\rho+3,\,5,\,\eta^2);
\end{array}$$

$$C_3(1,3) = C_3(3,1), \qquad D(1,3) = D(3,1);$$

$$\begin{array}{rl}
C_3(3,3), \quad D(3,3) = & \dfrac{a^3}{a'^3}\,\tfrac{5}{16}\eta^6 \\
5 \qquad & + \dfrac{a^5}{a'^5}\,\tfrac{35}{128}\eta^6(1 + \ldots) \\
\vdots \qquad & \\
2\rho+1 \qquad & + \dfrac{a^{2\rho+1}}{a'^{2\rho+1}}\,\dfrac{\Pi_{\frac{1}{2}}(\rho+\frac{3}{2})\,\Pi_{\frac{1}{2}}(\rho-\frac{1}{2})}{\Pi_{\frac{1}{2}}(\rho+2)\,\Pi_{\frac{1}{2}}(\rho-1)}\,\eta^6\,F(-2\rho+2,\,2\rho+3,\,7,\,\eta^2) \quad \&c.
\end{array}$$

IV.

[PDF p. 355; printed p. 333.] But for the planetary theory it is more suitable to arrange the expression of $D(j,j')$ according to the powers of η. We have, under the before-mentioned conditions, j not negative and not less in absolute magnitude than j',

$$D(j,j') = \sum_{\lambda=0}^{\infty} \frac{a'^{2\lambda}}{a^{2\lambda}}\, C_{j+2\lambda}(j,j'),$$

where λ extends from 0 to ∞; and, writing in the expression of $C_n(j,j')$, $j+2\lambda$ for n, we find

$$C_{j+2\lambda}(j,j') = \frac{2^{j+j'}\,\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j-j')+\lambda-\tfrac{1}{2}\big)}{\Pi_1\,\Pi_1\big(\tfrac{1}{2}(j-j')+\lambda\big)\,\Pi_1(j+j'+\lambda)}\,\eta^{j+j'}\,F(-2\lambda,\ 2j{+}2\lambda{+}1,\ j{+}j'{+}1,\ \eta^2),$$

or, substituting for the hypergeometric series its value,

$$C_{j+2\lambda}(j,j') = \sum_{\theta} \frac{2^{j+j'}\,\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j-j')+\lambda-\tfrac{1}{2}\big)}{\Pi_1\,\Pi_1\big(\tfrac{1}{2}(j-j')+\lambda\big)\,\Pi_1(j+j'+\lambda)}\,(-)^{\theta}\,\frac{\Pi 2\lambda\,\Pi(2j+2\lambda+\theta)}{\Pi(2\lambda-\theta)\,\Pi\theta\,\Pi(j+j'+\theta)\,\Pi(\ldots}$$

from $\theta=0$ to $\theta=\infty$. Substituting in the numerator for $\Pi 2\lambda$, $2^{2\lambda}\,\Pi\lambda\,\Pi_1(\lambda-\tfrac{1}{2})$, and in the denominator for $\Pi(2j+2\lambda)$, $2^{2j+2\lambda}\,\Pi(j+\lambda)\,\Pi(j+\lambda-\tfrac{1}{2})$, and reducing, this becomes

$$C_{j+2\lambda}(j,j') = \sum_{\theta} \frac{(-)^{\theta}\,2^{-j+j'}\,\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda+\theta-\tfrac{1}{2}\big)}{\Pi_1(j+\lambda)\,\Pi_1(j+\lambda-\tfrac{1}{2})\,\Pi_1\big(\tfrac{1}{2}(j-j')+\lambda\big)\,\Pi(j+j'+\theta)\,\Pi\lambda\,\Pi(2\lambda-\theta)}\,\eta^{j+j'+2\theta},$$

and substituting this expression in $D(j,j')$, we find

$$D(j,j') = \frac{1}{a^{j+j'}}\,\sum \frac{a^{2\lambda}}{a'^{2\lambda}}\,\eta^{j+j'+2\theta}\cdot\text{(the same kernel)},$$

$$\sum \frac{(-)^{\theta}\,2^{-j+j'}\,\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda+\theta-\tfrac{1}{2}\big)}{\Pi_1(j+\lambda)\,\Pi_1(j+\lambda-\tfrac{1}{2})\,\Pi_1\big(\tfrac{1}{2}(j-j')+\lambda\big)\,\Pi(j+j'+\theta)\,\Pi\lambda\,\Pi(2\lambda-\theta)}$$

from $\lambda=0$ to $\lambda=\infty$ and $\theta=0$ to $\theta=\infty$; which is the required expression. It is to be remarked that the series in $\big(\dfrac{a}{a'}\big)$ which multiplies $\eta^{j+j'+2\theta}$ is not in general expressible as a hypergeometric series. If, however, we attend only to the leading term in η, or write $\theta=0$, we find for $D(j,j')$ the value

$$\frac{1}{a^{j+j'}}\,\eta^{j+j'}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,\sum \frac{a^{2\lambda}}{a'^{2\lambda}}\,\frac{\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda-\tfrac{1}{2}\big)\,\Pi(2j+2\lambda)}{\Pi_1(j+\lambda)\,\Pi_1(j+\lambda-\tfrac{1}{2})\,\Pi_1\big(\tfrac{1}{2}(j-j')+\lambda\big)\,\Pi(j+j')\,\Pi\lambda},$$

[PDF p. 356; printed p. 334.] which may be simplified by putting in the numerator for $\Pi(2j+2\lambda)$,

$$2^{2j+2\lambda}\,\Pi_1(j+\lambda)\,\Pi_1(j+\lambda-\tfrac{1}{2}),$$

and in the denominator for $\Pi_1(j+j')$,

$$\frac{\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda\big)}{\Pi_1\big(\tfrac{1}{2}(j+j')-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j+j')\big)},$$

which is equal to

$$\frac{\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j+j')+\lambda\big)}{\Pi_1\big(\tfrac{1}{2}(j-j')-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j+j')\big)};$$

or finally we have, as regards the leading term in η,

$$D(j,j')=\eta^{j+j'}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,\frac{\Pi_1\big(\tfrac{1}{2}(j+j')\big)}{\Pi_1\,\Pi_1\big(\tfrac{1}{2}(j+j')-\tfrac{1}{2}\big)\,\Pi_1\big(\tfrac{1}{2}(j-j')\big)}\sum\frac{a^{2\lambda}}{a'^{2\lambda}}\,F\big(\tfrac{1}{2}(j+j')+\lambda,\ \tfrac{1}{2}(j+j')+\lambda-\tfrac{1}{2},\ \ldots$$

I remark that if in general

$$r^{\alpha+\beta}\,r'^{\alpha-\beta}\,(r^2+r'^2-2\,r\,r'\cos\vartheta)^{-\alpha-\frac{1}{2}}=\sum_i R_i^{\alpha,\beta}\cos i\vartheta,$$

then, writing $\frac{1}{2}(j-j')$ for β and $\frac{1}{2}(-j+j')$ for β', we have

$$D(j,j')=R_j^{\alpha,\beta}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,(\text{terms in }\eta),$$

and the last-mentioned expression for $D(j,j')$, as regards the leading term in η, becomes therefore

$$\eta^{j+j'}\,R_{j+j'}^{\alpha,\beta}\,(1-\eta^2)^{\frac{1}{2}(j-j')},$$

and we have in general

$$D(j,j')=\eta^{j+j'}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,R_{j+j'}^{\alpha,\beta}\,(\text{terms in }\eta).$$

As a verification, it may be noticed that for $j+j'=0$ we have $D(j,j')=(1-\eta^2)^j\,(R_0+$ terms in $\eta)$, and for $\eta=0$, $D(j,j')=R_j$, which is right, since the two sides each denote the coefficient of $\cos j\vartheta$ in $(r^2+r'^2-2\,r\,r'\cos\vartheta)^{-\frac{1}{2}}$. There is reason to believe that the expression for $D(j,j')$ might be further reduced so as to obtain in a convenient form the coefficients of the successive powers of η; but I have not yet accomplished this.

V.

[PDF p. 357; printed p. 335.] Considering now $D(j, j')$ as a given function of r and r', we must in the planetary theory write $r = a(1 + x)$, $r' = a'(1 + x')$, and develope in powers of x and x'. The general term is, of course,

$$\frac{a^\alpha\, a'^{\alpha'}}{\Pi a\, \Pi a'}\, \frac{d^{\alpha+\alpha'}\, D(j, j')}{da^\alpha\, da'^{\alpha'}},$$

where $D'(j, j')$ is what $D(j, j')$ becomes when a, a' are substituted for r, r'; and, writing $f + \mathfrak{C}$, $f' + \mathfrak{C}'$ for U, U', we see that in the planetary theory the terms to be developed are of the form

$$\frac{r^n\, r'^{n'}}{a^n\, a'^{n'}} \cos\bigl(jf + j'f' + j\mathfrak{C} + j'\mathfrak{C}'\bigr),$$

while, from what has preceded, the form for the lunar theory is

$$\frac{r^n\, r'^{n'}}{a^n\, a'^{n'}} \cos\bigl(jf + j'f' + j\mathfrak{C} + j'\mathfrak{C}'\bigr),$$

(n is always $= -n' - 1$, except in the terms $\cos(U - U')$); the values which have to be substituted being

$$r = a\ \mathrm{elqr}(e, g), \qquad x = \mathrm{elqr}(e, g) - 1,$$
$$f = \mathrm{elta}(e, g),$$
$$r' = a'\ \mathrm{elqr}(e', g'), \qquad x' = \mathrm{elqr}(e', g') - 1,$$
$$f' = \mathrm{elta}(e', g').$$

Suppose that in the former case the developments of

$$x^a \cos jf, \quad x^a \sin jf, \quad x'^{a'} \cos j'f', \quad x'^{a'} \sin j'f'$$

and in the latter case the developments of

$$\left(\frac{r}{a}\right)^n \cos jf, \quad \left(\frac{r}{a}\right)^n \sin jf, \quad \left(\frac{r'}{a'}\right)^{n'} \cos j'f', \quad \left(\frac{r'}{a'}\right)^{n'} \sin j'f',$$

are

$$\sum [\cos]^i \cos ig, \quad \sum [\sin]^i \sin ig, \quad \sum [\cos]^{i'} \cos i'g', \quad \sum [\sin]^{i'} \sin i'g',$$

where the summations extend from i or $i' = -\infty$ to ∞, and where the coefficients $[\cos]^i$, $[\sin]^i$ satisfy

$$[\cos]^{-i} = [\cos]^i, \qquad [\sin]^{-i} = -[\sin]^i,$$

and in like manner the coefficients $[\cos]^{i'}$, $[\sin]^{i'}$ satisfy

$$[\cos]^{-i'} = [\cos]^{i'}, \qquad [\sin]^{-i'} = -[\sin]^{i'}.$$

It is to be observed that $[\cos]^i$, $[\sin]^i$ are functions of e, and $[\cos]^{i'}$, $[\sin]^{i'}$

[PDF p. 358; printed p. 336.] functions of e'; the accents to the indices i and i' are sufficient to indicate this. Hence observing that

$$\begin{aligned}
\sum[\cos]^i \cos ig \cdot \sum[\cos]^{i'} \cos i'g' &= \sum\sum[\cos]^i\,[\cos]^{i'} \cos(ig+i'g'),\\
\sum[\cos]^i \cos ig \cdot \sum[\sin]^{i'} \sin i'g' &= \sum\sum[\cos]^i\,[\sin]^{i'} \sin(ig+i'g'),\\
\sum[\sin]^i \sin ig \cdot \sum[\cos]^{i'} \cos i'g' &= \sum\sum[\sin]^i\,[\cos]^{i'} \sin(ig+i'g'),\\
\sum[\sin]^i \sin ig \cdot \sum[\sin]^{i'} \sin i'g' &= \sum\sum[\sin]^i\,[\sin]^{i'} \cos(ig+i'g'),
\end{aligned}$$

we have for the products of $x^a\,x'^{a'}$, or as the case may be $\left(\frac{r}{a}\right)^n\left(\frac{r'}{a'}\right)^{n'}$, into

$$\begin{aligned}
\cos(jf+j'f'),\quad &\sum\sum\left([\cos]^i\,[\cos]^{i'}+[\sin]^i\,[\sin]^{i'}\right)\cos(ig+i'g'),\\
\sin(jf+j'f'),\quad &\sum\sum\left([\sin]^i\,[\cos]^{i'}+[\cos]^i\,[\sin]^{i'}\right)\sin(ig+i'g'),
\end{aligned}$$

and thence observing that

$$\begin{aligned}
\cos(j\mathfrak{C}+j'\mathfrak{C}')\sum\sum P^{i,i'}\cos(ig+i'g') &= \sum\sum P^{i,i'}\cos(ig+i'g'+j\mathfrak{C}+j'\mathfrak{C}'),\\
-\sin(j\mathfrak{C}+j'\mathfrak{C}')\sum\sum Q^{i,i'}\sin(ig+i'g') &= \sum\sum Q^{i,i'}\cos(ig+i'g'+j\mathfrak{C}+j'\mathfrak{C}'),
\end{aligned}$$

provided only that $P^{-i,-i'}=P^{i,i'}$, $Q^{-i,-i'}=Q^{i,i'}$, we find for the product of $x^a\,x'^{a'}$ or as the case may be $\left(\frac{r}{a}\right)^n\left(\frac{r'}{a'}\right)^{n'}$ into $\cos(jf+j'f'+j\mathfrak{C}+j'\mathfrak{C}')$ the expression

$$\sum\sum\left([\cos]^i\,[\cos]^{i'}+[\sin]^i\,[\sin]^{i'}+[\sin]^i\,[\cos]^{i'}+[\cos]^i\,[\sin]^{i'}\right)\cos(ig+i'g'+j\mathfrak{C}+j'\mathfrak{C}'),$$

or, finally, the expression

$$\sum\sum\left([\cos]^i+[\sin]^i\right)\left([\cos]^{i'}+[\sin]^{i'}\right)\cos(ig+i'g'+j\mathfrak{C}+j'\mathfrak{C}'),$$

which is the required development of the general term in multiple cosines of the mean anomalies.

VI.

Investigation of the coefficient $C_n(j,j')$.

It is possible that there might be some advantage in developing in the first instance according to the powers of $\cos H$, a process which, as it has been seen, leads very readily to the form of the general term; but the mode which I have adopted is to develope in the first instance according to the powers of η. I put therefore

$$\cos H = \cos(U-U') - 2\eta^2 \sin U \sin U',$$

and we have then for the reciprocal of the distance,

$$\left[r^2+r'^2-2\,r\,r'\cos(U-U')-r\,r'\,(-4\sin U\sin U')\,\eta^2\right]^{-\frac{1}{2}},$$

[PDF p. 359; printed p. 337.] which is to be developed in ascending powers of $\dfrac{1}{r'}$, and we have for the discrete general term of the development,

$$\frac{r^n}{r'^{n+1}}\, C_n(j,j')\, \cos(jU + j'U'),$$

which is the definition of $C_n(j,j')$, the coefficient the value of which is sought for. It has been seen that j, j' are each of them of the same parity with n (even or odd according as n is even or odd), and it has been seen also that it is sufficient to consider the case where j is not negative and not inferior in absolute magnitude to j'.

Expanding in powers of η, and putting as before

$$\Pi a = 1.2.3\ldots a, \qquad \Pi_1(a-\tfrac{1}{2}) = \tfrac{1}{2}.\tfrac{3}{2}\ldots(a-\tfrac{1}{2});$$

the general term is

$$\frac{\Pi_1(\chi-\frac{1}{2})}{\Pi_1\,\Pi_1(-\frac{1}{2})}\cdot\frac{\eta^{2\chi}\, r^{\chi}\, r'^{\chi}}{(r^2+r'^2-2\,r\,r'\cos(U-U'))^{\chi+\frac{1}{2}}}\,(-4\sin U\sin U')^{\chi},$$

where χ extends from 0 to ∞.

The factor $(-4\sin U\sin U')^{\chi}$ consists of a series of multiple cosines, and as usual it is assumed that the cosines to opposite arguments are made to occur with equal coefficients. The form of the general term is $\cos(\lambda U+\lambda' U')$, where λ, λ' have each of them the values χ, $\chi-2$, $\chi-4$,$\ldots -\chi$, that is, λ, λ' are each of them of the same parity with χ. Hence j, j' being as before of the same parity with each other, and ϑ being even or odd according as j, j' and χ are of the same or of opposite parities, the development of $(-4\sin U\sin U')^{\chi}$ will contain a series of terms $\cos\big[(j+\vartheta)U+(j'-\vartheta)U'\big]$, which (since the other factor contains only multiple cosines of $U-U'$) are the only terms which give rise to a term $\cos(jU+j'U')$. I represent the discrete term of $(-4\sin U\sin U')^{\chi}$ which contains the before-mentioned argument by

$$M_{\chi}^{j,j'}\cos\big[(j+\vartheta)U+(j'-\vartheta)U'\big].$$

On the before-mentioned assumption, j not negative and not less in absolute magnitude than j', we have $j+j'$ and $j-j'$ each of them not negative. We must have $j+\vartheta \ngtr \chi$ and $j'-\vartheta\ngtr\chi$, that is $\vartheta\ngtr(\chi-j)$ and $-\vartheta\ngtr(\chi-j')$, consequently

$$\vartheta \ngtr \chi - j, \qquad \vartheta \ngeq -(\chi-j'),$$

or $j'-j \nless -(\chi-j')$ or $2\chi\nless j+j'$. And this relation being assumed, ϑ extends from the inferior limit $-(\chi-j')$ to the superior limit $(\chi-j)$ by steps of two units, the extreme terms being

$$\vartheta = -(\chi-j'), \qquad M_{\chi}^{j,j'}\cos\big[(j+j'-\chi)U+\chi U'\big],$$

the coefficient of U increasing from $j+j'-\chi$ to χ, and that of U' diminishing from χ to $j+j'-\chi$ by steps of two units.

[PDF p. 360; printed p. 338.] Now expanding in ascending powers of $\dfrac{1}{r'}$, write

$$r^{\chi}\, r'^{\chi} \left\{r^2 + r'^2 - 2\, r\, r' \cos(U-U')\right\}^{-\chi-\frac{1}{2}} = \sum R^i_{\chi} \cos i(U-U'),$$

where i extends from $-\infty$ to $+\infty$, and $R^{-i}_{\chi} = R^i_{\chi}$; so that $R^i_{\chi} \cos i(U-U')$ is the discrete general term of the development. The term $R^i_{\chi} \cos i(U-U')$ in combination with the term $M^{j,j'}_{\chi} \cos[(j+\vartheta)U + (j'-\vartheta)U']$, gives rise to $R^i_{\chi}\, M^{j,j'}_{\chi}$; and restoring the multiplier which has been omitted, and giving to ϑ and χ the different admissible values, we find for the discrete general term containing $\cos(jU + j'U')$ the value

$$\frac{\Pi_1(\chi-\frac{1}{2})}{\Pi_1\, \Pi_1(-\frac{1}{2})}\, \eta^{2\chi} \sum\sum R^i_{\chi}\, M^{j,j'}_{\chi} \cos(jU + j'U'),$$

where ϑ extends from the inferior limit $\vartheta = -(\chi - j')$ to the superior limit $\vartheta = \chi - j$, by steps of two units, and χ extends from $\chi = \frac{1}{2}(j+j')$ to $\chi = \infty$. The portion of this containing $\dfrac{r^n}{r'^{n+1}} \cos(jU + j'U')$ is

$$\frac{r^n}{r'^{n+1}}\, C_n(j,j') \cos(jU + j'U'),$$

and we have therefore

$$C_n(j,j') = \sum\sum \frac{\Pi_1(\chi-\frac{1}{2})}{\Pi_1\, \Pi_1(-\frac{1}{2})}\, \eta^{2\chi}\, M^{j,j'}_{\chi}\, R^n_{\chi}.$$

There is some speciality in the case $j + j' = 0$, but the result just obtained subsists without variation. To find $M^{j,j'}_{\chi}$ we have

$$M^{j,j'}_{\chi} = \text{disct. coeff. } \cos\big[(j+\vartheta)U + (j'-\vartheta)U'\big] \text{ in } (-4\sin U \sin U')^{\chi},$$

and putting $\sin U = \dfrac{1}{2i}\big(v - \dfrac{1}{v}\big)$, $\sin U' = \dfrac{1}{2i}\big(v' - \dfrac{1}{v'}\big)$ where $i = \sqrt{-1}$ we have

$$(-4\sin U \sin U')^{\chi} = \Big(v - \frac{1}{v}\Big)^{\chi} \Big(v' - \frac{1}{v'}\Big)^{\chi},$$

and the function on the right hand contains the term

$$\frac{\Pi\chi}{\Pi f\, \Pi(\chi - f)}\, (-)^f\, \frac{\Pi\chi}{\Pi f'\, \Pi(\chi - f')}\, (-)^{f'}\, v^{\chi-2f}\, v'^{\chi-2f'},$$

or putting

$$\chi - 2f = j + \vartheta, \qquad \chi - 2f' = j' - \vartheta$$

and therefore

$$f = \tfrac{1}{2}(\chi - j - \vartheta), \qquad f' = \tfrac{1}{2}(\chi - j' + \vartheta),$$

which give integer values for f, f' since ϑ is even or odd according as j, j', and χ are of the same parity or of opposite parities, and replacing $v^{\chi-2f}\, v'^{\chi-2f'}$ by the half of its value $2\cos\big[(j+\vartheta)U + (j'-\vartheta)U'\big]$, the term is

$$\frac{2\,\Pi\chi\,\Pi\chi}{\Pi\big(\frac{1}{2}(\chi - j - \vartheta)\big)\, \Pi\big(\frac{1}{2}(\chi + j + \vartheta)\big)\, \Pi\big(\frac{1}{2}(\chi - j' + \vartheta)\big)\, \Pi\big(\frac{1}{2}(\chi + j' - \vartheta)\big)}\, (-)^{\chi - \vartheta - \frac{1}{2}(j-j')},$$

[PDF p. 361; printed p. 339.] and consequently,

$$M_\chi^{j,j'} = \frac{(-)^{-\chi-\frac{1}{2}(j-j')+\vartheta}\cdot 2\,(\Pi\chi)^2}{\Pi\big(\frac{1}{2}(\chi-j-\vartheta)\big)\,\Pi\big(\frac{1}{2}(\chi+j+\vartheta)\big)\,\Pi\big(\frac{1}{2}(\chi-j'+\vartheta)\big)\,\Pi\big(\frac{1}{2}(\chi+j'-\vartheta)\big)},$$

which is the expression for $M_\chi^{j,j'}$.

The expression for R_χ^i is found in a similar manner, viz., by substituting for $\cos(U-U')$ its exponential expression, by which means the function

$$r^\chi\,r'^\chi\,\big\{r^2+r'^2-2\,r\,r'\cos(U-U')\big\}^{-\chi-\frac{1}{2}}$$

breaks up into a pair of factors, each of which can be separately expanded; the result, i being positive, or zero, is

$$R_\chi^i = \sum \frac{r^{\chi+i+m}}{r'^{\chi+i+m+1}}\,\frac{\Pi_1(\chi+i+m-\frac{1}{2})\,\Pi_1(\chi+m-\frac{1}{2})}{\Pi_1(-\frac{1}{2})\,\Pi_1(i+m)\,\Pi(\chi-1)\,\Pi m\,\Pi(\chi-1)\,\Pi(\chi-m)}\left(\frac{r}{r'}\right)^{-2m-1}$$

from $m=0$ to $m=\infty$: this may also be written

$$R_\chi^i = \sum \frac{1}{r'^{n+1}}\,r^n\cdot\frac{\Pi_1(\chi-1)\,\Pi_1(\chi+m-\frac{1}{2})}{\Pi_1(-\frac{1}{2})\,\Pi_1(i+m)\,\Pi(\chi-1)\,\Pi m\,\Pi(\chi-m-1)},$$

writing now ϑ for i, and $\chi+\vartheta+2m=n$, that is $m=\frac{1}{2}(n-\chi-\vartheta)$ (n is of the same parity with j, j', and ϑ is even or odd according as j, j' and χ are of the same or opposite parities, m is therefore, as it should be, an integer), R_χ^n contains the term

$$\frac{r^n}{r'^{n+1}}\,\frac{\Pi_1(\frac{1}{2}(n+\vartheta)-\frac{1}{2})\,\Pi_1(\frac{1}{2}(n-\vartheta)-\frac{1}{2})}{\Pi(\frac{1}{2}(n+\vartheta)-\chi)\,\Pi_1(\frac{1}{2}(n-\vartheta)-\chi)\,\Pi(\chi-\frac{1}{2})\,\Pi(\chi-1)}\,K_\chi^{n,\vartheta},$$

if for shortness

$$K_\chi^{n,\vartheta} = \frac{\Pi_1\big(\frac{1}{2}(n+\vartheta)-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n-\vartheta)-\frac{1}{2}\big)}{\Pi\big(\frac{1}{2}(n-\chi+\vartheta)\big)\,\Pi\big(\frac{1}{2}(n-\chi-\vartheta)\big)\,\Pi(\chi-1)},$$

a formula which I assume to subsist as well for negative as for positive values of ϑ, so that $K_\chi^{n,-\vartheta}=K_\chi^{n,\vartheta}$. It is to be noticed that if $\chi>n$, then either $n-\chi+\vartheta$ or $n-\chi-\vartheta$ vanishes, and we have therefore $K_\chi^{n,\vartheta}=0$.

Substituting for R_χ^n its value we have

$$C_n(j,j') = \sum\sum \frac{\Pi_1(\chi-\frac{1}{2})}{\Pi_1\,\Pi_1(-\frac{1}{2})}\,\eta^{2\chi}\,M_\chi^{j,j'}\,K_\chi^{n,\vartheta},$$

where $M_\chi^{j,j'}$, $K_\chi^{n,\vartheta}$ have the values already obtained, and as before ϑ extends to $\vartheta = -(\chi-j')$ to $\vartheta=(\chi-j)$ by steps of two units, and χ (since $K_\chi^{n,\vartheta}$ vanishes for $\chi<n$) extends from $\chi=\frac{1}{2}(j+j')$ to $\chi=n$.

[PDF p. 362; printed p. 340.] To simplify; write $\chi = \frac{1}{2}(j+j') + u$, $\vartheta = -(\chi - j') + 2s$, $(\chi - j) - 2t$, so that $s + t = u$, s and t being any integer values (zero not excluded) which satisfy this relation, and lastly u extends from $u = 0$ to $u = n - \frac{1}{2}(j+j')$. We have

$$C_n(j,j') = \sum 2^{j+j'} \frac{\Pi_1\big(\frac{1}{2}(j+j') + u - \frac{1}{2}\big)}{\Pi_1\,\Pi_1(-\frac{1}{2})} \eta^{2(\frac{1}{2}(j+j')+u)}\, M_\chi^{j,j'}\, K_\chi^{n,\vartheta},$$

where

$$M_\chi^{j,j'} = (-)^u \frac{\Pi_1\big(\frac{1}{2}(j+j')+u\big)\,\Pi_1\big(\frac{1}{2}(j+j')+u\big)}{\Pi(s)\,\Pi(t)\,\Pi(j+s)\,\Pi(j'+t)},$$

$$K_\chi^{n,\vartheta} = \frac{\Pi_1\big(\frac{1}{2}(n+j)+s-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n+j')+t-\frac{1}{2}\big)}{\Pi_1\big(\frac{1}{2}(n-j)-s\big)\,\Pi_1\big(\frac{1}{2}(n-j')-t\big)\,\Pi\big(\frac{1}{2}(j+j')+u-1\big)},$$

and substituting these values, and observing that the result contains a factor $\dfrac{1}{\Pi\big(\frac{1}{2}(j+j')+u\big)\,\Pi_1\big(\frac{1}{2}(j+j}$ which may be replaced by $\dfrac{2^{j+j'+2u}}{\Pi(j+j'+2u)}$, the result is

$$C_n(j,j') = 2^{j+j'}\,\eta^{j+j'} \sum \frac{\Pi_1\big(\frac{1}{2}(j+j')+u-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(j+j')+u\big)}{\Pi_1\big(\frac{1}{2}(j+j')+u\big)\,\Pi_1\big(\frac{1}{2}(j+j')+u-\frac{1}{2}\big)}$$
$$\times \frac{\Pi_1\big(\frac{1}{2}(n+j)+s-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n+j')+t-\frac{1}{2}\big)}{\Pi(s)\,\Pi(j+s)\,\Pi_1\big(\frac{1}{2}(n-j)-s\big)\,\Pi(t)\,\Pi(j'+t)\,\Pi_1\big(\frac{1}{2}(n-j')-t\big)}\,\eta^{2u},$$

which is easily transformed into

$$C_n(j,j') = 2^{j+j'}\,\eta^{j+j'} \frac{\Pi_1\big(\frac{1}{2}(n+j)-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n+j')-\frac{1}{2}\big)}{\Pi(j+j')}\,S,$$

if for shortness

$$S = \sum \frac{(-)^{s+t}\,\eta^{2(s+t)}\,(j+j')!}{[\frac{1}{2}(j+j')+s-\frac{1}{2}]^{s+t}\,[\frac{1}{2}(j+j')+t-\frac{1}{2}]^{s+t}\,\Pi(s)\,\Pi(t)\,\Pi(j+s)\,\Pi(j'+t)},$$

$$s+t=u, \quad u = 0 \text{ to } u = n - \tfrac{1}{2}(j+j');$$

the value of the sum S is

$$S = (1-\eta^2)^{\frac{1}{2}(j-j')}\,F\big(-n+j,\, n+j+1,\, j+j'+1,\, \eta^2\big),$$

and we have consequently

$$C_n(j,j') = 2^{j+j'}\,\eta^{j+j'} \frac{\Pi_1\big(\frac{1}{2}(n+j)-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n+j')-\frac{1}{2}\big)}{\Pi_1\big(\frac{1}{2}(n-j)\big)\,\Pi_1\big(\frac{1}{2}(n-j')\big)\,\Pi(j+j')}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,F\big(-n+j,\, n+j+1,\, j+j'+1,\, \eta^2\big)$$

the required expression for $C_n(j,j')$. It only remains to prove the formula for S.

[PDF p. 363; printed p. 341.] For this purpose, observing the equation $s+t=u$, I form the equations

$$\frac{1}{(j+j'+2u)^u} = \frac{1}{[\frac{1}{2}(j+j')+u-\frac{1}{2}]^u[\frac{1}{2}(j+j')+u-1]^u},$$

$$[\tfrac{1}{2}(j+j')+u]^u = [\tfrac{1}{2}(j+j')+s]^u\,[\tfrac{1}{2}(j+j')+t]^u,$$
$$[\tfrac{1}{2}(j+j')+u]^u = [\tfrac{1}{2}(j+j')+s]^u\,[\tfrac{1}{2}(j+j')+t]^u,$$

and thence

$$\frac{2^u}{(j+j'+2u)^u} = \frac{[\frac{1}{2}(j+j')+s]^u\,[\frac{1}{2}(j+j')+t]^u}{[\frac{1}{2}(j+j')+s-\frac{1}{2}]^u\,[\frac{1}{2}(j+j')+t-\frac{1}{2}]^u},$$

and it is proper to remark that the summation as regards u may be continued indefinitely; for, if u exceed $n-\frac{1}{2}(j+j')$, then one at least of the relations $s>\frac{1}{2}(n-j)$, $t>\frac{1}{2}(n-j')$ must hold good, and at least one of the factorials $[\frac{1}{2}(n-j)]^s$, $[\frac{1}{2}(n-j')]^t$ will vanish. The two factors in the second sum are the general terms of two hypergeometric series. In fact, if we put

$$\alpha = -\tfrac{1}{2}n+\tfrac{1}{2}j, \qquad \beta = -\tfrac{1}{2}n+\tfrac{1}{2}j', \qquad \epsilon = \tfrac{1}{2}j+\tfrac{1}{2}j'+\tfrac{1}{2},$$

and therefore

$$\epsilon-\alpha = \tfrac{1}{2}n+\tfrac{1}{2}j'+\tfrac{1}{2}, \qquad \epsilon-\beta = \tfrac{1}{2}n+\tfrac{1}{2}j+\tfrac{1}{2},$$

and if, for shortness, we use $\{a\}^s$ to denote the factorial $a(a+1)\dots(a+s-1)$ of the increment positive unity, then we have

$$\alpha\sum\frac{\{\alpha\}^s\,\{\beta\}^s}{\{\epsilon+\frac{1}{2}\}^s\Pi s}\eta^{2s} = \sum\frac{1}{\Pi s\,\{\epsilon+\frac{1}{2}\}^s}\,\frac{\Pi(j+s)}{\Pi j}\cdot\frac{1}{\Pi(s)}\cdot\eta^{2s},$$

the two hypergeometric series being consequently

$$1+\frac{\alpha\cdot\beta}{1\cdot\epsilon}+\frac{\alpha\cdot\alpha+1}{1\cdot 2}\cdot\frac{\beta\cdot\beta+1}{\epsilon\cdot\epsilon+1}\,\eta^4+\&\text{c.}$$

[PDF p. 364; printed p. 342.] I assume the truth of the following remarkable theorem, [see 211] viz.:

"The series formed from the product of the two hypergeometric series,

$$F(\alpha,\, \beta,\, \epsilon+\tfrac{1}{2},\, \eta^2)\, F(\epsilon-\alpha,\, \epsilon-\beta,\, \epsilon+\tfrac{1}{2},\, \eta^2),$$

by multiplying the successive terms of the product by $1,\ \dfrac{1}{2},\ \dfrac{1\cdot 3}{2\cdot 4}$, &c. respectively, is

$$= (1-\eta^2)^{\alpha+\beta-\epsilon}\, F(2\alpha,\, 2\beta,\, 2\epsilon,\, \eta^2)."$$

Hence, observing that the general term of S is formed precisely in the manner in question, we have

$$S = (1-\eta^2)^{\alpha+\beta-\epsilon}\, F(2\alpha,\, 2\beta,\, 2\epsilon,\, \eta^2)$$

or, substituting for α, β, ϵ their values

$$S = (1-\eta^2)^{-\frac{1}{2}(n+\frac{1}{2})}\, F(-n+j,\, -n+j',\, j+j'+1,\, \eta^2),$$

which is the required value.

It is clear that α, β are interchangeable with $\epsilon-\alpha$, $\epsilon-\beta$; that is, we have

$$F(2\alpha,\, 2\beta,\, 2\epsilon,\, \eta^2) = F(2\epsilon-2\alpha,\, 2\epsilon-2\beta,\, 2\epsilon,\, \eta^2)$$

or

$$F(2\epsilon-2\alpha,\, 2\epsilon-2\beta,\, 2\epsilon,\, \eta^2) = (1-\eta^2)^{\alpha+2\beta-\epsilon-\epsilon}\, F(2\alpha,\, 2\beta,\, 2\epsilon,\, \eta^2),$$

which is a known property of hypergeometric series. The form

$$S = (1-\eta^2)^{-\frac{1}{2}(n+\frac{1}{2})}\, F\big(-n+j,\, n+j+1,\, j+j'+1,\, \eta^2\big)$$

is obviously less convenient than the one above mentioned, since the new expression is encumbered by a denominator $(1-\eta^2)^{\frac{1}{2}(j-j')}$, which really divides out, the finite hypergeometric series containing as a factor the square of such denominator. I have only noticed this for the verification which it affords by showing that j, j' may be interchanged.

VII.

The above expression of $C_n(j,j')$ is not, in its actual form, given in Hansen's memoir. The comparison with Hansen's formula is as follows: The formula (36), p. 329, is

$$C(n-2f',\, -(n-2f-2(f)) = H\cos^n \tfrac{1}{2}J\, \tan^{2f}\tfrac{1}{2}J\, F\big(-(2n-2f-2(f)),\, -2f,\, 2f+1,\, -\tan^2\tfrac{1}{2}J\big)$$

where

$$H = 2^{-n}\, \Pi(n-f)\, \Pi(n-f-(f))\, \Pi(f+(f))\, \Pi 2(f)$$

[PDF p. 365; printed p. 343.] and the formula (37), p. 330, is

$$F\big(-(2n-2f-2(f)),\,-2f,\,2(f)+1,\,-\tan^2\tfrac{1}{2}J\big)$$

$$=\cos^{-4n+4f+4(f)}\tfrac{1}{2}J\,F\big(-(2n-2f-2(f)),\,2f+2(f)+1,\,2(f)+1,\,\sin^2\tfrac{1}{2}J\big).$$

Combining these, and putting $\sin\frac{1}{2}J=\eta$, we have

$$C\big(n-2f,\,-(n-2f-2(f))\big)=H\eta^{2(f)}\,(1-\eta^2)^{-n+f+(f)}\,F\big(-(2n-2f-2(f)),\,2f+2(f)+1,\,2(f)+1,\,\eta^2\big),$$

and then, putting

$$n-2f=j',\qquad -n+2f+2(f)=j,$$

and for $C(j',j)$, $=C(j,j')$, writing $C_n(j,j')$, we have

$$C_n(j,j')=H\eta^{j+j'}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,F\big(-n+j,\,n+j+1,\,j+j'+1,\,\eta^2\big)$$

where

$$H=\frac{1}{2^n\,\Pi(n+j')\,\Pi(n-j')\,\Pi\frac{1}{2}(n+j)\,\Pi\frac{1}{2}(n-j)\,\Pi(j+j')},$$

or, finally, since

$$\Pi(n+j)=2^{n+j}\,\Pi_1\tfrac{1}{2}(n+j)\,\Pi_1\big(\tfrac{1}{2}(n+j)-\tfrac{1}{2}\big),$$

$$\Pi(n+j')=2^{n+j'}\,\Pi_1\tfrac{1}{2}(n+j')\,\Pi_1\big(\tfrac{1}{2}(n+j')-\tfrac{1}{2}\big),$$

we find

$$C_n(j,j')=2^{j+j'}\,\frac{\Pi_1\big(\frac{1}{2}(n+j)-\frac{1}{2}\big)\,\Pi_1\big(\frac{1}{2}(n+j')-\frac{1}{2}\big)}{\Pi_1\frac{1}{2}(n-j)\,\Pi_1\frac{1}{2}(n-j')\,\Pi(j+j')}\,\eta^{j+j'}\,(1-\eta^2)^{\frac{1}{2}(j-j')}\,F\big(-n+j,\,n+j+1,\,j+j'+1,\,\eta^2\big),$$

which is the formula of the present memoir.

215. A Supplementary Memoir on the Problem of Disturbed Elliptic Motion

[PDF p. 366; printed p. 344.]

[From the *Memoirs of the Royal Astronomical Society*, vol. XXVIII. pp. 217–234. Read January 12, 1860.]

The present memoir contains an investigation upon fundamentally different principles of a system of formula given in my "Memoir on the Problem of Disturbed Elliptic Motion," *Ast. Soc. Mem.* t. XXVII. pp. 1–29 (1858), [212]; it is shown how the formulae are deduced by means of the general equations of the *Mécanique Analytique*, from the expression for the *Vis viva* function T, in terms of the coordinates (the radius vector, the departure) of the body in its instantaneous orbit, viz., the ultimate form of this function is $T\,dt^2 = \frac{1}{2}(dr^2 + r^2\,d\phi^2)$, but T contains in the first instance terms, not identically vanishing, but which are to be equated to zero, thus furnishing equations of the problem which could not be obtained from the foregoing ultimate form of T. The investigation throws, I think, a further light upon the system of formulae, and completes the development which I was anxious to give of the dynamical problem. I have been a great deal indebted, in the composition of the memoir, to a correspondence some time ago with Professor Donkin on the general subject.

The word *orbit* is used to denote a great circle of the sphere, and it is assumed that in any orbit there is an origin of longitudes; the angular position of a body in reference to the orbit is determined by the longitude and latitude. It is ultimately assumed that the longitude is measured from an origin which is what I have called a *departure-point*; viz., in the general case of a variable orbit the departure-point is the point of intersection of the orbit by any orthogonal trajectory of the successive positions of the orbit: this includes the case of a fixed orbit, where the departure-point is simply a fixed point. As regards points in the orbit, the word *departure* may be used instead of longitude. In the present memoir the origin is in the first instance taken to be, not a departure-point, but an arbitrarily varying point of the orbit.

[PDF p. 367; printed p. 345.] The mutual position of any two orbits whatever, say the position of an orbit $x'y'$ and of the origin x' therein, in reference to the orbit xy and the origin x therein, is determined by

θ, the longitude of node,
ϖ, the departure of node,
ϕ, the inclination,

where the expression *departure of node* is used by way of distinction from longitude of node, the departure referring to the orbit $x'y'$ and origin x', the positions of which are to be determined, and the longitude to the orbit of reference xy and origin x. And this distinction will be preserved throughout. It should be recollected that it is not as yet assumed that the origins are departure-points.

Consider now a fixed orbit x_0y_0 and fixed origin x_0 therein, and suppose that the orbit x_1y_1 and origin x_1 therein are determined in reference to the orbit x_0y_0 and origin x_0 therein, by

θ', the longitude of node,
ϖ', the departure of node,
ϕ', the inclination,

the orbit x_2y_2 and origin x_2 therein, in reference to the orbit x_1y_1 and origin x_1 therein, by

Θ, the longitude of node,
$\varSigma$, the departure of node,
$\varPhi$, the inclination,

and, finally, the position of the body in reference to the orbit x_2y_2 and origin x_2 therein, by

r, the radius vector,
v, the longitude,
y, the latitude.

It is required to find the expression of T, the *Vis viva* function, or half square of the velocity. We may imagine the rectangular axes $x_0y_0z_0$, $x_1y_1z_1$, $x_2y_2z_2$, the positions of which are determined as above, and the rectangular axes $x_3y_3z_3$, that of x_3 passing through the body, that of y_3 lying in the orbit x_2y_2, 90° in advance of the body or

[PDF p. 368; printed p. 346.] passing through the pole of the latitude circle, and that of z_3 in the latitude circle 90° above the body. Or considering x_3, y_3, z_3 as points of the sphere, their positions in reference to the orbit x_2y_2 and origin x_2 therein are determined as follows, viz.,

for x_3,	longitude is v,	and latitude y,
y_3,	$v+90°$,	0,
z_3,	v,	$y+90°$,

and we may suppose that (x_0, y_0, z_0), (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3), are the coordinates of any point in reference to these sets of axes respectively, the coordinates of the body in reference to the axes $x_3y_3z_3$ being obviously $(r, 0, 0)$.

The displacements in the directions of the axes $x_3y_3z_3$ respectively of a point the coordinates of which are x_3, y_3, z_3, are as usual

$$\begin{aligned} &dx_3 + (-R\,y_3 + Q\,z_3)\,dt,\\ &dy_3 + (-P\,z_3 + R\,x_3)\,dt,\\ &dz_3 + (-Q\,x_3 + P\,y_3)\,dt, \end{aligned}$$

where the expressions of the rotations P, Q, R are as follows, viz.,

$$\begin{aligned} P\,dt = {} & \sin y\,dv\\ & + \cos y\big[\sin(v-\mathit{\Sigma})\sin\mathit{\Phi}\,d\Theta + \cos(v-\mathit{\Sigma})\,d\mathit{\Phi}\big]\\ & + \sin y\big[-d\mathit{\Sigma} + \cos\mathit{\Phi}\,d\Theta\big]\\ & + \big\{\cos y\cos(v-\mathit{\Sigma})\big\}\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big]\\ & + \big\{\cos y\sin(v-\mathit{\Sigma})\cos\mathit{\Phi} - \sin y\sin\mathit{\Phi}\big\}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big]\\ & + \big\{\cos y\sin(v-\mathit{\Sigma})\sin\mathit{\Phi} + \sin y\cos\mathit{\Phi}\big\}\big[-d\varpi' + \cos\phi'\,d\theta'\big], \end{aligned}$$

[PDF p. 369; printed p. 347.]

$$
\begin{aligned}
Q\,dt = dy & \\
& + \big[\cos(v-\mathit{\Sigma})\sin\mathit{\Phi}\,d\Theta - \sin(v-\mathit{\Sigma})\,d\mathit{\Phi}\big] \\
& - \sin(v-\mathit{\Sigma})\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big] \\
& + \cos(v-\mathit{\Sigma})\cos\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big] \\
& + \cos(v-\mathit{\Sigma})\sin\mathit{\Phi}\big[-d\varpi' + \cos\phi'\,d\theta'\big],
\end{aligned}
$$

$$
\begin{aligned}
R\,dt = \cos y\,dv & \\
& - \sin y\big[\sin(v-\mathit{\Sigma})\sin\mathit{\Phi}\,d\Theta + \cos(v-\mathit{\Sigma})\,d\mathit{\Phi}\big] \\
& + \cos y\big[-d\mathit{\Sigma} + \cos\mathit{\Phi}\,d\Theta\big] \\
& + \big\{-\sin y\cos(v-\mathit{\Sigma})\big\}\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big] \\
& + \big\{-\sin y\sin(v-\mathit{\Sigma})\cos\mathit{\Phi} - \cos y\sin\mathit{\Phi}\big\}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big] \\
& + \big\{-\sin y\sin(v-\mathit{\Sigma})\sin\mathit{\Phi} + \cos y\cos\mathit{\Phi}\big\}\big[-d\varpi' + \cos\phi'\,d\theta'\big].
\end{aligned}
$$

For the body itself we must write r, 0, 0, in the place of x_3, y_3, z_3; the displacements are therefore

$$dr, \quad rR\,dt, \quad rQ\,dt;$$

and the expression for the *Vis viva* function is

$$T\,dt^2 = \tfrac{1}{2}\big(dr^2 + r^2(Q^2+R^2)\,dt^2\big),$$

where Q, R have the above-mentioned values.

Let Ω be a function of the coordinates x_0, y_0, z_0 of a point, Ω will be a function of x_3, y_3, z_3, and also of the quantities θ', Θ, $\mathit{\Sigma}$, ϖ', ϕ', $\mathit{\Phi}$, which determine the positions of the axes $x_3y_3z_3$, and the complete differential of Ω will be

$$d\Omega = \frac{d\Omega}{dx_3}\big[dx_3 + (-R\,y_3 + Q\,z_3)\,dt\big] + \frac{d\Omega}{dy_3}\big[dy_3 + (-P\,z_3 + R\,x_3)\,dt\big] + \frac{d\Omega}{dz_3}\big[dz_3 + (-Q\,x_3 + P\,y_3)\,dt\big].$$

I assume for the present the truth of the following proposition, viz., that when Ω is a function of the coordinates of the body (in which case, as before, the values of x_3, y_3, z_3 are r, 0, 0), we have

$$\frac{d\Omega}{dx_3} = \frac{d\Omega}{dr}, \qquad \frac{d\Omega}{dy_3} = \frac{1}{r}\sec y\,\frac{d\Omega}{dv}, \qquad \frac{d\Omega}{dz_3} = \frac{1}{r}\,\frac{d\Omega}{dy}.$$

[PDF p. 370; printed p. 348.] Ω is here a function of r, v, y, θ', Θ, $\varSigma$, ϖ', ϕ', $\varPhi$, or, as we may express it, $\Omega = \Omega(r, v, y, \theta', \Theta, \varSigma, \varpi', \phi', \varPhi)$; and the expression for the complete differential, by what precedes, is

$$d\Omega = \frac{d\Omega}{dr}\,dr + R\,dt\sec y\,\frac{d\Omega}{dv} + Q\,dt\,\frac{d\Omega}{dy},$$

or, substituting for Q, R, their values, the expression is

$$\begin{aligned}
d\Omega = {} & \frac{d\Omega}{dr}\,dr \\
& + \sec y\,\frac{d\Omega}{dv}\Big\{\cos y\,dv - \sin y\big[\sin(v-\varSigma)\sin\varPhi\,d\Theta + \cos(v-\varSigma)\,d\varPhi\big] \\
& \qquad + \cos y\big[-d\varSigma + \cos\varPhi\,d\Theta\big] \\
& \qquad + \{-\sin y\cos(v-\varSigma)\}\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big] \\
& \qquad + \{-\sin y\sin(v-\varSigma)\cos\varPhi - \cos y\sin\varPhi\}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big] \\
& \qquad + \{-\sin y\sin(v-\varSigma)\sin\varPhi + \cos y\cos\varPhi\}\big[-d\varpi' + \cos\phi'\,d\theta'\big]\Big\} \\
& + \frac{d\Omega}{dy}\Big\{dy + \big[\cos(v-\varSigma)\sin\varPhi\,d\Theta - \sin(v-\varSigma)\,d\varPhi\big] \\
& \qquad - \sin(v-\varSigma)\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big] \\
& \qquad + \cos(v-\varSigma)\cos\varPhi\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big] \\
& \qquad + \cos(v-\varSigma)\sin\varPhi\big[-d\varpi' + \cos\phi'\,d\theta'\big]\Big\}
\end{aligned}$$

where, on the right-hand side, the terms involving the differentials of r, v, y, are, as they should be, $\dfrac{d\Omega}{dr}\,dr + \dfrac{d\Omega}{dv}\,dv + \dfrac{d\Omega}{dy}\,dy$.

I have given the complete expression, as it may be useful; but for the present purpose it will be sufficient to attend to the terms involving $\varSigma$, Θ, $\varPhi$. We have

$$\frac{d\Omega}{d\varSigma}+\frac{d\Omega}{d\Theta}+\frac{d\Omega}{d\varPhi} = \sec y\,\frac{d\Omega}{dv}\Big(-\sin y\sin(v-\varSigma)\sin\varPhi - \cos y\Big) + \frac{d\Omega}{dy}\big\{\cos(v-\varSigma)\sin\varPhi\,d\Theta - \sin(v-\varSigma)\,d\varPhi\big\},$$

and consequently

[PDF p. 371; printed p. 349.]

$$\frac{d\Omega}{d\varSigma} = -\frac{d\Omega}{dv},$$
$$\frac{d\Omega}{d\Theta} = \left\{\cos\varPhi - \tan y \sin(v-\varSigma)\sin\varPhi\right\}\frac{d\Omega}{dv} - \cos(v-\varSigma)\sin\varPhi\,\frac{d\Omega}{dy},$$
$$\frac{d\Omega}{d\varPhi} = -\tan y\cos(v-\varSigma)\,\frac{d\Omega}{dv} + \sin(v-\varSigma)\,\frac{d\Omega}{dy};$$

which, it is to be observed, are partial differential equations satisfied by Ω considered as a function of the form $\Omega = \Omega(r, v, y, \Theta, \varSigma, \varPhi)$; Ω is, in fact, a function of three quantities only (say the coordinates x_1, y_1, z_1); and it is clear, a priori, that when thus expressed as a function of six quantities, it must satisfy three partial differential equations.

I write in the second equation $\dfrac{1}{\sin\varPhi}\dfrac{d\Omega}{d\Theta}$ for $\dfrac{d\Omega}{d\Theta}$, and I then put $y = 0$; we thus have

$$\cos(v-\varSigma)\,\frac{d\Omega}{dy} = \cot\varPhi\,\frac{d\Omega}{dv} - \operatorname{cosec}\varPhi\,\frac{d\Omega}{d\Theta},$$

$$\sin(v-\varSigma)\,\frac{d\Omega}{dy} = \frac{d\Omega}{d\varPhi}.$$

I propose, as already mentioned, to apply the formulae to the demonstration of the results obtained in my Memoir on the Problem of Disturbed Elliptic Motion. It will be remembered that $x_0 y_0$ and x_0 denote a fixed orbit and fixed origin therein, $x_1 y_1$ and x_1 an arbitrarily varying orbit and origin therein. I assume, however, that x_1 is a departure-point, so that we have

$$d\varpi' + \cos\phi'\,d\theta' = 0.$$

The orbit $x_2 y_2$ will be taken to be the varying instantaneous orbit of the body itself, that is, we have constantly $y = 0$; and it is assumed that x_2 is a departure-point in this orbit. To conform to my former notation, I write $\flat$ instead of v; the position of the body in the varying instantaneous orbit is consequently determined by

r, the radius vector,

$\flat$, the departure.

The entire displacement of the body arises from the displacements dr and $r\,d\flat$ in the direction of the radius vector, and perpendicular to this direction, in the instantaneous orbit; that is, we have $Q = 0$, $R = d\flat$, and the expression for the *Vis viva* function is simply

$$T\,dt^2 = \tfrac{1}{2}(dr^2 + r^2\,d\flat^2),$$

and, putting in the foregoing expressions of Q and R, $y = 0$, and $\flat$ in the place of v, the equations $Q = 0$, $R\,dt = d\flat$, give

$$\begin{aligned}
&\cos(\flat-\varSigma)\sin\varPhi\,d\Theta - \sin(\flat-\varSigma)\,d\varPhi = \\
&\quad \sin(\flat-\varSigma)\left[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\right] \\
&\quad - \cos(\flat-\varSigma)\cos\varPhi\left[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\right] \\
&\quad - \sin\varPhi\left[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\right],
\end{aligned}$$

[PDF p. 372; printed p. 350.] Now considering r, $\flat$, y as the coordinates (y being ultimately put equal to zero) the general equations of motion are

$$\frac{d}{dt}\Big(\frac{dT}{dr'}\Big) - \frac{dT}{dr} = \frac{dV}{dr},$$

$$\frac{d}{dt}\Big(\frac{dT}{d\flat'}\Big) - \frac{dT}{d\flat} = \frac{dV}{d\flat},$$

$$\frac{d}{dt}\Big(\frac{dT}{dy'}\Big) - \frac{dT}{dy} = \frac{dV}{dy},$$

if for the moment r', $\flat'$, y' denote the differential coefficients of r, $\flat$, y. The expression of the force function V is $V = \dfrac{n^2a^3}{r} + \Omega$ where n^2a^3 denotes an absolute constant, the sum of the masses, and the disturbing function Ω has a contrary sign to that of the *Mécanique Céleste*. We may in the first and second equations write at once $T\,dt^2 = \frac{1}{2}(dr^2 + r^2\,d\flat^2)$, and these equations thus become

$$\frac{d}{dt}\,\frac{dr}{dt} - r\Big(\frac{d\flat}{dt}\Big)^2 = -\frac{n^2a^3}{r^2} + \frac{d\Omega}{dr},$$

$$\frac{d}{dt}\Big(r^2\frac{d\flat}{dt}\Big) = \frac{d\Omega}{d\flat}.$$

If in the third equation we take the general value of T and after the differentiation with respect to y', make the substitutions $y = 0$, $Q = 0$, $R\,dt = d\flat$, we find

$$\frac{dT}{dy'} = 0,$$

and the equation thus becomes

$$-\frac{dT}{dy} = \frac{d\Omega}{dy}.$$

But $T\,dt^2 = \frac{1}{2}\big\{dr^2 + r^2(Q^2+R^2)\,dt^2\big\}$, consequently

$$-\frac{d}{dy}\,T\,dt = -\frac{dT}{dy} = r^2\Big(Q\,\frac{dQ}{dy} + R\,\frac{dR}{dy}\Big)\,dt = r^2\,d\flat\,\frac{dR}{dy},$$

and from the value of R, putting after the differentiation $y = 0$, we find

$$\begin{aligned}\frac{dR}{dy}\,dt = &-\sin(\flat - \varSigma)\sin\varPhi\,d\Theta - \cos(\flat-\varSigma)\,d\varPhi\\ &-\cos(\flat-\varSigma)\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big]\\ &-\sin(\flat-\varSigma)\cos\varPhi\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big].\end{aligned}$$

[PDF p. 373; printed p. 351.] I write $r^2\,\dfrac{d\flat}{dt} = h$. We thus have

$$\frac{d}{dt}\,\frac{dr}{dt} - \left(\frac{d\flat}{dt}\right)^2 r = -\frac{n^2a^3}{r^2} + \frac{d\Omega}{dr},$$

$$r^2\,\frac{d\flat}{dt} = h,$$

and

$$dh = \frac{d\Omega}{d\flat}\,dt;$$

the remaining equations are

$$d\mathit{\Sigma} - \cos\mathit{\Phi}\,d\Theta = -\sin\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big],$$

$$\begin{aligned}\cos(\flat-\mathit{\Sigma})\sin\mathit{\Phi}\,d\Theta - \sin(\flat-\mathit{\Sigma})\,d\mathit{\Phi} &= \sin(\flat-\mathit{\Sigma})\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big]\\ &\quad - \cos(\flat-\mathit{\Sigma})\cos\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big],\end{aligned}$$

$$\begin{aligned}\sin(\flat-\mathit{\Sigma})\sin\mathit{\Phi}\,d\Theta + \cos(\flat-\mathit{\Sigma})\,d\mathit{\Phi} &= \frac{1}{h}\,\frac{d\Omega}{dy}\,dt\\ &\quad - \cos(\flat-\mathit{\Sigma})\big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big]\\ &\quad - \sin(\flat-\mathit{\Sigma})\cos\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big].\end{aligned}$$

Moreover we have

$$\cos(\flat-\mathit{\Sigma})\,\frac{d\Omega}{dy} = \cot\mathit{\Phi}\,\frac{d\Omega}{dv} - \operatorname{cosec}\mathit{\Phi}\,\frac{d\Omega}{d\Theta},$$

$$\sin(\flat-\mathit{\Sigma})\,\frac{d\Omega}{dy} = \frac{d\Omega}{d\mathit{\Phi}},$$

by means of which the preceding two equations are reduced to

$$\begin{aligned}d\Theta &= -\frac{\operatorname{cosec}\mathit{\Phi}}{h}\,\frac{d\Omega}{d\mathit{\Phi}}\,dt\\ &\quad - \cot\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big],\end{aligned}$$

$$\begin{aligned}d\mathit{\Phi} &= \frac{\cot\mathit{\Phi}}{h}\,\frac{d\Omega}{d\Theta}\,dt - \frac{\operatorname{cosec}\mathit{\Phi}}{h}\,\frac{d\Omega}{dv}\,dt\\ &\quad - \big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big],\end{aligned}$$

and we then have

$$d\mathit{\Sigma} = -\frac{\cot\mathit{\Phi}}{h}\,\frac{d\Omega}{d\mathit{\Phi}}\,dt - \operatorname{cosec}\mathit{\Phi}\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big].$$

[PDF p. 374; printed p. 352.] The entire system consequently is

$$\frac{d}{dt}\left(\frac{dr}{dt}\right) - r\left(\frac{d\flat}{dt}\right)^2 = -\frac{n^2a^3}{r^2} + \frac{d\Omega}{dr},$$

$$r^2\,\frac{d\flat}{dt} = h;$$

and

$$\begin{aligned}
dh &= \frac{d\Omega}{d\flat}\,dt,\\
d\Theta &= -\frac{\operatorname{cosec}\varPhi}{h}\,\frac{d\Omega}{d\varPhi}\,dt - \cot\varPhi\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big],\\
d\varSigma &= -\frac{\cot\varPhi}{h}\,\frac{d\Omega}{d\varPhi}\,dt - \operatorname{cosec}\varPhi\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big],\\
d\varPhi &= \frac{\cot\varPhi}{h}\,\frac{d\Omega}{d\Theta}\,dt - \frac{\operatorname{cosec}\varPhi}{h}\,\frac{d\Omega}{dv}\,dt - \big[\sin(\Theta-\varpi')\sin\phi'\,d\theta' + \cos(\Theta-\varpi')\,d\phi'\big],
\end{aligned}$$

where $\Omega = \Omega(r, \flat, \Theta, \varSigma, \varPhi)$. We may add the before-mentioned equation,

$$d\varSigma - \cos\varPhi\,d\Theta = -\sin\varPhi\big[\cos(\Theta-\varpi')\sin\phi'\,d\theta' - \sin(\Theta-\varpi')\,d\phi'\big].$$

To obtain the formulae of my Memoir before referred to, it would only be necessary to write $h = na^2\sqrt{1-e^2}$, and complete the solution by applying to the first equation the method of the variation of the elements; but I prefer leaving the system in its present form, as it is one which may perhaps be useful.

The formulae have especial reference to the lunar theory, and the orbit x_0y_0 denotes the fixed ecliptic, x_1y_1 the varying ecliptic, and x_2y_2 the instantaneous orbit of the moon. But if, instead of considering with Hansen the instantaneous orbit of the moon, the position of the moon is determined (as in the theories of Clairaut, Plana, and others) by

r, the radius vector,
v, the longitude,
y, the latitude,

where the longitude is measured on the variable ecliptic, then if, in the general expressions for P, Q, R, we first neglect the terms involving $d\theta'$, $d\varpi'$, $d\phi'$, and then write θ, ϖ, ϕ, in the place of Θ, $\varSigma$, $\varPhi$, we find

$$\begin{aligned}
P\,dt = {}& \sin y\,dv\\
&+ \cos y\big[\sin(v-\varpi)\sin\phi\,d\theta + \cos(v-\varpi)\,d\phi\big]\\
&+ \sin y\big[-d\varpi + \cos\phi\,d\theta\big],\\
Q\,dt = {}& dy\\
&+ \big[\cos(v-\varpi)\sin\phi\,d\theta - \sin(v-\varpi)\,d\phi\big],
\end{aligned}$$

[PDF p. 375; printed p. 353.]

$$\begin{aligned} R\,dt = \cos y\,dv \\ &- \sin y\big[\sin(v-\varpi)\sin\phi\,d\theta + \cos(v-\varpi)\,d\phi\big] \\ &+ \cos y\big[-d\varpi + \cos\phi\,d\theta\big], \end{aligned}$$

and with these values the expression for the *Vis viva* function is, as before,

$$T\,dt^2 = \tfrac{1}{2}\big(dr^2 + r^2(Q^2+R^2)\,dt^2\big).$$

The longitude may be measured from a departure-point, and then the expressions for P, Q, R, may be simplified by omitting the terms which contain $d\varpi + \cos\phi\,d\theta$.

Addition.

I have in the course of the memoir used some properties connected with the theory of rotation, but it is proper to give the analytical investigation.

Suppose that the rectangular coordinates (X, Y, Z) are connected with the rectangular coordinates $(X_{\prime}, Y_{\prime}, Z_{\prime})$ by the equations

$$\begin{aligned} X &= \alpha\,X_{\prime} + \beta\,Y_{\prime} + \gamma\,Z_{\prime}, \\ Y &= \alpha' X_{\prime} + \beta' Y_{\prime} + \gamma' Z_{\prime}, \\ Z &= \alpha'' X_{\prime} + \beta'' Y_{\prime} + \gamma'' Z_{\prime}, \end{aligned}$$

then we have

$$\begin{aligned} dX &= \alpha\,dX_{\prime} + \beta\,dY_{\prime} + \gamma\,dZ_{\prime} + X_{\prime}\,d\alpha + Y_{\prime}\,d\beta + Z_{\prime}\,d\gamma, \\ dY &= \alpha'\,dX_{\prime} + \beta'\,dY_{\prime} + \gamma'\,dZ_{\prime} + X_{\prime}\,d\alpha' + Y_{\prime}\,d\beta' + Z_{\prime}\,d\gamma', \\ dZ &= \alpha''\,dX_{\prime} + \beta''\,dY_{\prime} + \gamma''\,dZ_{\prime} + X_{\prime}\,d\alpha'' + Y_{\prime}\,d\beta'' + Z_{\prime}\,d\gamma'', \end{aligned}$$

and then, putting

$$\begin{aligned} p\,dt &= \gamma\,d\beta + \gamma'\,d\beta' + \gamma''\,d\beta'', \\ q\,dt &= \alpha\,d\gamma + \alpha'\,d\gamma' + \alpha''\,d\gamma'', \\ r\,dt &= \beta\,d\alpha + \beta'\,d\alpha' + \beta''\,d\alpha'', \end{aligned}$$

where p, q, r are the rotations round $X_{\prime}$, $Y_{\prime}$, $Z_{\prime}$, from $Y_{\prime}$ to $Z_{\prime}$, $Z_{\prime}$ to $X_{\prime}$, and $X_{\prime}$ to $Y_{\prime}$ respectively, we have

$$\begin{aligned} \alpha\,dX + \alpha'\,dY + \alpha''\,dZ &= dX_{\prime} + (-r\,Y_{\prime} + q\,Z_{\prime})\,dt, \\ \beta\,dX + \beta'\,dY + \beta''\,dZ &= dY_{\prime} + (-p\,Z_{\prime} + r\,X_{\prime})\,dt, \\ \gamma\,dX + \gamma'\,dY + \gamma''\,dZ &= dZ_{\prime} + (-q\,Z_{\prime} + p\,Y_{\prime})\,dt, \end{aligned}$$

where, considering XYZ as fixed axes, the left-hand sides are the displacements in the directions of the moveable axes $X_{\prime}Y_{\prime}Z_{\prime}$ arising from the variations of the coordinates $X_{\prime}Y_{\prime}Z_{\prime}$ and the variation of the position of these axes; and these displace-

[PDF p. 376; printed p. 354.] ments have therefore the values given by the right-hand sides. This is, of course, well known. It may be added that if the axes XYZ are moveable, and dX, dY, dZ denote the displacements in the directions of these axes, the formulae are still true.

Suppose that Ω is a function of X, Y, Z; say $\Omega = \Omega(X, Y, Z)$, we have

$$d\Omega = \frac{d\Omega}{dX}\,dX + \frac{d\Omega}{dY}\,dY + \frac{d\Omega}{dZ}\,dZ,$$

which may be written

$$\begin{aligned}
d\Omega = {} & \Big(\alpha\,\frac{d\Omega}{dX} + \alpha'\,\frac{d\Omega}{dY} + \alpha''\,\frac{d\Omega}{dZ}\Big)\big(dX_{,} + \alpha\,dY + \alpha''\,dZ\big) \\
& + \Big(\beta\,\frac{d\Omega}{dX} + \beta'\,\frac{d\Omega}{dY} + \beta''\,\frac{d\Omega}{dZ}\Big)\big(\beta\,dX + \beta'\,dY + \beta''\,dZ\big) \\
& + \Big(\gamma\,\frac{d\Omega}{dX} + \gamma'\,\frac{d\Omega}{dY} + \gamma''\,\frac{d\Omega}{dZ}\Big)\big(\gamma\,dX + \gamma'\,dY + \gamma''\,dZ\big),
\end{aligned}$$

and the coefficients of transformation α, β, γ, &c. being independent of X, Y, Z and $X_{,}$, $Y_{,}$, $Z_{,}$, the first factors are $\dfrac{d\Omega}{dX_{,}}$, $\dfrac{d\Omega}{dY_{,}}$, $\dfrac{d\Omega}{dZ_{,}}$, and we have

$$\begin{aligned}
d\Omega = {} & \frac{d\Omega}{dX_{,}}\big[dX_{,} + (-r\,Y_{,} + q\,Z_{,})\,dt\big] \\
& + \frac{d\Omega}{dY_{,}}\big[dY_{,} + (-p\,Z_{,} + r\,X_{,})\,dt\big] \\
& + \frac{d\Omega}{dZ_{,}}\big[dZ_{,} + (-q\,X_{,} + p\,Y_{,})\,dt\big],
\end{aligned}$$

which is a theorem assumed in the memoir.

The expressions of P, Q, R in the memoir depend upon the composition of several rotations. If, in Lamé's very convenient notation, we write

	$X_{,}$	$Y_{,}$	$Z_{,}$
X	α	β	γ
Y	α'	β'	γ'
Z	α''	β''	γ''

[PDF p. 377; printed p. 355.] to denote the before-mentioned relation between (X, Y, Z) and $(X_\prime, Y_\prime, Z_\prime)$, then the tables

	$X_\prime$	$Y_\prime$	$Z_\prime$
X	α	β	γ
Y	α'	β'	γ'
Z	α''	β''	γ''

	$X_\prime$	$Y_\prime$	$Z_\prime$
X	A	B	C
Y	A'	B'	C'
Z	A''	B''	C''

lead to the table

	$X_\prime$	$Y_\prime$	$Z_\prime$
X	$(\alpha, \beta, \gamma \succ A, A', A'')$	$(\alpha, \beta, \gamma \succ B, B', B'')$	$(\alpha, \beta, \gamma \succ C, C', C'')$
Y	$(\alpha', \beta', \gamma' \succ A, A', A'')$	$(\alpha', \beta', \gamma' \succ B, B', B'')$	$(\alpha', \beta', \gamma' \succ C, C', C'')$
Z	$(\alpha'', \beta'', \gamma'' \succ A, A', A'')$	$(\alpha'', \beta'', \gamma'' \succ B, B', B'')$	$(\alpha'', \beta'', \gamma'' \succ C, C', C'')$

where for shortness $(\alpha, \beta, \gamma \succ A, A', A'')$, &c., stand for $\alpha A + \beta A' + \gamma A''$, &c. these are of course the ordinary formulae for the composition of transformations from one set of rectangular axes to another. We may say that the coefficients of transformation from (X, Y, Z) to $(X_\prime, Y_\prime, Z_\prime)$ are (α, β, γ). And in dealing with several sets of coordinates, the coefficients of transformation may be distinguished by suffixes. I assume that the coefficients of transformation

from (x_0, y_0, z_0) to (x_1, y_1, z_1) are $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{01}$,

$$(x_1, y_1, z_1) \text{ to } (x_2, y_2, z_2) \quad (\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{12},$$

$$(x_2, y_2, z_2) \text{ to } (x_3, y_3, z_3) \quad (\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{23},$$

where $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{01}$, $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{12}$, $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{23}$ are to be considered as above mentioned. The relations p, q, r may be distinguished by suffixes in like manner, viz., $(p, q, r)_{01}$ will denote the rotations from (x_0, y_0, z_0) to (x_1, y_1, z_1), &c.

There is no difficulty in obtaining that

$$\begin{aligned}
p_{02} &= p_{01} + (\mathfrak{a}, \mathfrak{b}, \mathfrak{c} \succ \mathfrak{a}', \mathfrak{b}', \mathfrak{c}')_{12}\, (p, q, r)_{12}, \\
q_{02} &= q_{01} + (\mathfrak{a}', \mathfrak{b}', \mathfrak{c}' \succ \mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{12}\, (p, q, r)_{12}, \\
r_{02} &= r_{01} + (p, q, r)_{01}\, (p, q, r)_{12},
\end{aligned}$$

[PDF p. 378; printed p. 356.] and then by a repetition of the like transformation

$$p_{03} = p_{02} + (\mathfrak{a}, \mathfrak{b}, \mathfrak{c} \succ \mathfrak{a}', \mathfrak{b}', \mathfrak{c}')_{23}\,(p, q, r)_{23},$$

$$q_{03} = q_{02} + (\mathfrak{a}', \mathfrak{b}', \mathfrak{c}' \succ \mathfrak{a}, \mathfrak{b}, \mathfrak{c})_{23}\,(p, q, r)_{23},$$

$$r_{03} = r_{02} + (p, q, r)_{02}\,(p, q, r)_{23},$$

and in like manner for any number of systems, but for the present purpose this is enough, as p_{03}, q_{03}, r_{03} are in fact the P, Q, R of the memoir. And if θ', ϖ', ϕ', Θ, $\varSigma$, $\varPhi$, υ, y signify as before, then the coefficients of transformation from (x_0, y_0, z_0) to (x_1, y_1, z_1) are given by the table

	x_1	y_1	z_1
x_0	$\cos\varpi'\cos\theta' + \sin\varpi'\sin\theta'\cos\phi'$	$\sin\varpi'\cos\theta' - \cos\varpi'\sin\theta'\cos\phi'$	$\sin\theta'\sin\phi'$
y_0	$\cos\varpi'\sin\theta' - \sin\varpi'\cos\theta'\cos\phi'$	$\sin\varpi'\sin\theta' + \cos\varpi'\cos\theta'\cos\phi'$	$-\cos\theta'\sin\phi'$
z_0	$-\sin\varpi'\sin\phi'$	$\cos\varpi'\sin\phi'$	$\cos\phi'$

and the corresponding rotation coefficients are

$$\begin{aligned} p_{01} &= -\sin\varpi'\sin\phi'\,d\theta' + \cos\varpi'\,d\phi', \\ q_{01} &= \cos\varpi'\sin\phi'\,d\theta' + \sin\varpi'\,d\phi', \\ r_{01} &= -d\varpi' + \cos\phi'\,d\theta'. \end{aligned}$$

The coefficients of transformation from (x_1, y_1, z_1) to (x_2, y_2, z_2) are given by the table

	x_2	y_2	z_2
x_1	$\cos\varSigma\cos\Theta + \sin\varSigma\sin\Theta\cos\varPhi$	$\sin\varSigma\cos\Theta - \cos\varSigma\sin\Theta\cos\varPhi$	$\sin\Theta\sin\varPhi$
y_1	$\cos\varSigma\sin\Theta - \sin\varSigma\cos\Theta\cos\varPhi$	$\sin\varSigma\sin\Theta + \cos\varSigma\cos\Theta\cos\varPhi$	$-\cos\Theta\sin\varPhi$
z_1	$-\sin\varSigma\sin\varPhi$	$\cos\varSigma\sin\varPhi$	$\cos\varPhi$

and the corresponding rotation coefficients are

$$\begin{aligned} p_{12} &= -\sin\varSigma\sin\varPhi\,d\Theta + \cos\varSigma\,d\varPhi, \\ q_{12} &= \cos\varSigma\sin\varPhi\,d\Theta + \sin\varSigma\,d\varPhi, \\ r_{12} &= -d\varSigma + \cos\varPhi\,d\Theta. \end{aligned}$$

[PDF p. 379; printed p. 357.] and the coefficients of transformation from (x_2, y_2, z_2) to (x_3, y_3, z_3) are given by the table

	x_3	y_3	z_3
x_2	$\cos v \cos y$	$-\sin v$	$-\cos v \sin y$
y_2	$\sin v \cos y$	$\cos v$	$-\sin v \sin y$
z_2	$\sin y$	0	$\cos y$

and the corresponding rotation coefficients are given by

$$\begin{aligned} p_{23} &= 0, \\ q_{23} &= -dy, \\ r_{23} &= dv \cdot \sin y. \end{aligned}$$

From the last two tables it is easy to deduce the coefficients of transformation from (x_1, y_1, z_1) to (x_3, y_3, z_3), these are given by the table

	x_3	y_3	z_3
x_1	$\cos y\big[\cos(v-\mathit{\Sigma})\cos\Theta - \sin(v-\mathit{\Sigma})\sin\Theta\cos\mathit{\Phi}\big] + \sin y \sin\Theta \sin\mathit{\Phi}$	$-\sin(v-\mathit{\Sigma})\cos\Theta - \cos(v-\mathit{\Sigma})\sin\Theta\cos\mathit{\Phi}$	$\cdots$
y_1	$\cos y\big[\cos(v-\mathit{\Sigma})\sin\Theta + \sin(v-\mathit{\Sigma})\cos\Theta\cos\mathit{\Phi}\big] - \sin y \cos\Theta \sin\mathit{\Phi}$	$-\sin(v-\mathit{\Sigma})\sin\Theta + \cos(v-\mathit{\Sigma})\cos\Theta\cos\mathit{\Phi}$	$\cdots$
z_1	$\cos y \sin(v-\mathit{\Sigma})\sin\mathit{\Phi} + \sin y \cos\mathit{\Phi}$	$\cos(v-\mathit{\Sigma})\sin\mathit{\Phi}$	$\cdots$

It is not easy to form the values of P, Q, R. We have

$$\begin{aligned} P\,dt = {} & \sin y\, dv \\ & + \Big\{\cos y\big[\cos(v-\mathit{\Sigma})\cos\Theta - \sin(v-\mathit{\Sigma})\sin\Theta\cos\mathit{\Phi}\big] + \sin y \sin\Theta\sin\mathit{\Phi}\Big\}\big(-\sin\mathit{\Sigma}\sin\mathit{\Phi}\, d\Theta + \cos\mathit{\Sigma} \\ & + \Big\{\cos y\big[\cos(v-\mathit{\Sigma})\sin\Theta + \sin(v-\mathit{\Sigma})\cos\Theta\cos\mathit{\Phi}\big] - \sin y \cos\Theta\sin\mathit{\Phi}\Big\}\big(\cos\mathit{\Sigma}\sin\mathit{\Phi}\, d\Theta + \sin\mathit{\Sigma}\, d \\ & + \Big\{\cos y \sin(v-\mathit{\Sigma})\sin\mathit{\Phi} + \sin y\cos\mathit{\Phi}\Big\}\big(-d\mathit{\Sigma} + \cos\mathit{\Phi}\, d\Theta\big) \ldots \end{aligned}$$

[PDF p. 380; printed p. 358.]

$$
\begin{aligned}
&\quad + \cos v \cos y\,(-\sin\varpi' \sin\phi'\, d\theta' + \cos\varpi'\, d\phi') \\
&\quad + \sin v \cos y\,(\cos\varpi' \sin\phi'\, d\theta' + \sin\varpi'\, d\phi') \\
&\quad + \sin y\,(-d\varpi' + \cos\phi'\, d\theta'),
\end{aligned}
$$

$$
\begin{aligned}
Q\,dt &= -dy \\
&\quad + \big[-\sin(v-\varSigma)\cos\Theta - \cos(v-\varSigma)\sin\Theta\cos\varPhi\big]\big(-\sin\varSigma\sin\varPhi\, d\Theta + \cos\varSigma\, d\varPhi\big) \\
&\quad + \big[-\sin(v-\varSigma)\sin\Theta + \cos(v-\varSigma)\cos\Theta\cos\varPhi\big]\big(\cos\varSigma\sin\varPhi\, d\Theta + \sin\varSigma\, d\varPhi\big) \\
&\quad + \big[\cos(v-\varSigma)\sin\varPhi\big]\big(-d\varSigma + \cos\varPhi\, d\Theta\big) \\
&\quad - \sin v\,(-\sin\varpi' \sin\phi'\, d\theta' + \cos\varpi'\, d\phi') \\
&\quad + \cos v\,(\cos\varpi' \sin\phi'\, d\theta' + \sin\varpi'\, d\phi'),
\end{aligned}
$$

$$
\begin{aligned}
R\,dt &= \cos y\, dv \\
&\quad + \big[-\sin(v-\varSigma)\cos\Theta\cos\varPhi + \cos y\sin\Theta\sin\varPhi\big]\, d\Theta + \dots \\
&\quad + \big[\sin y\sin(v-\varSigma)\sin\varPhi\big]\, d\varPhi \\
&\quad + \cos y\big[-d\varSigma + \cos\varPhi\, d\Theta\big] \\
&\quad - \cos v \sin y\,(-\sin\varpi' \sin\phi'\, d\theta' + \cos\varpi'\, d\phi') \\
&\quad - \sin v \sin y\,(\cos\varpi' \sin\phi'\, d\theta' + \sin\varpi'\, d\phi') \\
&\quad + \cos y\,(-d\varpi' + \cos\phi'\, d\theta'),
\end{aligned}
$$

which are at once reducible into the before-mentioned form.

It only remains to prove the expressions for $\dfrac{d\Omega}{dx_3}$, $\dfrac{d\Omega}{dy_3}$, $\dfrac{d\Omega}{dz_3}$ assumed in the memoir. The relations between (x_2, y_2, z_2) and (x_3, y_3, z_3) give

$$
\begin{aligned}
\frac{d\Omega}{dx_3} &= \cos v\cos y\,\frac{d\Omega}{dx_2} + \sin v\cos y\,\frac{d\Omega}{dy_2} + \sin y\,\frac{d\Omega}{dz_2}, \\
\frac{d\Omega}{dy_3} &= -\sin v\,\frac{d\Omega}{dx_2} + \cos v\,\frac{d\Omega}{dy_2}, \\
\frac{d\Omega}{dz_3} &= -\cos v\sin y\,\frac{d\Omega}{dx_2} - \sin v\sin y\,\frac{d\Omega}{dy_2} + \cos y\,\frac{d\Omega}{dz_2};
\end{aligned}
$$

these formulae apply to any point whatever the coordinates of which to the one set of axes are (x_2, y_2, z_2) and to the other set (x_3, y_3, z_3) and in obtaining them the coefficients of transformation v, y are of course considered as independent of either set of coordinates. But the coordinates (x_2, y_2, z_2) of the body are

$$
x_2 = r\cos y\cos v, \qquad y_2 = r\cos y\sin v, \qquad z_2 = r\sin y;
$$

[PDF p. 381; printed p. 359.] and using the foregoing equations in order to introduce into the formulae the differential coefficients $\frac{d\Omega}{dr}, \frac{d\Omega}{dv}, \frac{d\Omega}{dy}$ in the place of $\frac{d\Omega}{dx_2}, \frac{d\Omega}{dy_2}, \frac{d\Omega}{dz_2}$, we find

$$\begin{aligned}
\frac{d\Omega}{dx_2} &= \cos v \cos y \frac{d\Omega}{dr} - \frac{1}{r} \sin v \sec y \frac{d\Omega}{dv} - \frac{1}{r} \cos v \sin y \frac{d\Omega}{dy}, \\
\frac{d\Omega}{dy_2} &= \sin v \cos y \frac{d\Omega}{dr} + \frac{1}{r} \cos v \sec y \frac{d\Omega}{dv} - \frac{1}{r} \sin v \sin y \frac{d\Omega}{dy}, \\
\frac{d\Omega}{dz_2} &= \sin y \frac{d\Omega}{dr} + \frac{1}{r} \cos y \frac{d\Omega}{dy}.
\end{aligned}$$

Substituting these values we have the required formulae

$$\frac{d\Omega}{dx_3} = \frac{d\Omega}{dr}, \qquad \frac{d\Omega}{dy_3} = \frac{1}{r} \sec y \frac{d\Omega}{dv}, \qquad \frac{d\Omega}{dz_3} = \frac{1}{r} \frac{d\Omega}{dy},$$

where on the right-hand side $\Omega = \Omega(r, v, y, \Theta, \mathit{\Sigma}, \mathit{\Phi}, \theta', \varpi', \phi')$.

2, Stone Buildings, W.C. 28th Dec., 1858.

216. Tables of the Developments of Functions in the Theory of Elliptic Motion

[PDF p. 382; printed p. 360.]

[From the *Memoirs of the Royal Astronomical Society*, vol. XXIX. (1861), pp. 191–306. Read January 12, 1859.]

The notation made use of is as follows; viz., r, f, a, e, g denote

r, the radius vector;
f, the true anomaly;
a, the mean distance;
e, the eccentricity;
g, the mean anomaly;

so that

$$\frac{r}{a} = \mathrm{elqr}(e, g),$$

and

$$f = \mathrm{elta}(e, g),$$

are known functions of e, g. Moreover, x denotes the periodic part of the quotient radius $\frac{r}{a}$, and y the equation of the centre, or periodic part of the true anomaly f; so that

$$\frac{r}{a} = 1 + x,$$

$$f = g + y,$$

and (x, y) are therefore also known functions of e, g.

[PDF p. 383; printed p. 361.] Formulae for the development in multiple cosines or sines, up to the terms in e^7, of the functions

$$\left(x^1, x^2, \ldots x^7\right),$$

$$\left(x^a, x^{a+1}, \ldots x^{a+7}\right)\begin{matrix}\cos\\ \sin\end{matrix} jf, \quad j = 1 \text{ to } 7,$$

$$\left(\left(\frac{r}{a}\right)^{+1}, \left(\frac{r}{a}\right)^{-1}, \ldots \left(\frac{r}{a}\right)^{+5}, \left(\frac{r}{a}\right)^{-5}\right), \quad \log\frac{r}{a}, \ldots$$

$$\left(\left(\frac{r}{a}\right)^{+1}, \left(\frac{r}{a}\right)^{-1}, \ldots \left(\frac{r}{a}\right)^{+5}, \left(\frac{r}{a}\right)^{-5}\right)\begin{matrix}\cos\\ \sin\end{matrix} jf, \quad j = 1 \text{ to } 5,$$

where j is an indeterminate symbol, are given by Leverrier in the *Annales de l'Observatoire de Paris*, tom. I. (1855), pp. 346–348. It occurred to me that it would be desirable to form tables such as are contained in the present memoir, of various functions of the forms

$$x^\alpha \begin{matrix}\cos\\ \sin\end{matrix} jf, \quad \left(\frac{r}{a}\right)^\alpha \begin{matrix}\cos\\ \sin\end{matrix} jf,$$

obtainable from Leverrier's formulae by assigning particular integer values to the symbol j. The calculations were performed under my superintendence by Messrs Greedy and Davis; and I have to acknowledge my obligation for a grant to defray the expenses, made to me out of the Donation Fund at the disposal of the Council of the Royal Society.

A cosine series is in general represented in the form

$$\sum [\cos]^i \cos ig,$$

where i extends from $-\infty$ to $+\infty$, and the coefficients $[\cos]^i$ satisfy the condition

$$[\cos]^{-i} = [\cos]^i;$$

and in like manner a sine series is represented in the form

$$\sum [\sin]^i \sin ig,$$

where i extends from $-\infty$ to $+\infty$, and the coefficients $[\sin]^i$ satisfy the condition

$$[\sin]^{-i} = -[\sin]^i, \quad \text{and in particular} \quad [\sin]^0 = 0.$$

Thus $[\cos]^i$ is a general symbol denoting the coefficient of $\cos ig$, in a cosine series considered as involving as well negative as positive multiples of the argument g, or, what is the same thing, $[\cos]^0$ is the constant term and $[\cos]^i$ is the half coefficient of $\cos ig$, when the series is considered as containing only positive multiples of the argument. And in like manner $[\sin]^i$ is a general symbol for the coefficient of $\sin ig$, in a sine series considered as involving as well negative as positive multiples of the argument g, or, what is the same thing, $[\sin]^i$ is the half coefficient of $\sin ig$, when the series is considered as containing only positive multiples of the argument.

[PDF p. 384; printed p. 362.] In the case of a pair of corresponding functions $x^m \cos jf$ and $x^m \sin jf$, or $\left(\frac{r}{a}\right)^m \cos jf$ and $\left(\frac{r}{a}\right)^m \sin jf$, one of them expanded in the form

$$\sum [\cos]^i \cos ig,$$

and the other in the form

$$\sum [\sin]^i \sin ig,$$

the sums and differences of the corresponding coefficients $[\cos]^i$, $[\sin]^i$, are represented by the notation

$$[\cos \pm \sin]^i,$$

and it is clear that we have

$$[\cos + \sin]^{-i} = [\cos - \sin]^i, \qquad [\cos \pm \sin]^0 = [\cos]^0.$$

These coefficients $[\cos \pm \sin]^i$ are for many purposes equally useful with the coefficients $[\cos]^i$, $[\sin]^i$, and they are here tabulated accordingly.

The functions tabulated are as follows:

$$\begin{array}{l}
\left(x^1, x^2, \ldots x^7\right) \quad , \quad [\cos], \\
\left(x^a, x^{a+1}, \ldots x^{a+7}\right)^{\cos}_{\sin} jf,\ j = 1 \text{ to } 7, \quad [\cos], [\sin], [\cos \pm \sin], \\
\left(\left(\frac{r}{a}\right)^{+1}, \left(\frac{r}{a}\right)^{-1}, \ldots \left(\frac{r}{a}\right)^{+5}, \left(\frac{r}{a}\right)^{-5}\right),\ \log \frac{r}{a}, \ldots, \quad [\cos], \\
\left(\left(\frac{r}{a}\right)^{+1}, \left(\frac{r}{a}\right)^{-1}, \ldots \left(\frac{r}{a}\right)^{+5}, \left(\frac{r}{a}\right)^{-5}\right)^{\cos}_{\sin} jf,\ j = 1 \text{ to } 5, \quad [\cos], [\sin], [\cos \pm \sin],
\end{array}$$

all the developments being carried up to e^7, the limit of the formulas from which they are deduced.

The calculations are based upon Leverrier's formulae for

$$\left(x^a, x^{a+1}, \ldots x^{a+7}\right)^{\cos}_{\sin} jf,$$

which are made use of under the following forms, which I call the Datum Tables

$$\left(x^a, x^{a+1}, \ldots x^{a+7}\right)^{\cos}_{\sin} jf.$$

[PDF p. 385; printed p. 363.]

Datum Tables of $(x^a, \ldots x^{a+7})\cos jf$ *and* $\cos jf$ *extracts.*

Cosine Table $\frac{1}{a}$ of	$\left[\frac{1}{a}\right]$ $\frac{e^a}{\lvert 0}$	$\frac{e^{a+1}}{\lvert 2 \rvert 0}$	$\frac{e^{a+2}}{\lvert 4 \rvert 0}$	$\frac{e^{a+3}}{\lvert 6 \rvert 0}$	$\left[\frac{2}{a}\right]$ $\frac{e^a}{\lvert 0}$	$\frac{e^{a+1}}{\lvert 2 \rvert 0}$	$\frac{e^{a+2}}{\lvert 4 \rvert 0}$	
$x^0\cos jf$	$+1$	$-8j^2$	$+28j^4-1280j^2$	$\ldots$	0	$-3j^2$	$+89j^2-110j^4$	$+18$
$x^1\cos jf$		$-2j^2$	$-3j^2+71j^4$	$\ldots$		-1		
$x^2\cos jf$	$+2$	$-2j^2$	$+1900j^2-\ldots$		-1	$+13j^2$	$-263j^4-\ldots$	
$x^3\cos jf$		$+2$	$-19j^2+1090j^2$		-9	$+579j^2-\ldots$	$-1125j^4+\ldots$	
$x^4\cos jf$		$+4$	$-49j^2+1502j^2-1320j^4$	$\ldots$		$+50j^2$		
$x^5\cos jf$			$+144$	$-1980j^2$			$+480$	
$x^6\cos jf$				$+441$				
$x^7\cos jf$				$+1400$				
$x^8\cos jf$				0				
Datum Table $(x^1,\ldots x^8)\cos jf\ldots$								

[*Numerical entries reproduce the printed Datum Tables; columns correspond to coefficients of successive powers of* e^a, e^{a+1}, *etc. in the cosine expansion of* $x^n\cos jf$.]

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Datum Table $(x^a, \dots x^{a+7}) \cos jf$, *continued.*

Cosine Table $\frac{1}{a}$ of	$\left[\frac{1}{a}\right]$ $\frac{e^a}{\lvert 2 \rvert 0}$	$\frac{e^{a+1}}{\lvert 4 \rvert 0}$	$\frac{e^{a+2}}{\lvert 6 \rvert 0}$	$\dots$	$\left[\frac{2}{a}\right]$ $\frac{e^a}{\lvert 2 \rvert 0}$	$\frac{e^{a+1}}{\lvert 4 \rvert 0}$	$\frac{e^{a+2}}{\lvert 6 \rvert 0}$
$x^0 \cos jf$	$+4j^2$	$-64j^2 + 296j^4$	$-2104j^2$	$\dots$	$+3j^2$	$-1100j^2 + \dots$	$+30163j^2 + \dots$
$x^1 \cos jf$		$+64j^2 - 1981j^2$	$-12247j^2 + \dots$	$\dots$		$+1742j^2 - \dots$	$\dots$
$x^2 \cos jf$	$+1$	$-2j^2$	$+440j^2$			$+613$	$-90209j^2$
$x^3 \cos jf$		$+5$	$-67j^2 + 1908j^2$		-12	$+845j^2 - \dots$	$+45980j^2 + \dots$
$x^4 \cos jf$			$+4$	$-1880j^2$			$+480j^2 - \dots$
$x^5 \cos jf$				-6	-12		$+540$
$x^6 \cos jf$				-96			$+540$
$x^7 \cos jf$				-3500			
$x^8 \cos jf$				0			
Datum Table $(x^1, \dots x^8) \cos jf \dots$							

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Datum Table $(x^a, \ldots x^{a+7}) \cos jf$, *concluded.*

Cosine Table $\frac{1}{a}$ of	$\left[\frac{1}{a}\right]$		$\left[\frac{2}{a}\right]$		$\left[\frac{3}{a}\right]$		
	$\frac{e^a}{\lvert 4\rvert 0}$	$\frac{e^{a+1}}{\lvert 6\rvert 0}$	$\frac{e^a}{\lvert 4\rvert 0}$	$\frac{e^{a+1}}{\lvert 6\rvert 0}$	$\frac{e^a}{\lvert 4\rvert 0}$	$\frac{e^{a+1}}{\lvert 6\rvert 0}$	
$x^0 \cos jf$	$+16j^2$	$-384j^2$	$+4096^a - 1180j^4$	$-50450j^2 - \ldots$	$+643j^4 + \ldots$	$+1883j^4 - \ldots$	$-$
$x^1 \cos jf$		-84	$-3137\ldots$	$-13125j^2 - \ldots$	$-1273j^2 + \ldots$	$+10193j^2 + \ldots$	$+$
$x^2 \cos jf$	$+48$	$-7008j^2$	$+1480j^2$	$-3360j^2 + \ldots$	$-127j^2 + \ldots$	$-1100j^2 + \ldots$	$+1$
$x^3 \cos jf$		$+96$	$-15080j^2 + \ldots$	$+1450j^2 + \ldots$	$+540j^2 + \ldots$	$-1422j^2 + \ldots$	$+$
$x^4 \cos jf$			$+75$	$-1242j^2 + \ldots$	$-540j^2 + \ldots$	$+1450j^2 + \ldots$	$-$
$x^5 \cos jf$				$+\ldots$	-100	$-1320j^2 + \ldots$	$-$
$x^6 \cos jf$					-7000	$+3800$	
$x^7 \cos jf$					-4200	$+700$	
$x^8 \cos jf$							
Datum Table $(x^1, \ldots x^8) \cos jf \ldots$							

[End of pp. 351–387 chunk; paper 216 continues in next chunk (tables of developments to e^7*).]*